From Human Markets to Machine Markets

Certify, Don't Identify: Safety and Identification in Markets with Adaptive Trading Agents

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What this document is. The intended final form of a three-paper thesis. Papers 1 and 2 are finished and appear as Chapters 2 and 3; Paper 3 (Chapter 4) is new, as are the introduction and conclusion. Every result in the new chapters carries a tag: PROVED IN THIS THESIS (with its proof), PROVED IN CH. 2/3 (proved in a paper chapter), TARGET: STATEMENT FIXED, PROOF TO BE WRITTEN (the statement the thesis will prove, with a proof plan), CONJECTURE, or PENDING DATA (an empirical test specified but not yet run). No number in the new chapters is invented; Appendix A lists the status of every main result.

Abstract

Electronic markets are feedback systems in which many adaptive trading algorithms respond to prices that their own trades move. Regulators often respond by asking to know which algorithms are trading. This thesis asks whether that knowledge is needed for safety, and answers that it generally is not. In the closed-loop markets studied here, prices reveal the agents only through a merged measure of their memories and weights; separating two agents whose memory rates differ by δ costs trading capital of order δ^{-4}, separating K unknown agents costs a number of experiments of order $(\Delta\eta)^{-(4K-2)}$, and with general memory the accuracy of any method improves only as $1/\log N$. Safety, by contrast, depends on the agents only through one number, the summed price response, which is certified with $\sigma^2\epsilon^{-2}\log(1/\epsilon)$ experiments whatever their number and spacing, provided memory is bounded, and cannot be certified at all without such a bound; the logarithm cannot be removed. A pre-trade guard built on the certificate is the largest safe admission rule and is minimax-optimal in lost capacity up to a logarithmic factor. A third paper organises these results under one framework in which every question an observer can ask is a functional of the agents' merged measure. For linear questions it determines the cost of certification exactly, up to a constant factor in accuracy, as the value of a polynomial approximation problem with penalized coefficients: a question is answerable at the parametric rate exactly when its kernel has square-summable coefficients, and the safety question sits on that boundary and pays exactly one logarithm. It extends the safety results to sector-bounded nonlinear agents, to several assets and to agents that keep learning, for which a certificate has a computable lifetime. In data, trader-class disclosures from Binance and the CFTC restrict a persistence that prices leave unrestricted, and in pre-registered tests the summed-response statistic forecasts adverse price excursions in thin crypto books and, more weakly, in thin US stocks; tests of a leverage or liquidation mechanism fail. Ten further registered tests, each run once, replicate the disclosure result on new days and on new symbols, and on three sets of markets never used before decide the safety question where the agents are not identified: with 430 days per market the margin is decided in 23 of 25 and in 24 of 26 markets, all safe, while the memories are identified in at most one. The decision rests mainly on price reversion; the summed response itself is no more precise than the separation of memories; a guard estimated from public data does not reduce realized tail losses; the estimated margin does not forecast the next day's largest adverse move or extreme moves, and does not rank damage on market-wide

stress days, but on 34 new markets it forecasts next-day variance beyond the Hawkes branching ratio, as registered, and on 36 more the size of the estimated closed-loop response scales with next-day losses at about half the predicted rate; the class persistence that disclosure reveals is, once corrected for bias, close to a random walk within the day; and an overshoot seen in a pilot after scheduled macro releases does not replicate on earlier releases. Eleven of their twenty-three primary hypotheses pass. The evidence supports the thesis in a narrower form than the theory: prices do not identify the agents, and with enough data they still answer the safety question for markets far from the boundary; nearer the boundary, disclosure or an outside shock that acts as an experiment is needed.

Statement of co-authorship

Chapter 2 reproduces the paper *Auditable Memory and Financial Identification Limits in Electronic Limit Order Books*, joint with Prashant Niroula. Chapter 3 reproduces the paper *From Human Markets to Machine Markets: Transient Amplification, an Exact Depth Condition, and a Maximal Pre-Trade Guard*, joint with Suboh Alkhushayni, Yuhao Chen, Mansi Bhavsar and Prashant Niroula. *[To be completed with the co-authors: the candidate's contribution to each chapter, and permission to include the papers.]* Chapter 4 (Paper 3), the introduction and the conclusion are the candidate's own work.

Contents

Abstract i

Statement of co-authorship iii

1 Introduction 1

1.1 The question . . . 1
1.2 The answer in one paragraph . . . 1
1.3 Three papers . . . 2
1.4 Status of the results . . . 3
1.5 Relation to existing work . . . 3
1.6 Reading guide . . . 4

2 Auditable memory and identification limits 6

2.1 Introduction . . . 7
2.2 Model and microfoundations . . . 12
2.3 The identification boundary . . . 16
2.4 Probing resources, memory resolution and money . . . 17
2.5 Account-level capital . . . 23
2.6 Unknown mixtures and passive data . . . 30
2.7 Robustness to nonlinear and adaptive trading rules . . . 40
2.8 What disclosure buys . . . 43
2.9 A spectral blind spot . . . 52
2.10 Empirical feedback and its measurement . . . 53
2.11 Implications for supervision and market design . . . 57
2.12 Conclusion and an open problem . . . 58
2.13 Appendix: Proofs . . . 62
2.14 Appendix: Notation . . . 69
2.15 Appendix: Further results . . . 69
2.16 Appendix: Reproducibility: source of every number . . . 72

3 Safety without identification: transient amplification and a maximal pre-trade guard **75**
3.1 Introduction . . . 75
3.2 The closed-loop market . . . 77
3.3 Spectral stability is not solvency . . . 80
3.4 When does a book amplify? An exact condition . . . 80
3.5 Identification and financial safety . . . 83
3.6 An identification-free pre-trade guard . . . 87
3.7 The value of disclosure for safe trading . . . 91
3.8 Who provides reversion? An equilibrium . . . 93
3.9 Evidence . . . 95
3.10 Tests of the information-safety results . . . 99
3.11 Policy implications . . . 100
3.12 Conclusion . . . 100
3.13 Appendix: Proofs . . . 102

4 Certify, don't identify: a theory of safety without identification **106**
4.1 Markets with adaptive agents . . . 107
4.2 Questions are functionals of the merged measure . . . 107
4.3 Observation contracts and the cost of a question . . . 108
4.4 The certification–identification dichotomy . . . 108
4.5 Questions that are not linear . . . 115
4.6 Horizon-limited amplification and natural experiments . . . 116
4.7 Passive data: certifying safety costs the inverse margin . . . 119
4.8 Uniform certification and the price of memory . . . 125
4.9 Contrarian agents . . . 130
4.10 Worked examples: how many experiments . . . 131
4.11 Disclosure changes the contract, not the question . . . 132
4.12 A robust safety bound: nonlinear and changing agents . . . 132
4.13 Certificates expire: learning agents . . . 135
4.14 Markets of trading algorithms that learn . . . 136
4.15 Agents who respond to the guard . . . 137
4.16 Relation to the statistics of functionals . . . 138
4.17 What the framework does not cover . . . 139

5 AI agents in the loop: supervising the market, not the models **140**
5.1 What the theory already says about AI agents . . . 140
5.2 Monoculture . . . 141
5.3 A two-number disclosure and a loop-level trigger . . . 142

5.4 Test B1: learning agents and a vendor model 143
5.5 What this chapter does not show 144

6 Conclusion and open problems **145**
6.1 The registered tests 146
6.2 Open problems 158

A Status of results and notation across chapters **161**
A.1 Status register 161
A.2 Notation across chapters 165

Bibliography for Chapters 1, 4 and 5 **166**

Chapter 1

Introduction

1.1 The question

Markets are moving from human traders to machines. Where prices were once set by people reading news and order books, they are now set mostly by algorithms that read prices, trade on them in milliseconds, and increasingly learn from the market while they trade in it. This transition changes what can be known about a market and what it takes to keep it safe, and it is the subject of this thesis.

Electronic markets are feedback systems. Trading algorithms watch prices, trade on what they see, and move the prices they watch. The algorithms are private, heterogeneous and changing, and a growing share of them learn from the market while they trade in it. Whoever is responsible for safety in such a market, a regulator, an exchange or a dealer, faces agents it cannot see.

The usual response is to ask to see inside. Under MiFID II, firms must flag algorithmic orders and identify the algorithm behind each decision; the US Consolidated Audit Trail records who stands behind every order. Behind these requirements sits an implicit claim: that a market cannot be kept safe without knowing who is trading in it.

This thesis tests that claim. Its central question is:

> *When can a market with unknown adaptive agents be certified safe without identifying the agents, and what does each kind of knowledge cost?*

1.2 The answer in one paragraph

The information that certifies safety is far smaller and far cheaper than the information that identifies the agents. Prices reveal the agents only through a merged measure of their memory rates and weights (Chapter 2). Recovering the agents from that measure is a moment problem whose cost explodes as agents become similar: the trading capital needed to separate two

memory rates δ apart grows as δ^{-4}, the number of experiments needed to separate K agents as $(\Delta\eta)^{-(4K-2)}$, and with general memory no method does better than accuracy $1/\log N$. Safety needs much less. The worst-case transient loss of a position depends on the agents only through one number, the summed price response, and that number is certified with $\sigma^2\epsilon^{-2}\log(1/\epsilon)$ experiments whatever the number and spacing of the agents, provided their memory is bounded (Chapter 3). Chapter 4 explains why, and shows that this count cannot be improved: every question an observer can ask is a functional of the merged measure, and for linear questions the cost is fixed exactly by a modulus of continuity, which is the value of a polynomial approximation problem. A question is answerable at the parametric rate exactly when its kernel has square-summable power-series coefficients; "is the market safe?" sits on that boundary and pays exactly one logarithm; "who is trading?" needs the fine structure of the measure and is expensive; questions that are singular at infinite memory cannot be answered from finite data at all. The safety half survives when agents are nonlinear, trade several assets and keep learning, at the price of certificates that expire. In data the identification half holds and replicates on new days. The safety half holds in theory and, in a limited form, in data: in registered tests on two sets of new markets, public data decide the safety question where they do not identify the agents, once the sample is long enough, but the decision rests mainly on price reversion, the summed response itself is no more precise than the separation of memories, and a guard estimated from public data is no better than a volatility limit. What makes certification attainable in practice is disclosure, or an outside shock that acts as a reset experiment (Chapter 6).

1.3 Three papers

The thesis consists of three papers, framed by this introduction and a conclusion.

Paper 1: identification limits (Chapter 2). In a linear cohort model whose reduced form arises from Kalman filtering and regularized online learning, prices identify only the merged rate–weight measure. Separating a known pair costs executed quantity of order δ^{-2} and capital of order δ^{-4}, with explicit constants and a finite-horizon certificate; K unknown cohorts cost $(\Delta\eta)^{-(4K-2)}$ experiments, matched by Prony's method; general completely monotone memory gives the logarithmic rate; passive data cost the same polynomial order. Class-tagged disclosure removes the barrier and is worth a factor $(\Delta\eta)^{-(4K-4)}$ in samples. Trader-class positions from Binance and the CFTC pin down memories that prices leave unrestricted.

Paper 2: safety without identification (Chapter 3). In a limit order book with adaptive cohorts, cumulative transient risk has the closed form $S = \tilde{\kappa}/(m - q\tilde{\kappa})$, a book overshoots if and only if depletion outweighs impact-weighted reversion, and a pre-trade guard on the summed impulse response is the largest safe admission rule. S is certified with $O(\sigma^2\epsilon^{-2}\log(1/\epsilon))$ experiments; a finite window certifies it only under a memory bound; the certified guard is

minimax-optimal in lost capacity up to a logarithmic factor; disclosure has a computable value for safe capacity; and reversion is under-supplied in equilibrium. In a pre-registered empirical program the statistic forecasts adverse excursions in thin books, while mechanism tests fail.

Paper 3: a theory of certification without identification (Chapter 4). Questions are treated as functionals of the merged measure and observation contracts as noise models on its moments. For linear questions the paper determines the cost of certification up to a constant factor in accuracy by a modulus of continuity and its dual, a polynomial approximation problem with penalized coefficients; shows that a question is parametric exactly when its kernel has square-summable coefficients; and shows that questions singular at infinite memory cannot be certified at all (Theorem 4.4.1 and Proposition 4.4.2, PROVED IN THIS THESIS). For the safety question the cost is exactly $\sigma^2\epsilon^{-2}\log(1/\epsilon)$, with explicit constants: the logarithm in Paper 2's certificate cannot be removed (Theorem 4.4.3, PROVED IN THIS THESIS). Questions that depend on finitely many responses are parametric, and the peak response, the question an operator might ask first, is known without data and carries no information about loss (Corollary 4.5.2, PROVED IN THIS THESIS). The rates of Papers 1 and 2 appear as cases of one picture. A robust safety bound then shows that the summed-gain guard stays safe for nonlinear agents whose rules, memories and weights change over time, given only a bound on their total sensitivity (Theorem 4.12.2, PROVED IN THIS THESIS), with a sharp version for several assets with positive cross-impact (Proposition 4.12.5, PROVED IN THIS THESIS), and a corollary gives the lifetime of a certificate when agents keep learning (Corollary 4.13.1, PROVED IN THIS THESIS); in a registered simulation with profit-driven learners the lifetime is close to tight when trend-following pays. Whether the guard survives agents who adapt to it is open: a discontinuous guard can leave the agents' game without an equilibrium.

1.4 Status of the results

1.5 Relation to existing work

The thesis connects four literatures that rarely meet. *System identification* (Ljung, 1999) asks how well a system's parameters can be recovered from input–output data; Paper 1 shows that in closed-loop markets the relevant parameters are zeros of the transfer function and are recoverable only at polynomial or logarithmic rates. *Mixture and moment problems* in statistics give the closest mathematical relatives: the optimal rate for estimating a k-atom mixing distribution is $n^{-1/(4k-2)}$ (Heinrich and Kahn, 2018; Wu and Yang, 2020), super-resolution has the same Vandermonde geometry (Candès and Fernandez-Granda, 2014; Moitra, 2015), and the estimation of smooth *functionals* of a distribution can be far easier than estimating the distribution itself (Lepski et al., 1999; Cai and Low, 2011). The theory of Paper 3 is the counterpart of that last observation for safety questions in feedback markets. *Robust and positive control* (Megretski

and Rantzer, 1997; Farina and Rinaldi, 2000; Rantzer, 2015) supplies the comparison arguments of Paper 3 and the ℓ_1 gain that bounds worst-case loss. *Partial identification* in econometrics (Manski, 2003) provides the stance: when the structure is not identified, ask which decisions are nonetheless determined. The closest stance in economics is the *sufficient-statistics* approach (Chetty, 2009), in which a welfare question depends on a few reduced-form quantities rather than on the full structure. The summed response plays that role here, with two differences: it is derived from the closed-loop dynamics rather than posited, and its cost of certification is computed exactly, including the long-memory case in which it cannot be certified at all. *Robust control* in economics (Hansen and Sargent, 2008) asks for decisions that survive model uncertainty; the guard is such a decision, certified from data rather than from a specified set of models. In finance, the closest work is on transient impact and resilience, liquidity spirals and automated-market crashes, reviewed in Chapters 2 and 3 (Brunnermeier and Pedersen, 2009; Kirilenko et al., 2017).

1.6 Reading guide

Chapters 2 and 3 are self-contained papers and keep their own notation, appendices and references. Chapter 4 can be read after either of them: it states the common framework, and each of its results points to the chapter where the special case is proved. Chapter 6 reports the registered tests of the thesis's empirical claims, including those that failed, and the open problems. Appendix A lists the status of every main result and maps the notation across chapters.

Table 1.1: Main results of the thesis and their status

Result	Where	Status
Prices identify only the merged measure	Thm 2.3.1	proved
Capital of order δ^{-4}; $(\Delta\eta)^{-(4K-2)}$ experiments; $1/\log N$	Ch. 2, §2.4–2.6	proved
Safety depends only on S; guard is maximal	Lemma 3.6.1	proved
S certified at the parametric rate; impossible without a memory bound	Thms 3.5.3, 3.5.4	proved
Cost of certification for any linear question: upper bound, truncation, impossibility	Thm 4.4.1(i)–(iii)	proved
Exact cost of any linear question (modulus and its dual)	Thm 4.4.1(iv)	proved
Parametric exactly for square-summable kernels	Prop. 4.4.2	proved
Safety costs exactly $\sigma^2\epsilon^{-2}\log(1/\epsilon)$	Thm 4.4.3	proved
Finitely dependent nonlinear questions; the peak	Prop. 4.5.1, Cor. 4.5.2	proved
Guard safe for nonlinear, time-varying agents	Thm 4.12.2	proved
Several assets (induced-norm and sharp bounds)	Props 4.12.4, 4.12.5	proved; comparison step in Lean
Certificate lifetime under learning	Cor. 4.13.1	proved
Guard under strategic adaptation	Remark 4.15.2	existence refuted
Learning-agent simulation	§4.14	simulation, registered
Horizon-limited amplification certified without a memory bound	Prop. 4.6.1	proved
Natural experiments: catch-up area = inverse stability margin	Prop. 4.6.2	proved
An overshoot certifies contrarian feedback	Prop. 4.6.3	proved; (i)–(iii) in Lean
Passive certification costs the inverse margin	Thms. 4.7.1, 4.7.2	proved (one root; several cohorts with finite memory)
No passive certificate without a memory bound; uniform cost $g(1)/M$	Thms. 4.8.1, 4.8.2	proved (the second given cited uniform limit theory)
Passive certificate of worst-case loss with contrarian agents	Thm. 4.8.4	proved
The tests' pooled price-and-flow estimator attains the cost $1/M$	Thm. 4.8.3	proved (fixed economies)
Monoculture: a shared model lowers the margin linearly in its adoption share	Prop. 5.2.1	proved
Simulated AI-agent market: early warning, value of a memory bound, learning accelerates monoculture	Ch. 5, Test B1	registered simulation; all four primaries passed
Contrarian agents: linear questions and the safety statistic	Prop. 4.9.1	proved; finite parts in Lean
The closed loop prices the margin into certification	Cor. 4.7.5	proved
Certifying a drifting margin: a floor of order $(s^2d)^{1/3}$	Prop. 4.7.4	proved; bounds in Lean
Ten registered tests (E1, D1, G1, V1, C2, O1, C3, F1, S1, R1)	§6.1	run: 11 of 23 primaries pass
Real agents on Hyperliquid (H1)	§6.1.4	prospective; opens 10 Dec 2026
Prospective test (E2)	§6.1.3	runs after 2 Feb 2027

Chapter 2

Auditable memory and identification limits

This chapter, the first paper of the thesis, reproduces the paper *Auditable Memory and Financial Identification Limits in Electronic Limit Order Books*, joint with Prashant Niroula. It establishes the identification half of the thesis: what prices reveal about the agents of a closed-loop market, and what it costs to learn more. Its notation is its own; Appendix A maps it to the notation of Chapter 4.

Summary. When trading algorithms condition on recent prices, price formation is a closed-loop system. In a linear cohort model with exponential memory, prices identify only a merged rate–weight measure. Separating two memory rates differing by δ requires executed quantity $\Theta(\delta^{-2})$ and trading capital $\Theta(\delta^{-4})$, with matching attainability, adaptive bounds and a finite-horizon certificate ($[61.41, 64.48]$ par units at the BTCUSDT calibration). Separating K unknown cohorts requires $\Theta((\Delta\eta)^{-(4K-2)})$ experiments; the merged memory distribution is recovered at the matching rate $N^{-1/(4K-2)}$ uniformly, even when rates collide, while individual weights are not recoverable there at all. General completely monotone memory reduces minimax accuracy to $\Theta(1/\log N)$. Passive data incur the same polynomial cost; online learners trained in the closed loop obey the same δ^{-4} scaling when their feedback weights match (slopes -4.01 to -3.97), whereas hard thresholds conceal cohorts. Class-tagged disclosure reduces sample complexity by $\Theta((\Delta\eta)^{-(4K-4)})$. Empirically, trader-class positions restrict persistence rates that prices leave unbounded: in a registered test on 30 Binance perpetuals never used before, with the price-only kernel matched to the class horizon, their intervals were narrower in 29 of 29 evaluable symbols (median factor 7.3), and across eight CFTC-regulated futures markets 9.1 to 71.4 times narrower; corrected for within-day bias, the Binance class persistence is close to a random walk, so these data restrict persistence without showing learning memory. Prices and flow pin down even the merged response in only 5 of 30 thin markets.

Reproducibility. Every number in the paper is produced by a script or notebook cell that writes a result file from public data or from a closed form; Appendix 2.16 maps each table, figure and number to its source. Every simulation number was re-run for this version. The specifications of the October 2026 follow-up tests (Sections 2.7 and 2.8.3) were committed to the repository before their test data were downloaded; because each specification reached the public repository in the same push as its results, that order is documented by the commit history rather than by an independent timestamp.

2.1 Introduction

Regulators increasingly require firms to say which algorithm sent each order. Under MiFID II, EU investment firms must flag algorithmic orders and identify the algorithm responsible for each investment and execution decision in their transaction reports (ESMA, 2016); the US Consolidated Audit Trail records who stands behind every order (SEC, 2012). These regimes are costly to build and to comply with, and they rest on an implicit claim: that a supervisor could not learn the same thing from public prices and order flow. This chapter tests that claim, and prices the answer.

The setting is a closed-loop market. Algorithms update on recent prices, their orders move prices, and prices feed back into the algorithms. We model K cohorts of algorithms that track the price with exponential memory, the reduced form of steady-state Kalman filtering and of regularized online gradient descent (Lemma 2.2.1), and ask what an outside observer can learn about them from market data. The answer is sharp. Prices reveal only a merged summary of the algorithms. Separating two algorithms whose memories differ by δ costs trading capital of order δ^{-4}, and separating K unknown algorithms costs a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$; with general memory, accuracy improves only as $1/\log N$, so halving the error squares the number of experiments. Class-tagged disclosure removes the barrier, and its value can be stated in samples: a factor of order $(\Delta\eta)^{-(4K-4)}$. In data, Binance's published long/short ratios by trader class, a coarse form of class tagging, pin down memories that prices and flow leave unrestricted in each of 30 futures markets, and the CFTC's mandated trader-class reports do the same in eight regulated U.S. futures markets.

1. An identification gap. Prices identify only the merged rate–weight measure $M = \sum_i q_i \eta_i \, \delta_{1-\eta_i}$ of the algorithms (Theorem 2.3.1). Cohorts that share a memory rate are observationally equivalent, whatever their identities or wealth. Nearby rates can be separated, but at a cost that rises quickly as they approach each other.

2. A resource gap. For a known symmetric pair separated by δ, price-only identification with deterministic probes requires executed quantity of order δ^{-2}, and a single pulse attains this order (Theorem 2.4.1). Because a probe moves the price against the prober, the quantity

order becomes a capital order δ^{-4}, whether capital is supplied by a counterparty under a margin contract (Theorem 2.4.3) or by the probe trader itself (Lemma 2.4.5). We then compute what an account actually needs. A spectral argument gives a lower bound on the trading loss of any flat round trip (Theorem 2.5.2); Wald's identity extends it to adaptive, sequential probing (Theorem 2.5.3); and a Rayleigh–Hölder dual bounds the finite-horizon optimum (Theorem 2.5.4). At the calibrated parameters the bracket is $[61.41, 64.48]$ par units. For K unknown cohorts the number of independent experiments is exactly of order $(\Delta\eta)^{-(4K-2)}$: a two-point lower bound (Theorem 2.6.4) is matched by Prony's method for every K (Theorem 2.6.5). The match needs separated rates for the weights, but not for the merged memory distribution, which a minimum-distance estimator recovers at the same rate uniformly over near-colliding economies (Theorem 2.6.6). For general completely monotone memory the minimax accuracy is of exact order $1/\log N$: a lower bound built from moment-matched cohort pairs (Lemma 2.6.7) is matched by a moment estimator solved as a linear program (Theorem 2.6.8), so the exponential-memory model is the optimistic case.

3. A measurement gap. Passive data do not escape the barrier. The closed-loop transfer function is minimum phase, so the price spectrum identifies the merged measure (Theorem 2.6.9), and natural excitation can separate a pair with zero controlled quantity (Proposition 2.15.2). But the information rate falls at the same polynomial order as for active probing, $(\Delta\eta)^{4K-2}$ for every number of cohorts K (Theorem 2.6.10), and in-loop price noise leaves the capital order of active probing intact up to horizons of that same order (Theorem 2.5.2(iv)). Only richer disclosure removes the dependence on rate spacing: tagged flow (Proposition 2.8.1) or audited equity (Proposition 2.4.4). Tagging reduces the sample requirement by a factor of order $(\Delta\eta)^{-(4K-4)}$ for rates (Theorem 2.8.2). On Binance, published long/short ratios by trader class restrict the persistence of class positioning, which prices and flow leave essentially unrestricted: at a matched eight-hour horizon the tagged intervals are narrower in all 30 futures markets studied, and in a registered test on 30 markets never used before in 29 of 29 (Section 2.8.3). Corrected for its within-day bias that persistence is close to a random walk, so the data show what prices cannot bound, not that the persistence is a learning memory. Calibrating the theorem to the uncorrected BTCUSDT class rates makes them worth about 333 times as much data as prices alone, an illustration rather than an estimate.

4. What feedback indices measure. Standard reduced-form feedback indices are dominated by filtering mechanics and clock alignment; a placebo pipeline reproduces the mechanical persistence exactly; measured feedback rises with market depth across 30 perpetuals; and two fee events that multiply trading volume by 2.6 and divide it by 4.9 leave measured feedback unchanged (Section 2.10).

5. A spectral blind spot. Even without identification, could a supervisor monitor risk through eigenvalues or volatility? No. Markets with identical eigenvalues, one-step volatility and stationary price variance can have arbitrarily large worst-case transient loss (Theorem 2.9.1). This motivates the open question with which the paper ends: what can be measured and controlled without identifying the algorithms? Chapter 3 takes it up.

What we do not claim. The identification results hold for the stated linear model and observation contracts. They are not universal limits on surveillance with participant-level records, which can be far more informative (Propositions 2.4.4 and 2.8.1). The matching upper bound holds on a class with separated rescaled rates and weights bounded away from zero. Capital figures depend on the observation-noise calibration, which we estimate rather than assume. The empirical results are reduced-form diagnostics; they do not identify structural learning parameters. Our pre-registered real-data test of the identification boundary failed its criterion on one of its three parts: on 42 days of data the merged response was precisely estimated in only 5 of 30 markets (Section 2.8.3). Binance's class ratios are positions, not tagged order flow, so they are the closest public analogue to the disclosure of Proposition 2.8.1, not an instance of it.

2.1.1 Related literature

Price formation and impact. Our price equation is a discrete linear impact model in the tradition of Kyle (1985), Glosten and Milgrom (1985) and Hasbrouck (1991), in which order flow moves prices and prices revert toward value. Transient impact and the cost of trading follow Almgren and Chriss (2001), Obizhaeva and Wang (2013), Gatheral (2010) and Bouchaud et al. (2004); Alfonsi et al. (2016) show that absence of price manipulation requires positive-definite decay kernels, which is the property Lemma 2.4.2 delivers in our cohort economy. Bouchaud et al. (2018) document order splitting and long memory in order flow, which is the main confounder in Section 2.10. Relative to this literature we treat the impact kernel not as a primitive but as the closed-loop output of adaptive traders, and ask what it reveals about them.

Algorithmic trading and feedback. Algorithms that react to recent prices can amplify or dampen moves. Brunnermeier and Pedersen (2009) and Shleifer and Vishny (1992) study liquidity spirals and fire sales, and Kirilenko et al. (2017) document the behaviour of automated intermediaries in the 2010 Flash Crash. Sornette et al. (2023) show that non-normal interactions can generate transient bubbles. Our contribution is on the measurement side: we show what a closed-loop observer can and cannot learn about the feedback, and that standard reduced-form indices are dominated by filtering and timing.

Learning algorithms and what can be inferred about them. A growing literature studies how learning algorithms trade and set prices: reinforcement-learning market makers and

informed traders can sustain wide spreads or collusive outcomes (Colliard et al., 2022; Dou et al., 2025), and empirical game-theoretic analysis treats the choice among simulated trading strategies as a game (Wellman, 2006). Empirically, algorithmic trading has been found to improve liquidity and price discovery on average (Hendershott et al., 2011; Brogaard et al., 2014). These studies simulate the algorithms or observe them through regulatory data; we ask what an outside observer can infer about them from market data alone, and answer in samples and in capital. Our stance is that of partial identification (Manski, 2003): when the structure is not identified, characterize what the data do determine, here the merged measure and, through it, the intensity-weighted distribution of the desks' information quality (Proposition 2.4.6).

Closest work. Three literatures are closest. Super-resolution and moment-based mixture estimation give rates for recovering clustered point sources and mixing measures from noisy moments (Heinrich and Kahn, 2018; Batenkov et al., 2021; Liu and Ammari, 2022; Wu and Yang, 2020; Kong and Valiant, 2017). Our K-cohort rate is, in that language, the square of the amplitude condition number $(\Delta\eta)^{-(2K-1)}$ for K clustered nodes (Heinrich and Kahn, 2018; Wu and Yang, 2020), and we do not claim the exponent as new. What changes when the mixture sits inside a closed-loop financial market is fourfold (Table 2.1): first, in mixture estimation and open-loop system identification the unknown rates are locations or poles of the observed law, whereas in a closed-loop market the cohorts feed back into the price, making the memory rates transmission *zeros* of the observable transfer function $H(z)$ that must first be extracted from the loop (Theorem 2.3.1); second, passive price observations incur the exact same $(\Delta\eta)^{-(4K-2)}$ sample-complexity barrier as active probing because $H(z)$ is minimum phase and the Whittle likelihood inherits the moment gap (Theorems 2.6.9 and 2.6.10); third, in a market statistical sample size translates into executed quantity $\Theta(\delta^{-2})$ and round-trip trading loss or margin capital $\Theta(\delta^{-4})$ via the self-impact Toeplitz quadratic form $\frac{1}{2}v^\top T_h v$ (Theorems 2.4.1–2.5.4); and fourth, class-tagged disclosure breaks the feedback loop cohort by cohort and cuts the sample requirement by $(\Delta\eta)^{-(4K-4)}$ (Proposition 2.8.1 and Theorem 2.8.2). Beyond the K-cohort case, we prove matching lower and upper bounds of exact logarithmic order $\Theta(1/\log N)$ for general completely monotone memory (Theorem 2.6.8) and price the resulting disclosure savings against audit-trail budgets.

Statistical identification. The identification results use two-point testing arguments (Le Cam, 1986; Tsybakov, 2009). The moment-matching construction of Section 2.6 is the dynamical analogue of the slow rates in finite-mixture estimation (Manole and Ho, 2022) and of super-resolution of near-colliding sources (Liu and Ammari, 2022; Batenkov et al., 2021); the general-memory result combines best-approximation duality (Bernstein, 1914; Rivlin, 1981) with the moment comparisons used by Kong and Valiant (2017) and Wu and Yang (2020), and its logarithmic rate parallels supersmooth deconvolution (Fan, 1991); recovery of exponentials connects to ESPRIT and matrix-pencil methods (Roy and Kailath, 1989; Hua and Sarkar, 1990).

Table 2.1: Comparison with finite-mixture estimation and open-loop identification

Feature	Finite-mixture estimation (Heinrich and Kahn, 2018; Wu and Yang, 2020)	Open-loop exponential fitting (Roy and Kailath, 1989; Batenkov et al., 2021)	Closed-loop market identification (this chapter)
Where rates live	Component locations / moments of density	Poles of impulse response	Transmission zeros of closed-loop $H(z)$ (Thm 2.3.1)
Active vs. passive data	i.i.d. draws only	Designed input vs. output spectrum	Same $\Theta((\Delta\eta)^{-(4K-2)})$ order via minimum-phase Whittle gap (Thm 2.6.10)
Economic unit of cost	Sample size N	Signal-to-noise ratio	Quantity $\Theta(\delta^{-2})$, trading loss and capital $\Theta(\delta^{-4})$ via $\frac{1}{2}v^\top T_h v$ (Thms 2.4.1–2.5.4)
Effect of class tagging	Observed component labels ($\Theta(1)$)	Separate sensor per mode	Breaks closed loop cohort-by-cohort; saves $\Theta((\Delta\eta)^{-(4K-4)})$ (Thm 2.8.2)

Minimum-phase spectral factorization and the Whittle likelihood are classical (Whittle, 1953). What is new here is the mapping from these statistical rates into executed quantity and money through a financially specified account, and the result that the relevant rates are zeros, not poles, of the observable transfer function.

Non-normal dynamics and supervision. Transient growth of stable non-normal systems is classical (Trefethen and Embree, 2005; Kreiss, 1962). We use it to show that supervision through eigenvalues and second moments of prices is blind to transient risk (Theorem 2.9.1), which motivates the identification-free control of Chapter 3. On the regulatory side, pre-trade risk controls for broker-dealers with market access are governed by SEC Rule 15c3-5 (SEC, 2010), and participant-level audit trails are the natural disclosure regime for Proposition 2.8.1. A large literature studies how pre- and post-trade transparency affects liquidity and trading costs (Pagano and Röell, 1996; Boehmer et al., 2005; Bessembinder et al., 2006); we ask a different question, what disclosure lets a supervisor learn about the algorithms that trade, and give the answer in samples. Crypto venues, whose market structure is studied by Makarov and Schoar (2020), are a natural laboratory for this question because they publish class-level positioning every five minutes, while the CFTC publishes it weekly for regulated futures.

Partial fractions, Gaussian testing, convolution inequalities and positive-system gains are established tools. Our contribution is to connect them into a quantitative chain from memory resolution, through probing resources, to money, with matching orders, explicit constants and a supervisory interpretation, and then to measure the relevant quantities in data.

Table 2.2: Map of results

Question	Answer	Where
What do prices identify?	The merged rate–weight measure only	Thm 2.3.1
Quantity to separate a known pair	$\Theta(\delta^{-2})$, exact error formula	Thm 2.4.1
Capital to separate a known pair	$\Theta(\delta^{-4})$, counterparty or own funds	Thm 2.4.3, Lem. 2.4.5
Constant at the calibration	$[61.41, 64.48]$ par units; $\approx 10^5$ USDT	Thms 2.5.2–2.5.4, Table 2.7
Does adaptivity help?	At most by a factor ≈ 0.49 in expected loss	Thm 2.5.3
Unknown mixtures	$(\Delta\eta)^{-(4K-2)}$, tight for every K	Thms 2.6.4, 2.6.5
General memory	Accuracy $\asymp 1/\log N$, lower and upper bounds	Lem. 2.6.7, Thm 2.6.8
Passive data	Identifies M; same polynomial cost for every K	Thms 2.6.9, 2.6.10, Prop. 2.15.2
Nonlinear and adaptive rules?	Settled online learners obey δ^{-4}; fast adaptation or thresholds break it	Table 2.15
What removes the barrier?	Tagged flow, audited equity	Props 2.4.4, 2.8.1
What is tagging worth?	Factor $(\Delta\eta)^{-(4K-4)}$; $\approx 333\times$ for BTCUSDT (illustrative)	Thm 2.8.2
Disclosure in dollars?	Annual probing loss exceeds CAT budget once $K \geq 3$ or $\Delta\eta \leq 0.04$	Table 2.17
Do class data restrict class persistence?	Yes: 30 of 30, and 29 of 29 new markets (registered); whether it is memory is open	Table 2.18
Can eigenvalues certify safety?	No	Thm 2.9.1
What do feedback indices measure?	Mostly filtering and timing; small real feedback	Tables 2.22, 2.24
Does feedback track activity?	No: fees move volume 2.6–4.9×, not $\hat{Q}$	Table 2.26

2.1.2 Outline

Section 2.2 sets out the model. Section 2.3 gives the identification boundary. Section 2.4 links memory resolution to quantity and money for a known pair. Section 2.5 computes account-level capital. Section 2.6 treats unknown mixtures and passive data. Section 2.7 checks the results against stylized nonlinear and adaptive trading rules. Section 2.8 studies disclosure and compares its cost in dollars with the U.S. Consolidated Audit Trail. Section 2.9 gives the spectral blind spot. Section 2.10 reports the empirical evidence. Section 2.12 concludes. Proofs are in Appendix 2.13; Appendix 2.15 collects additional results, and Appendix 2.16 maps every number to its source.

2.2 Model and microfoundations

Time is discrete. Let p_t be the price, p^* the fundamental value and $x_t = p_t - p^*$ the pricing error. Competitive dealers absorb net order flow with price impact $\kappa > 0$, and arbitrage pulls

the error toward zero with persistence $a \in [0, 1)$:

$$x_t = ax_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + w_t, \tag{2.1}$$

where v_t is an executed probe quantity and w_t a price innovation. Cohort i holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1 - \eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0, 1). \tag{2.2}$$

The ordering is *price first*: the price responds to the probe, then cohorts update. Write $r_i = 1-\eta_i$ for the memory rate and $w_i = q_i\eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{2.3}$$

and the transfer function from probes to prices, $H(z) = \sum_{j\geq 0} h_j z^{-j}$, is

$$H(z) = \frac{\kappa z}{z - a - \kappa G(z)} = \frac{\kappa z D(z)}{(z-a)D(z) - \kappa N(z)}, \qquad G = \frac{N}{D}, \quad D(z) = \prod_i (z - r_i), \tag{2.4}$$

with impulse response $h_0 = H(\infty) = \kappa, h_1, h_2, \ldots$; in particular $H(1) = \kappa/(1 - S) = \sum_j h_j$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = Jz_{t-1} + bv_t$, where, because cohorts update on the same period's price, the probe enters every state on impact through

$$b = \kappa\, (1, \eta_1, \ldots, \eta_K)^\top, \tag{2.5}$$

and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1 - \eta_1 + \eta_1 \kappa q_1 & \cdots & \eta_1 \kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K \kappa q_1 & \cdots & 1 - \eta_K + \eta_K \kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa \sum_i q_i. \tag{2.6}$$

In the *positive economy* ($q_i \geq 0$), J is entrywise nonnegative, its price row sums to S, and the economy is Schur stable iff $S < 1$.

Lemma 2.2.1 (Filtering and learning microfoundations)**.** *(i)* Kalman memory. *A desk that tracks a latent random-walk trend $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, from signals $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, using the steady-state Kalman filter, updates as $\hat{v}_{t+1} = (1 - \eta)\hat{v}_t + \eta r_t$ with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2\sigma_\nu^2} \in (0, 1). \tag{2.7}$$

(ii) Gradient memory. *Online gradient descent with L_2 regularization $\gamma \in (0,1)$, step size $\alpha > 0$ and loss $\frac{1}{2}H(x_t - \theta_t)^2$ gives $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H \in (0,1)$ and $q = \alpha H/\eta$. Tabular Q-learning with learning rate η has the same form.*

Remark 2.2.2 (Why exponential memory). Lemma 2.2.1 shows that two different design philosophies, Bayesian filtering and regularized online learning, produce the same observable reduced-form belief state $y_{i,t}$. When desk i chooses order flow $u_{i,t}$ subject to $I_{i,t} = (1-\delta_{\text{inv}})I_{i,t-1} + u_{i,t}$ and an inventory cap $|I_{i,t}| \le \bar{I}_i$ to maximize expected continuation return $\beta_i y_{i,t} u_{i,t}$ net of quadratic execution cost $\frac{1}{2}\lambda_{\text{tc},i} u_{i,t}^2$ and quadratic inventory holding cost $\frac{1}{2}\gamma_{\text{inv},i} I_{i,t}^2$, its interior first-order condition gives $u_{i,t}^* = q_i y_{i,t} - \psi_{\text{inv},i} I_{i,t-1}$ with market-response weight $q_i = \beta_i/(\lambda_{\text{tc},i} + \gamma_{\text{inv},i})$ and inventory mean-reversion $\psi_{\text{inv},i} = \gamma_{\text{inv},i}(1-\delta_{\text{inv}})/(\lambda_{\text{tc},i} + \gamma_{\text{inv},i})$, which shrinks toward zero as trading costs or inventory penalties tighten and destabilizes the market when the closed-loop spectral radius reaches one (`microfoundation_and_baselines.py`). This filtering-and-control recursion specifies an individual agent's decision rule, not a complete economic equilibrium (dealer–trader equilibrium is analysed in Chapter 3). The memory rate encodes the desk's signal-to-noise ratio or regularization, which is the private information a supervisor would like to recover; Section 2.3 shows why prices reveal it only in merged form.

Lemma 2.2.3 (Positive price-response geometry)**.** *In the positive, stable economy the impulse response is a nonnegative mixture of decaying exponentials,*

$$h_j = \sum_{\ell} \omega_\ell \lambda_\ell^j, \qquad \omega_\ell \ge 0, \quad \sum_{\ell} \omega_\ell = \kappa, \quad 0 \le \lambda_\ell < 1.$$

It is completely monotone and nonincreasing; its peak is the direct impact κ, even though J is non-normal. In price/memory coordinates the price-only gain is $\sup_t \|e_x^\top J^t\|_2 = 1$.

Numerically, $(-1)^n \Delta^n h_j \ge 0$ for $n \le 4$ held in all 892 stable random positive economies tested. Lemma 2.2.3 matters twice: it makes probing costs positive (Section 2.4), and it shows that in this economy transient amplification of the price response cannot arise from cohort feedback alone. Chapter 3 locates amplification in order-book depth instead.

2.2.1 Discussion of the assumptions

Linearity. The model is linear in prices, flows and memory states. Linearity is what makes the merged-measure equivalence exact and the testing problems Gaussian. Nonlinear strategies (for example threshold rules or position limits) can break the equivalence class, but they do so in regions of the state space that a small probe does not visit; for small probes the linearization around the steady state is the relevant object.

Price-first timing. We order the price update before the cohort update. Under the alternative lagged timing the transition matrix becomes $\left(\begin{smallmatrix} a & 1 \\ \eta Q & 1-\eta \end{smallmatrix}\right)$ and its eigenvalues differ, so stability

statements must be computed under the timing actually used. The identification results depend on the transfer function and survive either convention.

Exponential memory. Lemma 2.2.1 justifies exponential memory for filtering and regularized learning. Neural and reinforcement-learning policies are not exponential filters. Rules whose orders are smooth functions of an exponential-memory state are exponential filters to first order, so the results hold for small probes; Section 2.7 checks this in stylized simulations and finds where it fails, and how closely production algorithms follow such rules remains an empirical question we do not settle. Mixtures of exponentials can approximate any completely monotone kernel, so a power-law memory corresponds to a continuum of cohorts, for which Theorem 2.6.8 shows that accuracy improves only logarithmically in the data.

Algorithms that react to being probed. The probing results assume that the cohorts keep their memories while they are probed. An algorithm that detects a probe could pause, shorten its memory or trade against the probe. Four observations limit what this does to our conclusions. First, a reaction that all cohorts share does not separate them, and an algorithm that hides by mimicking another rate makes the hypotheses closer, so neither lowers the costs we derive; a reaction that differs across cohorts is itself a signature, but one the observer must model, which is a new identification problem rather than a free one. Second, the passive route needs no probe: Theorem 2.6.10 shows that natural price variation identifies the same separation at the same polynomial order in sample length, so an observer who fears detection pays in time, not in rate. Third, detection is itself a testing problem against the same noise: the generalized-eigenvector probe of Section 2.5 trades at most 1.25 par units in any period, and a probe sized within normal flow variation is hard to tell from noise, so the threat bites mainly for large single pulses, which our ledger shows are not optimal anyway. Fourth, what we cannot rule out is an algorithm designed to defeat surveillance, which turns identification into a game between observer and observed; our results are the benchmark against which such a game would be measured, and we leave it to future work.

Known primitives. The resource theorems take κ, a and the noise law as known. Unknown primitives can only make separation harder, so the lower bounds remain valid, while the upper bounds require estimating them, which is cheap relative to the memory rates because they enter at first order.

Observation contracts. Each result names what the observer sees. A price-only observer, an observer with executed quantities, one with noisy statements, one with audited equity and one with tagged flow face different experiments. The paper's central message is that these differences, not the observer's statistical skill, determine what can be learned.

2.3 The identification boundary

A central question in algorithmic market surveillance is whether an observer of executions and prices can identify the individual strategies active in the market.

Theorem 2.3.1 (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state x_t.*

(i) Equivalence class. *Two economies $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ and $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ produce identical price responses for every probe sequence if and only if their merged measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0, 1). \tag{2.8}$$

For $a > 0$ the minimal realization has dimension $1 + |\mathrm{supp}(M)|$, counting distinct rates with nonzero merged weight after signed cancellation; for $a = 0$ the price carries no state of its own and the dimension is $|\mathrm{supp}(M)|$. A cohort with $\eta = 1$ merges into the anchor a without adding a pole.

(ii) Separation complexity under average power. *Consider testing a single cohort with rate $\bar{\eta}$ against two cohorts with rates $\bar{\eta} \pm \delta$ and the same total weight, on a compact family that is uniformly Schur stable with rates bounded away from 0 and 1. Observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma_\nu^2)$ and use deterministic probes with $\frac{1}{T}\sum_{t<T} v_t^2 \le P$. Any test with worst-case error at most $\alpha \in (0, 1/4)$ needs*

$$T \;\ge\; \frac{4\sigma_\nu^2(1-2\alpha)^2}{K^2 P\, \delta^4} = \Omega(\delta^{-4}), \tag{2.9}$$

where K depends only on the family.

The memory rates are poles of G and *zeros* of H, not closed-loop poles. This has a practical consequence we verify in Section 2.6: pole-fitting methods applied directly to the price impulse response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$, $a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

Remark 2.3.2 (Scope). The result concerns the stated model and observation contract. Cohorts sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The δ^{-4} statement in (ii) is a lower bound on a declared family, with family-dependent constants. Individual algorithms may choose their rates strategically, but any unilateral change maps into the same observer-facing merged measure, which reinforces rather than circumvents the boundary.

2.4 Probing resources, memory resolution and money

Observation length alone cannot buy identification when the lifetime quantity of controlled executions is fixed. This section connects the merged-memory experiment to a financially specified input/output contract. Throughout, Φ is the standard normal distribution function.

2.4.1 Quantity

Theorem 2.4.1 (Sharp probing-resource orders for a known symmetric pair)**.** *Fix $0 \le a < 1$, $\kappa, q > 0$, $S = a + \kappa q < 1$ and $\eta \in (0,1)$. Compare a single cohort of weight q and rate η with two cohorts of weight $q/2$ and rates $\eta \pm \delta$, both in $[\epsilon, 1-\epsilon]$. All parameters are known. The price receives executed quantity through κv_t, starts from zero, has no process noise, and is observed with independent $N(0,\sigma^2)$ noise. Probes are deterministic and common to both alternatives. Let $d_T = (h^{\delta}_j - h^0_j)_{j<T}$ and $K_* = 2\kappa^2 q/[\epsilon^3(1-S)^2]$.*

(i) For every probe with $\sum_t |v_t| \le V$,

$$D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le \frac{K_*^2 V^2 \delta^4}{2\sigma^2}, \qquad e_{\mathrm{worst}} \ge \frac{1}{2}\Big(1 - \frac{K_* V \delta^2}{2\sigma}\Big)_+. \tag{2.10}$$

(ii) The optimal worst-case error over these probes and all classifiers is exactly

$$e^*_T(V) = \Phi\left(-\frac{V\|d_T\|_2}{2\sigma}\right), \tag{2.11}$$

attained by a pulse at the first input and the likelihood-ratio test. Waiting adds information but $\|d_T\|_2 \uparrow \|d_\infty\|_2 < \infty$, so it cannot make the error arbitrarily small at fixed V. Error at most $\alpha < 1/2$ requires $V \ge 2\sigma(1-2\alpha)/(K_*\delta^2)$ independently of T.

(iii) The first two taps coincide and $d_2 = -\kappa^2 q \delta^2$. A single pulse of size

$$A = \frac{2\sigma\Phi^{-1}(1-\alpha)}{\kappa^2 q \delta^2} \tag{2.12}$$

and the third observation achieve error exactly α. The necessary and sufficient quantity order is δ^{-2}.

(iv) Under average power $\sum_{t<T} v_t^2 \le PT$, the necessary horizon order is δ^{-4}, attained by the pulse whenever $T \ge \max\{3, \lceil A^2/P \rceil\}$.

(v) For a fixed monetary price exposure $\lambda \ne 0$, additive reserve L and cost envelope $c\sum|v_t|$, the sharp robust quantity allowance under budget $B > L$ is $V_B = (B-L)/(|\lambda|\kappa + c)$.

The identity in (iii) is verified to machine precision for $\delta \in \{0.02, 0.05, 0.10\}$: the difference is -1.2×10^{-4}, -7.5×10^{-4} and -3.0×10^{-3} at $\kappa = 1$, $q = 0.3$, and -3.2×10^{-4} at the calibration

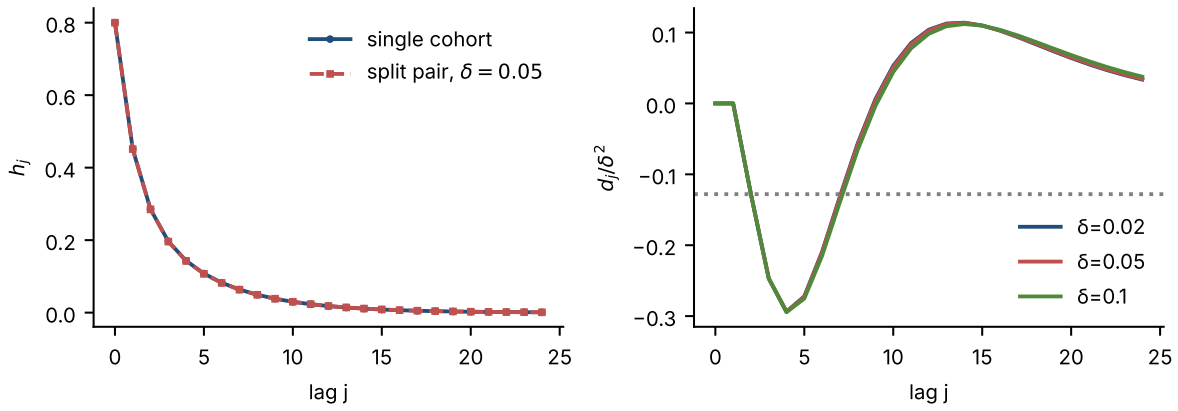


Figure 2.1: Left: impulse responses of the single cohort and the split pair ($\delta = 0.05$) are nearly indistinguishable. Right: the difference scaled by δ^2 collapses onto one curve, with first nonzero value $-\kappa^2 q$ at lag 2 (dotted). Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

Table 2.3: Exact known-pair error: formula versus simulation

Pulse V	Horizon T	$\Phi(-V\|d_T\|_2/2\sigma)$	Simulated error, H_0	Simulated error, H_δ
20	3	0.3745	0.3734	0.3762
40	3	0.2611	0.2599	0.2603
40	10	0.0029	0.0030	0.0027
80	20	1.8×10^{-10}	0	0

200,000 paths per hypothesis; $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

$\kappa = 0.8$, $q = 0.2$, $\delta = 0.05$. The theorem does not solve estimation of an unknown mixture (Section 2.6).

A Monte Carlo check of the exact error. Formula (2.11) is exact for the likelihood-ratio test. We simulated 200,000 noisy paths per hypothesis at the calibration ($\sigma = 0.01$, $\delta = 0.05$) for several pulse sizes and horizons (Table 2.3); simulated errors match the formula to Monte Carlo precision under both hypotheses.

Figure 2.1 shows why the problem is hard. The two economies have the same direct impact and the same first-lag response; they first differ at lag two, by an amount proportional to δ^2, and the whole difference sequence scales with δ^2. A test therefore needs the signal-to-noise ratio $V\|d\|/\sigma$ to grow like δ^{-2}, which is the quantity order, and because the adverse mark of a probe grows with the square of its size, capital grows like δ^{-4}.

2.4.2 Execution cost

A probe changes inventory as well as price. We therefore specify cash and execution timing. Price has units of currency per asset, and κ currency per squared quantity.

Lemma 2.4.2 (Fair execution)**.** *For a quote schedule refreshed after the uncontrolled state update, execution at the average price of a linear book costs, relative to the constant fundamental mark,*

$$C(v) = \tfrac{1}{2} v^\top T_h v + c\|v\|_1, \qquad (T_h)_{ij} = h_{|i-j|}. \tag{2.13}$$

In the positive stable economy T_h is positive definite, so no nonzero round trip produces a deterministic gain.

We verify the ledger identity (2.13) against an explicit cash account to 5.7×10^{-13} on 1,000 random probes, and the smallest eigenvalue of T_h over 1,000 random stable positive economies is 0.141. A direct-only proxy $\sum \frac{1}{2}\kappa v_t^2$ omits cross-time impact: for $v = (+1, -1)$ it gives 0.80 against a true Toeplitz loss of $h_0 - h_1 = 0.3488$.

2.4.3 Capital supplied by a counterparty

Theorem 2.4.3 (Sharp robust quantity cap from cash and margin)**.** *Use the positive, stable, reset economy with constant fundamental price $P_* > 0$ and no process noise. A designated dealer fills only nonnegative probes $v_t \geq 0$ at the average price $P^0_{t+1} + \kappa v_t/2$ of a linear schedule, holds inventory $I_{t+1} = I_t - v_t$ and cash $K_{t+1} = K_t + v_t(P^0_{t+1} + \kappa v_t/2) - cv_t$, starts with equity $B > 0$, and must satisfy $\mu|I_t|P_t \leq E_t$ at every time. For the robust contract admitting every probe with $\sum_t v_t \leq V$, the largest allowance guaranteeing margin is*

$$V_B = \frac{2B}{\beta + \sqrt{\beta^2 + 4(\mu + \frac{1}{2})\kappa B}}, \qquad \beta = c + \mu P_*, \tag{2.14}$$

equivalently $(\mu+\frac{1}{2})\kappa V^2 + (c+\mu P_*)V \leq B$. The cap holds across all positive cohort representations with $S < 1$, without resolving individual rates. An observer that receives only noisy prices and executed quantities has best error $\Phi[-V_B\|d_T\|_2/(2\sigma)]$ within this allowance; fixed error $\alpha < 1/2$ requires capital of order at least δ^{-4}, and a single positive pulse attains that order.

Even with zero initial inventory, the probe creates a quadratic adverse mark that a fixed initial price exposure would miss. Figure 2.2 illustrates the experiment.

Proposition 2.4.4 (Exact financial disclosure changes the experiment)**.** *In this noiseless-state account, exact disclosure of the third equity mark after a single positive pulse $0 < A \leq V_B$ distinguishes the known pair with zero error for every $\delta > 0$.*

Exact marked equity is extra information, not a free observation assumed by the price-only theorem. Real account marks have their own precision: with reports rounded to 0.01, a guaranteed change requires a larger trade (Section 2.8).

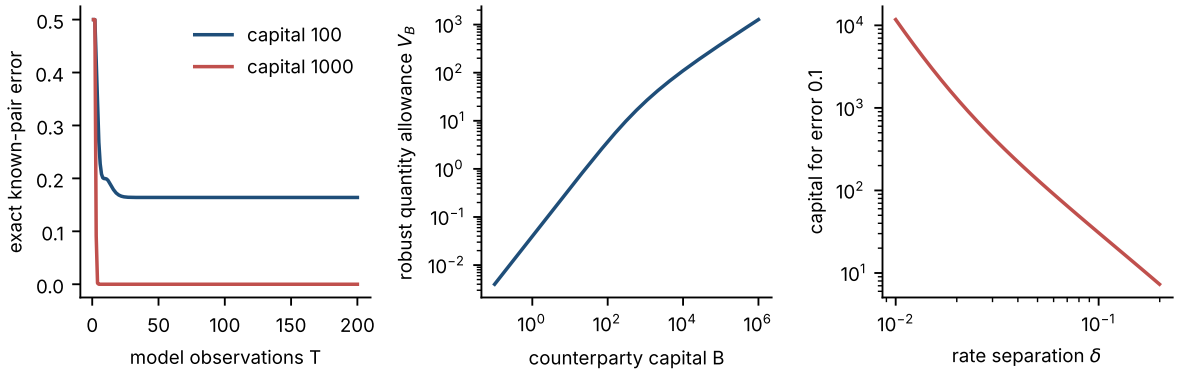


Figure 2.2: Known-pair experiment funded by the counterparty cash/margin contract. (a) Exact Gaussian error at fixed capital 100 and 1,000, separation $\delta = 0.05$. (b) The sharp robust quantity allowance (2.14). (c) Capital for error 0.1 with 200 observations. Parameters: $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, observation s.d. 0.003, reference price 100, margin 0.25, cost 0.02. No process noise.

2.4.4 Capital paid by the probe trader

Lemma 2.4.5 (Time-domain Toeplitz coercivity: capital order for funded probes)**.** *Use the same economy and execution law. The statistical observer receives only noisy prices and executed quantities. The probe trader starts with cash $B > 0$ and zero inventory; a finite signed sequence v changes inventory by $+v_t$ and cash by $K_{t+1} = K_t - v_t(P^0_{t+1} + \kappa v_t/2) - c|v_t|$, after which trading stops and the trader must remain solvent as the price returns to P_*. Put*

$$r = 1 - \epsilon(1 - S) < 1, \qquad m = \kappa \frac{1 - r}{1 + r} > 0, \qquad K_* = \frac{2\kappa^2 q}{\epsilon^3 (1 - S)^2}.$$

Every solvent strategy satisfies $D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le K_*^2 B \delta^4/(\sigma^2 m)$, so error at most $\alpha < 1/2$ requires

$$B \;\ge\; \frac{2\sigma^2 m (1 - 2\alpha)^2}{K_*^2 \delta^4}, \tag{2.15}$$

without a quantity cap and at any horizon. Conversely, for any margin fraction $0 < \mu \le 1$, the pulse A in (2.12) achieves error α and remains margin-solvent at every later mark with capital $B_A = \frac{\kappa}{2} A^2 + (c + \mu P_*) A$. Necessary and sufficient capital orders are δ^{-4}.

Lemma 2.4.5 is an order statement: an elementary time-domain coercivity argument for Toeplitz forms that establishes the δ^{-4} rate before the exact frequency-domain optimization of Section 2.5. Its explicit lower constant is correspondingly loose: at the calibrated parameters it evaluates to 4.9×10^{-4} par units at $\delta = 0.05$ with $\epsilon = 0.35$, five orders of magnitude below any design we can execute, and because $m/K_*^2 \propto \epsilon^7 (1 - S)^5$ it collapses near the boundary of the family. We therefore state it as a lemma; Section 2.5 computes the constant that matters, within 5% of a feasible design at every separation.

2.4.5 A worked example: two Kalman desks

To make the orders concrete, consider two desks that track the same trend with Kalman filters (Lemma 2.2.1(i)) and differ only in their signal-to-noise ratios: $\sigma_\zeta^2/\sigma_\nu^2 = 0.09$ and 0.16. Their gains are $\eta_1 = 0.2584$ and $\eta_2 = 0.3279$, so $\bar\eta = 0.2931$ and $\delta = 0.0348$. With equal weights, the economy is observationally the split pair of Theorem 2.4.1 around $\bar\eta$, and the response difference first appears at lag two, $d_2 = -\kappa^2 q\delta^2 = -1.55 \times 10^{-4}$ (verified).

A supervisor who wants to tell this market from a single desk with gain $\bar\eta$ at 5% error using a single pulse and the third observation needs $A = 2\sigma\Phi^{-1}(0.95)/(\kappa^2 q\delta^2) = 212$ units of executed quantity, and, under a 0.5% margin and 4 basis points of fees, single-pulse capital of 18,166 par units. The pulse is very inefficient: by the δ^{-4} scaling of Theorem 2.5.2, the optimal flat probe needs about $61.41 \times (0.05/0.0348)^4 \approx 262$ par units. Converting with the measured Roll noise (factor $(0.40/0.01)^2$), the optimal probe costs roughly 4×10^5 USDT and the naive pulse about 3×10^7 USDT. Two desks whose signal-to-noise ratios differ by almost a factor of two are therefore expensive, but not impossible, to tell apart by probing; three or more desks are much harder (Theorem 2.6.4).

2.4.6 What prices hide, in economic terms

The worked example states the barrier in signal-to-noise ratios. That is the general picture. Write $s = \sigma_\zeta^2/\sigma_\nu^2$ for a desk's signal-to-noise ratio, so that by (2.7) its memory rate is $1 - \eta(s)$ with

$$\eta(s) = \tfrac{1}{2}\big(\sqrt{s^2+4s} - s\big), \qquad \eta'(s) = \tfrac{1}{2}\Big(\frac{s+2}{\sqrt{s^2+4s}} - 1\Big) > 0,$$

and η is increasing and concave. With no inventory penalty, the first-order condition of the Remark on exponential memory gives the trading intensity $q_i = \beta_i/\lambda_{\mathrm{tc},i}$: what the desk expects to earn per unit of belief, over what it pays to trade.

Proposition 2.4.6 (Identification in economic primitives)**.** *Consider K Kalman desks (Lemma 2.2.1(i)) with signal-to-noise ratios $s_i \in [\underline{s}, \bar{s}] \subset (0, \infty)$ and intensities $q_i = \beta_i/\lambda_{\mathrm{tc},i}$, and write $c_- = \eta'(\bar{s})$ and $c_+ = \eta'(\underline{s})$.*

(i) Prices and aggregate order flow depend on the desks only through the measure $\sum_i q_i\, \eta(s_i)\, \delta_{1-\eta(s_i)}$. Desks are identified through their signal-to-noise ratios and intensities only; expected returns and trading costs enter only through their ratio, and two populations with the same intensity-weighted distribution of signal-to-noise ratios are observationally equivalent.

(ii) Fix a centre s_c and let the ratios lie in $s_c + \Delta s\,[-1, 1] \subset [\underline{s}, \bar{s}]$. Recovering the intensity-weighted distribution of signal-to-noise ratios to Wasserstein accuracy of order Δs, or its weights $q_i\eta(s_i)$, from N noisy reset experiments with worst-case error probability below $1/4$ requires $N \geq c\,\sigma^2(\Delta s)^{-(4K-2)}$; on separated configurations, Prony's method of

Theorem 2.6.5 composed with η^{-1} achieves this order. The constants are those of Theorems 2.6.4 and 2.6.5 multiplied by powers of c_- and c_+.

Proof. (i) By Lemma 2.2.1(i) desk i is a cohort with memory rate $1-\eta(s_i)$ and weight q_i, so Theorem 2.3.1 applies, and $\beta_i, \lambda_{\mathrm{tc},i}$ enter the economy only through q_i. (ii) On $[\underline{s}, \bar{s}]$, η is a C^1 bijection onto its image with $c_- \le \eta' \le c_+$. Lower bound: take the two economies of Theorem 2.6.4 with centre $r_c = 1-\eta(s_c)$ and rate spacing $\Delta\eta = c_-\Delta s$. By the mean value theorem their gains are images of ratios in $s_c + \Delta s\,[-1,1]$, so they are two populations of Kalman desks in the class, and the bound $N \ge c\,\sigma^2(\Delta\eta)^{-(4K-2)} = c\,c_-^{-(4K-2)}\sigma^2(\Delta s)^{-(4K-2)}$ applies to recovering their weights. Because η is c_+-Lipschitz, the Wasserstein distance between the two distributions of ratios is at least $1/c_+$ times that between the two distributions of rates, which Theorem 2.6.4 bounds below by a constant times $\Delta\eta = c_-\Delta s$. Upper bound: ratios in $s_c + \Delta s\,[-1,1]$ with spacing at least $\gamma\Delta s$ have gains in $\eta(s_c) + c_+\Delta s\,[-1,1]$ with spacing at least $c_-\gamma\Delta s$, a class of the form required by Theorem 2.6.5 with rescaled separation $\gamma c_-/c_+$; Prony's method recovers the rates and weights at that theorem's rate, and η^{-1}, which is $1/c_-$-Lipschitz, maps rate errors to ratio errors. □

The private information a supervisor would most like to know, how good each desk's information is, is exactly what prices hide. In the worked example the two desks' ratios differ by a factor of almost two, and separating them already costs hundreds of thousands of dollars of probing capital. The proposition takes intensities to vary independently of information quality, because they also reflect costs and capital. The next proposition asks what happens when the economics ties them together.

Proposition 2.4.7 (Intensity tied to information quality)**.** *Let each desk's intensity be a known function of its signal-to-noise ratio, $q = \varphi(s)$ with φ positive and C^2 on $[\underline{s}, \bar{s}]$ (for a mean–variance desk, $\varphi(s)$ increases with s), and let $n_i \ge 0$ desks have ratio s_i. Write $w(\eta) = \varphi(\eta^{-1}(\eta))\,\eta$ for the weight per desk as a function of its gain.*

(i) Unknown numbers of desks. *The merged measure is $\sum_i n_i\, w(\eta(s_i))\,\delta_{1-\eta(s_i)}$. Every positive measure with finitely many atoms in $[1-\eta(\bar{s}), 1-\eta(\underline{s})]$ is of this form, with $n_i = m_i/w(\eta_i)$ for an atom of mass m_i at $1-\eta_i$. So when the numbers of desks of each type are not known, tying intensity to information quality removes no economy, and every lower bound of Proposition 2.4.6(ii) holds unchanged.*

(ii) Known numbers of desks. *If the n_i are also known, the set of economies is smaller, but the single-versus-pair problem keeps its order: one type at gain $\bar{\eta}$ with $2n$ desks and two types at $\bar{\eta} \pm \delta$ with n desks each have taps that differ by $n\,\delta^2\,\partial_\eta^2[w(\eta)(1-\eta)^m]_{\bar{\eta}} + O(\delta^4)$, so separating them needs $N \ge c\,\sigma^2\delta^{-4}$ reset experiments, the order the symmetric pair has when intensities vary freely.*

(iii) Self-confirming configurations. *Suppose the ratio each desk perceives is the one implied by the closed-loop dynamics it trades in, through a map T from configurations $(s_1, \dots, s_K)$ to perceived ratios that is continuous on a box $\prod_i [\underline{s}, \bar{s}]$ and maps it into itself. Then a configuration with $T(s) = s$ exists, and parts (i)–(ii) and Proposition 2.4.6 apply at it.*

Proof. (i) Desk type i is a cohort with gain $\eta_i = \eta(s_i)$, memory rate $1 - \eta_i$ and total weight $n_i \varphi(s_i)$, so it contributes $n_i \varphi(s_i)\eta_i = n_i w(\eta_i)$ to the atom at $1 - \eta_i$. Since $w > 0$, any atom mass $m_i \geq 0$ is attained with $n_i = m_i / w(\eta_i)$. (ii) Write $g_m(\eta) = w(\eta)(1-\eta)^m$; the tap difference is $n[g_m(\bar{\eta} + \delta) + g_m(\bar{\eta} - \delta) - 2g_m(\bar{\eta})] = n\delta^2 g_m''(\bar{\eta}) + O(\delta^4)$, with $\sum_m \sup |g_m''|^2 < \infty$ on gains bounded away from zero, so the Euclidean distance between the two tap vectors is $O(n\delta^2)$ and the Kullback–Leibler divergence over N experiments is $O(Nn^2\delta^4/\sigma^2)$; Le Cam's two-point argument gives the bound. (iii) is Brouwer's fixed-point theorem. □

So the barrier does not depend on intensities varying freely: it survives any known link between intensity and information quality as long as the supervisor does not also know how many desks of each kind there are, and even then the simplest separation problem keeps its δ^{-4} order. Whether the full $(\Delta\eta)^{-(4K-2)}$ exponent survives with known numbers of desks is open. Part (iii) shows only that self-confirming configurations exist under continuity; it does not derive the map T from a model of the desks' learning.

2.4.7 Natural excitation and the cost of ambiguity

Natural price innovations can also separate the pair with zero controlled quantity: an exact finite-sample test on independent three-step blocks needs order δ^{-4} blocks (Internet Appendix, Proposition 2.15.2), the same order as the passive Whittle bound of Section 2.6. Non-identification also has a price on the quote side: when the relevant set of economies includes costlier ones, a robust dealer widens its quote, which Chapter 3 analyses; cohort non-identification alone does not establish that the worst relevant cost exceeds the informed one.

2.5 Account-level capital

Lemma 2.4.5 gives the order but not the constant. This section computes what a trader actually needs under a calibrated linear-impact environment, then converts it to Bitcoin dollars.

2.5.1 Observation design and the tri-region partition

An observer distinguishes two candidate mechanisms with rates θ_0 and $\theta_1 = \theta_0 + \delta$ over N periods, submitting probes $V = (V_1, \dots, V_N)^\top$ that generate inventory $I_t = \sum_{s \leq t} V_s$. Under

contract I_0 (active probing, Gaussian observation noise) $p_t = (h_i * V)_t + \epsilon_t$, $\epsilon_t \sim N(0, \sigma^2)$, so

$$D_{\mathrm{KL}}(P_0^{\otimes N} \| P_1^{\otimes N}) = \frac{1}{2\sigma^2} \sum_{t=1}^{N} \big((h_1 * V)_t - (h_0 * V)_t\big)^2. \tag{2.16}$$

The observer minimizes capital $C(N, V) = \sum_t (\lambda V_t^2 + \phi |V_t|) + \max_t \mu p_0 |I_t|$ subject to (i) separation $D_{\mathrm{KL}} \geq \frac{1}{2}(z_\alpha + z_\beta)^2$, (ii) caps $|V_t| \leq V_{\mathrm{eff}}$ and $|I_t| \leq I_{\max}$, and (iii) a solvency budget $\max_t \mathrm{Loss}_t(V) \leq B$. Here $V_{\mathrm{eff}} = \min(V_{\max}, D_{\max}/\kappa)$ combines resting depth and a price-displacement limit.

Proposition 2.5.1 (Tri-region partition and finite-session resolution floor)**.** *Under observation horizon T per pulse, (δ, B) space splits into three regions.*

(i) Feasible single pulse. *For $\delta \geq \delta_{\min}(T)$ and $B \geq C^*(\delta)$ one pulse within capacity suffices. For a simple rate shift ($\eta_1 = \eta_0 + \delta$) the response difference is $\Theta(\delta)$ and $C^* = \Theta(\delta^{-2})$; for symmetric splitting the first-order variation cancels, the difference is $\Theta(\delta^2)$ and $C^* = \Theta(\delta^{-4})$.*

(ii) Sequential bounded probing. *For $\delta < \delta_{\min}(T)$ the required quantity exceeds V_{eff}. Capped pulses each supply divergence $V_{\mathrm{eff}}^2 \|d_T\|^2/(2\sigma^2)$, so*

$$N_{\mathrm{pulses}}(T, \delta) = \Big\lceil \frac{4\sigma^2 z_\alpha^2}{V_{\mathrm{eff}}^2 \|d_T\|_2^2} \Big\rceil. \tag{2.17}$$

(iii) Infeasible with isolated pulses. *If the budget cannot finance sequential costs, peak inventory breaches $I_{\max}$, or a session deadline binds, the finite-session floor $\delta_{\min}(T_{\mathrm{session}})$ is the root of $V_{\mathrm{eff}} \|d_{T_{\mathrm{cycle}}}(\delta)\|_2 = 2\sigma z_\alpha / \sqrt{\lfloor T_{\mathrm{session}}/T_{\mathrm{cycle}} \rfloor}$, with small-$\delta$ approximation $\delta_{\min} = \Theta(T_{\mathrm{session}}^{-1/4})$.*

At the calibration $\kappa = 0.8$, $q = 0.2$, $\sigma = 0.01$, $V_{\max} = 50$, $D_{\max} = 5$ ($V_{\mathrm{eff}} = 6.25$), the single-tap floor is $\delta_{\min}(3) = \sqrt{2\sigma z_\alpha/(V_{\mathrm{eff}} \kappa^2 q)} = 0.2028$; extended observation lowers it to 0.1376 ($T = 4$), 0.0973 ($T = 10$), 0.0914 ($T = 20$) and 0.0905 ($T = 100$). At $\delta = 0.05$ the simplified one-leg benchmark (cost $\frac{1}{2}\kappa V_{\mathrm{eff}}^2 + \phi P_* V_{\mathrm{eff}} = 15.875$ per pulse plus initial margin 3.125) needs 271 pulses and 4,305.25 par units with $T = 3$, 58 pulses and 923.88 with $T = 4$, but only 12 pulses and 193.63 with $T = 20$ and 11 pulses and 177.75 with $T = 100$; optimizing the pulse size (6.06 rather than 6.25) lowers the $T = 20$ figure to 182.03. Most of the saving comes from observing the transient response rather than a single tap (Figure 2.3).

These are costs within an isolated-pulse design class, not minimum capital; the next subsection replaces the formula by a ledger.

2.5.2 Account-level capital and the gap to the lower bound

We run each candidate design through an explicit synthetic ledger at reference price $P_* = 100$: fills at the average linear-book price, taker fees of 4 basis points, marks at the prevailing price,

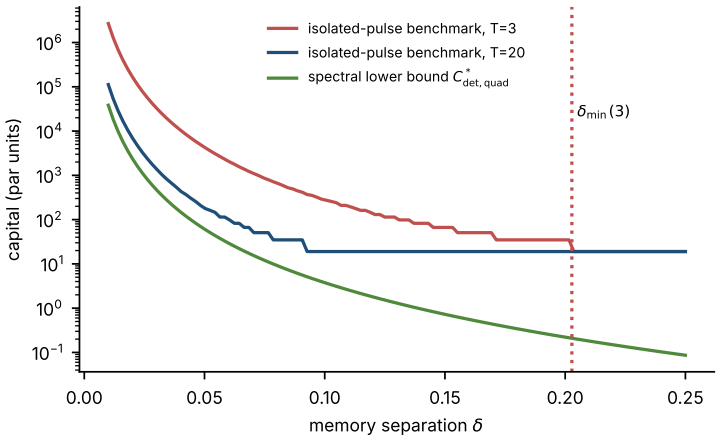


Figure 2.3: Capital against separation: isolated-pulse benchmarks with three and twenty observations per pulse, and the spectral lower bound of Theorem 2.5.2. The dotted line is the single-tap resolution floor $\delta_{\min}(3) = 0.2028$. Recomputed in this revision.

Table 2.4: Account-level initial capital at $\delta = 0.05$ (par units)

Design	C_0	Error	End inv.	Peak margin	Max $\lvert v_t \rvert$	Trades
12 buys, $T = 20$ apart, never unwound	229.13	0.0578	75.00	39.38	6.25	12
6 buy/sell round trips	189.45	0.0308	0.00	3.28	6.25	12
5 buy/sell round trips	157.89	0.0442	0.00	3.28	6.25	10
12 buys, each unwound after 20 steps	378.82	0.0040	0.00	3.28	6.25	24
Generalized-eigenvector probe ($L = 120$)	65.06	0.0500	0.00	2.74	1.24	120
Generalized-eigenvector probe ($L = 160$)	64.86	0.0500	0.00	2.17	1.09	160
Fee-refined probe ($L = 240$)	64.48	0.0500	0.00	—	1.30	240

The pulse designs place a pulse every 40 steps and close it 20 steps later; the fee-refined probe is re-optimized by sequential quadratic programming over $L = 240$. The certified lower bound of Theorem 2.5.2 is 61.41; the explicit constant of Lemma 2.4.5 is 4.9×10^{-4}.

and 0.5% maintenance margin. Initial capital is $C_0 = \max_t(\text{margin}_t - \text{equity}_t)$, maximized over both hypotheses, and the test uses the full price path including unwind legs.

Table 2.4 gives three conclusions. First, twelve buys that are never unwound fail the 5% target once the test uses the joint price path (divergence $4.95 < 5.41$) and end with 75 units of inventory. Second, if every unwind leg is itself informative, five or six round trips meet the target with zero terminal inventory. Third, pulses are not optimal: a generalized-eigenvector probe, which maximizes divergence per unit of fee-free round-trip cost, needs about 65 par units with maximum trade 1.25 and maximum displacement 2.23, well inside limits.

For flat fee-paying policies $v \in \mathbb{R}^L$ with $\sum_t v_t = 0$, the capital problem is

$$\min_{v} \max_{i \in \{0,1\}} L_i(v) + \sum_t \phi P_* |v_t| + \max_t \mu P_* |I_t| \quad \text{s.t.} \quad \|(H_1 - H_0)v\|_2 \geq 2\sigma z_{1-\alpha}, \tag{2.18}$$

where $L_i(v) = \frac{1}{2} v^\top T_{h_i} v$ is the exact Toeplitz round-trip loss. Zero-padding embeds horizon L

into $L+1$ at identical cost and information, so the minimum is non-increasing in L.

Theorem 2.5.2 (Spectral dissipation lower bound)**.** *Let $\hat{H}_i(z) = \sum_{j\geq 0} h_{i,j} z^{-j}$, $\Delta\hat{H} = \hat{H}_1 - \hat{H}_0$, and define the dissipation kernel*

$$g_i(\omega) = \operatorname{Re} \hat{H}_i(e^{i\omega}) - \kappa/2 > 0 \qquad \forall \omega \in [-\pi, \pi]. \tag{2.19}$$

(i) Toeplitz energy identity. *For any flat round trip, $L_i(v) = \frac{1}{2\pi}\int_{-\pi}^{\pi} g_i(\omega)|V(e^{i\omega})|^2 d\omega$.*

(ii) Lower bound. $\|(H_1 - H_0)v\|_2^2 \leq \sup_\omega \frac{|\Delta\hat{H}(e^{i\omega})|^2}{g_i(\omega)} L_i(v)$, *so every deterministic round trip with error at most α has*

$$L_i(v) \;\geq\; 4\sigma^2 z_{1-\alpha}^2 \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2} \;\equiv\; C^*_{\mathrm{det,quad}}(\delta). \tag{2.20}$$

(iii) Bracket. *At $\delta = 0.05$, $C^*_{\mathrm{det,quad}} = 61.14$ at $\omega^* = 0.2526$; adding the unavoidable fee $\phi P_* \cdot 2\sigma z / \sup_\omega |\Delta\hat{H}|$ gives the certified bound $C_{\mathrm{det}} \geq 61.41$. A feasible 240-step sequence attains 64.48, so the fee-paying optimum lies in* $[61.41, 64.48]$.

(iv) Process noise. *Let the price also receive the innovations $w_t \sim N(0, \sigma_w^2)$ of (2.1), independent of observation noise $\nu_t \sim N(0, \sigma_\nu^2)$, so that from the reset state the observer sees $\tilde{x}_t = (h_i * v)_t + \kappa^{-1}(h_i * w)_t + \nu_t$, $t < T$. For the known pair of Theorem 2.4.1, put $\sigma_{\mathrm{eff}}^2 = \sigma_\nu^2 + \sigma_w^2/4$, $\bar{M} = 1/(1-S)$ and $C_w = 2\sigma_w^4 \bar{M}^2 K_*^2(\sigma_\nu^2 + \sigma_w^2\bar{M}^2)/(\kappa^2\sigma_\nu^2\sigma_{\mathrm{eff}}^4)$. Every deterministic round trip whose test has errors α and β under the two hypotheses satisfies*

$$L_i(v) \;\geq\; 2\sigma_{\mathrm{eff}}^2 \big[\mathrm{kl}(\alpha, 1-\beta) - C_w T \delta^4\big]_+ \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{2.21}$$

with kl *as in Theorem 2.5.3. In-loop noise therefore leaves the δ^{-4} capital order intact for every horizon $T \leq \mathrm{kl}(\alpha, 1-\beta)/(2C_w\delta^4)$; at horizons of order δ^{-4} and longer, the noise itself separates the pair without a probe (Theorem 2.6.10).*

Positivity of g_i follows from Lemma 2.2.3: $g_i(\omega) = \frac{1}{2}\sum_\ell \omega_\ell \frac{1-\lambda_\ell^2}{1-2\lambda_\ell\cos\omega+\lambda_\ell^2} > 0$. We compute $C^*_{\mathrm{det,quad}}$ across separations (Table 2.5): 61.14 at $\delta = 0.05$, with $\delta^4 C^*_{\mathrm{det,quad}}$ between 3.67×10^{-4} and 3.84×10^{-4}, confirming the δ^{-4} law. The fee term, the feasible upper bounds and the adaptive bound of Theorem 2.5.3 are reported in the same table.

Theorem 2.5.3 (Adaptive and sequential probing)**.** *Let $(v_t)_{t\leq\tau}$ be any non-anticipating policy adapted to the price filtration, with an almost surely finite stopping time and $\sum_{t\leq\tau} v_t = 0$. Then*

$$\mathbb{E}_0[L_0(v)] \;\geq\; 2\sigma^2\,\mathrm{kl}(\alpha, 1-\beta) \inf_\omega \frac{g_0(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{2.22}$$

where $\mathrm{kl}(\alpha, 1-\beta) = \alpha\log\frac{\alpha}{1-\beta} + (1-\alpha)\log\frac{1-\alpha}{\beta}$. *Hence adaptation can lower expected fee-free*

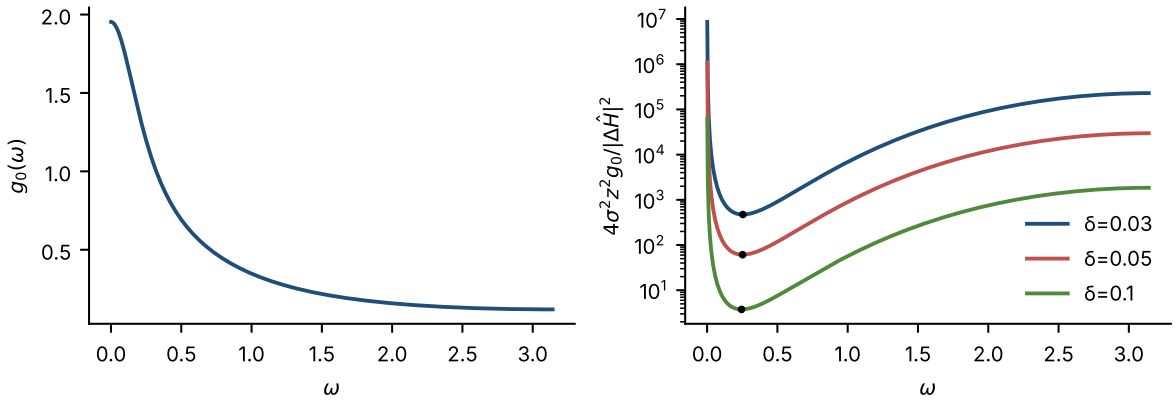


Figure 2.4: Left: the dissipation kernel $g_0(\omega)$ is strictly positive. Right: the capital ratio in (2.20); dots mark its minimum, near $\omega^* \approx 0.25$ for every separation. Recomputed in this revision.

Table 2.5: Capital lower bounds and feasible upper bounds across separations (par units)

δ	ω^*	$C^*_{\mathrm{det,quad}}$	$\delta^4 C^*_{\mathrm{det,quad}}$	Certified C_{det}	Feasible UB	Adaptive LB
0.02	0.2546	2397.85	3.84×10^{-4}	2399.58	2428.25	1176.04
0.03	0.2541	473.19	3.83×10^{-4}	473.97	484.50	232.51
0.05	0.2526	61.14	3.82×10^{-4}	61.41	64.48	30.22
0.08	0.2488	9.26	3.79×10^{-4}	9.37	10.38	4.64
0.10	0.2453	3.77	3.77×10^{-4}	3.83	4.44	1.91
0.15	0.2331	0.73	3.67×10^{-4}	0.76	1.01	0.38

Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, $\sigma = 0.01$, $\alpha = 0.05$. From `recompute_theory.py` and `recompute_p1_capital.py`; the certified column adds the fee lower bound to the fee-free infimum; the feasible column is the best flat probe found over $L \le 240$; the adaptive column adds the fee lower bound to the adaptive bound.

loss by at most the factor $\mathrm{kl}(\alpha, 1-\beta)/(2z^2_{1-\alpha}) \approx 0.4897$ *at* $\alpha = \beta = 0.05$*, and cannot alter the* δ^{-4} *order.*

At $\delta = 0.05$ the adaptive bound is 29.94, or 30.22 with the minimal fee. Pairing it with the feasible deterministic 64.48 gives a certified ratio of at least 0.464. In 20,000 simulated sequential probability-ratio tests, which split the optimal probe into ten copies of equal information, mean loss was 39.5 under H_0, between the two bounds, but losses vary with the stopping time: the 95th and 99th percentiles of required account capital were 91.7 and 128.4, and funding a trader with only the mean loss gives a 37.5% insolvency rate before the test ends. Expected-loss savings are not solvency savings.

Theorem 2.5.4 (Finite-horizon Rayleigh–Hölder duality)**.** *For a horizon of* $L \ge 4$ *trading periods, let* $E \in \mathbb{R}^{L\times(L-1)}$ *be the first-difference basis of flat probes* ($\sum_t v_t = 0$)*,* $Q_i = \frac{1}{2}T_{h_i}$ *(so that* $L_i(v) = v^\top Q_i v$*),* $A_{\mathrm{diff}} = (H_1 - H_0)^\top (H_1 - H_0)$*, and let* $G_i = E^\top Q_i E \succ 0$ *and* $M = E^\top A_{\mathrm{diff}} E \succeq 0$ *be the reduced execution-loss and information matrices, with* G_i*-orthonormal*

Table 2.6: Finite-horizon capital at $\delta = 0.05$ (par units)

Horizon L	Quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$	Full bound $C_{\mathrm{RH}}(L)$	Best feasible probe	Gap to $C_{\mathrm{det}} = 61.41$
80	62.55	64.72	65.02	5.9%
120	61.84	63.39	64.59	5.2%
160	61.54	62.46	64.50	5.0%
240	61.32	61.76	64.48	5.0%
320	61.25	—	≤ 64.48	$\leq 5.0\%$
480	61.19	—	≤ 64.48	$\leq 5.0\%$

Quadratic bound $c_{\mathrm{req}}/\lambda_L$ from the generalized eigenproblem of Theorem 2.5.4 on flat probes (equivalently $c_{\mathrm{req}}/(2\lambda'_L)$ with λ' computed against T_{h_0} rather than $\frac{1}{2}T_{h_0}$); it converges to the infinite-horizon value 61.14. The full bound minimizes (2.24) over 3,600 values of θ (`hardening/rh_bound.py`); at $L = 80$ and 120 a direct numerical minimization of loss plus fee under the separation constraint gives 64.76 and 64.43, above the bound. Feasible probes: top generalized eigenvector refined by sequential quadratic programming on the full ledger; at $L = 320$ and 480 the $L = 240$ probe padded with zeros is feasible at the same cost, so the best feasible value can only be lower.

generalized eigenvectors e_j and eigenvalues $0 \leq \lambda_1 \leq \cdots \leq \lambda_{L-3} < \lambda_{L-2} \leq \lambda_{L-1}$; put $c_{\mathrm{req}} = 4\sigma^2 z_{1-\alpha}^2$ and $c_{\mathrm{fee}} = \phi P_*$. For $\theta \in [0, \pi)$ let $d(\theta) = \cos\theta\, e_{L-2} + \sin\theta\, e_{L-1}$, $\lambda(\theta) = \lambda_{L-2}\cos^2\theta + \lambda_{L-1}\sin^2\theta$, $s_\theta = \mathrm{sign}(Ed(\theta))$, and

$$F_{\mathrm{dom}}(\theta) = c_{\mathrm{fee}}\sqrt{\frac{c_{\mathrm{req}}}{\lambda(\theta)}}\,\|Ed(\theta)\|_1, \qquad W_\perp(\theta) = c_{\mathrm{fee}}^2 \sum_{j \leq L-3} \frac{(s_\theta^\top E e_j)^2}{1 - \lambda_j/\lambda(\theta)},$$
$$\gamma(\theta) = \frac{\lambda(\theta)\,\lambda_{L-3}}{(\lambda(\theta) - \lambda_{L-3})\,c_{\mathrm{req}}}. \tag{2.23}$$

Then every flat probe meeting the separation constraint of (2.18) satisfies

$$L_i(v) + c_{\mathrm{fee}}\|v\|_1 \;\geq\; \min_\theta \Big\{ \frac{c_{\mathrm{req}}}{\lambda(\theta)} + \Psi(\theta) \Big\} \;\equiv\; C_{\mathrm{RH}}(L) \;\geq\; \frac{c_{\mathrm{req}}}{\lambda_{L-1}}, \tag{2.24}$$

$$\Psi(\theta) = \min_{x \geq 0} \Big\{ x^2 + \Big[F_{\mathrm{dom}}(\theta)\big(1 - \gamma(\theta)x^2\big)_+ - \sqrt{W_\perp(\theta)}\,x \Big]_+ \Big\} \;\geq 0,$$

a piecewise-quadratic minimization in closed form; when $\gamma F_{\mathrm{dom}} < 1$ and the minimizer has $\gamma x^2 \leq 1$, $\Psi = \max\big(0, F_{\mathrm{dom}} - W_\perp/(4(1 - \gamma F_{\mathrm{dom}}))\big)$.

The bound is computable for any horizon. Table 2.6 reports its quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$, which decays toward the infinite-horizon value 61.14 at rate $O(L^{-1})$, the full bound $C_{\mathrm{RH}}(L)$, and the best feasible fee-paying probe we find at each horizon. At $L = 80$ the full bound is 64.72 against a best probe of 64.76 on the same loss-plus-fee objective, so the dual is essentially tight at short horizons; at longer horizons the fee term of the dual weakens and the bracket widens to about 5%. The fee of unpadded full-support eigenvectors grows as $\Theta(\sqrt{L})$, which is why zero-padded or L_1-localized probes are needed at long horizons.

Table 2.7: Capital lower bound under alternative observation-noise calibrations (USDT)

Noise calibration	$\delta = 0.02$	$\delta = 0.05$	$\delta = 0.10$
Roll estimator, median (0.40 USDT)	3,874,696	99,193	6,200
One-second price-change s.d., median (1.98 USDT)	93,688,369	2,398,422	149,902
Earlier assumption (4 USDT, upper scenario)	383,812,500	9,825,600	614,100

$C(\sigma, \delta) = 61.41 \times (\sigma/0.01)^2 \times (0.05/\delta)^4$, with σ the unrounded median. Noise estimated on 11 days of checksum-verified Binance BTCUSDT one-second klines.

2.5.3 Robustness: heteroskedastic impact and probing protocols

Corollary 2.5.5 (Invariance under impact-induced heteroskedasticity)**.** *Suppose probing inflates innovation variance as* $\sigma^2_{i,t}(v) = \sigma^2(1 + \xi[(h_i * v)_t]^2)$ *and slippage scales as* $\kappa_{\text{eff}} = \kappa(1 + \xi_0|v_t|)$*. For capacity-bounded probes,* $D_{\text{KL}}(P_0^v \| P_1^v) \le \frac{1+O(\xi^2 V_{\text{eff}}^2)}{2\sigma^2} \|(H_1 - H_0)v\|_2^2 = O(\|v\|_2^2 \delta^4)$*, and convex impact only raises dissipation. Required capital remains* $\Theta(\delta^{-4})$*.*

Four probing protocols (unconstrained single pulse, repeated capped pulses, a sequential test with dynamic throttling, and a predetermined generalized-eigenvector trajectory) all show capital slopes near -4 in $\log \delta$: -3.90, -3.84, -3.47 and -3.81 over $\delta \in [0.03, 0.12]$, where the sequential slope is flattened by the 1.7 pulses needed at $\delta = 0.12$ (Appendix Table 2.28). At equal observation horizon, sequential stopping saves 14–49% of capital relative to repeated capped pulses, but most of the apparent saving from the single-tap baseline comes from observing the transient response, not from adaptivity.

2.5.4 From par units to Bitcoin dollars

Capital scales as $\sigma^2 \delta^{-4}$, so the dollar magnitude hinges on the observation noise. Treating the 4 USDT ten-second price volatility as observation noise would give 9.8 million USDT at $\delta = 0.05$, but that conflates price volatility with measurement error. We now estimate observation noise from Binance BTCUSDT one-second closes on January 10–20, 2023, with the Roll covariance estimator $\hat{\sigma} = \sqrt{-\text{cov}(\Delta p_t, \Delta p_{t-1})}$ (Roll, 1984). Daily estimates range from 0.09 to 0.77 USDT, median 0.40 USDT.

At realistic noise, separating rates that differ by 0.05 costs about 9.9×10^4 USDT, which is not prohibitive for a well-capitalized desk (Table 2.7). The barrier becomes large only for finer separation (3.9 million USDT at $\delta = 0.02$) and, through Theorem 2.6.4, when several cohorts are unknown. Two factors we do not model would raise these figures: walking several levels of a real book costs more than the linear top-of-book impact assumed here, and impact-induced heteroskedasticity raises the noise a probe must overcome. The capital figures are therefore lower bounds for the stated model.

Table 2.8: Daily observation-noise estimates, Binance BTCUSDT one-second closes

Date	Roll $\hat{\sigma}$ (USDT)	One-second change s.d. (USDT)	Mean price (USDT)
2023-01-10	0.152	0.679	17,288
2023-01-11	0.092	0.891	17,456
2023-01-12	0.232	2.292	18,343
2023-01-13	0.348	1.762	19,075
2023-01-14	0.773	3.511	20,818
2023-01-15	0.482	1.790	20,778
2023-01-16	0.629	2.357	21,036
2023-01-17	0.457	1.976	21,194
2023-01-18	0.695	2.541	21,122
2023-01-19	0.402	1.316	20,850
2023-01-20	0.316	2.038	21,306
Median	0.402	1.976	

The Roll estimator measures the negative first-order autocovariance of price changes, which reflects bid-ask bounce and recording noise rather than price volatility.

2.6 Unknown mixtures and passive data

Theorem 2.4.1 concerns a *known* pair. A supervisor faces an unknown number of cohorts with unknown rates and weights. This section asks how fast the cost of recovering them grows.

2.6.1 Why naive pole fitting fails

Because the memory rates are zeros of H (Theorem 2.3.1), extracting poles directly from the price impulse response by matrix-pencil or Prony methods targets closed-loop eigenvalues. A four-step inversion repairs this: (1) decouple $g_t = h_{t+1} - ah_t$; (2) form a Hankel matrix of g and compute a minimal realization by Kung's eigensystem realization; (3) compute its transmission zeros, which are the roots of the denominator of G; (4) project reconstructed flow onto the recovered memory filters to obtain the weights (Appendix 2.15.3). Without noise this recovers $K = 0$, $K = 1$ and well-separated two-cohort systems exactly (Appendix 2.15.3); with noise, per-cohort weights become unreliable once rates are close. The reason is information-theoretic, as the rest of this section shows.

2.6.2 Local information: the Cramér–Rao benchmark

Proposition 2.6.1 (Cramér–Rao benchmark)**.** *Let θ collect the weights $w_1, \dots, w_K$ and rates $\eta_1, \dots, \eta_K$. Suppose the impulse response $h_t(\theta)$, $t \le T$, is observed with i.i.d. Gaussian error from N reset probes of size V_{probe} against innovation noise σ_p, so the error has standard deviation $\sigma_h = \sigma_p/(V_{\text{probe}}\sqrt{N})$. With Jacobian $J_{tj} = \partial h_t/\partial\theta_j$ and $F_{\text{unit}} = J^\top J$, any regular unbiased estimator has $\operatorname{Var}(\hat{w}_k) \ge \sigma_h^2[F_{\text{unit}}^{-1}]_{kk}$, and a 95% interval of half-width* tol *requires*

$$N^* \ge \Big(\frac{1.96}{\text{tol}}\Big)^2 \Big(\frac{\sigma_p}{V_{\text{probe}}}\Big)^2 [F_{\text{unit}}^{-1}]_{kk}. \tag{2.25}$$

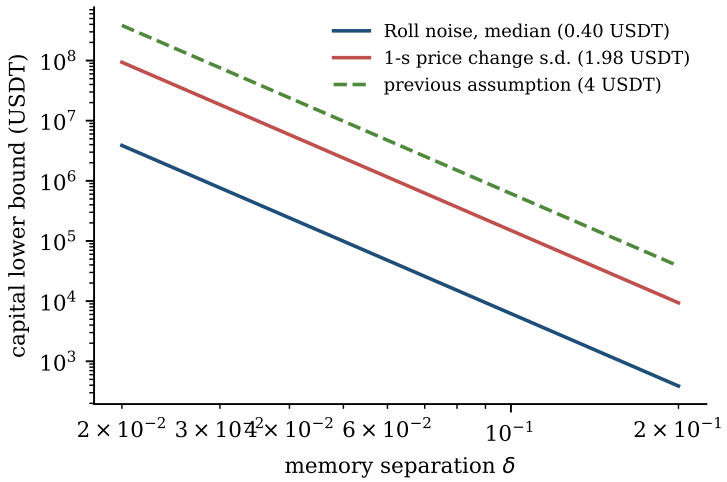


Figure 2.5: Capital lower bound against memory separation for three observation-noise calibrations.

Table 2.9: Cramér–Rao sample sizes for multi-cohort weight recovery ($T = 120$ taps)

Scenario	K	N^* probes	σ_{crit}	ML success ($f = 1$)	RMSE/CRB
$\eta = 0.30$	1	0.1	6.05×10^{-3}	94%	1.02
$\Delta\eta = 0.30$	2	47.8	2.31×10^{-4}	94%	1.01
$\Delta\eta = 0.20$	2	597	6.55×10^{-5}	96%	0.98
$\Delta\eta = 0.10$ ($\delta = 0.05$ pair)	2	40,374	7.96×10^{-6}	93%	1.03
$\Delta\eta = 0.05$	2	2,620,320	9.88×10^{-7}	95%	1.03
spacing 0.10	3	17,349,791	3.84×10^{-7}	98%	1.22
spacing 0.20	3	27,038	9.73×10^{-6}	98%	0.88

Tolerance ± 0.01 at 95%. σ_{crit} is the single-probe noise at which one probe suffices; operational single-probe noise is 1.6×10^{-3}. ML columns: least squares over all weights and rates from six generic starts (never the true values), 200 replications per row at σ_{crit}; success means every weight within ± 0.01. Fitted log–log slope of N^* in $\Delta\eta$: -6.04.

For $K = 2$ with spacing $\Delta\eta$, $[F_{\text{unit}}^{-1}]_{kk} = \Theta((\Delta\eta)^{-6})$, so $N^* = \Theta((\Delta\eta)^{-6})$.

At the critical noise level maximum likelihood attains the bound: its RMSE for the hardest weight is between 0.88 and 1.22 times the Cramér–Rao value and every weight is within tolerance in 93% to 98% of replications (Table 2.9), so the critical noise levels, not estimator failure, are what make close cohorts expensive. Under ordinary surveillance budgets ($N \leq 50$) unbiased estimators cannot resolve close weights.

2.6.3 A global two-point bound for any estimator

Proposition 2.6.1 applies to regular unbiased estimators. Could biased, ridge, shrinkage or Bayesian estimators escape the $(\Delta\eta)^{-6}$ scaling? No.

Theorem 2.6.2 (Global two-point bound and matching upper bound, $K = 2$)**.** *Fix total weight $q = 0.20$, mean rate $\bar{\eta} = 0.40$, target weight separation $\tau = 0.01$, and $\Delta\eta \in (0, 8/11)$. Let*

$p_{\mp} = \frac{1}{2} \mp \tau/q$ *and*

$$\theta_0 = (qp_-, qp_+, \bar{\eta} - p_+\Delta\eta, \bar{\eta} + p_-\Delta\eta), \qquad \theta_1 = (qp_+, qp_-, \bar{\eta} - p_-\Delta\eta, \bar{\eta} + p_+\Delta\eta). \tag{2.26}$$

(i) Label cohorts in increasing order of rate, so that weight vectors are compared rate by rate (as unordered sets the two weight vectors coincide). Both economies have spacing $\Delta\eta$*, weight separation* $\|w(\theta_1) - w(\theta_0)\|_\infty = 2\tau$ *in this labelling, and equal rate moments* $M_m = \sum_k w_k \eta_k^m$ *for* $m \in \{0, 1, 2\}$*, while*

$$M_3(\theta_1) - M_3(\theta_0) = 4\tau p_- p_+ (\Delta\eta)^3. \tag{2.27}$$

Hence $\|h(\theta_1) - h(\theta_0)\|_2 = \Theta(\tau(\Delta\eta)^3)$*.*

(ii) For any measurable estimator (including biased, ridge, James–Stein or MAP) that reports weights in the same rate order, the supremum being over the class of two-cohort economies with total weight q*, mean rate* $\bar{\eta}$ *and spacing* $\Delta\eta$*,*

$$\sup_\theta P_\theta(\|\hat{w} - w(\theta)\|_\infty \geq \tau) \;\geq\; 1 - \Phi\Big(\frac{\sqrt{N}\|h(\theta_1) - h(\theta_0)\|_2}{2\sigma_h}\Big), \tag{2.28}$$

so failure probability at most 0.05 *requires* $N \geq 4\sigma_h^2 z_{0.95}^2 / \|h(\theta_1) - h(\theta_0)\|_2^2 = \Omega((\Delta\eta)^{-6})$*. The same bound holds without any labelling convention for the permutation-invariant loss*

$$d_\infty(\hat{\theta}, \theta) = \min_{\pi \in S_2} \max_k \big(|\hat{w}_{\pi(k)} - w_k| + |\hat{\eta}_{\pi(k)} - \eta_k|\big)$$

in place of $\|\hat{w} - w(\theta)\|_\infty$*, whenever* $\Delta\eta \geq 2\tau q/(q + 2\tau)$*, because then* $d_\infty(\theta_0, \theta_1) = \min\{2\tau(1 + \Delta\eta/q), (1 + 2\tau/q)\Delta\eta\} \geq 2\tau$*.*

(iii) Conversely, on the known-midpoint family a linear estimator that projects the averaged impulse response on the normalized third-moment direction and inverts (2.27) has error at most $|\hat{M}_3 - M_3|/(2p_-p_+(\Delta\eta)^3)$*, giving* $N = O(\tau^{-2}(\Delta\eta)^{-6}\log(1/\delta_{\text{fail}}))$*. On the separated class with weights and rates bounded away from the boundary, Toeplitz deconvolution followed by a Prony–Hankel root step has Hankel determinant* $g_1 g_3 - g_2^2 = w_1 w_2 \eta_1 \eta_2 r_1 r_2 (\Delta\eta)^2 > 0$ *for the deconvolved kernel* $g_m = \sum_i w_i \eta_i r_i^m$*, and perturbation analysis gives the same order.*

We recomputed the two-point quantities exactly (Table 2.10): the moment gap equals $4\tau p_- p_+ (\Delta\eta)^3$ to machine precision, and the Le Cam requirement at $\Delta\eta = 0.10$ is 5,927.6 resets. The worst-case failure rates of four re-implemented estimators all lie above the lower bound (Table 2.10).

Remark 2.6.3 (The hard directions carry no risk). The two economies of Theorem 2.6.2 have the same total weight $\sum_i q_i = q$, so their summed price response $\sum_t h_t = \kappa/(1 - a - \kappa \sum_i q_i)$

Table 2.10: Two-point sample complexity for two-cohort weight recovery

$\Delta\eta$	ΔM_3	$\|\Delta h\|_2$	N^*_{LeCam} (95%)	N^*_{CRB}
0.30	2.67×10^{-4}	1.98×10^{-3}	7.1	47.8
0.20	7.92×10^{-5}	5.60×10^{-4}	88.2	597.0
0.14	2.72×10^{-5}	1.89×10^{-4}	775.3	5,267.9
0.10	9.90×10^{-6}	6.84×10^{-5}	5,927.6	40,374.2
0.08	5.07×10^{-6}	3.49×10^{-5}	22,740.5	155,048.9
0.06	2.14×10^{-6}	1.47×10^{-5}	128,333.0	875,732.3
0.05	1.24×10^{-6}	8.50×10^{-6}	383,859.1	2,620,321.2
0.04	6.34×10^{-7}	4.35×10^{-6}	1,466,361.5	10,012,662.4

Worst-case failure at $\Delta\eta = 0.10$*,* $N = 50$*, 1,000 trials per hypothesis*: bound 44.1%; ML 94.5%, ridge 47.5%, positive-part shrinkage 100%, heavy prior 73.9%.

$\tau = 0.01$, $\sigma_h = 1.6 \times 10^{-3}$, $a = 0.5$, $\kappa = 0.8$, 121 taps. Recomputed columns from `recompute_theory.py`. Estimators re-implemented in `recompute_p1_estimators.py`: ML is least squares from four starts; ridge and heavy prior are posterior modes with prior standard deviations $(0.05, 0.10)$ and $(0.005, 0.02)$ for (weights, rates) around the symmetric centre of the two hypotheses; positive-part shrinkage pulls the ML weights toward their mean. Failure means some weight is off by at least τ.

is identical, 2.353 at the calibration of Table 2.10: the directions in which prices are least informative about the cohorts are directions in which cumulative price risk does not move. The summed response is a linear functional of the response and is estimated at the parametric rate whatever the spacing of the memories; Chapter 3 turns this into a certification theorem, a minimax-optimal pre-trade guard and a decision value of disclosure (Chapter 3). Identification and safety ask different questions of the same data.

2.6.4 Many unknown cohorts

Theorem 2.6.4 (K-cohort two-point lower bound)**.** *Fix $K \geq 2$, a center $r_c \in (0,1)$ and total merged weight $W > 0$. There exist two K-cohort economies with rates in $r_c + \Delta\eta[-1,1]$, positive weights, and merged measures whose moments of order $0, \dots, 2K-2$ coincide while the moment of order $2K-1$ differs by a constant times $(\Delta\eta)^{2K-1}$. Their impulse responses differ by $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, so recovering the weights from N independent noisy reset experiments with worst-case error probability below $1/4$ requires*

$$N \;\geq\; c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

The same requirement holds for the permutation-invariant loss $W_1(\hat{M}, M)$ between estimated and true merged measures at accuracy $c_K W \Delta\eta$, where $2c_K > 0$ is the Wasserstein distance between the two normalized quadrature rules, so no relabelling of atoms escapes it.

The construction uses the K-point Gauss and Gauss–Radau rules of the same weight function: both are positive, have K atoms and integrate polynomials of degree $2K-2$ exactly. For $K = 2$ it reproduces the order of Theorem 2.6.2. Table 2.11 lists the two rules: their low

Table 2.11: Gauss and Gauss–Radau rules on $[-1,1]$ used in Theorem 2.6.4

K	Gauss nodes / Gauss–Radau nodes	Max gap, orders $< 2K-1$	Gap at order $2K-1$
2	± 0.5774 / $-1,\ 0.3333$	6×10^{-17}	0.2222
3	$0,\ \pm 0.7746$ / $-1,\ -0.2899,\ 0.6899$	2×10^{-16}	0.0533
4	$\pm 0.3400,\ \pm 0.8611$ / $-1,\ -0.5753,\ 0.1811,\ 0.8228$	4×10^{-16}	0.0131

All Gauss–Radau weights are positive.

Table 2.12: Numerical check of Theorem 2.6.4

	$K=2$	$K=3$	$K=4$
Theoretical exponent of N^*	-6	-10	-14
Fitted log–log slope (small $\Delta\eta$)	-5.96	-9.96	-14.01

Le Cam bound $N^* \propto \sigma^2/\|h^A - h^B\|_2^2$ with $\sigma = 1$, $a = 0.5$, $\kappa = 1$, total merged weight 0.10, rates centered at 0.70, responses over 400 lags.

moments agree to machine precision and their moment of order $2K-1$ differs by 0.222, 0.0533 and 0.0131 for $K = 2, 3, 4$. The shrinking gap explains why constants deteriorate with K on top of the exponent.

Table 2.12 and Figure 2.6 confirm the order numerically.

Theorem 2.6.5 (Matching upper bound for every K)**.** *Fix $K \geq 2$, a compact set $\mathcal{S} \subset \{s \in [-1,1]^K : s_1 < \cdots < s_K\}$ of ordered rescaled rates with $\min_{i \neq j} |s_i - s_j| \geq \gamma > 0$, and weights in $[\omega_{\min}, \omega_{\max}]$ with $\omega_{\min} > 0$. Consider the class of K-cohort economies with rates $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, and N independent reset experiments whose responses carry Gaussian noise with variance σ^2. There are constants $c_0, C > 0$ depending only on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ such that, whenever $\epsilon_N \equiv \sigma N^{-1/2} (\Delta\eta)^{-(2K-1)} \leq c_0/\sqrt{\log(2/\beta)}$, Prony's method applied to the first $2K$ estimated kernel values, with its output atoms listed in increasing order of rate, recovers the weights with*

$$\max_i |\hat{w}_i - w_i| \ \leq\ C\sqrt{\log(2K/\beta)}\ \epsilon_N$$

with probability at least $1-\beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice for any fixed accuracy and confidence, and together with Theorem 2.6.4 the minimax sample requirement is $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$.

Proof. The kernel values $k_m = \sum_i w_i r_i^m$ are the power moments of the merged measure. The closed-loop recursion $h_t = a h_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ (plus κ at $t = 0$) can be solved for $k_0, \ldots, k_{2K-1}$ from $h_0, \ldots, h_{2K}$ with $h_0 = \kappa \neq 0$; the map is smooth with Lipschitz constant independent of $\Delta\eta$, so the estimated kernel values have error $O_P(\sigma N^{-1/2})$. Centring at r_c is a fixed triangular linear map, $\mu_m = \sum_{j \leq m} \binom{m}{j} (-r_c)^{m-j} k_j$, and $\mu_m = (\Delta\eta)^m t_m$ with $t_m = \sum_i w_i s_i^m$. The map $F : (s, \omega) \mapsto (t_0, \ldots, t_{2K-1})$ does not depend on $\Delta\eta$; its Jacobian is a confluent Vandermonde matrix with determinant $\prod_i \omega_i \prod_{i<j} (s_i - s_j)^4$, bounded away from zero on the compact

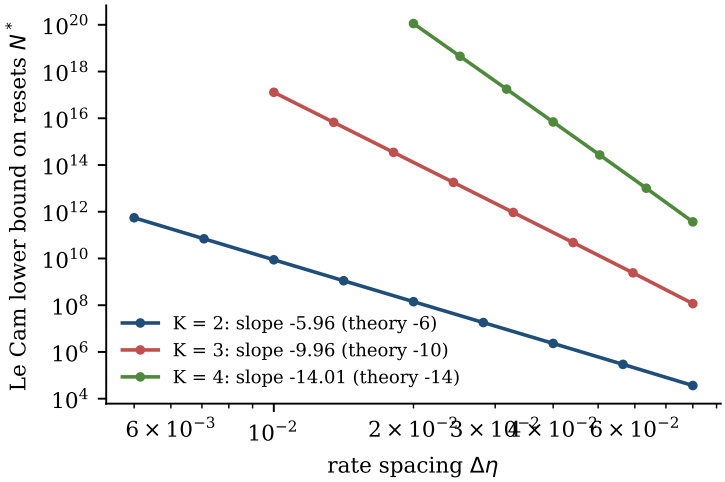


Figure 2.6: Two-point lower bound on the number of reset experiments needed to separate K-cohort economies, as a function of the rate spacing.

class. By the quantitative inverse function theorem there are $\rho_0 > 0$ and $L < \infty$, uniform over the class, such that F^{-1} exists and is L-Lipschitz on the ball of radius ρ_0 around every point of $F(\text{class})$. An error e in μ_m is an error $e(\Delta\eta)^{-m}$ in t_m, largest at $m = 2K - 1$. The kernel errors are Gaussian with standard deviation of order $\sigma N^{-1/2}$, so a union bound over the $2K$ coordinates gives $\|\hat{t} - t\|_\infty \le C_1 \sqrt{\log(2K/\beta)}\, \epsilon_N$ with probability at least $1 - \beta$. When this is below ρ_0, $\hat{t}$ lies in the ball where F^{-1} is L-Lipschitz. The moment map is invariant to relabelling the atoms, so it is injective only on ordered nodes; on the class of ordered, γ-separated nodes it is injective, and the inverse function theorem applies there. Prony's method computes the algebraic inverse of the moment map up to the order of the atoms, so, listing its output in increasing order of rate (which the error bound below keeps unambiguous once $C\sqrt{\log(2K/\beta)}\, \epsilon_N < \gamma/2$), its output is $F^{-1}(\hat{t})$ and the stated bound follows with $C = LC_1$ and $c_0 = \rho_0/C_1$. The matching statement requires the class to contain the Gauss and Gauss–Radau pair of Theorem 2.6.4 (ordered nodes with separation at least the smaller of their two minimum gaps, and weights within $[\omega_{\min}, \omega_{\max}]$ after scaling to total weight W); for any such class the orders match, with constants that depend on the class. □

Table 2.13 confirms the order numerically. For numerical stability the estimator works in centred and rescaled coordinates (the estimator is the same; only its floating-point evaluation changes). The argument parallels the stability analysis of super-resolution for clustered complex nodes (Batenkov et al., 2021), here for real nodes, and it reproduces the ESPRIT slope of -6.15 reported for $K = 2$ in the earlier version.

Table 2.13: Sample requirement of Prony's method against the lower-bound order

	$K=2$	$K=3$	$K=4$	$K=5$
Lower-bound order $-(4K-2)$	-6	-10	-14	-18
Fitted slope of N^* in $\Delta\eta$ (Prony)	-5.94	-10.00	-13.93	-18.34

Rates $0.7+\Delta\eta\, s$ with s the K-point Gauss nodes, total weight 0.1, $\Delta\eta \in [0.03, 0.2]$; N^* is the number of experiments for a weight RMSE of 10% of the smallest weight, from 400 simulated estimates per point.

2.6.5 Near-colliding rates: a uniform rate for the merged measure

Theorem 2.6.5 needs the rates to stay γ-separated and every weight to stay away from zero. Neither restriction can be dropped for weight recovery: two atoms of weights w and $W-w$ at rates r and $r+\Delta$ and two of weights $W-w$ and w at the same rates have merged measures that converge to the same measure as $\Delta \to 0$, so no estimator recovers the individual weights uniformly over near-colliding economies. What can be recovered uniformly is the merged measure itself, in the permutation-invariant Wasserstein distance.

Theorem 2.6.6 (Uniform rate without separation)**.** *Fix $K \geq 1$, a memory bound $\bar{r} < 1$ and $0 < m_{\min} \leq B$. Let $\mathcal{M}_K$ be the merged measures with at most K atoms in $[0, \bar{r}]$ and total mass $k_0 \in [m_{\min}, B]$, with no separation condition and no lower bound on individual weights, and write $\bar{M} = M/k_0$ for the normalized memory distribution. From N reset experiments as in Theorem 2.6.5, let $\hat{M}$ be any element of $\mathcal{M}_K$ whose first $2K$ kernel values are closest in the sup norm to the estimated ones. There is C depending only on $(K, \bar{r}, a, \kappa, B)$ such that, with probability at least $1-\beta$,*

$$W_1(\hat{\bar{M}}, \bar{M}) \;\leq\; C\Big(\frac{\sigma\sqrt{\log(4K/\beta)}}{m_{\min}\sqrt{N}}\Big)^{1/(2K-1)} \qquad \text{uniformly over } \mathcal{M}_K.$$

Hence $N = O(\sigma^2 m_{\min}^{-2}\epsilon^{-(4K-2)})$ experiments give accuracy ϵ, and with the Wasserstein statement of Theorem 2.6.4 the minimax rate for the merged memory distribution is $N^{-1/(4K-2)}$, uniformly over near-colliding economies.

Proof. As in the proof of Theorem 2.6.5, the kernel values $k_0, \ldots, k_{2K-1}$ are a Lipschitz function of the first $2K+1$ responses on the bounded set of responses the class produces, so a union bound gives $\max_{t<2K} |\bar{k}_t - k_t| \leq e = C_1\sigma\sqrt{\log(4K/\beta)/N}$ with probability at least $1-\beta$. The class is compact and the moment map continuous, so a closest element $\hat{M}$ exists, and $\max_t |k_t(\hat{M}) - k_t(M)| \leq 2e$ by the triangle inequality. Since all atoms lie in $[0,1]$, $|k_t| \leq k_0$, and the normalized moments $m_t = k_t/k_0$ satisfy $|m_t(\hat{M}) - m_t(M)| \leq 2e/k_0 + (|\hat{k}_t|/\hat{k}_0)(2e/k_0) \leq 4e/m_{\min}$. The affine map $r \mapsto 2r/\bar{r} - 1$ carries both normalized measures to distributions on $[-1,1]$ with at most K atoms, multiplies W_1 by $2/\bar{r}$ and the differences of moments of order below $2K$ by at most $(1+2/\bar{r})^{2K-1}$. The moment comparison inequality for K-atomic distributions on $[-1,1]$

(Wu and Yang, 2020), $W_1 \le c_K \delta^{1/(2K-1)}$ when the first $2K-1$ moments differ by at most δ, then gives the bound with $C = c_K(\bar{r}/2)(1+2/\bar{r})(4C_1)^{1/(2K-1)}$. □

The theorem separates two questions that Theorem 2.6.5 answers together. Which memory distribution do the participants have? That has the same polynomial cost whether or not their rates collide. Which participant has which weight? That becomes impossible as rates collide, at any sample size. A supervisor who needs the first can have it at the rate of Theorem 2.6.4; one who needs the second has to be told the number of cohorts and a separation, or has to obtain tagged data (Section 2.8).

2.6.6 General memory: the minimax rate is logarithmic

Exponential memory is a modelling choice. Suppose instead that the feedback kernel is completely monotone, $k_m = \int_{[0,\rho]} r^m \, d\mu(r)$ with $\rho < 1$, where μ is any nonnegative measure of total mass W: a finite or infinite mixture of exponential memories with merged weight W. Write $\mathcal{M}(\rho, W)$ for this class, assume the closed loop is uniformly stable, $s \equiv a + \kappa W/(1-\rho) < 1$, and measure accuracy by

$$d(\mu, \mu') = \sup\Big\{ \int f \, d(\mu - \mu') : \ f \text{ 1-Lipschitz on } [0,\rho], \ f(0) = 0 \Big\},$$

which equals the Wasserstein-1 distance between measures of equal mass and also compares estimates whose mass is not exactly W. Let $E_L(|x|)$ denote the error of the best uniform approximation of $|x|$ on $[-1,1]$ by polynomials of degree L, and let $c_* = \inf_{L \ge 1} L\, E_L(|x|)$. Bernstein (1914) showed that $L\, E_L(|x|)$ converges to a positive constant (0.2802; Varga and Carpenter, 1985), so $c_* > 0$; numerically $c_* = 1/4$, attained at $L = 2$ (Table 2.14).

Lemma 2.6.7 (Moment-matched cohort pairs)**.** *For every $n \ge 2$ there are two probability measures on $[0,\rho]$, each with exactly n atoms, whose moments of orders $0, \dots, 2n-2$ coincide and whose Wasserstein-1 distance is at least $\rho\, E_{2n-2}(|x|) \ge c_* \rho/(2n-2)$.*

The atoms are the points where the best approximation of $|r - \rho/2|$ by polynomials of degree $2n-2$ attains its maximal error with alternating signs, and the weights are those of the divided difference on these points, which annihilates every polynomial of degree $2n-2$ (Appendix 2.13). Each measure is an n-cohort economy.

Theorem 2.6.8 (Minimax rate for general memory)**.** *Let N independent reset experiments observe the impulse response with Gaussian noise of variance σ^2 per tap, and let $B(\rho) = (1+\sqrt{2})(1+2/\rho)$ and $C_h = \kappa^2/(1-s)^2$.*

(i) Lower bound. *Let $\Lambda_N = \log(N/\sigma^2) + \log\big(2C_h^2 W^2/((1-\rho)\rho)^2\big)$. Whenever $\Lambda_N \ge 4\log(1/\rho)$, every estimator $\hat{\mu}$ satisfies*

$$\sup_{\mu \in \mathcal{M}(\rho,W)} P_\mu\Big(d(\hat{\mu}, \mu) \ge \frac{c_*\, \rho W \log(1/\rho)}{\Lambda_N} \Big) \ \ge \ \frac{1}{4e},$$

and the bound already holds over n-cohort economies, for a finite n that grows with N.

(ii) Upper bound. *With a and κ known, invert the averaged response into kernel values $\hat{k}_0, \dots, \hat{k}_L$, with $L+1 = \lfloor \log(N/\sigma^2)/(4 \log B(\rho)) \rfloor$, and let $\hat{\mu}$ be any measure on $[0,\rho]$ minimizing $\max_{j \le L} |\int r^j d\hat{\mu} - \hat{k}_j|$ (a linear program). For every $\beta \in (0,1)$ there is N_0 depending only on $(\rho, W, a, \kappa, \beta)$ such that for $N/\sigma^2 \ge N_0$,*

$$\sup_{\mu \in \mathcal{M}(\rho,W)} P_\mu\Big(d(\hat{\mu}, \mu) > \frac{3\pi\rho W \log B(\rho)}{\log(N/\sigma^2)} \Big) \;\le\; \beta.$$

Hence the minimax accuracy is of exact order $1/\log(N/\sigma^2)$, and reaching accuracy w requires and suffices $N/\sigma^2 = \exp(\Theta(1/w))$.

Proof sketch. (i) Put the measures of Lemma 2.6.7, scaled to mass W, into the model. Their difference annihilates polynomials of degree $2n-2$, so the generating function of the kernel difference is $\int (1-rz)^{-1} d(\mu_A - \mu_B) = z^{2n-1} \int r^{2n-1}(1-rz)^{-1} d(\mu_A - \mu_B)$; on $|z| = 1$ it is at most $2W\rho^{2n-1}/(1-\rho)$, and Parseval bounds $\|k^A - k^B\|_2$ by the same quantity. The response difference is $H_A H_B$ times the shifted kernel difference, so $\|h^A - h^B\|_2 \le C_h \|k^A - k^B\|_2$. Choosing the smallest n for which the Kullback–Leibler divergence of the N-sample experiments is at most one, the Bretagnolle–Huber form of Le Cam's lemma gives the bound with half the Wasserstein separation of the lemma. (ii) For a 1-Lipschitz f with $f(0) = 0$, Jackson's theorem gives a polynomial q of degree L with $\sup|f - q| \le \pi\rho/(4(L+1))$, and expanding q in Chebyshev polynomials of $2r/\rho - 1$ bounds the sum of its monomial coefficients by $\rho B(\rho)^{L+1}$. Hence $d(\hat{\mu}, \mu) \le \pi\rho(W+\delta)/(2(L+1)) + 2\delta\rho B(\rho)^{L+1}$, where δ is the largest kernel error. The kernel is recovered from the response by a series reciprocal, $K(z) = (1 - az - \kappa/H(z))/(\kappa z)$, whose error is at most $2((1+s)/\kappa)^2(L+2)$ times the largest response error, so $\delta = O_P(\sigma N^{-1/2} L\sqrt{\log L})$. The choice of L makes $\delta B(\rho)^{L+1} = O_P((\sigma^2/N)^{1/4} L\sqrt{\log L})$, negligible against the first term. Full proofs are in Appendix 2.13. □

Theorem 2.6.8 replaces a polynomial barrier with a logarithmic one. With exponential memory and K known cohorts, halving the error multiplies the required number of experiments by a constant; with general memory, halving the error squares it. The lower bound holds already for finite mixtures, so knowing that the memory is a finite mixture of exponentials does not help unless the number of cohorts and their spacing are also known. The constants of the two bounds differ, by a factor that grows as $\rho \to 1$, so the dependence on how long memory lasts is not pinned down. The source of the gap is visible in the proof: as $\rho \to 1$ the moment-matched pairs separate their kernels more slowly (ρ^{2n-1} decays slowly), which weakens the lower bound through $\log(1/\rho)$, while the upper bound pays $\log B(\rho)$, which stays bounded away from zero, for the conditioning of polynomial approximation on $[0,\rho]$. Logarithmic rates of this kind are familiar from deconvolution with supersmooth errors (Fan, 1991) and from moment methods for

Table 2.14: General memory: moment-matched cohort pairs and the moment estimator

A. Best approximation of $\lvert x\rvert$						
L	2	4	6	10	20	30
$L\,E_L(\lvert x\rvert)$	0.2500	0.2705	0.2756	0.2785	0.2797	0.2800
B. Pairs of Lemma 2.6.7						
Atoms per measure n	2	4	6	8	10	12
$W_1(\mu_A,\mu_B)$, $\rho=0.9$	0.2250	0.1037	0.0669	0.0494	0.0391	0.0324
$n\,W_1$, $\rho=0.9$	0.450	0.415	0.402	0.395	0.391	0.388
$\lVert k^A-k^B\rVert_2$, $\rho=0.9$	4.41×10^{-2}	1.03×10^{-3}	4.08×10^{-5}	1.96×10^{-6}	1.05×10^{-7}	5.95×10^{-9}
$\lVert k^A-k^B\rVert_2$, $\rho=0.99$	7.41×10^{-2}	5.84×10^{-3}	1.07×10^{-3}	2.76×10^{-4}	8.33×10^{-5}	2.76×10^{-5}
C. Linear-programming estimator, continuous memory distribution, $\rho=0.9$						
$\log_{10} N$	4	8	12	16	20	24
Moments used L	2	3	5	6	7	8
Error $d(\hat\mu,\mu)$	0.00878	0.00759	0.00432	0.00445	0.00382	0.00399
Error $\times\log(N/\sigma^2)$	0.202	0.245	0.179	0.226	0.229	0.276

A: Remez algorithm; the limit is Bernstein's constant 0.2802. B: moments of orders $0,\dots,2n-2$ agree to 10^{-61} (60-digit arithmetic); every W_1 exceeds $\rho E_{2n-2}(\lvert x\rvert)$. C: true merged measure Beta(2,3) on $[0,0.9]$ with mass 0.1; kernel values observed with noise $\sigma N^{-1/2}$, $\sigma=10^{-3}$; L from the rule in Theorem 2.6.8(ii); measure fitted on a 401-point grid; 20 replications. Kind: recomputed (`general_minimax.py`).

mixing distributions (Kong and Valiant, 2017; Wu and Yang, 2020); here they arise from the closed loop of a market.

Table 2.14 checks the construction and the estimator. The pairs of Lemma 2.6.7 have exactly n atoms, agree in all moments below order $2n-1$ to 10^{-61}, and remain at Wasserstein distance about $0.4/n$, while the distance between their kernels shrinks by a factor of about 0.2 per atom at $\rho=0.9$. The linear-programming estimator applied to a continuous memory distribution has error times $\log(N/\sigma^2)$ between 0.18 and 0.28 for N from 10^4 to 10^{24}: a 10^{20}-fold increase in data halves the error. The exponential-memory model of the previous subsections, where the cost is polynomial in $1/\Delta\eta$, is therefore the optimistic case: relaxing the parametric form makes identification from prices much harder, not easier.

2.6.7 Passive data

Theorem 2.6.9 (Passive identification: minimum phase)**.** *In any stable economy with $\lvert r_i\rvert<1$ and nonzero merged weights, the zeros of H are exactly the memory rates r_i, and H is minimum phase. With a single price innovation, the price spectrum identifies H up to scale, hence the memory rates; with κ and a known, as in Theorem 2.3.1, it identifies the merged measure.*

The result holds for signed weights (mean-reverting cohorts): in 1,733 stable random economies with weights of both signs, the computed zeros equalled the memory rates and lay inside the unit disk. With a second independent shock in the cohorts' memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

Theorem 2.6.10 (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of a single innovation with variance $\sigma^2 > 0$, for θ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0,1)$, and let $D(\theta_1\|\theta_0)$ be the Kullback–Leibler divergence rate per observation.*

(i) For a single cohort at $\bar\eta$ against the symmetric pair at $\bar\eta \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.

(ii) For the moment-matched two-cohort economies of Theorem 2.6.2, $D = \Theta(\tau^2(\Delta\eta)^6)$.

(iii) For every $K \geq 2$, for the K-cohort Gauss and Gauss–Radau economies of Theorem 2.6.4, $D = \Theta((\Delta\eta)^{4K-2})$.

Hence any test based on T passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$, $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ and $T = \Omega((\Delta\eta)^{-(4K-2)})$ respectively, and the likelihood-ratio test attains these orders as $T \to \infty$: passive data cost the same polynomial order as active probing for every number of cohorts.

The proof (Appendix 2.13) combines the Whittle form of the divergence rate with the resolvent identity $H_1 - H_0 = H_1 H_0 (G_1 - G_0)$: the two spectra differ by the leading unmatched moment, of order δ^2 for the pair and $\tau(\Delta\eta)^3$ for the moment-matched economies, and the divergence rate is the square of that difference. Numerically, the fitted slopes of the required sample length are -4.21 and -5.93, against the theoretical -4 and -6; for part (iii) the fitted slopes of the Whittle divergence rate in $\Delta\eta$ are 5.91, 9.98 and 14.05 for $K = 2, 3, 4$, against 6, 10 and 14 (`hardening/p1_checks.py`).

Passive surveillance is possible in principle but costs the same polynomial order as active probing. Proposition 2.15.2 is the exact finite-sample version for the known pair.

2.7 Robustness to nonlinear and adaptive trading rules

The theorems of Sections 2.3–2.6 use the linear exponential-memory recursion (2.2). Real desks cap positions, trade only beyond thresholds, adapt their gains and learn online. A small probe of size A moves the state by $O(A)$, so any rule whose order is a twice-differentiable function of an exponential-memory state is linear to first order and differs from it by $O(A^2)$; the theorems should then hold for small probes and fail for large ones. We check this, and locate where it fails, in simulations of five stylized rules at $a = 0.5$, $\kappa = 0.8$, total weight $q = 0.2$ and mean rate $\bar\eta = 0.4$, over 240 periods after a pulse probe.

The rules are: (1) *momentum with a position limit*, the cohort's order a smooth saturating (tanh) function of its memory state $y_{i,t}$, normalized to unit slope at the operating point; (2) a *threshold rule* that trades only when $|y_{i,t}| > \tau = 0.05$, in two versions, an exact dead zone and one averaged over background state variation of standard deviation 0.04; (3) an *adaptive-gain*

rule in the form of recursive least squares with forgetting factor $1-\eta_i$, whose gain falls when the regressor is large; (4) an *online neural readout*, a linear readout of a fixed two-layer tanh feature map with eight hidden units, trained by regularized stochastic gradient steps of size η_i; and (5) a *tabular value rule* that updates a 21-bin table at rate η_i, in two versions, kernel-smoothed across bins and a single-bin lookup. These are stylized: each is built so that its small-signal dynamics are an exponential filter with rate η_i, and they are not calibrated to any production system.

Two checks follow the theory. *Test A (merged measure):* one cohort with weight q and position limit 1 against two cohorts with weights $q/2$ at the same rate and limits 0.55 and 1.65, so the two economies have the same merged measure; the raw response gap $\|X^{(1)}(A)-X^{(2)}(A)\|_2$ should be $O(A^2)$ (slope 2 in $\log A$) and the probe-normalised gap $\|X^{(1)}(A)-X^{(2)}(A)\|_2/A$ should be $O(A)$ (slope 1 in $\log A$) over $A \in [0.002, 0.064]$. *Test B (δ^{-4} law):* one cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$, $\delta \in [0.015, 0.12]$, at probe size $A = 0.005$; the normalized response difference $\|\Delta h\|_2$ should scale as δ^2 and the certified capital of Theorem 2.5.2 as δ^{-4}. The saturation threshold A_{sat} is the smallest probe at which $\|\Delta h\|_2$ at $\delta = 0.05$ departs by more than 10% from its small-probe value.

To check that these conclusions do not depend on hand-built rules, we pre-registered a second experiment (`P1_trained_learners.json`, SHA-256 `3e8bf30cbfb6...`) in which online learners are trained from scratch inside the closed-loop market for $T_{\text{burn}} = 5{,}000$ periods of flow and noise innovations; their learned parameters are saved and reloaded into the probing simulator without rescaling. Four learning rules are used: (i) recursive least squares with forgetting factor $\lambda = 0.995$, learning its return-forecast slope; (ii) a two-layer tanh network with eight hidden units whose readout is trained by regularized online stochastic gradient descent on trading profit; (iii) a scalar policy-gradient rule trained on trading profit; and (iv) a tabular Q-learner with 21 states and 21 actions, TD(0) Bellman backups ($\gamma = 0.90$) and Boltzmann action choice. Each learner acts on an exponential-memory state $y_{i,t}$ with rate η_i: training sets its response gain and curvature, not its memory rate, so the experiment asks whether learned policies preserve the merged-measure and δ^{-4} predictions, not whether learning produces exponential memory. The specification fixed four predictions: (a) a single exponential fits each trained learner's small-probe impulse response with $R^2 \geq 0.90$; (b) two trained populations with the same merged measure differ by at most 1% in relative L_2 response at $A = 0.005$; (c) the certified capital to separate trained populations whose memory rates differ by 2δ scales with log–log slope in $[-4.15, -3.85]$; and (d) we report where equivalence and the δ^{-4} law fail when learning continues during the probe. A first implementation of this experiment did not pass the trained parameters to the probing simulator; the results below come from the corrected implementation (`validate_trained_learners_hardened.py`), which reloads the saved parameters.

Table 2.15 and Figure 2.7 show both where the linear reduction holds and where it breaks down. In Panel A, for small probes every smooth or dithered rule behaves like the linear model: economies with the same merged measure have raw response gap $O(A^2)$ (slopes 2.011

Table 2.15: Merged-measure equivalence and the δ^{-4} law under stylized nonlinear rules and closed-loop trained online learners

Rule	Test A raw ($O(A^2)$)	Test A norm. ($O(A)$)	Test B slope ($\|\Delta h\|_2$)	Test B slope (capital)	Capital at $\delta = 0.05$	A_{sat} / R^2_{exp}
Panel A. Stylized nonlinear and adaptive rules (fixed operating point)						
Linear benchmark	—	—	2.007	−4.001	61.42	—
1. Momentum (tanh limit)	2.015	1.015	2.007	−4.001	61.47	0.834
2. Threshold rule, averaged	2.043	1.043	2.007	−4.001	61.42	0.049
3. Adaptive gain (RLS)	2.064	1.064	2.007	−4.001	61.46	0.751
4. Online neural readout	2.011	1.011	2.007	−4.001	61.48	0.751
5. Tabular value, smoothed	2.039	1.039	2.007	−4.001	61.44	1.271
Panel B. Learners trained in the closed loop, saved and reloaded (frozen during the probe; shared training stream)						
6. Recursive least squares	3.000	2.000	2.008	−4.014	54.58	$R^2 = 0.993$
7. Two-layer network (SGD)	1.878	0.878	2.007	−4.012	65.19	$R^2 = 0.994$
8. Policy gradient on profit	3.000	2.000	2.004	−4.006	58.05	$R^2 = 0.995$
9. Tabular Q-learner (TD(0))	2.108	1.108	1.994	−3.973	186,821	$R^2 = 1.000$
Panel C. Where the trained-learner results change (range over the four learners)						
Independent training streams	rel. gap $\leq 10^{-5}$		0.13 to 0.92	−1.75 to −0.38	—	(c) fails
Learning during the probe	rel. gap $\leq 1.1 \times 10^{-3}$		1.99 to 2.01	−4.02 to −3.98	—	(c) holds

Exact dead zone ($\tau = 0.05$): $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe moves the state past τ only for $A > 0.139$.
Single-bin value lookup: $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe leaves the central bin only for $A > 0.149$.

Panel A: deterministic simulations around the operating point; Test A reports the raw response-gap slope in $\log A$ ($O(A^2)$) and the probe-normalised gap slope $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ ($O(A)$, plotted in Figure 2.7). Panel B: learners trained for 5,000 periods, parameters saved and reloaded without rescaling and frozen during the probe; the two compared populations are trained on the same innovation stream; relative equivalence gaps at $A = 0.005$ are at most 10^{-5}. A raw Test A slope above 2 means the gap is of higher order than A^2. Panel C: the same learners trained on independent innovation streams (first row) or updated at learning rate 0.35 during the probe (second row). Capital in par units, certified bound of Theorem 2.5.2 (61.42 for the linear model); the Q-learner settles on a small response gain (0.022), so separating it costs far more capital. Kind: recomputed (Panel A: `recompute_p1_learners.py` → `R_P1_learners.json`; Panels B–C: `validate_trained_learners_hardened.py` → `R_P1_trained_learners_hardened.json`).

to 2.064) and probe-normalised gap $O(A)$ (slopes 1.011 to 1.064), while the response difference and certified capital scale as δ^2 (slope 2.007) and δ^{-4} (slope −4.001), with capital at $\delta = 0.05$ within 0.1% of the linear 61.42. In Panel B, predictions (a)–(c) pass for all four trained learners: a single exponential explains $R^2 \geq 0.993$ of each small-probe impulse response, populations with the same merged measure differ by at most 10^{-5} in relative response at $A = 0.005$, and certified capital scales with slopes −4.014 to −3.973, inside the registered band. The level of capital differs across learners because training sets the response gain.

Panel C shows where this changes. When the two populations are trained on independent innovation streams, prediction (c) fails: the capital slopes are −1.75, −0.97, −1.62 and −0.38. Independently trained populations end with learned gains that differ by sampling error of order $T_{\text{burn}}^{-1/2}$, so they differ in merged weight as well as in memory rate, and a difference in weight is cheaper to detect than an $O(\delta^2)$ difference in rates. The δ^{-4} law therefore prices the separation of memory rates between populations whose feedback weights match; it is not a claim about telling

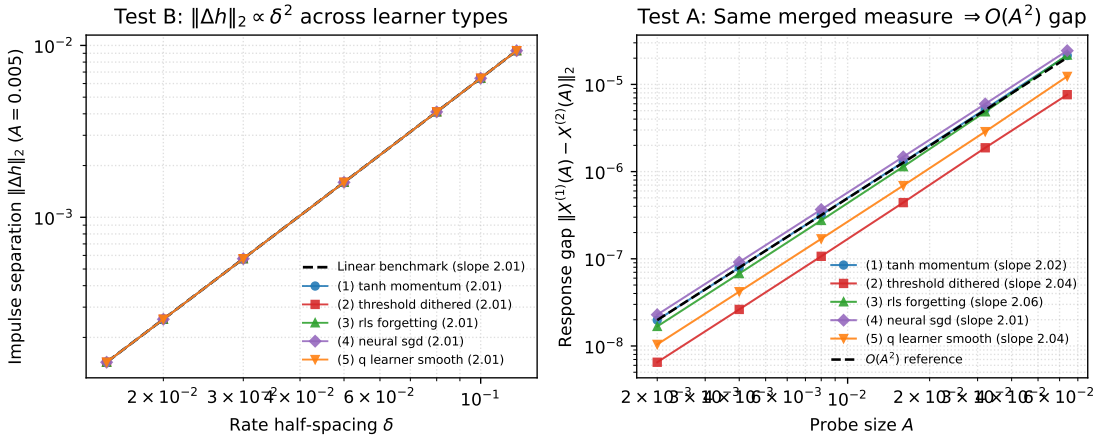


Figure 2.7: Left: Test B, probe-normalised response difference $\|\Delta h\|_2$ against the rate half-spacing δ at probe size $A = 0.005$, five stylized rules and the linear benchmark. Right: Test A, probe-normalised response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ between two economies with the same merged measure against the probe size A, with an $O(A)$ reference line (corresponding to an $O(A^2)$ raw gap).

apart any two trained populations. When learning continues during the probe at rate 0.35, the law survives (slopes -4.02 to -3.98) and the merged-measure gap rises only to 1.1×10^{-3}, so the breakdown we anticipated under prediction (d) did not occur in this design.

The linear theory fails in two further places. First, saturation, adaptive gains and the neural readout leave the small-signal regime once probes exceed $A_{\text{sat}} \in [0.751, 1.271]$ par units, and the dithered threshold rule leaves it already at $A_{\text{sat}} = 0.049$, the threshold itself. Second, rules with hard discontinuities are different in kind: an exact dead zone ($\tau = 0.05$) or a single-bin lookup does not respond at all to probes smaller than 0.139 and 0.149 par units ($\|\Delta h\|_2 = 0$), concealing those cohorts from small probes altogether. Neither smooth nonlinearity nor hard thresholds offers a cheaper route to separating memory rates than the linear benchmark; populations whose learned weights differ are told apart by their weights, not their rates.

2.8 What disclosure buys

Because active probing is costly and passive data are slow, the natural policy question is what a supervisor gains from disclosure. We compare six observation contracts at $\delta = 0.05$.

- I_0: public prices only.
- I_1: prices and aggregated executed flow.
- I_2: prices and one independent noisy report of dealer marked equity per reset ($\sigma_E = 0.5$).
- I_3: prices and exact audited equity (Proposition 2.4.4), also with reports rounded to 0.01.

Table 2.16: Disclosure contracts at $\delta = 0.05$ (pulse-design capital, par units)

Contract	Channel	Pulses	Benchmark	Saving vs. $T = 20$
I_0: prices, $T = 3$	public prices	271	4,305.25	−2,123.5%
I_0^*: prices, $T = 20$	full price path	12	193.62	0.0%
I_1: prices and signed flow	public flow	12	193.62	0.0%
I_2: noisy equity ($\sigma_E = 0.5$)	statement	12	193.62	0.0%
I_3: exact equity, 5-unit lot	audited equity	1	12.70	93.4%
I_3: exact equity, 0.01 rounding	rounded report	1	15.52	92.0%
I_4: timestamps only	no likelihood	12	193.62	0.0%
I_5: audit registry	internal states	0	0.00	100.0%

Closed-form values. Noisy equity adds information linearly in the number of resets, $D_{\mathrm{KL}} = N\Delta_E^2/(2\sigma_E^2)$, and saves only 1.5% against the $T = 3$ baseline. The equity difference is $\Delta E = \kappa^2 q\delta^2 V^2$; with reports rounded to 0.01, a guaranteed change needs $V \geq 5.59$.

- I_4: algorithm update timestamps without the update values.
- I_5: a supervisory audit registry with access to internal algorithm states.

Table 2.16 shows that disclosure is not one thing. Timestamps without update values carry no information about δ in this model. Noisy periodic statements barely help. Exact audited equity collapses the requirement to a single trade, limited only by lot sizes and report precision, and an audit registry removes the trading requirement altogether.

The cleanest form of useful disclosure is flow tagged by participant class.

Proposition 2.8.1 (Tagged flow removes the barrier)**.** *Let the price be driven by i.i.d. innovations of variance σ_ξ^2 in a positive, stable economy, with stationary autocovariances $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^{|s|}$ ($c_\ell \geq 0$, $0 \leq \lambda_\ell < 1$ the closed-loop modes). Suppose the observer receives prices and each cohort's flow $\tilde{u}_{i,t} = u_{i,t} + \varepsilon_{i,t}$, $u_{i,t} = q_i y_{i,t}$, with i.i.d. measurement noise of variance σ_f^2 independent of prices. Then $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with $e_{i,t} = \varepsilon_{i,t} - r_i \varepsilon_{i,t-1}$, and the just-identified IV estimator of $\beta_i = (r_i, w_i)$ with instruments $Z_t = (x_{t-1}, x_t)^\top$ satisfies $\sqrt{T}(\hat{\beta}_i - \beta_i) \to N(0, V_i)$,*

$$V_i = \Sigma_i^{-1} \Omega_i \Sigma_i^{-\top}, \qquad \Sigma_i = \begin{pmatrix} w_i A_0 & \gamma_x(1) \\ w_i A_1 & \gamma_x(0) \end{pmatrix}, \qquad \Omega_i = \sigma_f^2 \big[(1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top) \big], \quad (2.29)$$

where $A_m = \sum_{s \geq 0} r_i^s \gamma_x(s+m)$, $\Gamma_0 = \left(\begin{smallmatrix} \gamma_x(0) & \gamma_x(1) \\ \gamma_x(1) & \gamma_x(0) \end{smallmatrix}\right)$ and $\Gamma_1 = \mathbb{E}[Z_t Z_{t-1}^\top]$. The Jacobian determinant is

$$\det \Sigma_i = w_i \sum_{\ell,m} c_\ell c_m \frac{1 - \lambda_\ell \lambda_m}{1 - r_i \lambda_m} \; > \; 0. \quad (2.30)$$

V_i depends only on cohort i's own (w_i, r_i), σ_f^2 and the price autocovariances, which are bounded and nondegenerate uniformly on a uniformly stable family whatever the rate spacing; so the merged weights are estimated at rate $T^{-1/2}$ with a constant that does not depend on $|r_i - r_j|$.

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about w_i does not involve the other cohorts' rates; the closed form (2.30) matches the simulated Jacobian

(0.1330 against 0.1322 in a two-cohort economy with 4×10^5 periods; `hardening/p1_checks.py`). In simulation ($T = 5{,}000$, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between 5.8×10^{-4} and 6.6×10^{-4}). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem.

2.8.1 How much disclosure is worth

Proposition 2.8.1 says that tagging removes the barrier; the next result says by how much.

Theorem 2.8.2 (Value of class-tagged disclosure)**.** *In the class of Theorem 2.6.5, let N_p be the number of experiments needed with prices alone and N_t the number needed with each class's flow observed with noise σ_f. To recover the weights to fixed accuracy, $N_p/N_t \asymp (\sigma/\sigma_f)^2(\Delta\eta)^{-(4K-2)}$; to resolve the rates to accuracy proportional to their spacing, $N_p/N_t \asymp (\sigma/\sigma_f)^2(\Delta\eta)^{-(4K-4)}$.*

Proof. $N_p \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ by Theorems 2.6.4 and 2.6.5. With tagged flow each class is a single-exponential problem whose Fisher information for (w_i, r_i) does not depend on the other classes, so $N_t \asymp \sigma_f^2$ for the weights and $N_t \asymp \sigma_f^2(\Delta\eta)^{-2}$ for rates resolved to a fraction of $\Delta\eta$. □

With the same noise and lags for both observers, the Cramér–Rao requirements give fitted slopes of -6.08 for prices and -4.34 for the ratio at $K = 2$ (theory -6 and -4). Calibrated to the class memories estimated below, prices alone would need about 333 times as many days as the published class ratios to separate the top-trader and all-account memories in BTCUSDT (rates 0.9889 and 0.9986 per five minutes; about 6,000 days against 18), and about 19 times as many in ETHUSDT (rates 0.9793 and 0.9972). These rates come from day-demeaned regressions, which bias persistence down (Section 2.8.3), so the factors are illustrative.

2.8.2 Disclosure in dollars

Theorem 2.8.2 states the value of class-tagged disclosure in samples. Theorems 2.5.2 and 2.6.4 let us translate that value into a flow of round-trip trading losses and set it beside the official operating budgets of the U.S. Consolidated Audit Trail (CAT). Two adjustments make the comparison exact and like-for-like:

(i) *Exact K-cohort moment-gap constant.* Rather than extrapolating from Cramér–Rao ratios, we evaluate the spectral round-trip trading loss bound of Theorem 2.5.2,

$$L^*_{\mathrm{rt}}(K, \Delta\eta) \;=\; \frac{2\sigma^2 z^2_{1-\alpha}}{\sup_\omega\big(|\Delta\hat{H}_{K,\Delta\eta}(e^{i\omega})|^2/g_0(\omega)\big)},$$

directly for the K-point Gauss versus Gauss–Radau economies of Theorem 2.6.4 and Table 2.11, whose moments below order $2K-1$ coincide and whose $(2K-1)$-th moment gap

Table 2.17: Price-only round-trip probing loss per market and annualised weekly-renewal cost against official CAT budgets

	Loss / probe (USDT)		Annual (52×, USDT)		Mkts ($228.3M / $1.7B)	
Economy ($\Delta\eta$)	Roll	1-sec	Roll	1-sec	Roll	1-sec
$K = 2$, known (0.20)	6,200	149,901	322,400	7,794,852	708.1 / 5,273.0	29.29 / 218.09
$K = 2$, known (0.10)	99,192	2,398,422	5,157,984	124,717,944	44.3 / 329.6	1.83 / 13.63
$K = 2$, known (0.04)	3,874,696	93,688,369	201,484,192	4,871,795,188	1.1 / 8.4	0.05 / 0.35
$K = 2$, Radau (0.20)	4,292	103,788	223,184	5,396,976	1,022.9 / 7,617.0	42.30 / 314.99
$K = 2$, Radau (0.10)	283,488	6,854,612	14,741,376	356,439,824	15.5 / 115.3	0.64 / 4.77
$K = 3$, Radau (0.20)	804,403	19,450,107	41,828,956	1,011,405,564	5.5 / 40.6	0.23 / 1.68
$K = 3$, Radau (0.10)	1,137,325,880	27,500,017,407	59,140,945,760	1,430,000,905,164	0.0 / 0.0	0.00 / 0.00
$K = 4$, Radau (0.20)	156,632,086	3,787,291,895	8,144,868,472	196,939,178,540	0.0 / 0.2	0.00 / 0.01

Separation variable $\Delta\eta$ throughout ($\Delta\eta = 2\delta$ for the symmetric known pair). Gauss–Radau rows use the exact K-cohort quadrature pair of Theorem 2.6.4 and Table 2.11 (moment gaps $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, $\Delta_4 = 0.013061$ at $W = 0.08$, $r_c = 0.60$). Loss per probe is the spectral round-trip trading loss bound of Theorem 2.5.2 at the median BTCUSDT Roll noise (0.40 USDT) and median one-second price-change standard deviation (1.98 USDT); annual loss multiplies by 52 weekly renewals. The final columns divide the revised CAT LLC 2025 operating budget ($228.3M; Nov. 2024 filing $248.8M, SEC Release No. 34-101901, p. 4) and the SEC 2016 annual industry estimate ($1.7B, Release No. 34-79318, p. 363) by the annual probing loss per market. Probing cost is a lower bound on the cost of learning from prices, not a social cost. Kind: recomputed (`recompute_p1_cat_costs.py` → `R_P1_cat_disclosure_dollars.json`).

has the exact constant $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, and $\Delta_4 = 0.013061$ at spacing $\Delta\eta$ (with $\Delta\eta = 2\delta$ for the symmetric known pair).

(ii) *Flow against flow.* A single probe loss L^*_{rt} is a one-time certification cost, whereas an audit trail is an ongoing annual expenditure. Because algorithm populations turn over and regulatory surveillance is periodic, we compare annual CAT operating costs against the annualised round-trip trading loss $52 \times L^*_{\mathrm{rt}}$ under weekly certification renewal (matching the CFTC weekly reporting cycle of Section 2.8.4). We benchmark against three primary official figures: the CAT LLC 2025 operating budget of $248.8 million in the November 2024 filing (SEC Release No. 34-101901, Dec. 12, 2024, p. 4), revised to $228.3 million in May 2025; the SEC's 2016 CAT NMS Plan Approval Order estimate of $2.4 billion in initial industry build costs and $1.7 billion per year in ongoing industry costs (SEC, 2016, Release No. 34-79318, Nov. 15, 2016, 81 FR 84696 at 84801, p. 363); and the SEC's Rule 613 proposing and adopting releases estimating $4.0 billion initial and $2.1 billion annual ongoing costs (SEC, 2012, Release No. 34-62174, May 26, 2010, 75 FR 32556 at 32598, 32600, pp. 147, 153; Release No. 34-67457, July 18, 2012, 77 FR 45722 at 45790–45795, pp. 220–238).

Probing cost is a lower bound on the cost of learning from prices, not a social cost. Within that interpretation, Table 2.17 shows how rapidly the polynomial exponent $(\Delta\eta)^{-(4K-2)}$ and the shrinking Gauss–Radau moment-gap constant Δ_K turn the comparison in favour of disclosure. For a known pair at coarse separation $\Delta\eta = 0.20$ ($\delta = 0.10$), weekly price-only renewal costs $0.32 million per market per year at Roll noise and $7.79 million at one-second noise, so matching

Table 2.18: Memory rates with and without class-tagged data

Symbol	Days	Top position $\hat{r}$ [95% CI]	All accounts $\hat{r}$ [95% CI]	Width ratio	Classes differ
BTCUSDT	48	0.9889 [0.9842, 0.9935]	0.9986 [0.9955, 1.0005]	121.7	yes
ETHUSDT	48	0.9793 [0.9616, 0.9907]	0.9972 [0.9934, 0.9997]	55.6	yes
CELRUSDT	48	0.9884 [0.9830, 0.9922]	0.9882 [0.9804, 0.9914]	81.6	no
CVXUSDT	33	0.9822 [0.9732, 0.9908]	0.9829 [0.9694, 0.9900]	45.9	no

Pre-registered test on fresh days, 30 USD-M perpetuals, 2025–2026

(i) width ratio > 10	30 of 30 (range 27.6–124.0)
(ii) price-only rate interval wider than 0.25	30 of 30 (range 0.65–0.99)
(iii) merged response $\sum_m k_m$ identified	5 of 30
Classes differ	11 of 30

Top panel: training days 2023–2024; width ratio is the mean price-only rate-interval width over the mean tagged width. The price-only intervals for both rates cover most of $[0, 0.99]$. Bottom panel: specification registered before the fresh days were downloaded (SHA-256 `65acda361861...`); a symbol passes if all three of the first rows hold, and the test required 80% of symbols to pass.

the \$228.3 million CAT LLC operating budget requires 708.1 or 29.29 markets. At $\Delta\eta = 0.10$ ($\delta = 0.05$) the unknown $K = 2$ Gauss–Radau pair costs \$14.74 million (Roll) or \$356.44 million (one-second) per market per year, matching the CAT LLC budget in 15.5 or 0.64 markets and the SEC's \$1.7 billion industry figure in 115.3 or 4.77 markets. Once a supervisor must separate $K = 3$ unknown cohorts at $\Delta\eta = 0.20$, annual probing loss rises to \$41.83 million (Roll) and \$1.01 billion (one-second) per market (5.5 and 0.23 markets at the CAT LLC budget; 40.6 and 1.68 markets at \$1.7 billion), and at $K = 3, \Delta\eta = 0.10$ or $K = 4, \Delta\eta = 0.20$ a single market's annual probing loss exceeds both the CAT LLC operating budget and the SEC's total annual industry estimate. The renewal frequency is an assumption: break-even market counts scale inversely with it, so monthly renewal multiplies every count in the last two columns by 4.3 and renewal on each of 252 trading days divides them by 4.8.

2.8.3 Disclosure in real data

Binance publishes, every five minutes, long/short ratios for its top-trader group, by position and by account, and for all accounts. These are class-level positions, the closest public analogue to the tagged data of Proposition 2.8.1. We treat a class's reported net position as the observable counterpart of its cohort state: it obeys the same one-lag recursion class by class, so the block-diagonal information structure of Proposition 2.8.1 carries over. The mapping is an assumption, because the reports are positions rather than flows and returns stand in for the unobserved deviation x_t. For each class c we estimate $n_{c,t} = r_c n_{c,t-1} + w_c \,\mathrm{ret}_t$, with $n = (R-1)/(R+1)$ from the ratio R. The price-only alternative regresses signed five-minute taker flow on twelve lagged five-minute returns and fits two exponentials to the kernel; it estimates the memory rates without class information. Intervals come from a day bootstrap with 200 draws.

Table 2.18 gives three results. First, tagged class data restrict persistence rates that prices

and flow leave essentially unrestricted: with the one-hour price-only kernel, the price-only intervals are 28 to 124 times wider in every one of 30 symbols on fresh days, and cover most of the admissible range. Part of this ratio is mechanical, because a series of positions identifies its own persistence directly while the price-only route must fit two exponentials to a weakly identified kernel. Restricting attention to the five symbols in which the price-only fit does identify the merged response (ACH, API3, OGN, ZEN and DASH) does not change the conclusion: the price-only rate intervals are still 0.64 to 0.92 wide and the ratios run from 27.6 to 62.5 (median 47.5). We therefore read the ratios as a measure of how little prices reveal about class memory, not as a horse race between two estimators of equal standing. The ratios also overstate the gap for a second reason: the price-only kernel spans one hour (twelve five-minute lags), while the tagged half-lives are several hours, so the price-only fit cannot see memories of that length. An exploratory check on the 30 symbols and 105 days of the follow-up test refits the price-only kernel with 48 and 96 lags (four and eight hours) and compares both routes in one unit, the half-life. With 96 lags the price-only slow half-life (median 5.6 hours) reaches the tagged scale (median 4.2 hours). The tagged intervals remain narrower in all 30 symbols, but by a median factor of 7.5 in log half-life rather than 28 to 124, and the price-only interval still reaches a bound of the fit in 25 of 30. The horizon-matched comparison was then registered as a primary hypothesis of a test on 30 symbols never used in this project, before its data were downloaded, and passed: the tagged interval was narrower in 29 of 29 evaluable symbols (median factor 7.3, smallest 2.6), and the price-only interval reached a bound of the fit in 26. A registered secondary of the same test qualifies what the tagged estimate measures. Day-demeaned regressions bias persistence down; corrected by a split-panel jackknife, the median tagged rate is 1.0008 per five minutes, and the corrected interval lies below one in only 7 of 87 class fits. Within a day the class ratios are close to a random walk, so the half-lives of a few hours reported here are largely produced by removing each day's mean. The evidence therefore supports a narrower claim than identification of class memory: positioning data restrict a persistence parameter that prices cannot bound. Whether that persistence reflects learning memory or slow turnover of positions, which theory's tagged contract (class-tagged flow) would distinguish, is not established by these data. Second, the classes differ where theory says the information should be: in BTCUSDT and ETHUSDT the top-trader group's positioning has a half-life of about three to five hours against about one to two days for all accounts, while in thin symbols the classes are often indistinguishable. Third, the pre-registered test fails its 80% criterion, on the part of the prediction that concerns what prices *do* reveal: the merged response is significantly different from zero in only 5 of 30 symbols. Theorem 2.6.9 says the merged measure is identified; on 42 days of thin-market data it is not estimated precisely. This is a failure of the prediction as registered, and we report it as one. A post hoc reading, not tested here, is that identification is an asymptotic statement and 42 days of passive thin-market data are far from that regime. Two follow-ups were registered before their data were downloaded, and each ran once. On 105 untouched days in the same 30 markets,

price-only intervals were again at least ten times wider than class-tagged ones in all 30 symbols (median ratio 74.9 with the one-hour kernel), but the merged response was significantly different from zero in only 9 symbols, and pinning it down was at least ten times cheaper than separating the memory rates in only 10 of 30 (23 were required): that prediction failed. On 58 further days, the advantage of tagging grew with the sample in 22 of 29 symbols, the number required, but the tagged intervals shrank more slowly than the parametric rate (20 of 29 in the predicted range, 22 required), consistent with drift in class persistence over 21 months. Both results are reported in full in the companion thesis.

To test directly whether the memory rates identified by class-tagged positions improve out-of-sample return predictions after large order-flow shocks, we pre-registered a dedicated out-of-sample test (`P1_disclosure_oos.json`, SHA-256 `f0182ce50978...`) before downloading the 22nd-of-month data for January 2025–September 2026 across 19 thin perpetuals (6,137 top-decile five-minute flow shocks on 2024 training days; 4,579 shocks on untouched 2025–2026 test days). On the training sample we estimate each symbol's two memory rates $\hat{r}_1, \hat{r}_2$ either from the class-tagged positions or from the price-only two-exponential fit, construct the filtered memory states $y_{k,t} = \hat{r}_k y_{k,t-1} + (1 - \hat{r}_k)\mathrm{ret}_t$, and forecast the 30-minute post-shock cumulative return $R_{t+1\to t+6}$ out of sample relative to an AR(1) return-and-flow baseline. The primary hypothesis fails: on the untouched test shocks, the price-only memory states yield $\Delta R^2 = -0.199$ percentage points relative to the baseline, the class-tagged memory states yield $\Delta R^2 = -1.288$ points, and the tagged-minus-price-only difference is $\Delta R^2 = -1.089$ points (95% date-block bootstrap CI $[-2.605, +0.376]$, Clark–West $t = 0.96$, $p = 0.1684$). Likewise, adding the level and one-hour change of the top-trader and all-account positions to the hourly tail-risk controls of Chapter 3 leaves out-of-sample R^2 unchanged (-0.19 points, CI $[-0.99, 0.83]$).

After this test failed, we registered a follow-up (`P1_disclosure_state_oos.json`, SHA-256 `5136cc6b846d...`) on untouched 2nd-and-9th-of-month data for January 2025–September 2026 across the same 19 thin perpetuals (12,046 top-decile five-minute flow shocks on 24 training dates in 2024; 9,875 shocks on 42 test dates; test opened once). It asks the narrower question that Theorem 2.3.1 and Proposition 2.8.1 answer directly: can prices and flow track the class states themselves? A two-exponential price-and-flow filter ($\alpha_{\text{fast}} = 0.50$, $\alpha_{\text{slow}} = 0.10$ on five-minute returns and signed flow, plus shock return, shock flow and symbol fixed effects) is compared with a filter that adds the current class states and their one-bar changes. Both registered primaries pass by a wide margin (Table 2.19): for the 30-minute-ahead divergence between top-trader and all-account positions, out-of-sample R^2 is -4.8% with prices and flow and 46.0% with class data; for the top-trader state it is 4.4% and 54.8%.

The gain is almost entirely persistence. In an exploratory decomposition on the same test shocks (`hardened_empirical_audits.py` → `R_hardened_empirical_audits.json`), an AR(1) on the current level alone accounts for 100.5% and 99.1% of the improvement, and when the target is the 30-minute *change* in positions the class data add -0.26 and $+1.41$ points. Because

Table 2.19: Pre-registered out-of-sample cohort-state trajectory test on untouched 2nd-and-9th-of-month days (`P1_disclosure_state_oos`), 2025–2026

30-min post-shock target (19 thin)	R^2_{pf}	R^2_{tag}	ΔR^2 (pts) [95% CI]	CW t (p)	Res.
Prim. 1: Divergence d_{t+6} (9,875 shocks)	−4.766	45.962	+50.728 [+45.291, +55.918]	21.27 ($< 10^{-16}$)	pass
Prim. 2: Top state $n_{\text{top},t+6}$ (9,875 shocks)	4.441	54.845	+50.404 [+47.226, +53.724]	21.85 ($< 10^{-16}$)	pass
Sec.: Divergence d_{t+6}, all (210,144 bars)	−0.586	37.053	+37.639 [+36.232, +39.107]	72.75 ($< 10^{-16}$)	—
Sec.: Top state $n_{\text{top},t+6}$, all (210,144 bars)	0.186	46.779	+46.593 [+45.555, +47.643]	80.44 ($< 10^{-16}$)	—

Out-of-sample R^2 in percent. Trained on 24 dates in 2024 (12,046 shocks, 120,384 bars); evaluated once on 42 untouched test dates (2nd and 9th of each month, Jan 2025–Sep 2026); 2,000-draw date-block bootstrap 95% CIs; Holm $p < 10^{-16}$ across both primaries. Spec SHA-256 `5136cc6b846d...` (`R_P1_disclosure_state_oos.json`).

the class filter observes the series it forecasts, this is a weak test: it confirms that prices and flow cannot reconstruct where the classes are positioned, which Theorem 2.3.1 implies, but it does not show that class data reveal how their positions will move, and neither registered test finds that class data improve return forecasts.

2.8.4 A second disclosure regime: the CFTC's trader-class reports

Binance publishes its class ratios voluntarily, for one venue. The U.S. Commodity Futures Trading Commission requires large traders in regulated futures markets to report their positions and publishes them weekly by class: leveraged funds, asset managers and dealers in financial futures, and managed money, producer/merchants and swap dealers in commodities. This is class-tagged disclosure mandated by law. We registered a test (SHA-256 `6f1d6f35ae88...`) before downloading any data, in eight markets that span asset classes: CME Bitcoin and Ether, the E-mini S&P 500, 10-year Treasury notes, the euro, the yen, WTI crude oil and Henry Hub natural gas, using every weekly report through October 2026 (282 to 1,039 weeks per market). For each class, $n_{c,t} = (\text{long} - \text{short})/\text{open interest}$ follows $n_{c,t} = \alpha_c + r_c n_{c,t-1} + w_c \,\text{ret}_t$, the weekly version of the cohort recursion. The price-only alternative fits a two-exponential kernel to twelve weekly return lags. Intervals come from a 13-week moving-block bootstrap. The registered hypotheses were: (H1, primary) the fast class's memory rate is pinned to an interval narrower than 0.25 while both price-only intervals are wider, in at least 75% of markets; (H2) the fast class trend-follows, $w_c > 0$, in at least 75%; (H3) the fast class has the shorter memory in at least half.

The primary hypothesis holds in all eight markets (Table 2.20). The class reports pin each fast class's memory rate to an interval between 0.013 and 0.106 wide, while weekly prices leave both memory rates anywhere in the admissible range; the width ratios run from 9 to 71. The ratio partly reflects how little weekly returns reveal, which is the point of the theory rather than an artefact of it: with prices alone the memories are not located at all. The classes differ as the cohort model allows. Leveraged funds and managed money have the shorter memory in six of eight markets, with half-lives of 6 to 18 weeks against months to years for asset managers and

Table 2.20: Trader-class memory from the CFTC Commitments of Traders, eight regulated futures markets

Market	Weeks	Fast $\hat{r}$ [95% CI]	Slow $\hat{r}$	$\hat{r}_{\text{fast}} - \hat{r}_{\text{slow}}$ [95% CI]	$\hat{w}_{\text{fast}}$	Ratio
Bitcoin (CME)	434	0.951 [0.928, 0.969]	0.986	−0.035 [−0.062, −0.010]	−	23.7
Ether (CME)	282	0.890 [0.826, 0.932]	0.960	−0.070 [−0.143, −0.011]	−	9.1
E-mini S&P 500	1,039	0.914 [0.884, 0.937]	0.954	−0.040 [−0.071, −0.012]	−	18.3
10-year T-note	1,039	0.992 [0.985, 0.998]	0.993	−0.000 [−0.006, +0.007]	(0)	71.4
Euro	1,035	0.961 [0.947, 0.972]	0.996	−0.034 [−0.051, −0.022]	+	37.3
Yen	1,035	0.955 [0.939, 0.968]	0.984	−0.029 [−0.047, −0.013]	+	34.0
WTI crude oil	1,039	0.961 [0.946, 0.974]	0.999	−0.037 [−0.053, −0.023]	+	35.2
Natural gas	1,039	0.981 [0.969, 0.990]	0.980	+0.001 [−0.014, +0.017]	+	44.0

Pre-registered hypotheses: H1 (primary) 8 of 8, pass; H2 4 of 8, fail; H3 6 of 8, pass.

Fast class: leveraged funds (financial futures) or managed money (commodities); slow class: asset managers or producer/merchants. $\hat{w}_{\text{fast}}$: sign of the response to returns, + or − when the 95% interval excludes zero, (0) otherwise. Ratio: mean width of the two price-only rate intervals over the width of the fast-class interval; the price-only intervals span 0.95–0.96 of the admissible range $[0, 0.99]$ in every market. Weekly returns from Binance spot (Bitcoin, Ether) and Yahoo Finance (others; the Treasury return is minus the change in the 10-year yield). Specification SHA-256 `6f1d6f35ae88...`; test opened once. Kind: real data (Part 11).

producers; in Treasuries and natural gas the two memories are indistinguishable.

The trend-following hypothesis fails. The fast class buys after rises in the euro, the yen, crude oil and natural gas, but sells after rises in Bitcoin, Ether and the S&P 500 futures. The first half matches the finding of Moskowitz et al. (2012) that speculators' positions tend to go with past returns. We did not predict this. One candidate explanation is that leveraged-fund positions in financial futures are dominated by the short futures leg of cash–futures basis trades, which a single memory per class cannot represent. A post-hoc check is consistent with it but not decisive (Part 13, `R6`): weekly position changes of leveraged funds and asset managers are strongly negatively correlated in Bitcoin, Ether, the S&P 500 and Treasuries (-0.37 to -0.67) but not in the euro or yen (-0.04 and -0.05), and holding the asset managers' change fixed removes the negative response in Bitcoin and halves it in Ether and the S&P 500. Because class positions sum to zero across all reporting categories, some negative correlation is mechanical, so this is a signature, not a test. The failure sharpens rather than weakens the case for class tagging: class labels are not strategies, and a supervisor who wants to know how a class responds to prices needs its positions, which the price process does not reveal.

Together with Table 2.18, the evidence now covers two disclosure regimes, one voluntary and one mandated by law, in crypto and in regulated equity-index, rate, currency and commodity futures. In both, class-tagged positions restrict persistence rates that prices leave unrestricted; whether that persistence is learning memory or slow turnover of positions, these data do not say.

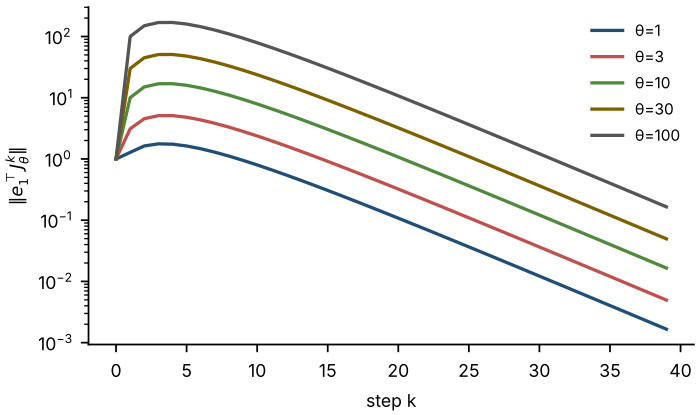


Figure 2.8: Price-row transient gain of the spectral-blindness family J_θ: eigenvalues $(0.8, 0.7)$, one-step price variance 1 and stationary price variance 5 are the same for every θ.

2.9 A spectral blind spot

Even without identification, could a supervisor monitor risk through eigenvalues or volatility?

Theorem 2.9.1 (Spectral blindness)**.** *For any eigenvalues $0 < \lambda_2 < \lambda_1 < 1$, one-step price variance $s_1 > 0$ and stationary price variance V large enough, there is a family of two-state markets $J_\theta = \left(\begin{smallmatrix} \lambda_1 & \theta \\ 0 & \lambda_2 \end{smallmatrix}\right)$ with shock covariance $\mathrm{diag}(s_1, s_2(\theta))$ such that all members share the eigenvalues, the one-step price variance and the stationary price variance, while the worst-case transient gain $\sup_k \|e_1^\top J_\theta^k\|$ grows without bound in θ.*

At $\lambda = (0.8, 0.7)$, $s_1 = 1$ and $V = 5$, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 for $\theta = 1, 3, 10, 30, 100$, with stationary variance 5.000 in every case; the cumulative gain rises from 18.3 to 1,667.7. No monitor that depends only on eigenvalues and second moments of the price can be uniformly safe (Figure 2.8).

Two clarifications prevent over-reading. First, the cohort economy of Section 2.2 is itself non-normal, yet its price output does not amplify (Lemma 2.2.3). Large weighted gains can be manufactured by output weights alone: with $C_w = (1, 160)$, a price-and-memory functional peaks at 160.003 at $t = 0$, which is the static weight, not dynamic growth (Appendix 2.15.5). Second, in a network of M identical cohort economies coupled by symmetric nonnegative cross-impact, the coupled operator stays symmetric in the coordinates of Lemma 2.2.3; with cross-impact 0.05, the spectral radius rises from 0.822 ($M = 1$) to 0.973 ($M = 10$) while the Kreiss constant stays at one, as it must for a stable symmetric operator (`recompute_oct20.py`). Non-normality is necessary for transient amplification of the price, not sufficient; where amplification comes from is the subject of Chapter 3.

2.10 Empirical feedback and its measurement

The theory says what prices can reveal. The empirical work has two roles, and neither is a test of the identification bounds themselves, which concern controlled experiments no outside researcher can run. First, Section 2.8.3 tests the theory's policy prediction directly: that class-tagged data pin down memories that prices leave unrestricted. Second, this section asks what the reduced-form feedback indices a supervisor would compute from public data actually measure, and how much of them is feedback rather than filtering, timing or trading activity.

2.10.1 Estimand and filtering mechanics

In theory $x_t = p_t - p_t^*$ is the deviation from an unobserved fundamental. Empirical work replaces p_t^* by a trailing filter. Let p_b be a log price proxy on uniform bars of length Δ and $m_b = (1-\alpha)m_{b-1} + \alpha p_b$ an exponential moving average with $\alpha = 2/(W/\Delta + 1)$. The empirical state is

$$x_b^{\mathrm{EMA}} \equiv p_b - m_b = (1-\alpha)(p_b - m_{b-1}) = \frac{1-\alpha}{1-(1-\alpha)L}\Delta p_b. \tag{2.31}$$

Because $x_{b+1}^{\mathrm{EMA}} = (1-\alpha)(p_{b+1} - m_b)$, a martingale price gives

$$\mathbb{E}[x_{b+1}^{\mathrm{EMA}} \mid \mathcal{F}_b] = (1-\alpha)x_b^{\mathrm{EMA}} : \tag{2.32}$$

the filtered state is an AR(1) with mechanical persistence $a_0 = 1-\alpha$, which equals $29/31 \approx 0.9355$ for $W = 300$ seconds and $\Delta = 10$ seconds.

Let f_t be signed flow. The empirical VAR(1)

$$\begin{pmatrix} x_{t+1}^{\mathrm{EMA}} \\ f_{t+1} \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix} \begin{pmatrix} x_t^{\mathrm{EMA}} \\ f_t \end{pmatrix} + e_{t+1} \tag{2.33}$$

has reduced-form coefficients, not authenticated impact or learner parameters. When denominators are nonzero we use the scale-invariant indices

$$\hat{Q} = \frac{\hat{J}_{12}\hat{J}_{21}}{1-\hat{J}_{22}}, \qquad \hat{S} = \hat{J}_{11} + \hat{Q}, \qquad \hat{L} = \frac{\hat{Q}}{1-\hat{J}_{11}}. \tag{2.34}$$

Equation (2.32) implies that $\hat{J}_{11}$ must be compared with $1-\alpha$, not with zero: persistence that a naive reading attributes to feedback can be pure filtering.

2.10.2 Data

2.10.3 Clock alignment dominates reduced-form feedback

We estimate (2.33) on ten-second bars built from one-second Binance klines across a 96-date, eight-pair spot panel (the 10th and 24th of each month, 2021–2024) under three constructions

Table 2.21: Samples

Sample	Coverage	Size	Source
Binance spot panel, 8 pairs	96 dates, 2021–2024	66.3M one-second bars	Binance archive
Binance spot fee-event windows, 8 pairs	±20 days around 2022-07-08 and 2023-03-22	640 pair-days	Binance archive
Binance USD-M perpetuals, 30 symbols	90 dates, 2023–2026	2,565 symbol-days	Binance archive
Binance BTCUSDT one-second panel	72 dates, 2021–2023	6.2M one-second bars	Binance archive
Binance BTCUSDT noise-calibration days	January 10–20, 2023	11 days	Binance archive

Binance files are verified against the published SHA-256 checksums (every file used). Sampled dates are the 10th and 24th of each month.

Table 2.22: Multi-year Binance spot panel: reduced-form indices under three timing constructions

Construction	$\hat{Q}$	Date interval	$\hat{S}$	Date interval	$\hat{Q} > 0$
Forward-paired flow (look-ahead)	0.0081424	[0.0072965, 0.0089882]	0.92074	[0.91665, 0.92482]	87.4%
Observed-flow EMA	0.0003733	[0.0002351, 0.0005116]	0.92072	[0.91640, 0.92504]	62.4%
Observed, 30 flow lags	0.0002580	[0.0001415, 0.0003745]	0.91738	[0.91167, 0.92308]	59.1%
Observed raw return	−0.0001555	[−0.0003732, 0.0000622]	−0.04051	[−0.05078, −0.03025]	49.2%
Placebo random walk	0.0000016	[−0.0000020, 0.0000052]	0.93510	[0.93485, 0.93534]	51.0%

768 study units (96 dates × 8 pairs), 6,634,752 ten-second transitions; equal-weight asset means within date, then date means with Student intervals. Mechanical persistence $29/31 = 0.93548$.

(Table 2.22), and run the identical pipeline on a placebo random walk with each pair-day's volatility. Pairing the state with *next-bar* flow (a look-ahead) gives $\hat{Q} = 0.0081$; aligning flow with the bar in which it is observed lowers this by more than an order of magnitude, to 0.00037; and replacing the EMA state by raw returns gives an interval that includes zero. Mean $\hat{J}_{11}$ under observed flow is 0.9203, below the mechanical 0.9355, and the placebo reproduces the mechanical value (0.9351) with $\hat{Q}$ indistinguishable from zero.

2.10.4 Feedback beyond filtering mechanics

We re-estimate feedback on a separate sample and add a placebo. We use Binance BTCUSDT one-second klines on 72 sampled days in 2021–2023. The state x_t is the log price deviation from its EMA (span 30 one-second bins), f_t is next-second signed taker flow, and a VAR(1) gives $\hat{Q}$. Because the filter alone produces persistence, we run the identical pipeline on a placebo: a random walk with the same one-second volatility, independent of the real flow.

Table 2.24 gives two findings. Feedback, in the sense of a loop gain the filter cannot produce, is detectable: the real $\hat{Q}$ lies far outside the placebo band. It is also small, and the price persistence that a naive reading would attribute to feedback is almost entirely mechanical (0.9364 against 0.9355). Because this construction pairs the state with next-second flow, part of $\hat{Q}$

Table 2.23: Asset means under forward-paired and observed flow

Pair	Forward-paired $\hat{Q}$	Observed $\hat{Q}$	$\hat{Q}_{30}$	Observed $\hat{S}$
BTC/USDT	0.0141870	0.0006073	0.0004891	0.93464
ETH/USDT	0.0130885	0.0005675	0.0002549	0.93511
SOL/USDT	0.0087763	0.0002129	0.0001100	0.91966
ADA/USDT	0.0067465	0.0003928	0.0003546	0.91452
DOGE/USDT	0.0058956	0.0002464	0.0001432	0.92496
BNB/USDT	0.0057420	0.0004479	0.0002762	0.89533
XRP/USDT	0.0057034	0.0002249	0.0001508	0.92362
LTC/USDT	0.0049997	0.0002870	0.0002853	0.91792

96 dates per pair. $\hat{Q}_{30}$ controls 30 ten-second bars of lagged flow. Placebo $|\hat{Q}| \leq 1.1 \times 10^{-5}$ for every pair.

Table 2.24: Feedback beyond filtering mechanics (Binance BTCUSDT, 72 days, 2021–2023)

	Real prices	Placebo (independent random walk)
Mean loop gain $\hat{Q}$	0.0045	0.0000009
95% CI (day bootstrap)	[0.0034, 0.0056]	[−0.0000006, 0.0000023]
Mean price persistence $\hat{J}_{11}$	0.9364	0.9357
Mechanical value $1 - 2/31$		0.9355

Placebo 95% band across days: $[-0.0000095, 0.0000156]$. This construction pairs the state with next-second flow; it is not comparable in level with the observed-flow ten-second index of Table 2.22.

reflects the timing effect of Table 2.22; the placebo shows only that the filter does not produce it.

2.10.5 Order splitting and the estimation of feedback

Persistent order flow makes lagged-flow regressions inconsistent. Suppose flow follows $q_t = \gamma y_t + u_t$ with memory $y_t = (1-\eta)y_{t-1} + \eta \Delta p_{t-1}$, order splitting $u_t = \rho u_{t-1} + \nu_t$, and $\Delta p_t = \kappa q_t + \lambda z_t + e_t$ with an observed exogenous price shock z_t. Lagged returns are correlated with u_{t-1}, so OLS of flow on returns is biased even when $\gamma = 0$. Quasi-differencing ($q_t - \rho q_{t-1} = \gamma(y_t - \rho y_{t-1}) + \nu_t$) or instruments that move y through prices but are independent of u ($z_{t-1}, z_{t-2}, z_{t-3}$) restore consistency. We compare estimators over 500 replications with $\eta = 0.3$ and $\rho = 0.8$ (Table 2.25).

OLS is badly biased. Cochrane–Orcutt with the true ρ, 2SLS and LIML have correct size and coverage; Cochrane–Orcutt with an estimated ρ under-covers (Cochrane and Orcutt, 1949; Staiger and Stock, 1997). These are properties of estimators in a simulated market. On real data, a 2024 sub-panel of 192 spot units shows why the correction matters little at our bar length: with one VAR lag, the flow-equation residuals have essentially no first-order serial correlation (median $\hat{\rho} = -0.0002$), and Cochrane–Orcutt quasi-differencing moves the mean loop gain from 3.15×10^{-4} to 3.21×10^{-4} (Part 13, `R7`, post hoc). Order splitting shows up instead at longer lags, which the 30-lag control of Table 2.23 addresses. A real-data instrumental-variables estimate needs an instrument whose exclusion can be defended, which we do not claim to have; funding-

Table 2.25: Feedback estimators under order splitting (simulation, 500 replications)

Estimator	Coverage at $\gamma = 0.2$	Rejection at $\gamma = 0$	Bias	RMSE
OLS	0.000	1.000	0.892	0.893
Cochrane–Orcutt, true ρ	0.944	0.054	0.001	0.039
Cochrane–Orcutt, estimated ρ	0.916	0.080	0.002	0.043
2SLS	0.938	0.044	−0.002	0.116
LIML	0.938	0.044	−0.003	0.117

Newey–West standard errors (10 lags), $T = 3{,}000$. Median first-stage $F = 224$; a placebo instrument (future z) has $F = 0.62$. Estimators outside 93–97% coverage give invalid inference.

Table 2.26: Zero fees on BTCUSDT: activity and measured feedback

	Zero fees introduced (2022-07-08)	Zero fees removed (2023-03-22)	Pooled
Log quote volume, $\hat{\beta}$	0.945 (0.074)	1.599 (0.080)	1.272 (0.084)
Observed-flow $\hat{Q}$, $\hat{\beta}$	0.00058 (0.00032)	−0.00091 (0.00060)	−0.00016 (0.00036)
p-value	0.078	0.140	0.650
Pair-days	320	320	640

$\hat{\beta}$ is the effect of zero fees on BTCUSDT relative to seven control pairs; date-clustered standard errors in parentheses. For the second event ZeroFee equals one before March 22, 2023.

settlement times, for example, may affect order flow directly.

Controlling for 30 ten-second bars of lagged flow lowers $\hat{Q}$ further (Table 2.23, column $\hat{Q}_{30}$), consistent with persistent order splitting.

2.10.6 Feedback across 30 perpetual markets

Chapter 3 estimates the same loop gain on one-minute bars of 30 Binance USD-M perpetuals spread across the liquidity ranking (2023–2024 training days). Measured feedback rises with depth: its Spearman correlation with log depth within 1% of the mid is +0.70 [0.41, 0.87], from about 0.012 in BTCUSDT and ETHUSDT to 0.0016 on average in the thinnest third, and the placebo pipeline gives $\hat{Q} \approx 0$ for every symbol. Feedback in this reduced-form sense is a feature of deep, algorithmically traded books.

2.10.7 Two fee events

Binance set spot trading fees on BTC pairs to zero on July 8, 2022 and reinstated them on March 22, 2023. Each event changes the cost of trading BTCUSDT while leaving the seven other pairs unchanged, and the second reverses the first. For every pair and every day within 20 days of each event we compute daily quote volume and the observed-flow $\hat{Q}$, and estimate $Y_{it} = \alpha_i + \gamma_t + \beta\,\text{ZeroFee}_{it} + e_{it}$ with pair and date fixed effects and date-clustered errors, separately for each event and pooled (Table 2.26).

The fee change moved activity enormously: zero fees multiplied BTCUSDT volume relative to the controls by $e^{0.945} \approx 2.6$ when they began, and their removal divided it by $e^{1.599} \approx 4.9$.

Measured feedback did not follow. The two point estimates have opposite signs, neither is significant, and the pooled effect is -0.00016 with an interval of $[-0.00087, 0.00055]$, against a sample mean of 0.00037. A strong first stage with no second-stage response says that $\hat{Q}$ is not a by-product of trading activity, and the reversal design removes the main weakness of a single treated event. With one treated asset the interval remains wide, so this is evidence that $\hat{Q}$ does not scale with activity, not a proof of invariance.

2.11 Implications for supervision and market design

What a supervisor can learn from public data. Prices and anonymous flow reveal the merged measure M, the closed-loop impulse response and the reduced-form operator, and nothing finer. A supervisor who wants to know *which* algorithms drive a feedback loop cannot get there from public data at reasonable cost when their memories are close. This is not a failure of econometric technique: Theorem 2.6.2 holds for every estimator, biased or not.

What probing costs. At realistic observation noise, an exchange or a large desk could separate two well-separated memory rates for about 10^5 USDT of capital. Probing is therefore feasible for coarse questions and infeasible for fine ones. Because capital grows as δ^{-4} and experiments as $(\Delta\eta)^{-(4K-2)}$, halving the separation multiplies the cost by 16 for a known pair and by 2^{4K-2} for K unknown cohorts.

What disclosure buys. Disclosure regimes differ sharply in value (Table 2.16). Timestamps without update values are worthless for this purpose. Noisy periodic statements are nearly worthless. Audited equity and, above all, flow tagged by participant class remove the barrier entirely (Proposition 2.8.1). This gives a quantitative argument for audit-trail regimes that record the participant class of each order, and a quantitative argument against relying on periodic statements for the same purpose. Calibrated to Binance's published class ratios, separating the memories of the top-trader group and of all accounts in BTCUSDT would take prices alone about 333 times as long (Theorem 2.8.2). The class reports that the CFTC already requires have the same property in regulated markets (Table 2.20), so the argument concerns the granularity and frequency of existing disclosure, not a new kind of reporting.

When audit trails already exist. The US Consolidated Audit Trail (SEC Rule 613) gives the SEC and the self-regulatory organizations a record of every order in NMS stocks and listed options from origination to execution or cancellation, linked to an anonymized customer identifier and to the broker-dealer that handled it; since 2025 it no longer collects the names, addresses and years of birth of natural persons (SEC, 2012, 2016, 2024, 2025). Our results bear on such regimes in three ways. First, they price what the record is worth for the question of who drives a feedback loop: participant-level records remove a barrier that otherwise costs round-trip trading loss of

order δ^{-4} and samples of order $(\Delta\eta)^{-(4K-4)}$ (Proposition 2.8.1, Theorem 2.8.2); set against the CAT LLC 2025 operating budget ($248.8 million in November 2024, revised to $228.3 million in May 2025) and the SEC's 2016 annual industry estimate ($1.7 billion), the annualised weekly-renewal probing loss that disclosure makes unnecessary is small for two well-separated classes and exceeds the entire CAT budget once $K \geq 3$ at $\Delta\eta \leq 0.10$ or $K = 4$ at $\Delta\eta = 0.20$ (Section 2.8.2, Table 2.17). For this purpose a stable account identifier is what matters and names are not needed, which is a quantitative reason to keep the account linkage while removing personal data. Second, an audit trail records accounts, not decision rules. One account can run several algorithms and one algorithm can trade through several accounts, so an observer with account-level records who wants rule-level memories faces our problem within each account's flow, with fewer cohorts but the same exponent; MiFID II's algorithm identifier (ESMA, 2016) is the closer analogue of the tagged flow in Proposition 2.8.1. Third, the audit trail does not reach the markets we study. Crypto perpetual futures are outside it, futures regulated by the CFTC have separate reporting, and its data are not available to exchanges, other participants or researchers. For these observers the price-only and coarse-disclosure contracts of this chapter are the relevant ones.

What to monitor instead. Because transient loss can be arbitrarily large at fixed eigenvalues and volatility (Theorem 2.9.1), supervision that wants to control risk rather than identify participants should monitor quantities that bound transient loss directly: the summed absolute impulse response and operator confidence sets, both of which are identified from public data. These are cheap where identification is expensive (Remark 2.6.3); Chapter 3 shows that the summed response is certified at the parametric rate from controlled experiments, and only under a declared bound on memory length from a finite window (Chapter 3).

How to measure feedback. Empirical feedback indices should (i) compare persistence with the mechanical value of the filter, (ii) align flow with the bar in which it is observed, (iii) control for order-splitting persistence or use instruments with a defended exclusion, and (iv) report placebo pipelines. Each of these changes the sign or the order of magnitude of published-style estimates in our data.

2.12 Conclusion and an open problem

Prices reveal a merged summary of algorithmic feedback. Recovering the algorithms behind it costs executed quantity of order δ^{-2} and trading capital of order δ^{-4} for a known pair, with a certified constant of about 61 par units at the calibration, and a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$ for K unknown cohorts, passively as well as actively; in simulations of stylized nonlinear rules and of online learners trained in the closed loop the same costs hold when the competing populations' feedback weights match, populations whose learned weights differ

are told apart by those weights instead, and hard thresholds hide cohorts from small probes altogether. Without the exponential form, accuracy improves only as $1/\log N$, with matching lower and upper bounds. Class-tagged disclosure removes the barrier, and its value can be stated in samples and in dollars: a sample factor of order $(\Delta\eta)^{-(4K-4)}$, about 333 for the trader classes Binance already publishes in BTCUSDT, and an annualised weekly-renewal probing loss that matches the $228.3 million CAT LLC operating budget across 15.5 (Roll) or 0.64 (one-second) markets for an unknown two-cohort Gauss–Radau pair at $\Delta\eta = 0.10$ and exceeds it in a single market once $K \geq 3$ at $\Delta\eta = 0.10$; the CFTC's class reports restrict persistence rates that prices leave unrestricted in all eight regulated markets we examined, while a pre-registered out-of-sample test finds that class-tagged memory states do not improve 30-minute post-shock return forecasts ($\Delta R^2 = -1.09$ points). At realistic noise the capital barrier for a moderate separation is about 10^5 USDT. Measured feedback in crypto markets is real but small, rises with depth, does not move when fees change trading volume severalfold, and is easily confused with filtering and timing artefacts. Finally, eigenvalues and volatility cannot certify safety, because transient loss can be arbitrarily large at fixed spectrum and variance.

This leaves an open question for supervision: *what can be measured and controlled without identifying the algorithms?* The impulse response $g_m = c^\top J^m b$, the operator J and finite-sample confidence sets for it are identified from public data even when cohorts are not. Chapter 3 uses exactly this interface to build an identification-free pre-trade control.

Data and code availability. All data are public: Binance archives (`https://data.binance.vision`), verified against published checksums. The pre-registered specification of the disclosure test, its SHA-256 hash and its test-set unlock counter are in the replication package. Every number in this chapter is produced by a notebook cell or script that writes a result file (Appendix 2.16).

Use of AI tools. The authors used Claude (Anthropic), an AI assistant, to write and debug analysis code, run checks of derivations and numerical results, and draft and edit text. Every number in the paper is produced by code the authors executed and archived in the replication package. The authors reviewed all content and take full responsibility for it.

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2.13 Appendix: Proofs

Lemma 2.2.1. (i) For $v_{t+1} = v_t + \zeta_{t+1}$ and $r_t = v_t + \nu_t$, the steady-state prior variance P solves $P = P\sigma_\nu^2/(P+\sigma_\nu^2) + \sigma_\zeta^2$, i.e. $P^2 - \sigma_\zeta^2 P - \sigma_\zeta^2\sigma_\nu^2 = 0$, whose positive root is $P = \big(\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}\big)/2$. The gain is $\eta = P/(P+\sigma_\nu^2) \in (0,1)$, which simplifies to (2.7), and the update $\hat v_{t+1} = \hat v_t + \eta(r_t - \hat v_t)$ has the stated form. (ii) $\nabla_\theta \tfrac12 H(x_t-\theta)^2 = -H(x_t-\theta)$, so $\theta_{t+1} = (1-\gamma)\theta_t + \alpha H(x_t - \theta_t) = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H$ and $q = \alpha H/\eta$.

Lemma 2.2.3. Omit zero-weight cohorts. For $a > 0$ put $b_i = \kappa q_i$ and

$$T = \operatorname{diag}\big(\sqrt{a}, \sqrt{b_1/\eta_1}, \ldots\big), \qquad u = \big(\sqrt{a}, \sqrt{b_1\eta_1}, \ldots\big)^\top.$$

The price-first transition becomes the symmetric matrix $J' = TJT^{-1} = \operatorname{diag}(0, r_1, \ldots) + uu^\top$, the input transforms to κu and $J'e_x = \sqrt{a}\,u$, so $h_j = (\kappa/a)e_x^\top (J')^{j+1} e_x$. The spectral decomposition of J' gives nonnegative weights summing to $h_0 = \kappa$, and Schur stability puts the eigenvalues in $[0,1)$. The case $a = 0$ follows by continuity. A nonnegative mixture of decaying geometric sequences is completely monotone. Since J is nonnegative with row sums at most one, every price row of J^t has ℓ_1, hence ℓ_2, norm at most one, attained at $t = 0$.

Theorem 2.3.1. With zero initial states, $y_{i,t} = \sum_{s\le t} \eta_i r_i^{t-s} x_s$, so $\sum_i q_i y_{i,t} = \sum_{s\le t} k_{t-s} x_s$ and every response is determined by (a, κ, k), hence by M. Conversely, equal responses for all probes imply equal H, hence equal G, and partial fractions of G identify the distinct poles with nonzero residues w_i, i.e. M. After cancellation, $H = \kappa z D(z)/[(z-a)D(z) - \kappa N(z)]$ with $D = \prod(z - r_i)$ and $N(z) = \sum_i w_i z \prod_{j\ne i}(z - r_j)$, as in (2.4); at a retained r_i the denominator equals $-\kappa N(r_i) = -\kappa w_i r_i \prod_{j\ne i}(r_i - r_j) \ne 0$ (memory rates are positive because $\eta_i < 1$), so no root of D cancels. The remaining numerator factor z cancels if and only if the denominator

vanishes at 0; since $N(0) = 0$ the denominator there equals $-aD(0)$, which is nonzero for $a > 0$ and zero for $a = 0$. Hence the McMillan degree, which is the dimension of a minimal realization, is $1 + |\mathrm{supp}(M)|$ for $a > 0$ and $|\mathrm{supp}(M)|$ for $a = 0$ (numerically, the rank of the Hankel matrix of $(h_j)_{j\geq 1}$ is $K + 1$ and K respectively for $K = 1, 2, 3$; `hardening/p1_checks.py`). For (ii), direct algebra gives

$$G_\delta(z) - G_0(z) = -\frac{qz(z-1)\delta^2}{(z-1+\bar{\eta})\big((z-1+\bar{\eta})^2 - \delta^2\big)}.$$

On $|z| = 1$ each factor has modulus at least ϵ. Uniform stability gives $M_H = \sup |H_\theta| < \infty$, and the identity $H_\delta - H_0 = H_\delta H_0(G_\delta - G_0)$ gives an ℓ_2 gain at most $K\delta^2$. The mean difference of a power-constrained probe has squared norm at most $K^2\delta^4 PT$; the equal-noise KL is at most $K^2\delta^4 PT/(2\sigma_\nu^2)$, and Pinsker's inequality with total variation at least $1 - 2\alpha$ gives (2.9).

Theorem 2.4.1. In either positive economy $J \geq 0$ and $\|J\|_\infty \leq 1 - \epsilon(1 - S) < 1$, so the response is nonnegative with $\max h = \kappa$ and $\sum h = \kappa/(1 - S)$. From the display above, $|G_\delta - G_0| \leq 2q\delta^2/\epsilon^3$ on the circle; combined with the resolvent identity and positivity this gives $\|H_\delta - H_0\|_\infty \leq K_*\delta^2$, and Parseval bounds $\|h^\delta - h^0\|_2$ by the same constant. Young's inequality gives $\|\mu_\delta - \mu_0\|_2 \leq K_*\delta^2\|v\|_1$, and Gaussian KL with Pinsker gives (2.10). For (ii), the mean difference is a lower-triangular convolution matrix times v; each column is a shifted copy of a prefix of d_T, so $\|\mu_\delta - \mu_0\|_2 \leq V\|d_T\|_2$ with equality for a pulse at $t = 0$. For two known equal-covariance Gaussian laws the minimax error is $\Phi(-\|\mu_\delta - \mu_0\|_2/(2\sigma))$, attained by the midpoint likelihood-ratio test. For (iii), the merged measures have equal mass and their first moments differ by $-q\delta^2$, which enters h_2 with factor κ^2 while h_0, h_1 coincide; a pulse A separates the third-observation means by $A\kappa^2 q\delta^2$. Part (iv) follows from the ℓ_2 gain bound, and (v) from superposition and $\max_j h_j = \kappa$, which give $\sup_t |\lambda x_t| \leq |\lambda|\kappa V$, attained by a single pulse.

Lemma 2.4.2. After the uncontrolled update, the pre-probe displacement at step j is $\sum_{i<j} h_{j-i}v_i$; average linear-book execution adds $\kappa v_j/2$. Summing signed execution costs gives $\frac{1}{2}v^\top T_h v$. Each kernel $\lambda^{|i-j|}$ is a positive-definite AR(1) covariance for $0 \leq \lambda < 1$, and a nonnegative mixture with positive total weight is positive definite. For a round trip the fundamental cash term cancels.

Theorem 2.4.3. Let $U_t = \sum_{j<t} v_j$. Positive responses give $0 \leq x_t \leq \kappa U_t$ and $I_t = -U_t$. Summing cash and marking inventory, $B - E_t = U_t x_t - \sum_{j<t} v_j x_{j+1}^0 - \frac{\kappa}{2}\sum_{j<t} v_j^2 + cU_t$. Its quadratic part $v^\top A_t v$ has diagonal $h_{t-1-i} - \kappa/2$ and off-diagonal $\frac{1}{2}(h_{t-1-i} + h_{t-1-j} - h_{j-i})$ for $i < j$; monotonicity of h bounds every entry by $\kappa/2$. Hence $B - E_t \leq \kappa U_t^2/2 + cU_t \leq \kappa V^2/2 + cV$, while position value is at most $V(P_* + \kappa V)$; the stated inequality guarantees margin. A single pulse attains both adverse bounds at its first response, so no larger allowance is robust. Solving

the quadratic gives (2.14). The quantity requirement $V_B \gtrsim \delta^{-2}$ then forces $B \gtrsim \delta^{-4}$, and the pulse A funded with $(\mu + \frac{1}{2})\kappa A^2 + (c + \mu P_*)A$ attains it.

Proposition 2.4.4. The initial cash receipt is common to both alternatives and the dealer then holds $-A$. At time three $E_{3,r} = B + \kappa A^2/2 - cA - A^2 h_{r,2}$, so $E_{3,\delta} - E_{3,0} = A^2\kappa^2 q\delta^2 > 0$.

Lemma 2.4.5. Marking residual inventory at P_* gives $E_\infty = B - \frac{1}{2}v^\top T_h v - c\sum|v_t| \geq 0$. For each mode $\lambda \leq r$ the Toeplitz kernel has spectral density $(1-\lambda^2)/(1-2\lambda\cos\omega+\lambda^2) \geq (1-r)/(1+r)$, so $v^\top T_h v \geq m\|v\|_2^2$ and $\|v\|_2^2 \leq 2B/m$. The transfer bound gives mean-difference squared norm at most $K_*^2\delta^4\|v\|_2^2$, hence the KL bound; Pinsker gives (2.15). For one pulse, cash becomes $B_A - A(P_* + \kappa A/2) - cA$ and inventory A; at later marks $E_t = \mu A P_* + A^2 h_{t-1}$ and $\mu|I_t|P_t = \mu A P_* + \mu A^2 h_{t-1}$, so margin holds for $\mu \leq 1$.

Proposition 2.5.1. Parts (i) and (ii) follow from (2.16) with $\Delta h = \Theta(\delta)$ for a rate shift and $\Theta(\delta^2)$ for symmetric splitting, and from additivity of KL over independent pulse cycles. Part (iii) solves the session constraint for δ; substituting $d_T(\delta) \approx \delta^2 \tilde{d}_T$ gives the fourth-root approximation.

Theorems 2.5.2 and 2.5.3. For a flat round trip, telescoping the cash account gives $L_i(v) = \frac{1}{2}v^\top T_{h_i} v = \frac{1}{2\pi}\int g_i(\omega)|V(e^{i\omega})|^2 d\omega$, with $g_i > 0$ by Lemma 2.2.3. Parseval gives $\|(H_1 - H_0)v\|_2^2 = \frac{1}{2\pi}\int |\Delta\hat{H}|^2|V|^2 \leq \sup_\omega(|\Delta\hat{H}|^2/g_i)L_i(v)$, and the Neyman–Pearson requirement $\|(H_1 - H_0)v\|_2 \geq 2\sigma z_{1-\alpha}$ gives (2.20). For adaptive policies, Doob's optional stopping theorem applied to the log-likelihood-ratio martingale and the data-processing inequality give $\mathbb{E}_0[\Lambda_\tau] \geq \mathrm{kl}(\alpha, 1-\beta)$, while $\mathbb{E}_0[\Lambda_\tau] = \mathbb{E}_0\sum_t(\Delta\mu_t)^2/(2\sigma^2) \leq \sup(|\Delta\hat{H}|^2/g_0)\mathbb{E}_0[L_0]/(2\sigma^2)$, which gives (2.22).

Part (iv). Let G_θ be the $T \times T$ lower-triangular Toeplitz section of H_θ/κ, so the observation covariance is $\Sigma_\theta = \sigma_w^2 G_\theta G_\theta^\top + \sigma_\nu^2 I$. A finite section of a bounded analytic symbol has norm at most its supremum on the circle, so $\|G_\theta\| \leq \sup_\omega |H_\theta|/\kappa \leq \bar{M}$ because $|H_\theta(e^{i\omega})| \leq \sum_j h_j = \kappa/(1-S)$. The inverse of a lower-triangular Toeplitz section is the section of the reciprocal series; κ/H_θ is analytic on and outside the unit circle because H_θ is minimum phase (Theorem 2.6.9), and Lemma 2.2.3 gives $\mathrm{Re}\, H_\theta(e^{i\omega}) = \sum_\ell \omega_\ell(1-\lambda_\ell\cos\omega)/(1-2\lambda_\ell\cos\omega+\lambda_\ell^2) \geq \sum_\ell \omega_\ell/(1+\lambda_\ell) \geq \kappa/2$, so $\|G_\theta^{-1}\| \leq 2$ and $\Sigma_\theta \succeq \sigma_{\mathrm{eff}}^2 I$ (the smallest singular value of G_θ is at least 0.522 in 762 random positive economies and sections; `hardening/p1_checks.py`). The Gaussian divergence splits into a mean and a covariance part. The mean part is $\frac{1}{2}\Delta\mu^\top\Sigma_1^{-1}\Delta\mu \leq \|(H_1 - H_0)v\|_2^2/(2\sigma_{\mathrm{eff}}^2)$. For the covariance part, the eigenvalues λ_k of $A = \Sigma_1^{-1/2}\Sigma_0\Sigma_1^{-1/2}$ are at least $c = \sigma_\nu^2/(\sigma_\nu^2 + \sigma_w^2\bar{M}^2)$, and $\lambda - 1 - \log\lambda \leq (\lambda-1)^2/\lambda$, so it is at most $\|A - I\|_F^2/(2c) \leq \|\Sigma_0 - \Sigma_1\|_F^2/(2c\,\sigma_{\mathrm{eff}}^4)$. Writing $\Sigma_0 - \Sigma_1 = \sigma_w^2(\Delta G\, G_0^\top + G_1\Delta G^\top)$ and $\|\Delta G\|_F^2 \leq T\|h^\delta - h^0\|_2^2/\kappa^2 \leq TK_*^2\delta^4/\kappa^2$ (proof of Theorem 2.4.1) gives a covariance part of at most $C_w T\delta^4$. The data-processing inequality turns errors (α, β) into $D_{\mathrm{KL}} \geq \mathrm{kl}(\alpha, 1-\beta)$, and part (ii) bounds $\|(H_1 - H_0)v\|_2^2$ by $\sup_\omega(|\Delta\hat{H}|^2/g_i)L_i(v)$, which gives (2.21).

Theorem 2.5.4. Write a feasible flat probe as $v = Eu$ and expand $u = \sum_j c_j e_j$ in the G_i-orthonormal eigenbasis, so $L_i(v) = \sum_j c_j^2$ and the separation constraint reads $\sum_j \lambda_j c_j^2 \geq c_{\mathrm{req}}$. Write the top-two component as $(c_{L-2}, c_{L-1}) = r(\cos\theta, \sin\theta)$ with $r \geq 0$; replacing θ by $\theta + \pi$ flips d and s_θ and leaves (2.23) unchanged, so $\theta \in [0, \pi)$ suffices. Put $L_0(\theta) = c_{\mathrm{req}}/\lambda(\theta)$ and $D = L_i(v) - L_0(\theta)$. The constraint gives $r^2\lambda(\theta) \geq c_{\mathrm{req}} - \sum_{j \leq L-3} \lambda_j c_j^2$, hence

$$D = r^2 + \sum_{j \leq L-3} c_j^2 - L_0(\theta) \;\geq\; \sum_{j \leq L-3} \Big(1 - \frac{\lambda_j}{\lambda(\theta)}\Big) c_j^2 \;\geq\; \Big(1 - \frac{\lambda_{L-3}}{\lambda(\theta)}\Big) \sum_{j \leq L-3} c_j^2 \;\geq 0,$$

and $r^2 \geq L_0(\theta) - \frac{\lambda_{L-3}}{\lambda(\theta)} \sum_{j \leq L-3} c_j^2 \geq L_0(\theta)\big(1 - \gamma(\theta) D\big)$. By Hölder's inequality with the sign vector s_θ, $c_{\mathrm{fee}} \|Eu\|_1 \geq c_{\mathrm{fee}} r \|Ed(\theta)\|_1 + c_{\mathrm{fee}} \sum_{j \leq L-3} c_j \, s_\theta^\top E e_j$, and Cauchy–Schwarz with weights $1 - \lambda_j/\lambda(\theta)$ bounds the second term below by $-\sqrt{D\, W_\perp(\theta)}$. Using $r \geq \sqrt{L_0(1 - \gamma D)_+} \geq \sqrt{L_0}\,(1 - \gamma D)_+$ and the fact that the fee is nonnegative, with $x = \sqrt{D}$,

$$L_i(v) + c_{\mathrm{fee}} \|v\|_1 \;\geq\; L_0(\theta) + x^2 + \Big[F_{\mathrm{dom}}(\theta)(1 - \gamma(\theta) x^2)_+ - \sqrt{W_\perp(\theta)}\, x\Big]_+ \;\geq\; L_0(\theta) + \Psi(\theta).$$

Minimizing over the unknown θ gives (2.24), and $\Psi \geq 0$, $L_0(\theta) \geq c_{\mathrm{req}}/\lambda_{L-1}$ give the last inequality. On the piece $\gamma x^2 \leq 1$ with $\gamma F_{\mathrm{dom}} < 1$ the bracket is a convex quadratic in x with minimum $F_{\mathrm{dom}} - W_\perp/(4(1 - \gamma F_{\mathrm{dom}}))$, which gives the closed form.

Corollary 2.5.5. Since $\Delta h_t = O(\delta^2)$, the variance difference $\sigma_{1,t}^2 - \sigma_{0,t}^2 = \sigma^2 \xi[(h_1 + h_0) * v]_t [\Delta h * v]_t = O(\delta^2)$, and the variance term of the Gaussian KL, $\frac{1}{2} \sum (x - 1 - \log x)$ with $x = \sigma_{0,t}^2/\sigma_{1,t}^2$, is quadratic in that difference. Convex impact only increases dissipation.

Proposition 2.6.1. Standard Cramér–Rao for a Gaussian mean model with Jacobian J. For $K = 2$ the columns of J for the two rates become collinear as $\Delta\eta \to 0$; expanding to third order, the smallest singular value of J is of order $(\Delta\eta)^3$, so $[F^{-1}]_{kk} = \Theta((\Delta\eta)^{-6})$.

Theorem 2.6.2. Expanding $M_m(\theta_1) - M_m(\theta_0)$ under (2.26) gives zero for $m \leq 2$ and $4\tau p_- p_+ (\Delta\eta)^3$ for $m = 3$. The map from moments to impulse responses is analytic with nonsingular derivative at the common limit, so $\|\Delta h\|_2 = \Theta(\tau(\Delta\eta)^3)$. Le Cam's two-point lemma with N independent Gaussian experiments gives (2.28). For the permutation-invariant loss, the identity matching of θ_0 and θ_1 costs $2\tau + (p_+ - p_-)\Delta\eta = 2\tau(1 + \Delta\eta/q)$ per atom and the swapped matching costs $\max(2p_+, 2p_-)\Delta\eta = (1 + 2\tau/q)\Delta\eta$; when both are at least 2τ, any estimate within τ of one economy in d_∞ is farther than τ from the other by the triangle inequality, and the two-point argument is unchanged. For (iii), the projection estimator of M_3 is unbiased with variance $\sigma_h^2/(N\|\nabla M_3 h\|^2)$, and sub-Gaussian concentration gives the sample size; the Prony–Hankel determinant identity follows from $g_m = w_1\eta_1 r_1^m + w_2\eta_2 r_2^m$.

Theorem 2.6.4. Let (s^A, ω^A) and (s^B, ω^B) be the K-point Gauss and Gauss–Radau rules for the uniform weight on $[-1, 1]$, normalized to unit mass. Both are positive and exact for polynomials of degree $2K-2$, and their moments of degree $2K-1$ differ by $g_K \neq 0$. With $r = r_c + \Delta\eta\, s$ and weights $W\omega$, the raw moments $\sum w_i r_i^m$ are polynomials in $\Delta\eta$ that agree below degree $2K-1$, so the kernels differ by $O((\Delta\eta)^{2K-1})$ with nonzero leading coefficient; analyticity of the kernel-to-response map gives $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, and Le Cam's lemma gives the bound. For the W_1 version, the two merged measures are the images of the fixed normalized rules under $s \mapsto r_c + \Delta\eta\, s$ scaled to mass W, so their distance is $W\Delta\eta$ times the distance $2c_K > 0$ between the rules, and the same two-point argument applies with the metric loss.

Lemma 2.6.7. Let $L = 2n-2$ and let p^* be the best uniform approximation of $|x|$ on $[-1, 1]$ by polynomials of degree L, with error $E = E_L(|x|)$. By Chebyshev's equioscillation theorem (Rivlin, 1981) there are points $t_0 < \cdots < t_{L+1}$ at which $|x| - p^*(x)$ equals $\pm E$ with alternating signs. The divided-difference weights $w_i = 1/\prod_{j\neq i}(t_i - t_j)$ alternate in sign and satisfy $\sum_i w_i p(t_i) = 0$ for every polynomial p of degree at most L; multiplying by -1 if necessary, w_i has the sign of $|t_i| - p^*(t_i)$ for every i. Exactly n weights are positive and n negative, and the positive and negative parts have equal mass because constants are annihilated. Normalizing each part to unit mass and mapping $x \mapsto \rho(x+1)/2$ gives n-atom probability measures π_A (positive part) and π_B (negative part) on $[0, \rho]$ with equal moments of orders $0, \dots, L$. For the 1-Lipschitz function $f(r) = |r - \rho/2| = (\rho/2)|x|$,

$$\int f\, d(\pi_A - \pi_B) = \frac{\rho}{2} \int \big(|x| - p^*(x)\big)\, d(\pi_A - \pi_B) = \frac{\rho}{2}(E + E) = \rho E,$$

because p^* integrates to the same value under both measures, $|x| - p^* = E$ on the support of π_A and $-E$ on that of π_B. By Kantorovich–Rubinstein duality $W_1(\pi_A, \pi_B) \geq \rho E \geq c_* \rho / L$.

Theorem 2.6.8. Write $K(z) = \sum_m k_m z^m$, $D = 1 - az - \kappa z K$ and $H = \kappa/D$, which is the generating function of the recursion $h_t = a h_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ with $h_0 = \kappa$. Since $k_m \geq 0$ and $\sum_m k_m = \int (1-r)^{-1} d\mu \leq W/(1-\rho)$, the ℓ_1 norm of $az + \kappa z K$ is at most $s < 1$, so $\|H\|_1 \leq \kappa/(1-s)$ and $\|1/H\|_1 = \|D\|_1/\kappa \leq (1+s)/\kappa$.

(i) Let $\mu_A = W\pi_A$ and $\mu_B = W\pi_B$ with π_A, π_B from Lemma 2.6.7; both lie in $\mathcal{M}(\rho, W)$. With $\nu = \mu_A - \mu_B$, $\sum_m (k_m^A - k_m^B) z^m = \int (1 - rz)^{-1} d\nu(r) = z^{2n-1} \int r^{2n-1} (1 - rz)^{-1} d\nu(r)$, because the first $2n-1$ terms of the geometric series integrate to zero. On $|z| = 1$ the modulus is at most $2W\rho^{2n-1}/(1-\rho)$, so Parseval gives $\|k^A - k^B\|_2 \leq 2W\rho^{2n-1}/(1-\rho)$. Next, $H_A - H_B = z(K_A - K_B) H_A H_B$, so by Young's inequality $\|h^A - h^B\|_2 \leq \|H_A\|_1 \|H_B\|_1 \|k^A - k^B\|_2 \leq C_h \|k^A - k^B\|_2$. The Kullback–Leibler divergence between the N-sample Gaussian experiments is $N\|h^A - h^B\|_2^2/(2\sigma^2) \leq (N/\sigma^2)\, 2C_h^2 W^2 \rho^{2(2n-1)}/(1-\rho)^2$, which is at most one as soon as $2(2n-1)\log(1/\rho) \geq \Lambda_N - 2\log(1/\rho)$. Take n the smallest integer with this property

and $n \geq 2$; then $2n-2 < \Lambda_N/(2\log(1/\rho))$ when $\Lambda_N \geq 4\log(1/\rho)$. Since d is a norm on signed measures and $d(\mu_A,\mu_B) = W\,W_1(\pi_A,\pi_B) \geq Wc_*\rho/(2n-2) > 2c_*\rho W\log(1/\rho)/\Lambda_N$, the Bretagnolle–Huber form of Le Cam's lemma (Tsybakov, 2009, Lemma 2.6 and Theorem 2.2) gives $\max_{j\in\{A,B\}} P_j(d(\hat\mu,\mu_j) \geq c_*\rho W\log(1/\rho)/\Lambda_N) \geq \frac{1}{4}e^{-1}$.

(ii) *Approximation.* Let f be 1-Lipschitz on $[0,\rho]$ with $f(0)=0$, so $|f| \leq \rho$. The function $g(x) = f(\rho(x+1)/2)$ is $(\rho/2)$-Lipschitz on $[-1,1]$, and Jackson's theorem (Rivlin, 1981) gives a polynomial p of degree L with $\sup|g-p| \leq (\pi/2)(\rho/2)/(L+1) \leq \rho$, hence $\sup|p| \leq 2\rho$. Its Chebyshev coefficients satisfy $|b_j| \leq 2\sup|p| \leq 4\rho$. The polynomial $q(r) = p(2r/\rho-1)$ approximates f with the same error, and its monomial coefficients satisfy $\sum_j |q_j| \leq \sum_{j\leq L}|b_j|\,\|T_j(2r/\rho-1)\|_{\mathrm{coef}} \leq 4\rho\sum_{j\leq L}B(\rho)^j \leq \rho B(\rho)^{L+1}$, where $\|\cdot\|_{\mathrm{coef}}$ is the sum of absolute monomial coefficients, $\|T_j\|_{\mathrm{coef}} \leq (1+\sqrt{2})^j$, substituting $y = \alpha r - 1$ multiplies the coefficient sum of y^i by at most $(1+\alpha)^i$, and $B(\rho)-1 \geq 4$. *Moment comparison.* If $\max_{j\leq L}|\int r^j d\hat\mu - k_j| \leq 2\delta$, then $\hat\mu$ has mass at most $W+2\delta$ and

$$\int f\,d(\hat\mu-\mu) = \int (f-q)\,d(\hat\mu-\mu) + \sum_{j\leq L} q_j\Big(\int r^j d\hat\mu - k_j\Big) \leq \frac{\pi\rho(W+\delta)}{2(L+1)} + 2\delta\rho B(\rho)^{L+1}.$$

Kernel recovery. From $\kappa z K = 1 - az - \kappa/H$, the kernel values $k_0,\dots,k_L$ are the coefficients of $(1-az-\kappa/H)/(\kappa z)$, which depend on $h_0,\dots,h_{L+1}$ only. Work in the algebra of power series modulo z^{L+2} with the ℓ_1 norm, which is submultiplicative. If the averaged response is $\hat H = H+E$ and $\|1/H\|_1\|E\|_1 \leq \frac{1}{2}$, then $1/\hat H = (1/H)(1+E/H)^{-1}$ and $\|1/\hat H - 1/H\|_1 \leq 2\|1/H\|_1^2\|E\|_1$, so $\max_j|\hat k_j - k_j| \leq \delta \equiv 2((1+s)/\kappa)^2(L+2)\max_{t\leq L+1}|E_t|$. Each E_t is Gaussian with variance σ^2/N, so with probability at least $1-\beta$, $\max_t|E_t| \leq \sigma\sqrt{2\log(2(L+2)/\beta)/N}$. The true μ is feasible for the linear program with value at most δ, so the minimizer satisfies the moment condition above. *Choice of L.* With $L+1 = \lfloor \log(N/\sigma^2)/(4\log B(\rho))\rfloor$, $B(\rho)^{L+1} \leq (N/\sigma^2)^{1/4}$, so $\delta B(\rho)^{L+1} = O((\sigma^2/N)^{1/4}L\sqrt{\log(L/\beta)})$ and $\delta/(L+1)$ are $o(1/\log(N/\sigma^2))$, while $\pi\rho W/(2(L+1)) \leq 2\pi\rho W\log B(\rho)/(\log(N/\sigma^2) - 4\log B(\rho))$. For N/σ^2 larger than a threshold depending on (ρ,W,a,κ,β) the side condition $\|1/H\|_1\|E\|_1 \leq \frac{1}{2}$ holds on the event and the total is below $3\pi\rho W\log B(\rho)/\log(N/\sigma^2)$, uniformly over $\mathcal{M}(\rho,W)$. The final statement combines (i) and (ii).

Theorem 2.6.10. For stationary Gaussian processes with spectral densities $f_j \propto |H_j(e^{i\omega})|^2$, the divergence rate is

$$D = \frac{1}{4\pi}\int_{-\pi}^{\pi}\big(\varrho - 1 - \log\varrho\big)\,d\omega, \qquad \varrho = f_1/f_0 = |H_1/H_0|^2.$$

By the resolvent identity, $H_1/H_0 - 1 = H_1\Delta G$ with $\Delta G = G_1 - G_0$, and uniform stability bounds $|H_1|$ on the circle, so $u \equiv \varrho - 1 = 2\,\mathrm{Re}(H_1\Delta G) + O(|\Delta G|^2)$. Write $G(z) = \sum_k w_k\phi(z,\eta_k)$, with w_k the cohort weights (any factor of η_k is absorbed in ϕ), where ϕ is analytic in η on a neighbourhood of $[\eta_{\min},\eta_{\max}]$ uniformly for $|z|=1$ (its pole $z = 1-\eta$ stays inside the disk).

Expanding ϕ around $\bar\eta$, $\Delta G = \sum_{j\ge0} \frac{\partial_\eta^j \phi(z,\bar\eta)}{j!}\,\Delta\tilde M_j$, where $\tilde M_j = \sum_k w_k(\eta_k-\bar\eta)^j$. For the pair, $\Delta\tilde M_0 = \Delta\tilde M_1 = 0$ and $\Delta\tilde M_2 = q\delta^2$; for the moment-matched economies, $\Delta\tilde M_j = 0$ for $j \le 2$ and $\Delta\tilde M_3 = 4\tau p_- p_+(\Delta\eta)^3$ by (2.27); for the K-cohort Gauss and Gauss–Radau economies with rates $\bar\eta + \Delta\eta\, s$ and weights $W\omega$, both rules integrate polynomials of degree $2K-2$ exactly, so $\Delta\tilde M_j = W(\Delta\eta)^j \sum_k(\omega_k^A (s_k^A)^j - \omega_k^B (s_k^B)^j)$ vanishes for $j \le 2K-2$ and equals $\pm W g_K (\Delta\eta)^{2K-1} \neq 0$ at $j = 2K-1$. Higher terms are smaller by a factor of the spacing: $|\Delta\tilde M_j| \le 2W(\Delta\eta)^j$, and analyticity of ϕ in η on a disk of radius $\varrho > 0$ around $\bar\eta$, uniformly on the circle, gives $|\partial_\eta^j\phi|/j! \le C\varrho^{-j}$, so for $\Delta\eta < \varrho/2$ the tail beyond the first nonzero term is $O((\Delta\eta)^{j_0+1})$. Hence $\sup_\omega |u| = O(\delta^2)$, $O(\tau(\Delta\eta)^3)$, respectively $O((\Delta\eta)^{2K-1})$, and $u = c(\omega)\epsilon + o(\epsilon)$ with $\epsilon = \delta^2$, $\tau(\Delta\eta)^3$ or $(\Delta\eta)^{2K-1}$ and $c(\omega)$ proportional to $2\,\mathrm{Re}\big(H_0\,\partial_\eta^{j_0}\phi(\cdot,\bar\eta)\big)/j_0!$, $j_0 = 2$, 3 or $2K-1$. The function c does not vanish identically: $H_0\partial_\eta^j\phi$ is analytic outside the disk and non-constant, and an analytic function whose real part vanishes on the circle is an imaginary constant. For $|u| \le \frac12$, $u^2/4 \le u - \log(1+u) \le u^2$, so $D \asymp \int c(\omega)^2 d\omega\, \epsilon^2$, which gives (i)–(iii). The sample requirements follow from $D(P_1^T \| P_0^T) = TD + O(1)$ for stationary Gaussian processes with smooth spectra and the Bretagnolle–Huber inequality (Tsybakov, 2009, Lemma 2.6); attainment follows from Stein's lemma for the likelihood-ratio test.

Theorem 2.6.9. In the delay variable $\lambda = z^{-1}$, $H = \kappa\prod_i(1-r_i\lambda)/\big[(1-a\lambda)\prod_i(1-r_i\lambda) - \kappa\lambda\sum_i w_i\prod_{j\neq i}(1-r_j\lambda)\big]$, which is (2.4) divided through by z^{K+1}; its zeros are the memory rates, which do not cancel by the argument for Theorem 2.3.1, and stability puts the poles inside the disk. A rational spectral density $\sigma^2|H(e^{i\omega})|^2$ has a unique minimum-phase factor up to a unimodular constant, which is H; Theorem 2.3.1 then gives the merged measure.

Proposition 2.8.1. With tagged flow, $u_{i,t} = r_i u_{i,t-1} + w_i x_t$ holds exactly and involves only cohort i, so $\tilde u_{i,t} = r_i\tilde u_{i,t-1} + w_i x_t + e_{i,t}$ with the MA(1) error $e_{i,t}$. Prices are independent of the measurement noise, so $\mathbb{E}[Z_t e_{i,t}] = 0$ and the IV estimator is consistent when $\Sigma_i = \mathbb{E}[Z_t(\tilde u_{i,t-1}, x_t)]$ is nonsingular. Since $u_{i,t-1} = w_i\sum_{s\ge0} r_i^s x_{t-1-s}$, $\mathbb{E}[x_{t-1}u_{i,t-1}] = w_i A_0$ and $\mathbb{E}[x_t u_{i,t-1}] = w_i A_1$, which gives Σ_i. The price is a nonnegative mixture of geometric responses to its innovation (Lemma 2.2.3), so $\gamma_x(s) = \sum_\ell c_\ell\lambda_\ell^s$ with $c_\ell = \sigma_\xi^2\kappa^{-2}\omega_\ell\sum_m \omega_m/(1-\lambda_\ell\lambda_m) \ge 0$; substituting, $\det\Sigma_i = w_i(\gamma_x(0)A_0 - \gamma_x(1)A_1) = w_i\sum_{\ell,m} c_\ell c_m(1-\lambda_\ell\lambda_m)/(1-r_i\lambda_m)$, each term positive. The scores $Z_t e_{i,t}$ are one-dependent with $\mathbb{E}[Z_t Z_t^\top e_{i,t}^2] = \sigma_f^2(1+r_i^2)\Gamma_0$ and $\mathbb{E}[Z_t Z_{t-1}^\top e_{i,t}e_{i,t-1}] = -r_i\sigma_f^2\Gamma_1$, so their long-run variance is Ω_i, and the central limit theorem for m-dependent sequences gives (2.29). On a uniformly stable family the modes and weights lie in a compact set with $\lambda_\ell \le \bar\lambda < 1$, so $\det\Sigma_i \ge w_{\min}(1-\bar\lambda^2)\gamma_x(0)^2 > 0$ and $\|\Omega_i\|$ is bounded, uniformly in the spacing.

Theorem 2.9.1. With J_θ upper triangular the eigenvalues are fixed. The stationary covariance solves $P = J_\theta P J_\theta^\top + \mathrm{diag}(s_1, s_2)$, and P_{11} is affine in s_2 with positive slope, so for each θ a unique

$s_2(\theta) \geq 0$ gives $P_{11} = V$ when V exceeds the value at $s_2 = 0$. The one-step price variance is s_1 throughout, and $e_1^\top J_\theta e_2 = \theta$ gives $\sup_k \|e_1^\top J_\theta^k\| \geq |\theta|$.

2.14 Appendix: Notation

Symbol	Meaning
x_t	pricing error (price minus fundamental)
a	price persistence (anchor), $0 \leq a < 1$
κ	price impact per unit of executed quantity
$q_i,\ \eta_i,\ r_i$	cohort weight, adaptation rate, memory rate $r_i = 1 - \eta_i$
$w_i = q_i\eta_i$	merged weight
$M = \sum_i w_i \delta_{r_i}$	merged rate–weight measure
$k_m,\ G(z)$	feedback kernel and its transform
$H(z),\ h_j$	closed-loop transfer function and impulse response, $h_0 = \kappa$
$J,\ S$	state transition matrix and $S = a + \kappa \sum q_i$
$v_t,\ V$	executed probe quantity and total quantity $\sum \lvert v_t \rvert$
d_T	impulse-response difference between hypotheses, first T taps
σ	observation-noise standard deviation
$\delta,\ \Delta\eta$	half-separation of a known pair; spacing of unknown cohorts
K_*	transfer-difference constant $2\kappa^2 q/[\epsilon^3(1-S)^2]$
T_h	Toeplitz matrix $(h_{\lvert i-j \rvert})$ of execution cost
$g_i(\omega)$	dissipation kernel $\mathrm{Re}\,\hat{H}_i(e^{i\omega}) - \kappa/2$
$B,\ \mu,\ c,\ \phi$	capital, maintenance-margin fraction, proportional cost, taker fee
V_{eff}	per-trade execution capacity $\min(V_{\max}, D_{\max}/\kappa)$
$\hat{Q},\ \hat{S}$	reduced-form loop gain and persistence indices

2.15 Appendix: Further results

2.15.1 A finite-window price-surveillance bound

Theorem 2.15.1 (Finite-window abstention)**.** *Observe $x_t = sx_{t-1} + w_t$, $w_t \sim N(0, \sigma^2)$, $x_0 = 0$, $t \leq T$, with $s \in (0, 2)$. Let a possibly randomized rule Ψ output Stable, Unstable or Abstain, with $\sup_{s<1} P_s(\Psi = \text{Unstable}) \leq \alpha$ and $\sup_{s\geq 1} P_s(\Psi = \text{Stable}) \leq \alpha$, $\alpha < 1/2$. Choose $0 < c_0 < 1$ with $c_0 e^{c_0}/(2\sqrt{2}) < 1 - 2\alpha$. On $B_T = (1 - c_0/T, 1 + c_0/T)$,*

$$\inf_{s \in B_T} P_s(\Psi = \text{Abstain}) \ \geq\ 1 - 2\alpha - \frac{c_0 e^{c_0}}{2\sqrt{2}} > 0,$$

and for every finite T the minimax worst-case error of binary rules over $s \in (0,2)$ is at least $1/2$.

Proof. Pair $s_0 \in (1 - c_0/T, 1)$ with $s_1 = s_0 + c_0/T$. Then $D_{\mathrm{KL}}(P_{s_0} \| P_{s_1}) = \frac{(s_1 - s_0)^2}{2\sigma^2} \sum_t \mathbb{E} x_{t-1}^2 \le c_0^2 e^{2c_0}/4$, so Pinsker bounds total variation by $c_0 e^{c_0}/(2\sqrt{2})$. Error control gives $P_{s_0}(\text{Stable}) \le \alpha + \text{TV}$ and $P_{s_0}(\text{Unstable}) \le \alpha$, which yields the abstention bound. For binary rules, continuity in total variation and Le Cam's two-point bound give worst-case error tending to $1/2$. □

The interval B_T is an abstention layer that any error-controlled monitor must respect in the scalar Gaussian case; richer disclosure can remove it.

2.15.2 Passive distinction of a known pair in finite samples

Proposition 2.15.2 (Passive distinction with zero controlled quantity)**.** *Keep the known pair, but add independent uniform innovations $w_t \in [-b, b]$ of variance $s_w^2 = b^2/3$ at each price update, observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma^2)$, use independent three-step reset blocks and no probes, with $P_* > 3b$. Let $b_1 = a + \kappa q \eta$, $b_{2,0} = b_1^2 + \kappa q \eta (1 - \eta)$ and $b_{2,\delta} = b_{2,0} - \kappa q \delta^2$. In each block the covariance of the first and third observations differs by $\Delta = \kappa q s_w^2 \delta^2$. With*

$$R = \max_{r \in \{0, \delta\}} \Big\{ (s_w^2 + \sigma^2) \big[s_w^2 (1 + b_1^2 + b_{2,r}^2) + \sigma^2 \big] + s_w^4 b_{2,r}^2 \Big\},$$

the midpoint test of the average product has worst-case error at most α whenever $n \ge 4R/(\alpha \Delta^2)$ blocks are observed: zero controlled quantity and order δ^{-4} blocks.

The Chebyshev bound is conservative but the order is real. With $b = 0.03$, observation s.d. 0.01 and $\delta = 0.10$, the bound asks for 3.9×10^7 blocks at $\alpha = 0.1$; in simulation the midpoint test still errs 32–38% of the time with 2×10^5 blocks. Independent resets and known innovation laws are extra conditions. The proposition does not show that anonymous public records identify participants; it shows that passive excitation can accumulate memory information while the probe account stays untouched.

Proof. In each block $x_1 = w_1$, $x_2 = b_1 w_1 + w_2$ and $x_3 = b_{2,r} w_1 + b_1 w_2 + w_3$, so $|x_t| \le 3b$. Observation noise changes variances but not the covariance $s_w^2 b_{2,r}$. The product variance is the product of marginal variances plus the squared covariance plus a fourth-cumulant term $b_{2,r}^2 (\mathbb{E} w^4 - 3 s_w^4) \le 0$ for uniform innovations, so it is at most R. Chebyshev bounds each midpoint error by $4R/(n\Delta^2)$. □

2.15.3 Blind inversion of the closed loop

1. *Impulse decoupling:* $g_t = h_{t+1} - a h_t$.
2. *Minimal realization:* form the Hankel matrix $H_{ij} = g_{i+j-1}$ and run Kung's eigensystem realization with a truncated SVD.

3. *Transmission zeros:* compute the zeros of the realization, which under known (κ, a) isolate the roots of the denominator of G.

4. *Weights:* with recovered rates, project $u_t = (x_t - ax_{t-1})/\kappa$ on the reconstructed filters $y_{k,t-1}$.

Without noise each step is exact: the realization has the order of the true system, its transmission zeros are the memory rates (Theorem 2.3.1), and the projection returns the weights, so the procedure removes the pole-shift bias of fitting poles directly to the price response. With noise, per-cohort recovery degrades as rates move together at the rate given by Theorems 2.6.4 and 2.6.5, which is why we do not report it as an estimator of individual cohorts.

2.15.4 Probing protocols

Table 2.28: Probing protocols at observation horizon $T = 4$ per cycle

Protocol	δ	Executed vol.	Max trade	Error (SE)	Capital
A: single pulse (unconstrained)	0.03	131.5	131.5	0.047 (0.003)	6,985.91
	0.05	47.3	47.3	0.054 (0.004)	921.73
	0.08	18.5	18.5	0.053 (0.004)	146.73
	0.12	8.2	8.2	0.044 (0.003)	31.45
B: repeated capped pulses	0.03	2,768.8	6.2	0.050 (0.003)	7,035.75
	0.05	362.5	6.2	0.054 (0.004)	923.88
	0.08	56.2	6.2	0.047 (0.003)	146.00
	0.12	12.5	6.2	0.038 (0.003)	34.88
C: sequential test, capped	0.03	1,410.2	6.2	0.045 (0.003)	3,585.14
	0.05	195.6	6.2	0.038 (0.003)	500.00
	0.08	35.9	6.2	0.032 (0.003)	94.30
	0.12	10.5	6.2	0.011 (0.002)	29.85
D: generalized eigenvector	0.03	2,359.7	3.7	0.050 (0.003)	2,632.77
	0.05	314.8	3.7	0.048 (0.003)	353.67
	0.08	49.2	3.7	0.046 (0.003)	57.53
	0.12	9.9	3.8	0.045 (0.003)	13.56

2,000 replications per hypothesis and cell. Log–log capital slopes: −3.90, −3.84, −3.47, −3.81. Protocols A and B reproduce the earlier version exactly. C and D are re-implemented because the earlier code was not retained: C is a Wald test with capped pulses; D repeats the top generalized eigenvector of a flat $(T + 1)$-step probe, scaled to a maximum trade of 3.75, and is cheaper than the earlier D. Kind: [RECOMPUTED] (`recompute_p1_protocols.py`).

2.15.5 Coordinate-weighted output

For $J = \left(\begin{smallmatrix} a & \kappa q \\ \eta a & 1-\eta+\eta\kappa q \end{smallmatrix}\right)$ with $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.3$ ($\rho(J) = 0.8224$, non-normal), the price output has $\sup_t \|e_x^\top J^t\| = 1$. The weighted output $C_w = (1, \lambda/\text{MMR}) = (1, 160)$ with $\lambda = 0.8$ and MMR $= 0.005$ has $\sup_t \|C_w J^t\| = \|C_w\| = 160.003$, attained at $t = 0$, then 122.32, 98.32, 80.79 [RECOMPUTED IN CLOSED FORM]. The factor 160 is a static coordinate weight; $C_w z$ is not a margin quantity because z contains no inventory. Margin must be computed on a cash-and-inventory ledger.

2.16 Appendix: Reproducibility: source of every number

The replication package contains numbered Colab notebooks and stand-alone scripts; its README lists them in run order. "Part N" below and in table notes refers to notebook N (Part 1: theorem checks and spot tests; Part 2: BTCUSDT futures depth; Parts 4–13: the cross-section, replications, fresh-day tests, second venues, the CFTC test, the certified guard and post-hoc checks). Result identifiers such as `X6` or `U6` name the JSON file that the notebook writes. "Kind" distinguishes closed-form or simulation results (recomputed), results from public market data (real data), and audits.

Result	Source		Kind
Complete monotonicity, 892 economies	`L26_complete_monotonicity`		recomputed
$d_2 = -\kappa^2 q \delta^2$	`P1.7-T1_third_tap`		recomputed
Ledger identity; T_h positive definite	`R2-T1`, `R2-T3`		recomputed
Figure 2.2	`make_figs2.py`		recomputed
$\delta_{\min}(3)$, Table 2.5 recomputed columns, 61.14, 29.94	`recompute_theory.py` `R_P1_theory_constants.json`	→	recomputed
Pulse benchmarks (271, 58, 12, 11 pulses), Figures 2.1–2.3	`make_figs3.py` `R_P1_pulse_benchmarks.json`	→	recomputed
Table 2.3, Table 2.11, worked example, passive simulation	`recompute_extra.py` `R_P1_extra.json`	→	recomputed
Daily noise (Table 2.8)	`P1.3-T1_sigma_obs`		real data
Table 2.10 recomputed columns	same		recomputed
K-cohort slopes (Table 2.12)	`T1_K_cohort_rate`		recomputed
ESPRIT slope −6.15	`P1.2-T1_esprit_rate`		recomputed
Minimum phase, 1,733 economies; rank 6 of 6	`T2a`, `T2b`		recomputed
Passive slopes −4.21, −5.93	`T3_passive_rate`		recomputed
Tagged-flow slope 0.011	`T4_disclosure`		recomputed
Spectral blindness gains	`T5_spectral_blindness`		recomputed
Noise and capital grid (Table 2.7)	`P1.3-T1_sigma_obs`		real data
Loop gain and placebo (Table 2.24)	`P1.6-T1_ema_placebo`		real data
Estimator table (Table 2.25)	`P1.5-T1`, `T2`, `T3`		recomputed
Data integrity (111 files)	`R2-T9_data_integrity`		audit
Tables 2.4, 2.5 (certified, feasible, adaptive columns)	`recompute_p1_capital.py` `R_P1_capital_ledger.json`	→	recomputed
Table 2.16; resolution floors; Table 2.28	`recompute_p1_protocols.py` `R_P1_protocols.json`	→	recomputed

Result	Source	Kind
SPRT capital figures	`recompute_p1_sprt.py` → `R_P1_sprt.json`	recomputed
Table 2.9 (N^*, σ_{crit})	`recompute_crb.py` → `R_P1_crb.json`	recomputed
Table 2.9 (ML columns), Table 2.10 (estimators)	`recompute_p1_estimators.py`	recomputed
Symmetric multi-asset network (Section 2.9)	`recompute_oct20.py`	recomputed
Proof audit: K_* bound and d_2 identity (Theorem 2.4.1), moment gaps (Theorem 2.6.2), spectral-blindness example (Theorem 2.9.1)	`recompute_audit.py` → `R_audit_oct2026.json`	recomputed
Theorem 2.6.5, Table 2.13	`kcohort_upper.py` → `R_P1_kcohort_upper.json`	recomputed
Lem. 2.6.7, Thm 2.6.8, Tab. 2.14	`general_minimax.py`	recomputed
Theorem 2.8.2 numbers	`disclosure_value.py` → `R_P1_disclosure_value.json`	recomputed
Tables 2.22, 2.23, 2.26	Part 8: `V2_spot_panel`, `V3_fee_events`	real data
Table 2.18, top panel	Part 4: `X9_disclosure_metrics`	real data
Table 2.18, bottom panel; class-position forecasts	Part 6: `Z6_C_identification`, `Z6_secondary`; `P6_fresh_days.json` (65acda36…)	real data
Table 2.20	Part 11: `C3_cot_data`, `C4_cot_disclosure`; `P11_cot.json` (6f1d6f35…)	real data
Basis-trade signature; real-data Cochrane–Orcutt	Part 13: `R6`, `R7`	real data (post hoc)
Feedback across 30 perpetuals	Part 4: `X3_T6_cross_section`	real data
Theorem 2.5.4 full bound (Table 2.6), direct minimization	`hardening/rh_bound.py`, `rh_check.py` → `R_P1_rh_explicit.json`	recomputed
Minimal dimension at $a = 0$; Toeplitz sections; passive slopes for $K \leq 4$; IV determinant	`hardening/p1_checks.py` → `R_P1_hardening_checks.json`	recomputed

Result	Source	Kind
Stylized nonlinear and trained online rules (Table 2.15, Figure 2.7)	`recompute_p1_learners.py` → `R_P1_learners.json` (Panel A); `validate_trained_learners_hardened.py` → `R_P1_trained_learners_hardened.json` (Panels B–C); specification `P1_trained_learners.json` (3e8bf30c…)	recomputed
Disclosure in dollars (Table 2.17)	`recompute_p1_cat_costs.py` → `R_P1_cat_disclosure_dollars.json`	recomputed
Out-of-sample post-shock return test (Section 2.8.3)	`recompute_prereg_oct21_22nd.py` → `R_P1_disclosure_oos.json`; `P1_disclosure_oos.json` (f0182ce5…)	real data
Out-of-sample cohort-state trajectory test (Table 2.19)	`recompute_prereg_oct21_followup.py` → `R_P1_disclosure_state_oos.json`; `P1_disclosure_state_oos.json` (5136cc6b…)	real data

Chapter 3

Safety without identification: transient amplification and a maximal pre-trade guard

This chapter, the second paper of the thesis, condenses the paper *From Human Markets to Machine Markets: Transient Amplification, an Exact Depth Condition, and a Maximal Pre-Trade Guard*, joint with Suboh Alkhushayni, Yuhao Chen, Mansi Bhavsar and Prashant Niroula. It keeps the theorems the rest of the thesis uses, with their proofs, and the empirical program in summary, with every registered test and its outcome in Table 3.4. The full version, with every table, robustness check and the source of every number, is in the replication package (`thesis/ch3_paper2_full.tex`).

Summary. Eigenvalues and volatility do not bound the losses a market produces before it settles. In a limit order book with adaptive traders, cumulative transient risk is $S = \tilde{\kappa}/(m - q\tilde{\kappa})$, long-run impact over reversion net of momentum; a book overshoots its direct impact if and only if $\varphi\varepsilon > (1-a)\kappa$; and a pre-trade guard on the summed impulse response is the largest safe admission rule and does not depend on how the traders split into cohorts. The summed response is certified with $\Theta(\sigma^2\epsilon^{-2}\log(1/\epsilon))$ experiments whatever the cohort spacing, while identifying the cohorts costs polynomially more, and the certified guard is minimax-optimal in lost capacity up to a logarithm. Of 25 registered empirical tests, 15 pass: six-hour estimates of the statistic forecast the size of the next hour's adverse excursion in thin Binance perpetuals beyond a full volatility baseline and VPIN, more weakly in thin US stocks, and not on Coinbase or in the deepest book; tests of a liquidation mechanism fail.

3.1 Introduction

On October 10, 2025, crypto derivatives markets liquidated an estimated $19 billion of leveraged positions in a single day (21Shares, 2025); on May 6, 2010, US equity prices fell and recovered by

several percent within minutes (Kirilenko et al., 2017). In both episodes the losses that mattered were transient: positions were closed out on the path, before prices settled. Algorithmic markets are closed-loop systems, and a common shortcut equates their stability with eigenvalues inside the unit circle. For a leveraged intermediary that is the wrong object: a market can be spectrally stable and still push a dealer through its maintenance margin within seconds.

The chapter asks three questions. When does a limit order book amplify a flow shock? How can a dealer or a venue control transient loss without identifying who is trading, which Chapter 2 shows is expensive from public data? And what information certifies a safe trading decision, at what cost? The answers, in a linear book model, are these.

Spectral stability does not bound transient loss. The safe shock radius of a stable market is the loss budget over its peak transient gain, and that gain can be arbitrarily large at any fixed spectral radius (Theorem 3.3.1).

When a book amplifies. A book with depletion and refill overshoots its direct impact if and only if $\varphi\varepsilon > (1-a)\kappa$; momentum alone cannot make a stable book overshoot; and cumulative risk has the closed form $S = \tilde{\kappa}/(m - q\tilde{\kappa})$, which needs no depletion (Theorems 3.4.2 and 3.4.3).

The maximal guard. Under a declared risk contract, worst-case loss is inventory times the shock bound times the summed absolute impulse response, so a guard on that sum admits exactly the safe actions (Lemma 3.6.1); a guard on the peak response admits unsafe ones. The guard survives estimation error (Theorem 3.6.2) and needs no identification of the traders (Proposition 3.2.2).

Certifying is cheaper than identifying. Two markets can agree on their early responses and differ threefold in risk (Example 3.5.1). A finite window certifies the summed response under a memory bound and certifies nothing without one (Theorem 3.5.3); with noisy experiments the cost of certification is $\Theta(\sigma^2\epsilon^{-2}\log(1/\epsilon))$ whatever the cohort structure, against $(\Delta\eta)^{-(4K-2)}$ for identification (Theorem 3.5.4). The guard built on the certificate has a closed form and is minimax-optimal (Theorem 3.6.4); disclosure is worth most when memory is long, and a noisy proxy can be worth less than nothing (Theorem 3.7.1).

Who controls the risk. Reversion is a public good: dealers under-provide it, momentum traders over-provide feedback, and the corrective rebate is a function of the guard's own statistic (Proposition 3.8.1).

Evidence. Twenty-five registered primaries were tested with hashed specifications and test sets opened once; fifteen pass (Table 3.4). The deepest book, BTCUSDT, sits on the stable side of the amplification condition and fails a registered forecasting test. In thin books the

statistic forecasts the size of the next hour's adverse excursion in every registered forecasting test on Binance, on new symbols and on untouched days, by up to 2.8 points of R^2 beyond a full volatility baseline; about a fifth as much in thin US stocks; negligibly on Bybit; not at all on Coinbase. We have not identified a deeper mechanism: the effect does not run through open-interest liquidations, and it is carried by slow reversion relative to impact, which is what the statistic measures. As a risk control the certified guard lowers tail loss against a static limit but not against a volatility cap at matched exposure.

What we do not claim. The information-safety theorems concern declared observation contracts (controlled reset experiments, exact windows, audited data), not finite-sample guarantees for observational order-book records. Cohorts are any traders whose demand is a linear filter of past prices; nothing attributes an effect to algorithms rather than people. The forecasting result is reduced-form, and its size depends on the venue.

Related work. Models of transient impact and resilience (Obizhaeva and Wang, 2013; Bouchaud et al., 2004, 2018) take the decay of impact as given; we derive when depletion, refill and momentum make a book overshoot and give cumulative impact in closed form. Liquidity spirals (Brunnermeier and Pedersen, 2009; Shleifer and Vishny, 1992) need funding constraints; our amplification does not, and our guard is the control that bounds it. Sornette et al. (2023) show that non-normal interactions create transient bubbles. The guard's worst-case identity is the ℓ_1 gain of ℓ_1-optimal control (Vidyasagar, 1986; Dahleh and Pearson, 1987; Dahleh and Diaz-Bobillo, 1995); what is new is its use as an admission rule, its invariance to the cohort coordinates and its identification from prices. The early-warning benchmark is VPIN (Easley et al., 2012; Andersen and Bondarenko, 2014); forecast evaluation uses Clark and West (2007) with the volatility baselines of Corsi (2009) and J.P. Morgan/Reuters (1996), and we pre-register following Nosek et al. (2018) and Harvey (2017). Rule 15c3-5 requires pre-trade controls on credit and capital (SEC, 2010); the guard is a candidate with a provable maximality property.

3.2 The closed-loop market

3.2.1 Price formation

Time is discrete with bars of length Δ. Let p_t be the price, $p_t^* = p_{t-1}^* + \xi_t$ the unobserved fundamental and $x_t = p_t - p_t^*$ the pricing error. Competitive dealers post aggregate depth $D > 0$, so price impact is $\kappa = 1/D$, and fundamental arbitrage removes a fraction $\kappa\varphi_f$ of the error each period. With algorithmic demand u_t and liquidity noise, the pricing error follows

$$x_{t+1} = a x_t + \kappa u_t + w_{t+1}, \qquad a = 1 - \kappa\varphi_f \in [0, 1). \tag{3.1}$$

Algorithmic cohorts trade $u_t = \sum_i q_i y_{i,t}$ with exponential memory $y_{i,t+1} = (1-\eta_i)y_{i,t} + \eta_i x_{t+1}$, a reduced form that arises when an adaptive desk updates its belief by steady-state Kalman filtering or regularized online gradient descent and maximizes expected continuation return $\beta_i y_{i,t} u_{i,t}$ minus quadratic execution cost $\frac{1}{2}\lambda_{\mathrm{tc},i} u_{i,t}^2$ and inventory penalty $\frac{1}{2}\gamma_{\mathrm{inv},i} I_{i,t}^2$ ($I_{i,t} = (1-\delta_{\mathrm{inv}})I_{i,t-1} + u_{i,t}$, $|I_{i,t}| \le \bar{I}_i$), yielding $u_{i,t}^* = q_i y_{i,t} - \psi_{\mathrm{inv},i} I_{i,t-1}$ with $q_i = \beta_i/(\lambda_{\mathrm{tc},i} + \gamma_{\mathrm{inv},i})$ and $\psi_{\mathrm{inv},i} = \gamma_{\mathrm{inv},i}(1-\delta_{\mathrm{inv}})/(\lambda_{\mathrm{tc},i} + \gamma_{\mathrm{inv},i})$ (Chapter 2; this single-agent filtering-and-control rule is distinct from the joint dealer–trader equilibrium of Section 3.8). The baseline equations set $\psi_{\mathrm{inv},i} = 0$, the limit in which inventory holding penalties are small relative to execution costs ($\gamma_{\mathrm{inv},i} \ll \lambda_{\mathrm{tc},i}$) or inventory is reset at the bar horizon; Proposition 3.2.3 below relaxes this and gives the closed-form summed response when cohorts carry inventory across bars with $\psi_{\mathrm{inv},i} > 0$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_{t+1} = Jz_t + bw_{t+1}$, where $b = (1, \eta_1, \ldots, \eta_K)^\top$ because cohorts update on the same period's price (a unit of flow enters through κb), and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1-\eta_1+\eta_1\kappa q_1 & \cdots & \eta_1\kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K\kappa q_1 & \cdots & 1-\eta_K+\eta_K\kappa q_K \end{pmatrix}. \tag{3.2}$$

3.2.2 Stability and what an observer can use

Proposition 3.2.1 (Stability with one cohort)**.** *With one cohort (q, η) and $a = 1 - \kappa\varphi_f$, the market is Schur stable if and only if*

$$q < \varphi_f, \qquad \kappa < \frac{4-2\eta}{(2-\eta)\varphi_f - \eta q} \quad \text{when } (2-\eta)\varphi_f > \eta q, \qquad |1-\kappa\varphi_f|(1-\eta) < 1. \tag{3.3}$$

Higher impact κ (thinner depth) lowers price persistence a; feedback is destabilizing only when cohort aggressiveness q reaches the arbitrage strength φ_f.

Chapter 2 shows that prices identify only the merged rate–weight measure $M = \sum_i q_i \eta_i \delta_{1-\eta_i}$ of the cohorts. A pre-trade control must therefore depend on the market only through objects that are the same for every cohort representation of M.

Proposition 3.2.2 (Input–output invariance)**.** *Two populations with the same merged measure generate the same transfer function $H(z) = \sum_k g_k z^{-k} = \kappa z/[z - a - \kappa G(z)]$, $G(z) = \sum_i q_i\eta_i z/(z-1+\eta_i)$, and the same impulse coefficients $g_k = \kappa\, c_{\mathrm{price}}^\top J^k b$ for every $k \ge 0$. Under any invertible change of state coordinates $z \mapsto Tz$, the Euclidean gain $\|CT^{-1}(TJT^{-1})^k\|_2$ depends on T, whereas the price path*

$$\Delta p_{t+k} = \sum_{j\ge 1} g_{k-1+j} u_{t-j} + \sum_{m=0}^{k-1} g_{k-1-m} u_{t+m}$$

and its worst case over $\|u\|_\infty \le \bar{u}$ depend only on $\{g_k\}$ and are invariant to both T and cohort splitting.

Numerically, splitting a cohort of 1,000 random economies into two with the same memory rate leaves every impulse coefficient unchanged up to rounding ($\max_k |g_k(\mathcal{E}^A) - g_k(\mathcal{E}^B)| \le 2.5 \times 10^{-12}$ over 200 lags; `recompute_oct20.py`). Proposition 3.2.2 is the reason we build the guard on impulse coefficients and avoid state norms in latent coordinates. When a state norm is needed, we fix units by observable calm-market innovation standard deviations, which is what makes the spot statistic of Section 3.9 unit-free.

Proposition 3.2.3 (Summed response with carried cohort inventory)**.** *Suppose each cohort i carries inventory $I_{i,t} = (1 - \delta_{\text{inv}})I_{i,t-1} + u_{i,t}$ with $\delta_{\text{inv}} \in (0, 1]$ and submits order flow $u_{i,t} = q_i y_{i,t} - \psi_i I_{i,t-1}$ with $q_i \ge 0$ and $\psi_i \ge 0$. Then:*

(i) The closed-loop transfer function from a flow impulse to price is $H_{\text{inv}}(z) = \kappa z / \big[z - a - \kappa G_{\text{inv}}(z)\big]$ with

$$G_{\text{inv}}(z) = \sum_{i=1}^{K} \frac{q_i \eta_i z(z - 1 + \delta_{\text{inv}})}{(z - 1 + \eta_i)(z - 1 + \delta_{\text{inv}} + \psi_i)},$$

which is invariant to splitting any cohort that shares (η_i, ψ_i). Whenever the augmented state matrix is Schur stable, the signed cumulative price response is

$$S(\psi) = \sum_{t=0}^{\infty} h_t(\psi) = H_{\text{inv}}(1) = \frac{\kappa}{1 - a - \kappa \Phi_{\text{eff}}(\psi)}, \qquad \Phi_{\text{eff}}(\psi) \equiv \sum_{i=1}^{K} \frac{\delta_{\text{inv}}}{\delta_{\text{inv}} + \psi_i} q_i \le \Phi. \tag{3.4}$$

(ii) If $0 \le \psi_i \le 1 - \delta_{\text{inv}}$ and $\delta_{\text{inv}} \ge \eta_i$ for each i and $a + \kappa \Phi_{\text{eff}}(\psi) < 1$, every open-loop feedback tap $k_m(\psi)$ and hence every closed-loop impulse coefficient $h_t(\psi)$ is nonnegative, so the worst-case ℓ_1 gain equals the signed sum, $\|h(\psi)\|_1 = \sum_{t=0}^{\infty} |h_t(\psi)| = S(\psi)$. When ψ_i is large enough that inventory payback reverses the sign of late taps, $\|h(\psi)\|_1 > S(\psi)$.

Inventory mean-reversion $\psi_i > 0$ acts as an automatic stabilizer that attenuates each cohort's long-run feedback weight from q_i to $\frac{\delta_{\text{inv}}}{\delta_{\text{inv}} + \psi_i} q_i$ (across 500 random stable economies (3.4) matches the simulated state-space impulse sum to 1.78×10^{-15}, and in the monotone regime $\|h(\psi)\|_1 = S(\psi)$ to 1.78×10^{-15}, reducing S from 2.7778 at $\psi = 0$ to 2.5000 at the calibrated worked example; in the sign-reversal regime $\|h(\psi)\|_1 = 2.0960 > S(\psi) = 1.7241$; `recompute_p2_step2_theory.py`).

The model is linear: it describes the transient response to flow shocks of ordinary size, which is what a pre-trade control must bound, not regime changes such as exchange outages. Every input to the guard (impulse coefficients, an operator estimate and its confidence radius, inventory and margin) is observable to a dealer or a venue.

3.3 Spectral stability is not solvency

Theorem 3.3.1 (Safe radius, Kreiss bounds and the failure of the spectral radius)**.** *Let $z_t = J^t z_0$ with $\rho(J) < 1$, loss budget $B > 0$ and loss output C.*

(i) The largest initial-state radius with $\sup_t \|CJ^t z_0\|_2 \le B$ for every $\|z_0\|_2 \le r$ is

$$r^*_{\text{exact}} = \frac{B}{M_C}, \qquad M_C \equiv \sup_{t\ge 0} \|CJ^t\|_2. \tag{3.5}$$

(ii) With $C = I$ and the Kreiss constant $\mathcal{K}(J) = \sup_{|z|>1}(|z|-1)\|(zI-J)^{-1}\|_2$, the Kreiss matrix theorem gives $\mathcal{K}(J) \le \sup_t \|J^t\|_2 \le e\,n\,\mathcal{K}(J)$ for $J \in \mathbb{R}^{n\times n}$, hence

$$\frac{B}{e\,n\,\mathcal{K}(J)} \;\le\; r^*_{\text{exact}} \;\le\; \frac{B}{\mathcal{K}(J)}. \tag{3.6}$$

(iii) For any $\rho \in (0,1)$ and any $B > 0$ there is a family J_θ with $\rho(J_\theta) = \rho$ for all θ and an output $C = (1,0)$ such that $r^*_{\text{exact}}(J_\theta) \to 0$ as $\theta \to \infty$.

For $J_\theta = \left(\begin{smallmatrix} 0.8 & \theta \\ 0 & 0.4 \end{smallmatrix}\right)$ the spectral radius is 0.8 throughout, while the price-row gain rises from 1 to 120 as θ goes from 0 to 100 and the safe radius falls by the same factor. Chapter 2 shows more: markets that share their eigenvalues, one-step price variance and stationary price variance can differ arbitrarily in worst-case transient gain, so neither eigenvalue nor volatility monitors bound transient loss. In the cohort economy with nonnegative weights the price response is a nonnegative mixture of decaying exponentials and peaks at the direct impact (Chapter 2, positive-geometry lemma): non-normality is necessary for amplification but not sufficient, and adaptive momentum alone does not supply it. Depth depletion does.

3.4 When does a book amplify? An exact condition

Transient path risk takes two forms. *Peak overshoot* is a response that rises above its direct impact; *cumulative amplification* is a summed response much larger than direct impact. The data support the second form, not the first.

3.4.1 A book with depletion and refill

Let x_t be the price displacement, d_t the signed depth imbalance (bid minus ask depth near the mid, relative to total) and u_t signed order flow. Flow moves the price directly and depletes one side of the book, which refills at rate r; depth imbalance moves the price:

$$x_t = a\,x_{t-1} + \varphi\,d_{t-1} + \kappa\,u_t, \tag{3.7}$$

$$d_t = (1-r)\,d_{t-1} + \varepsilon\,u_t, \tag{3.8}$$

with $\kappa > 0$, $a \in [0,1)$ and $r \in (0,1]$. Write $b = 1 - r$. With the sign convention $d = (B - A)/(B + A)$, a buy that consumes ask depth raises d, so $\varepsilon > 0$ means *depletion* of the side that is hit, and $\varphi > 0$ means that a bid-heavy book pushes the price up. The impulse response to a unit flow shock at $t = 0$ is $h_0 = \kappa$ and

$$h_k = a\, h_{k-1} + \varphi\varepsilon\, b^{k-1}, \qquad k \geq 1. \tag{3.9}$$

Definition 3.4.1. The book *amplifies* if $\max_k h_k > h_0$: the price keeps moving in the direction of the shock after the shock has passed.

3.4.2 Peak overshoot

Add momentum traders whose flow $f_t = (1-\eta)f_{t-1} + \eta q\, x_{t-1}$ enters both (3.7) and (3.8) alongside u_t; the two-state model is the case $q = 0$. Write $C_2 \equiv \varphi\varepsilon - (1-a)\kappa$ and $C_3 \equiv \varphi\varepsilon + \kappa^2\eta q - (1-a)\kappa$.

Theorem 3.4.2 (Peak overshoot: depletion versus momentum)**.** *(i)* Two-state condition. *Without momentum ($q = 0$) the book amplifies if and only if $C_2 > 0$; when $C_2 \leq 0$ the response peaks at impact,* $\max_k h_k = \kappa$.

(ii) Shape. *Let $q = 0$, $0 < a < 1$ and $0 < b < 1$. If $a \neq b$, $h_k = Aa^k + Bb^k$ with $B = \varphi\varepsilon/(b-a)$ and $A = \kappa - B$; if $a = b$, $h_k = a^{k-1}(a\kappa + \varphi\varepsilon k)$ for $k \geq 1$. When the book amplifies, h viewed as a function of real $k \geq 0$ has a single maximum, at*

$$k^* = \frac{\log\bigl(-A\log a/(B\log b)\bigr)}{\log(b/a)} \quad (a \neq b), \qquad k^* = \frac{1}{\log(1/a)} - \frac{a\kappa}{\varphi\varepsilon} \quad (a = b),$$

the second being the limit of the first as $b \to a$; the peak lag is $\lfloor k^* \rfloor$ or $\lceil k^* \rceil$ (and is 1 when $k^* < 1$, because $h_1 > h_0$). The cumulative response is $\sum_k h_k = \kappa/(1-a) + \varphi\varepsilon/[(1-a)r]$ in all cases, and faster refill lowers both the peak and the cumulative response. When $a = 0$ ($h_k = \varphi\varepsilon b^{k-1}$) or $r = 1$ ($h_k = a^{k-1}(a\kappa + \varphi\varepsilon)$) the response peaks at lag 1.

(iii) One-period margin with momentum. *$h_1 - h_0 = C_3$, so $C_3 > 0$ implies that the book amplifies.*

(iv) Momentum alone cannot amplify. *If $\varphi\varepsilon = 0$, $q \geq 0$ and the system is stable, then $h_k = c_1\lambda_1^k + c_2\lambda_2^k$ with residues $c_1, c_2 \geq 0$, $c_1 + c_2 = \kappa$ and $|\lambda_2| \leq \lambda_1 = \rho < 1$, so $h_k \leq \kappa\rho^k < \kappa$ for every $k \geq 1$; in particular $C_3 < 0$.*

(v) A sufficient condition for no overshoot. *For any signs of φ, ε and q, if*

$$\bar{C}_3 \equiv |\varphi\varepsilon|\Bigl(1 + \frac{\kappa|q|}{r}\Bigr) + \kappa^2|q| - (1-a)\kappa \;\leq\; 0,$$

then $\sup_k |h_k| = \kappa$. When $\varphi\varepsilon \geq 0$ and $q \geq 0$, $C_3 > 0 \Rightarrow$ amplification $\Rightarrow \bar{C}_3 > 0$.

Proof. (i) *If.* $h_1 - h_0 = a\kappa + \varphi\varepsilon - \kappa = C_2 > 0$. *Only if.* Suppose $C_2 \le 0$. If $\varphi\varepsilon < 0$, then by (3.9) $h_k = a^k\kappa + \varphi\varepsilon \sum_{j<k} a^{k-1-j} b^j \le a^k\kappa \le \kappa$. If $\varphi\varepsilon \ge 0$, let $\Delta_k = h_k - h_{k-1}$. Then $\Delta_1 = C_2 \le 0$ and, from (3.9), $\Delta_{k+1} = a\Delta_k + \varphi\varepsilon(b^k - b^{k-1}) = a\Delta_k - \varphi\varepsilon\, r\, b^{k-1} \le a\Delta_k$, so by induction $\Delta_k \le 0$ for all k and h peaks at $h_0 = \kappa$.

(ii) Solving (3.9) gives the two forms. For $a \neq b$, $\frac{d}{dk}h = Aa^k \log a + Bb^k \log b$ has at most one zero because $(b/a)^k$ is monotone, which gives k^*; for $a = b$, $\frac{d}{dk}a^{k-1}(a\kappa + \varphi\varepsilon k) = a^{k-1}[(a\kappa + \varphi\varepsilon k)\log a + \varphi\varepsilon]$ vanishes once, at the stated k^*, and expanding the first formula in $b - a$ gives the same limit. A unimodal function of real k attains its integer maximum at $\lfloor k^* \rfloor$ or $\lceil k^* \rceil$. Summing the geometric series gives the cumulative response, which falls in r; the peak falls in r because h_k is decreasing in r term by term for $\varphi\varepsilon > 0$.

(iii) With a unit shock at $t = 0$, $x_0 = \kappa$, $d_0 = \varepsilon$ and $f_1 = \eta q \kappa$, so $h_1 = a\kappa + \varphi\varepsilon + \kappa^2\eta q$.

(iv) Without the depth channel the state (x_t, f_t) evolves by $M = \left(\begin{smallmatrix} a+\kappa\eta q & \kappa(1-\eta) \\ \eta q & 1-\eta \end{smallmatrix}\right)$, which is entrywise nonnegative, and $h_k = \kappa(M^k)_{11}$. The off-diagonal product is nonnegative, so the eigenvalues are real, and by the Perron–Frobenius theorem $\lambda_2 \le m_{11} \le \lambda_1 = \rho$. Writing $(M^k)_{11} = c_1'\lambda_1^k + c_2'\lambda_2^k$ with $c_1' = (m_{11} - \lambda_2)/(\lambda_1 - \lambda_2) \in [0, 1]$ and $c_2' = 1 - c_1'$ gives the residues $c_j = \kappa c_j' \ge 0$ and $(M^k)_{11} \le \rho^k$. Since $m_{11} \le \rho < 1$, $C_3 = \kappa(m_{11} - 1) < 0$.

(v) Suppose $|x_j| \le \kappa$ for $j < k$. Then $|f_j| \le |q|\kappa$ for $j \le k$ and $|d_{k-1}| \le |\varepsilon|\big(b^{k-1} + \sum_{j\ge1} b^{k-1-j}|q|\kappa\big) \le |\varepsilon|(1 + |q|\kappa/r)$, so $|x_k| \le a\kappa + |\varphi\varepsilon|(1 + \kappa|q|/r) + \kappa^2|q| \le \kappa$. Induction from $x_0 = \kappa$ completes the proof. □

Interpretation. The left side $\varphi\varepsilon$ of C_2 is the extra price pressure that a unit of flow creates one period later through the book it leaves behind. The right side $(1 - a)\kappa$ is the part of the direct impact that reverts in one period. The book amplifies exactly when the depth channel adds more than mean reversion removes. Refill r does not enter the condition; it governs how long an overshoot lasts.

The converse of part (iii) fails with two slow modes: near the stability boundary, or with slow refill and slow momentum, a book can have $C_3 < 0$ and still peak far above its impact, because the depth channel creates a negative residue on an intermediate mode. No condition on the one-period margin alone is exact.

3.4.3 Cumulative transient risk

Theorem 3.4.3 (Summed response in closed form)**.** *Let $\tilde\kappa = \kappa + \varphi\varepsilon/r$ (long-run impact: direct plus depth) and $m = 1 - a$ (reversion). In the stable three-state model,*

$$S \equiv \sum_{k\ge0} h_k = \frac{\tilde\kappa}{m - q\tilde\kappa},$$

and when $\varphi\varepsilon \ge 0$ and $q \ge 0$ the system is stable only if $m > q\tilde\kappa$. When all $h_k \ge 0$, S is the summed absolute response that enters the guard of Section 3.6.

Proof. $S = e_1^\top (I - M_3)^{-1}(\kappa, \varepsilon, 0)^\top$ with M_3 the transition matrix of (x, d, f). Direct computation gives $\det(I - M_3) = \eta\,[\,r(m - \kappa q) - \varphi\varepsilon q\,]$ and the stated ratio; the formula matches the simulated sums to a relative error of 7×10^{-14} in 3,000 stable draws. If $\varphi\varepsilon \geq 0$ and $q \geq 0$, M_3 is entrywise nonnegative and stability requires $\det(I - M_3) > 0$, i.e. $m > q\tilde{\kappa}$. □

Cumulative transient risk is long-run impact divided by reversion net of momentum. Two practical consequences follow. First, the statistic that matters for losses is the summed response, which grows without bound as reversion falls to $q\tilde{\kappa}$, even when no single lag overshoots (Theorem 3.4.2(iv)). Second, C_3 is best read as a one-period amplification margin: it moves with m, q and $\tilde{\kappa}$ in the same direction as S and is what we estimate in Sections 3.9 and 3.9.2. Both statistics are invariant in sign to rescaling flow.

3.5 Identification and financial safety

Chapter 2 shows that identifying the cohorts from prices is expensive. A pre-trade control does not need them; it needs a bound on the risk they create. This section asks what information certifies a safe decision, what it costs, and how much safe trading is lost without it.

3.5.1 Observation and risk contracts

We work in the positive cohort economy (3.2) with $q_i \geq 0$ and known a and κ. Its merged measure $M = \sum_i q_i \eta_i \delta_{r_i}$, $r_i = 1 - \eta_i$, has kernel $k_m = \int r^m\, dM$, and the price response to a unit of flow obeys $h_0 = \kappa$, $h_{t+1} = a h_t + \kappa \sum_{m \leq t} k_m h_{t-m}$. All $h_t \geq 0$ (Chapter 2, positive-geometry lemma), and the summed response is

$$S = \sum_t h_t = \frac{\kappa}{1 - a - \kappa\Phi}, \qquad \Phi = \int \frac{dM(r)}{1 - r} = \sum_i q_i, \tag{3.10}$$

with stability exactly when $a + \kappa\Phi < 1$. A *memory bound* $\bar{r} < 1$ states that every memory rate satisfies $r_i \leq \bar{r}$.

Risk contract. A position I is held fixed, each period's flow shock is bounded by $\bar{u}$, losses are marked to market at every date, and the horizon is unbounded. By Lemma 3.6.1 the worst-case loss is $|I|\bar{u}S$, so the safe capacity, the largest shock bound with loss at most the budget $\mathcal{B}$, is $\mathcal{B}/(|I|S)$. Section 3.6 adds admission, margin and fills. All statements below are deterministic worst cases under this contract; none is a probability-of-ruin statement.

Observation contracts. We consider four: (O1) N independent reset experiments, each returning $h_0, \ldots, h_{T-1}$ with independent Gaussian errors of standard deviation σ per tap, the controlled-probe contract of Chapter 2; (O2) an exact window $h_0, \ldots, h_n$; (O3) class-tagged flows; and (O4) audited exposure, which reveals Φ. The theorems concern these contracts. Observational public records of a live book are a different contract, for which we offer evidence

(Section 3.9) but no finite-sample guarantee.

Given data D, let $\Theta(D)$ be the set of economies compatible with it. A decision is *certifiably safe* if its worst-case loss is within budget for every economy in $\Theta(D)$, *certifiably unsafe* if it exceeds the budget for every one, and *undecidable* otherwise. This classification is a definition, not a result; it is the finance version of the data-informativity view of control without identification (van Waarde et al., 2020) and of set-membership estimation (Milanese and Vicino, 1991). The results below make it explicit for this economy.

3.5.2 Two markets that look the same

Example 3.5.1 (An exact ambiguity witness). Let $a = 0.4$ and $\kappa = 1$. Market A has one cohort ($q = 0.08$, $\eta = 0.5$); market B has two ($q = 0.02/0.95$, $\eta = 0.95$; $q = 0.4$, $\eta = 0.05$). Their merged measures have the same mass and first moment, so in rational arithmetic their price responses coincide at lags 0, 1 and 2 (1, 11/25, 267/1250) and first differ at lag 3. Their summed responses are 1.92 and 5.59, so for a unit position and budget the safe shock bound is 0.52 in A and 0.18 in B.

Proposition 3.5.2 (Unavoidable loss of safe capacity)**.** *Let two economies have safe capacities $v_0 < v_1$ and data laws P_0, P_1. If a data-driven rule $A(D) \in [0, v_1]$ admits an unsafe level in economy 0 with probability at most β, $P_0(A > v_0) \le \beta$, then*

$$\mathbb{E}_1[v_1 - A(D)] \;\ge\; (v_1 - v_0)\,\big[1 - \beta - \mathrm{TV}(P_0, P_1)\big]_+.$$

The bound is attained when $P_0 = P_1$ by the rule that admits v_1 with probability β and v_0 otherwise.

Proof. Let $E = \{A \le v_0\}$. Then $P_0(E) \ge 1 - \beta$ and $P_1(E) \ge P_0(E) - \mathrm{TV}$. On E the capacity given up in economy 1 is at least $v_1 - v_0$, and it is nonnegative elsewhere. □

The argument is the two-point method of statistical decision theory (Le Cam, 1986); its use here is to price missing information in units of safe trading. In Example 3.5.1, a supervisor who sees the first three coefficients and must admit an unsafe level in B with probability at most 5% gives up at least $0.95 \times 0.34 = 0.32$ units of safe shock capacity in A.

3.5.3 What a finite window certifies

Lags 1 to n of the response determine the first n moments of M exactly, through $k_t = (h_{t+1} - a h_t - \kappa \sum_{m<t} k_m h_{t-m})/\kappa^2$. The summed response depends on M only through $\Phi = \int f\, dM$ with $f(r) = (1 - r)^{-1}$. Certifying it from a window is therefore an extremal moment problem.

Theorem 3.5.3 (Finite-window certification)**.** *Observe $h_0, \dots, h_n$ exactly under a memory bound $\bar{r} < 1$, and suppose the implied moment vector $(k_0, \dots, k_{n-1})$ lies in the interior of the*

moment space $\mathcal{M}_n([0,\bar{r}])$ (equivalently, the Hankel matrices of the moment sequence and of its localization by $r(\bar{r}-r)$ are positive definite). Let μ_- and μ_+ be its lower and upper principal representations on $[0,\bar{r}]$ (the extremal measures of the Markov–Krein theorem, unique in the interior; μ_+ has an atom at $\bar{r}$), and $\Phi_\pm = \int f\, d\mu_\pm$. On the boundary of the moment space the compatible measure is unique and $\Phi_- = \Phi_+$.

(i) Sharp interval. *The summed responses of stable economies compatible with the data form the interval $[S(\Phi_-), S(\Phi_+)]$ if $a + \kappa\Phi_+ < 1$, and $[S(\Phi_-), \infty)$ otherwise. Both endpoints are attained by economies with at most $\lfloor n/2 \rfloor + 1$ cohorts.*

(ii) Geometric convergence. *$\Phi_+ - \Phi_- \le 2k_0 E_{n-1}$, where $E_{n-1} = (2/\bar{r})\,\varrho^{n-1}/(x_0^2-1)$ is the error of the best uniform approximation of f on $[0,\bar{r}]$ by polynomials of degree $n-1$, $x_0 = 2/\bar{r} - 1$ and $\varrho = x_0 - \sqrt{x_0^2-1} < 1$. A window of length $n = O\big(\log(1/\epsilon)/\log(1/\varrho)\big)$ certifies Φ to precision ϵ, and $\log(1/\varrho) \sim 2\sqrt{(1-\bar{r})/\bar{r}}$ as $\bar{r} \to 1$.*

(iii) No certificate without a memory bound. *If memory rates may lie anywhere in $[0,1)$, then for every n and every window whose moment vector is interior to the moment space, compatible stable economies have arbitrarily large summed responses: no finite window certifies any finite risk bound.*

Part (i) applies the Markov–Krein theorem (Krein and Nudel'man, 1977; Karlin and Studden, 1966), part (ii) Chebyshev's best approximation of $1/(x_0 - x)$ (Achieser, 1956); the proofs are in Appendix 3.13. In an eight-cohort economy with memory bound 0.95 and true summed response 5.56, lags 0–3 certify nothing, lags 0–6 give $[5.01, 7.20]$ and lags 0–8 give $[5.30, 6.00]$; for fixed data the upper end grows without limit as the memory bound approaches one, as part (iii) says. The endpoints are proved, not only computed: polynomial bounds whose sign conditions are verified in interval arithmetic with outward rounding turn them into certified intervals (Rump, 2010).

3.5.4 What certification costs, and what identification costs

Theorem 3.5.4 (Certification cost of the summed response)**.** *Consider the class $\mathcal{K}(\bar{r},\bar{s})$ of positive economies with memory bound $\bar{r}$ and $a + \kappa\Phi \le \bar{s} < 1$, observed under contract (O1). Let $\bar{\lambda} = 1 - (1-\bar{r})(1-\bar{s})$ and $\bar{h}_t$ the average of the N responses.*

(i) A valid certificate. *For every economy in the class, $h_t \le \kappa\bar{\lambda}^t$, and*

$$\hat{S}_T^+ = \sum_{t<T} \bar{h}_t + z_{1-\gamma}\,\sigma\sqrt{T/N} + \frac{\kappa\bar{\lambda}^T}{1-\bar{\lambda}}$$

satisfies $P(S \le \hat{S}_T^+) \ge 1-\gamma$. With $T_\epsilon = \lceil \log(2\kappa/((1-\bar{\lambda})\epsilon))/\log(1/\bar{\lambda}) \rceil$ and $N \ge 16 z_{1-\gamma}^2 \sigma^2 T_\epsilon/\epsilon^2$, the excess $\hat{S}^+ - S$ is at most ϵ with probability at least $1-2\gamma$. Certification to precision ϵ therefore costs $N = O(\sigma^2\epsilon^{-2}\log(1/\epsilon))$.

(ii) A matching lower bound. *Fix an interior single-cohort economy θ_0 and let $C_0 = \|\partial h/\partial q\|_2/(\partial S/\partial q)$ at θ_0. Any procedure that is valid at level $1-\gamma$ on the class and overshoots S by less than 2ϵ with probability at least $1/2$ at θ_0 needs $N \geq \big((1-2\gamma)\sigma/(2C_0\epsilon)\big)^2(1+o(1))$ as $\epsilon \to 0$.*

(iii) The logarithm is necessary. *There is $c > 0$ depending only on $(a, \kappa, \bar{r}, \bar{s})$ such that any procedure that is valid at level $1-\gamma$, $\gamma < 1/4$, on the class and overshoots S by at most ϵ with probability at least $1-\gamma$ at every economy of the class needs $N \geq c\,\sigma^2\epsilon^{-2}\log(1/\epsilon)$ for all small ϵ, however many responses each experiment records. With part (i), the cost of certifying S is $\Theta(\sigma^2\epsilon^{-2}\log(1/\epsilon))$.*

(iv) Independence of the cohort structure. *None of these bounds depends on the number of cohorts or on the spacing of their memories, whereas identifying them costs $N \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ with exponential memory and $\exp(\Theta(1/w))$ experiments for accuracy w with general memory (Chapter 2).*

The certificate in part (i) is elementary: it is a sum of averaged taps, a Gaussian margin and a tail that the memory bound controls. Part (iv) shows that its logarithm is not an artefact of truncation: no procedure, however it weights the observed responses, certifies the summed response at the parametric rate, because $1/(1-r)$, the kernel of the total feedback, has a pole at infinite memory and any polynomial that approximates it on the memory range must carry large coefficients. Part (ii) is the parametric bound at a fixed economy; part (iv) is the worst case over the class. The content is in the comparison. In contrast to the $\mathcal{H}_\infty$ norm, which Tu et al. (2018) show is no easier to estimate than the model itself, the summed response of a positive economy is a linear functional of the response, so certifying it costs only one logarithm more than a parametric problem, while the cohort parameters cost polynomially more. Table 3.1 makes this concrete on Chapter 2's hardest two-cohort pairs. Those pairs have *identical* summed responses, because they share the total weight $\sum_i q_i$: the directions in which prices are least informative are directions in which the risk a guard must bound does not move. Identifying their weights to ± 0.01 needs from 7 to 1.5 million reset experiments as the spacing falls from 0.30 to 0.04 (log–log slope -6.06), while certifying the summed response to 1% needs 11 to 18 at every spacing (slope $+0.23$).

Two refinements are in the full version. If the noise in the taps of one experiment is correlated, the certificate of part (i) can under-cover (75% at nominal 95% in simulation), while a Student-t certificate on the summed taps and a Hotelling region over the window keep their level (Hotelling, 1931); and experiments can be run one at a time with an anytime-valid version of the certificate that stops once the data decide.

Table 3.1: Identifying the cohorts versus certifying the risk they create

Rate spacing $\Delta\eta$	0.30	0.20	0.14	0.10	0.08	0.06	0.05	0.04
Identify weights (± 0.01), N	7	88	775	5,928	22,741	128,333	383,859	1,466,362
Certify S (1%), N	18	14	13	12	11	11	11	11

The two-cohort moment-matched pairs of Chapter 2 ($q = 0.2$, mean rate 0.4, $\tau = 0.01$, $a = 0.5$, $\kappa = 0.8$, per-tap noise 1.6×10^{-3}). Identification: Le Cam requirement at 95% (Chapter 2, Table on two-point sample complexity). Certification: sufficient N from Theorem 3.5.4(i) with memory bound equal to the largest rate in the pair and $\bar{s} = a+\kappa q+0.02$. The two economies of each pair have the same summed response, 2.353. Kind: recomputed (`research_extension/information_safety`, test T03).

3.6 An identification-free pre-trade guard

3.6.1 Maximality

A dealer with inventory I chooses an admission level $\alpha \in [0, 1]$ (the fraction of incoming algorithmic flow it accepts, or the depth it quotes). The closed-loop operator is $J(\alpha) = J_0 + \alpha B_F$ with price impulse response $g_k(\alpha)$. Shocks satisfy $|u_t| \le \bar{u}$. Over horizon H under a reset initial state $z_0 = 0$ (or after subtracting a known or directly observed initial drift $b_t = CJ(\alpha)^t z_0$ from the remaining budget), the shock-driven price change is $\sum_{k<H} g_k(\alpha) u_{t-k}$, and the dealer has a loss budget $\mathcal{B}$. The contract is: inventory $I(\alpha)$ is held fixed over the horizon, the initial state is reset ($z_0 = 0$) or its drift is observed, each period's shock is bounded by $\bar{u}$, and loss is the mark-to-market loss at any date in the horizon; because the partial sums $\sum_{k<h} |g_k|$ increase in h, the worst date is the last. Margin, hedging, fills and non-zero initial states are treated in the full version.

Lemma 3.6.1 (Summed-gain guard is maximal)**.** *The worst-case loss over horizon H is*

$$L_{\max}(\alpha) = |I(\alpha)|\, \bar{u} \sum_{k<H} |g_k(\alpha)|.$$

Hence the guard "admit α iff $L_{\max}(\alpha) \le \mathcal{B}$" is necessary and sufficient for bounded-shock solvency: it admits every safe α and no unsafe one. Replacing $\sum_k |g_k|$ by $\max_k |g_k|$ can admit unsafe levels; replacing it by $H \max_k |g_k|$ is safe but conservative.

Proof. By Hölder's inequality $|\sum_k g_k u_{t-k}| \le \bar{u} \sum_k |g_k|$, with equality for $u_{t-k} = \bar{u}\,\mathrm{sign}(g_k)$, which is admissible. Multiplying by $|I(\alpha)|$ gives $L_{\max}$, so the admission set equals the safe set. Since $\max_k |g_k| \le \sum_k |g_k| \le H \max_k |g_k|$, the peak guard under-states loss and the H-scaled guard over-states it. □

The identity is the classical ℓ_1 gain of a convolution operator (Vidyasagar, 1986; Dahleh and Pearson, 1987; Dahleh and Diaz-Bobillo, 1995); we use it rather than claim it. The finance content is that the dealer's admission level changes the operator, so the guard is a rule over an action set; that its single input is invariant to how the traders split into cohorts and is

identified from prices; and that the same statistic prices the corrective rebate in equilibrium (Proposition 3.8.1). The guard coincides with a brute-force adversary's maximal safe level in 200 of 200 random states, and the peak-gain guard admitted an unsafe level in 2,000 of 2,000. An infinite-horizon gain is computed on a finite horizon with a contraction-block tail bound.

3.6.2 Robustness to estimation error

Theorem 3.6.2 (Robust gain)**.** *Let* $M = \sup_k \|J^k\|$ *and* $S = \sum_k \|J^k\|$*. If* $\|E\| \le \epsilon$ *and* $\epsilon S < 1$*, then* $\sup_k \|(J+E)^k\| \le M/(1-\epsilon S)$*.*

Proof. Let $P_k = (J+E)^k$. Expanding, $P_k = J^k + \sum_{j<k} J^{k-1-j} E P_j$. With $m_k = \|P_k\|$, $m_k \le M + \epsilon \sum_{j<k} \|J^{k-1-j}\| m_j \le M + \epsilon S \max_{j<k} m_j$. Since $m_0 = 1 \le M$, induction gives $m_k \le M/(1-\epsilon S)$ for all k. □

Theorem 3.6.2 is deterministic: it holds whenever $\|E\| \le \epsilon$. A statistical guarantee follows only through a confidence set for the operator, and then holds with that set's coverage probability, not with certainty. The bound held in all 30,000 random perturbations of 100 non-normal operators, with median tightness 0.50 (realized gain over bound; maximum 0.57). In a simulated window estimator of the kind used in Section 3.9 (600 transitions), a residual-bootstrap 95% ball for the operator covered the true J in 94.7% of 300 replications, with median radius 0.086; but $\epsilon S < 1$ held in only 8% of windows. The certificate is valid but often unavailable at realistic sample sizes; longer windows or pooled estimates are needed to use it in practice.

The guard needs the summed gain, which is invariant to cohort splitting (Proposition 3.2.2) and certified with $\Theta(\sigma^2 \epsilon^{-2} \log(1/\epsilon))$ experiments whatever the spacing of the memories (Theorem 3.5.4); it does not need the algorithms' identities, which are expensive (Chapter 2). In a two-cohort economy with rates $0.7 \pm \Delta$, the relative error of the estimated summed gain is flat in Δ (log–log slope -0.15), while the error of the cohort weights grows as $\Delta^{-3.07}$.

3.6.3 An uncertainty-aware guard

The guard of Lemma 3.6.1 assumes the operator is known. With data, the relevant question is which admission levels are safe for every economy the data allow. We answer it for a venue gate (a venue that admits a fraction of incoming algorithmic flow) in the positive cohort economy, where admitting a fraction α of the algorithmic flow scales every cohort's weight, $M \mapsto \alpha M$, so that $S(\alpha) = \kappa/(1 - a - \kappa\alpha\Phi)$.

Lemma 3.6.3 (Soundness of compatible-set guards)**.** *If a data-dependent set* $\mathcal{C}(D)$ *contains the true economy with probability at least* $1-\gamma$*, then every action whose worst-case loss is within budget for all economies in* $\mathcal{C}(D)$ *is within budget for the true economy with probability at least* $1-\gamma$*, including an action selected from the data. If no action qualifies, the guard certifies nothing: it halts new admission and flags the state, because existing inventory can be unsafe even with no new trading.*

Theorem 3.6.4 (Optimal uncertainty-aware admission)**.** *Under the reset or observed-drift contract ($z_0 = 0$, or known drift subtracted from $\mathcal{B}$), let position I, shock bound $\bar{u}$ and budget $\mathcal{B}$ be given, and let Φ^+ be an upper bound on Φ valid with probability at least $1-\gamma$ (for example $\Phi^+ = (1-a-\kappa/\hat{S}^+)/\kappa$ from Theorem 3.5.4, or Φ_+ from Theorem 3.5.3). Put $c_I = 1-a-\kappa|I|\bar{u}/\mathcal{B}$.*

(i) Closed form and interval. *If $c_I \geq 0$, the levels that are safe for every $\Phi \leq \Phi^+$ form the interval $[0, \alpha^*]$ with $\alpha^* = \min\{1,\ c_I/(\kappa\Phi^+)\}$.*

(ii) Safety. *With probability at least $1-\gamma$, every admitted level is safe for the true economy.*

(iii) Maximality. *Every level above α^* is unsafe for an economy with $\Phi = \Phi^+$; when Φ^+ is attained by a compatible economy (as for Φ_+ in Theorem 3.5.3), no rule using the same information can admit more without admitting an unsafe level.*

(iv) Infeasibility. *If $c_I < 0$, no level is safe, including $\alpha = 0$: the guard halts admission and the position must fall to $|I| \leq \mathcal{B}(1-a)/(\kappa\bar{u})$.*

(v) Minimax capacity loss. *On the class of Theorem 3.5.4 with Φ bounded away from zero, the certificate-based guard gives up $\alpha^\circ - \alpha^* = O\big(\sigma\sqrt{\log N/N}\big)$ of the oracle capacity $\alpha^\circ = \min\{1, c_I/(\kappa\Phi)\}$ with probability at least $1-2\gamma$, and every rule that admits an unsafe level with probability at most $\beta < 1$ gives up at least $c\,\sigma/\sqrt{N}$ in expectation at some economy of the class. The guard is minimax-optimal in lost capacity up to a logarithmic factor.*

Monotonicity of $S(\alpha)$ in α and Φ gives (i)–(iv); (v) combines the certificate's excess with Proposition 3.5.2 applied to two economies whose capacities differ by order $\sigma/\sqrt{N}$ (Appendix 3.13). In 500 random economies the closed form matched a brute-force search over 1,001 admission levels, computed from the state matrices rather than from (3.10), in every case; every safe set was an interval; and the 28 infeasible states were flagged (test T06). Across 18 designs the certificate covered the true summed response at least at its nominal level (minimum 93.6% at nominal 90%), with a median excess of 3–15% when the memory bound is 0.6 or 0.9 and about 230% when it is 0.97, where the tail term dominates (test T04).

When admission also changes the book, deepening it and speeding reversion as well as adding feedback, risk can first rise and then fall with the admission level, and the safe levels need not form an interval: certifying the largest safe level then no longer certifies those below it. The full version certifies the whole safe set by interval branch and bound (Hansen and Walster, 2004), with soundness and completeness; in a book where admission raises feedback linearly and liquidity quadratically the certified safe set is $[0, 0.053] \cup [0.178, 1]$.

Why the peak guard is unsafe in a real book. At the median BTCUSDT estimates of Section 3.9, price impact is persistent ($\hat{a} \approx 0.97$ per minute), so the summed gain over 15 minutes is about 13 times the peak. A guard calibrated to the peak would admit inventory about 13

times larger than is safe against a sustained adversarial flow of the declared size, although that book does not amplify. Persistence alone makes peak-based limits unsafe.

The guard on a dealer's balance sheet. The full version carries the guard through maintenance margin, hedging and worst-case fills, and proves forward solvency at every step at which an admissible level exists; when none exists the guard halts new fills and flags the state, because existing inventory can still breach margin. In a matched-shock Monte Carlo inside the model, with every rule calibrated to the same trading capacity, the guard has no forced liquidations against 31.6% with no control, 31.4% under a trailing volatility limit and 20.3% under an inventory cap. With a hidden cohort state the drift of future prices is not identification-invariant under a Euclidean bound, but it is under the natural interval bound on the cohorts' memories:

Proposition 3.6.5 (Uniformly safe guard under interval-bounded unknown hidden drift)**.** *In any positive cohort economy with $\Phi = \sum_{i=1}^{K} q_i \le \Phi^+$, observed current pricing error x_0 and unknown hidden cohort states satisfying $\|y_0\|_\infty \le \bar{y}_0$, let admission $\alpha \in [0,1]$ scale cohort flow as $M \mapsto \alpha M$ with $a + \alpha\kappa\Phi^+ < 1$. Then:*

(i) Identification-free drift envelope. *For every compatible cohort decomposition $(q_i, \eta_i)_{i=1}^{K}$ with $\sum_i q_i = \Phi$, the worst-case cumulative unforced drift over $\|y_0\|_\infty \le \bar{y}_0$ is*

$$\sup_{\|y_0\|_\infty \le \bar{y}_0} \sum_{t=1}^{\infty} |e_x^\top J(\alpha)^t z_0| \;=\; \frac{a|x_0| + \alpha\kappa\Phi\bar{y}_0}{1 - a - \alpha\kappa\Phi}, \tag{3.11}$$

attained when $y_{i,0} = \operatorname{sign}(x_0)\bar{y}_0$ for all i (or $y_{i,0} = \pm\bar{y}_0$ when $x_0 = 0$).

(ii) Closed-form drift-adjusted guard. *Under shock bound $\bar{u}$, inventory I and budget $\mathcal{B}$, the set of admission levels that are safe for every compatible economy with $\Phi \le \Phi^+$ and every hidden state $\|y_0\|_\infty \le \bar{y}_0$ is $[0, \alpha^*_{\text{drift}}]$ (empty when $(1-a)\mathcal{B} < |I|(a|x_0| + \kappa\bar{u})$), where*

$$\alpha^*_{\text{drift}} \;=\; \min\left\{1,\; \left[\frac{(1-a)\mathcal{B} - |I|(a|x_0| + \kappa\bar{u})}{\kappa\Phi^+(\mathcal{B} + |I|\bar{y}_0)}\right]_+\right\}. \tag{3.12}$$

(iii) Exact capacity cost of unknown state. *When $x_0 = 0$ and the reset guard $\alpha^* = \min\{1, c_I/(\kappa\Phi^+)\}$ of Theorem 3.6.4 is interior ($\alpha^* \in (0,1)$), the capacity given up to protect against unknown hidden cohort states $\|y_0\|_\infty \le \bar{y}_0$ is exactly*

$$\alpha^* - \alpha^*_{\text{drift}} \;=\; \alpha^* \,\frac{|I|\bar{y}_0}{\mathcal{B} + |I|\bar{y}_0}. \tag{3.13}$$

At the Table 3.1 calibration ($a = 0.4$, $\kappa = 1$, $\Phi^+ = 0.4$, $|I| = 1$, $\bar{u} = 0.48$, $\mathcal{B} = 1$, $x_0 = 0$) with $\bar{y}_0 = 0.25$, the reset guard admits $\alpha^* = 0.3000$ whereas the drift-adjusted guard admits

$\alpha^*_{\text{drift}} = 0.2400$, an exact capacity cost of 0.0600, and commits zero budget breaches across 500 random multi-cohort economies (`recompute_p2_step2_theory.py`).

Re-optimization by the traders does not undo the guard: when a venue throttles admission, filtering cohorts lower their adaptation rates, and in the positive cohort economy every finite-horizon summed response falls with them.

3.7 The value of disclosure for safe trading

Chapter 2 measures disclosure in samples needed to identify the cohorts. For a pre-trade control the relevant value is the safe trading capacity it unlocks.

Theorem 3.7.1 (Decision value of disclosure)**.** *Under the risk contract of Section 3.5.1, compare an observation contract O with a richer contract O' that reveals everything O reveals and more, both with exact data.*

(i) Monotonicity. *The compatible sets are nested, $\Theta_{O'} \subseteq \Theta_O$, so $\Phi^+_{O'} \le \Phi^+_O$ and the admitted level of Theorem 3.6.4 satisfies $\alpha^*_{O'} \ge \alpha^*_O$.*

(ii) Strictness. *$\alpha^*_{O'} > \alpha^*_O$ if and only if $\alpha^*_O < 1$, $c_I > 0$, and O' excludes every compatible economy with $\Phi = \Phi^+_O$. Under the window contract that economy is the upper principal representation of Theorem 3.5.3: disclosure has decision value exactly when it rules out a cohort at the memory bound with the extremal weight.*

(iii) Long memory. *Under the window contract with memory bound $\bar{r}$, $\alpha^*_O \to 0$ as $\bar{r} \to 1$ for any fixed window (Theorem 3.5.3(iii)), whereas audited exposure, which reveals Φ, admits the oracle capacity α°. The decision value of audited disclosure therefore tends to the whole oracle capacity as memory lengthens.*

(iv) Net value. *With value w per unit of admitted capacity and disclosure cost $c(O')$, the net value $w(\alpha^*_{O'} - \alpha^*_O) - c(O')$ is zero or negative whenever O already identifies Φ (a window whose moment vector lies on the boundary of the moment space) or the budget does not bind.*

(v) Noisy proxies. *With noisy data the certificates of two contracts are confidence bounds, not nested sets, and a richer contract can admit less. Under class-tagged flows measured with noise over T periods and estimated as in Chapter 2, class i's exposure $\hat{\Phi}_i = \hat{w}_i/(1-\hat{r}_i)$ has delta-method standard error*

$$\operatorname{se}(\hat{\Phi}_i) = \frac{1}{(1-r_i)\sqrt{T}}\Big[V_{ww} + 2\Phi_i V_{wr} + \Phi_i^2 V_{rr}\Big]^{1/2},$$

with V the asymptotic covariance of $(\hat{w}_i, \hat{r}_i)$, so the tagged bound $\sum_i \hat{\Phi}_i + z_{1-\gamma}\text{se}$ admits less than the public certificate whenever its margin exceeds $\Phi^+_O - \Phi$. Intersecting the two

Table 3.2: Admitted capacity under three observation contracts (simulation)

Memory rate of the slow class	0.60	0.90	0.97	0.995
Public responses (certificate, Theorem 3.5.4)	0.640	0.623	0.601	0.544
Class-tagged flows, 10% noise	0.662	0.655	0.631	0.507
Audited exposure (oracle)	0.667	0.667	0.667	0.667
Public ∩ tagged, each at 97.5% (test T15)	0.661	0.653	0.626	0.537
Value of audited exposure over public	0.027	0.044	0.066	0.122

Two classes ($q = 0.06$ with the stated memory rate, $q = 0.12$ with rate 0.5), $a = 0.4$, $\kappa = 1$, position 1, shock bound 0.48, budget 1; public: 500 reset experiments, per-tap noise 0.05, certificate at 95% with the true memory bound; tagged: 2,000 periods of each class's flow with 10% measurement noise, two-stage least squares as in Chapter 2, 95% delta-method bound; 300 replications. Entries are mean admitted fractions α^*. The intersection row comes from an independent run of the same design (seed 15), whose public and tagged rows are 0.639/0.662, 0.625/0.655, 0.600/0.631 and 0.549/0.506. Kind: recomputed (tests T07 and T15).

certificates, each at level $1-\gamma/2$*, gives a bound valid at level* $1-\gamma$ *that admits at least as much as either certificate at level* $1-\gamma/2$*, which restores monotonicity.*

The theorem is set inclusion plus the closed form of Theorem 3.6.4; its content is the location of the value. Disclosure is worth most where public certificates are weakest, in markets whose participants have long memories. In simulation (test T07; Table 3.2), with two classes and one of them slow, audited exposure adds 0.027 of admitted capacity to a public certificate when the slow class's memory rate is 0.6 and 0.122 when it is 0.995, out of an oracle capacity of 0.667. Class-tagged flows measured with 10% noise add 0.02–0.03 at memory rates up to 0.97 but *lower* capacity at 0.995, because a class's exposure $q = w/(1-r)$ is badly conditioned when r is near one: disclosure of a noisy proxy can be worth less than nothing. Part (v) quantifies this. In a second simulation of the same design (test T15), the standard error of the slow class's exposure rises from 0.0004 to 0.035 as $1-r$ falls from 0.4 to 0.005, roughly in proportion to $(1-r)^{-1}$ over the last decade, while the public certificate's margin grows only with the square root of its horizon. At $r = 0.995$ the tagged bound admits 0.506 against 0.549 for the public one; the intersection admits 0.537, close to the better of the two, and covers the true exposure in 99% of replications (Table 3.2). Chapter 2's evidence that Binance's and the CFTC's class positions pin down memories that prices leave unrestricted concerns class-level positions, which are proxies, not records of individual algorithms.

Pricing versus guarding. A conservative bound is the right input for a guard and the wrong input for a price. The corrective rebate of Proposition 3.8.1 depends on S^2; computed from an upper bound that overstates S by 10% it over-subsidizes reversion and loses 2.4–2.8% of first-best welfare, and by 50%, 70–80% (test T12). Guards should use certified upper bounds; rebates should use point estimates.

3.8 Who provides reversion? An equilibrium

Theorem 3.4.3 says that transient risk is governed by $S = \tilde{\kappa}/(m - q\tilde{\kappa})$, and its ingredients are chosen by participants: dealers decide how hard to lean against displacements, which sets reversion m, and momentum traders how strongly to chase them, which sets q.

3.8.1 Setup

There are N dealers. Dealer i supplies reversion intensity $s_i \geq 0$ by absorbing order flow against the displacement, so $m = m_0 + \sum_i s_i$, where m_0 is reversion supplied by other means. Leaning earns a spread v per unit of intensity at quadratic inventory cost, so the private value is $vs_i - \frac{c}{2}s_i^2$. By Lemma 3.6.1 the worst-case transient loss of a position of size Q facing flow shocks bounded by U is $QU\,S$, so we take aggregate expected tail loss to be $\Lambda S(m, q)$ with $\Lambda > 0$. Dealers bear a share $\theta < N$ of it, so each bears θ/N; leveraged users and their counterparties bear the rest. Momentum traders choose aggregate intensity q with private value $bq - \frac{e}{2}q^2$ and bear none of the tail loss. Long-run impact $\tilde{\kappa}$ is taken as given.

3.8.2 Results

Proposition 3.8.1 (Equilibrium transient risk)**.** *Let $g(m) \equiv -\partial S/\partial m = \tilde{\kappa}/(m - q\tilde{\kappa})^2$.*

(i) Under-provision of reversion. *The symmetric Nash equilibrium solves $m - m_0 = Nv/c + \theta\Lambda g(m)/c$, the planner's optimum solves $m - m_0 = Nv/c + N\Lambda g(m)/c$, and $m^{\mathrm{eq}} < m^{\mathrm{SP}}$, so $S^{\mathrm{eq}} > S^{\mathrm{SP}}$.*

(ii) Fewer dealers, more risk. *$\partial m^{\mathrm{eq}}/\partial N = (v/c)/(1 - \theta\Lambda g'(m)/c) > 0$, so equilibrium reversion rises and S^{eq} falls with the number of dealers.*

(iii) Over-provision of momentum. *Momentum traders choose $q = b/e$; the planner chooses $q^{\mathrm{SP}} < b/e$, solving $b - eq = \Lambda\tilde{\kappa}^2/(m - q\tilde{\kappa})^2$.*

(iv) The corrective price is identified. *The first best for reversion is implemented by a rebate per unit of reversion*

$$\tau^* = \Big(1 - \frac{\theta}{N}\Big)\Lambda\, g(m) = \Big(1 - \frac{\theta}{N}\Big)\Lambda\,\frac{S^2}{\tilde{\kappa}},$$

which depends on the market only through S and $\tilde{\kappa}$.

(v) Endogenous depth. *Suppose in addition that each dealer posts depth $D_i \geq 0$ at private value $v_D D_i - \frac{c_D}{2}D_i^2$, that long-run impact is $\tilde{\kappa} = 1/D$ with $D = \sum_i D_i$, and that momentum traders size their flow to the book, $q = \psi D$ with ψ fixed. Then $S = \Gamma/D$ with $\Gamma \equiv S/\tilde{\kappa} = 1/(m - \psi)$, which does not depend on depth. Symmetric equilibrium depth solves $D = Nv_D/c_D + \theta\Lambda\Gamma/(c_D D^2)$ and the planner's $D = Nv_D/c_D + N\Lambda\Gamma/(c_D D^2)$, so $D^{\mathrm{eq}} < D^{\mathrm{SP}}$;*

D^{eq} increases with N, so $S^{\mathrm{eq}} = \Gamma/D^{\mathrm{eq}}$ falls with the number of dealers while $S/\tilde{\kappa}$ does not move. The rebate per unit of depth $\tau_D^* = (1-\theta/N)\Lambda S\tilde{\kappa}$ implements the first-best depth.

Proof. (i) Dealer i's first-order condition is $v - cs_i + (\theta/N)\Lambda g(m) = 0$; the planner's is $v - cs + \Lambda g(m) = 0$. Summing over dealers gives the two fixed-point equations. The right-hand sides are decreasing in m because g is decreasing on $m > q\tilde{\kappa}$, and the planner's lies above the dealers' for every m when $\theta < N$, so its crossing with the line $m - m_0$ lies to the right. (ii) follows by implicit differentiation, since $g' < 0$. (iii) $\partial S/\partial q = \tilde{\kappa}^2/(m - q\tilde{\kappa})^2 > 0$ is the marginal external cost of momentum that traders ignore. (iv) The rebate closes the wedge between the two first-order conditions, and $g = S^2/\tilde{\kappa}$ by Theorem 3.4.3. (v) With $q\tilde{\kappa} = \psi$, Theorem 3.4.3 gives $S = \tilde{\kappa}/(m - \psi) = \Gamma/D$ and $\partial S/\partial D_i = -\Gamma/D^2$. Dealer i's first-order condition for depth is $v_D - c_D D_i + (\theta/N)\Lambda\Gamma/D^2 = 0$ and the planner's $v_D - c_D D_i + \Lambda\Gamma/D^2 = 0$; summing gives the two fixed points. In each, the left side D increases and the right side decreases in D, so each has a unique positive solution, the planner's to the right of the dealers' because $\theta < N$; and $D - \theta\Lambda\Gamma/(c_D D^2) = N v_D/c_D$ has a left side increasing in D, so D^{eq} increases with N. The rebate closes the wedge $(1-\theta/N)\Lambda\Gamma/D^2 = (1-\theta/N)\Lambda S\tilde{\kappa}$. □

At $v = 0.002$, $c = 0.5$, $\theta = 0.3$, $\Lambda = 10^{-4}$, $m_0 = 0.005$, $q = 0.02$ and $\tilde{\kappa} = 1$, equilibrium S falls from 27.9 with one dealer to 6.8 with forty, and the planner's S is 1.5–2.7 times lower throughout; the rebate computed from S alone coincides with the first-order wedge to machine precision. With $m = 0.03$, $b = 0.3$ and $e = 20$, momentum traders choose $q = 0.015$ against a planner's 0.0062.

Interpretation. Reversion is a public good: each dealer who leans against a displacement lowers the tail exposure of every leveraged position in the market but is paid only for his own fills. Momentum is the mirror image. Part (iv) closes the loop with the guard: the policy price of the externality is a function of the same summed response that the guard bounds, and Proposition 3.2.2 and Theorem 3.5.4 show that this quantity is identified from public data without identifying the traders. A venue that rebates reversion in proportion to $S^2/\tilde{\kappa}$, or a supervisor that sets pre-trade limits by S, prices the externality with an observable.

Across the liquidity terciles of 30 Binance perpetuals the summed response per dollar of flow rises steeply as depth falls (Spearman correlation with log depth -0.94), as part (ii) predicts, but the unit-free multiple $S/|h_0|$ is the same in all three terciles (medians 46.4, 47.3 and 46.9): thinner books carry more cumulative risk because their impact is larger, not because they revert more slowly. That is the pattern of part (v), which we added after seeing the comparison, so the match is a consistency check, not a test.

3.9 Evidence

The empirical program tests the theory's content in order-book data. Every test fixed its specification, hashed it and committed it before its test data were opened, and every test set was opened once except where Table 3.4 says otherwise. Data are public: Binance USD-M futures and spot archives (one-minute klines with taker volume, depth within ±1–5% of the mid at about 30-second intervals, open interest), verified against published checksums and screened by validation rules fixed in advance; Bybit trade and order-book archives; Coinbase trades; and consolidated US trades. The out-of-sample criterion is a one-sided Clark–West test *and* a date-block bootstrap interval for the gain in R^2 that excludes zero.

3.9.1 The deepest book does not amplify

In BTCUSDT futures on 718 training days, the estimated one-period margin $\hat{C}_3$ is negative on 93.5% of days and the median implied response peaks at impact. Local projections show no overshoot once future flow is held fixed. A registered test of whether six-hour estimates of $\hat{C}_3$ forecast the next hour's maximum adverse excursion beyond volatility, flow and depth controls fails (+0.018 points of R^2, interval including zero), and a volatility flag classifies tail hours far better. In spot data, a unit-free measure of transient amplification on five crash days is indistinguishable from one and from a no-feedback null, and a registered forecasting test fails. Binance's depth bands appear to show replenishment of the side being hit; on a fixed price grid built from Bybit's full order books the sign reverses, so the replenishment is an artefact of band re-centring and depletion is present, but it is not large enough to produce an overshoot. Where the theory predicts nothing, we find nothing.

3.9.2 Thin books: the summed response forecasts the size of excursions

Thirty Binance perpetuals were fixed from one 2024 training day before any statistic was computed, spanning BTCUSDT to symbols with under $5M of daily volume. The amplifying regime is more frequent in thinner books (16.1% of symbol-days with $\hat{C}_3 > 0$ in the thin tercile against 9.4% in the thick one), although the average margin is negative everywhere. Measured feedback is larger in *deeper* books: feedback and amplification are different objects.

The result holds in every registered forecasting test on thin Binance books (Table 3.3). It strengthens as the baseline gets harder: beyond a full volatility-forecasting baseline that already explains 18.6% of the variance, $\hat{C}_3$ adds 2.8 points of R^2, the volatility benchmarks themselves add nothing beyond the simpler controls, and the summed response S, the guard's own statistic, adds 2.1 points. The decomposition matches the theory: a larger momentum term predicts larger excursions, a larger reversion term smaller ones, and the depth term nothing; most of the variation is in impact-weighted reversion, as $S = \tilde{\kappa}/(m - q\tilde{\kappa})$ implies. It is not price impact (with four impact controls it keeps about 80% of its gain, and impact alone forecasts nothing), not

Table 3.3: Registered forecasting tests in thin Binance books, 2025–2026

Test (what changes)	ΔR^2 (pts) [CI]	Clark–West t
Ten thin symbols, volatility and flow controls	+1.216 [+0.273, +7.987]	8.53
Ten new thin symbols drawn by a seeded rule	+0.845 [+0.303, +6.371]	11.66
Fresh days, 20 thin symbols, price-impact controls in the baseline	+1.583 [+0.588, +3.237]	9.46
Fresh days, full volatility-forecasting baseline (HAR, EWMA, hour effects)	+2.787 [+1.425, +4.944]	12.59
the guard's statistic S_{60} instead of $\hat{C}_3$ (secondary)	+2.098 [+1.165, +3.597]	—
Untouched days, model frozen at its training estimates	+4.196 [+1.551, +8.233]	—
Untouched thin symbols	+0.727 [+0.377, +1.428]	—
Untouched thin symbols, beyond the baseline plus VPIN	+0.562 [+0.274, +1.159]	—

Outcome: the next hour's maximum adverse excursion; candidate estimated from the previous six hours. Date-block bootstrap intervals (97.5% for the fresh-day and untouched-day primaries, 95% otherwise). Every row is a registered primary except the S_{60} row. Full tables and secondaries in the full version.

tick size (it survives tick controls, the removal of days at a non-training tick and of any single symbol; unit-free versions work on fresh days), not order-flow toxicity (it keeps about 73% of its gain beyond VPIN, a comparison made post hoc and then confirmed in a registered test), and not order-flow persistence (a placebo with the same flow persistence but broken alignment with prices adds nothing). In economic terms, within a volatility quintile the top quintile of $\hat{C}_3$ has next-hour excursions about 26 basis points larger than the bottom quintile, enough to roughly double the hourly liquidation probability of a 20× position.

3.9.3 Other venues

The result belongs to thin markets, not to one venue, but its size depends on the venue. In 23 thin US stocks a registered test passes at about a fifth of the Binance size (+0.50 points beyond the full baseline), carried again by the reversion term; mid-size stocks show a gain too. On Bybit the statistic is absent with trades alone (a registered failure) and, with order-book depth, meets its registered criterion with an effect about one eightieth of Binance's; on Coinbase spot the registered test fails. On both crypto venues where it fails, thin pairs trade sparsely (11% of Bybit minutes have no trade, against 0.1% on Binance), which leaves the series the statistic is estimated from much noisier.

3.9.4 The guard as a risk control

The guard's theorems concern worst-case loss under a certified operator; whether its statistic helps size positions in practice is a separate question, and the answer is: modestly, and not instead of volatility. A six-hour estimate scaled into a volatility limit does not improve it (a registered failure), and limits built on the estimate alone are worse than a static limit. With an

operator certified from three days of data, the scaled limit lowers tail loss per unit of position by 2.4% (a narrow registered pass). Used as the theorem uses it, as a one-sided throttle, the certified guard lowers tail loss against a static limit by 8% on untouched days and 23% on untouched symbols (registered passes), but not against a volatility cap at matched activity. Over 18-hour holding periods, an uncapped version failed its registered capacity test; a capped throttle of the kind Rule 15c3-5 requires, registered after that failure, lowers the forced-liquidation rate of a 20× position by 7.4 percentage points against a volatility cap, mostly because it admits 11% less exposure; scaled to the same mean exposure the two rules are indistinguishable.

3.9.5 What the effect is not: the mechanism tests

Two registered tests asked whether the forecasting content runs through forced liquidations, measured by five-minute open-interest contractions with adverse price moves. Both fail: $\hat{C}_3$ does not forecast them, and they mediate none of its forecasting gain (-6.8%). A leverage interaction, specified in advance but not hash-registered, also fails. A follow-up registered after these failures finds that $\hat{C}_3$ forecasts price persistence over the next 15 to 60 minutes and that this persistence accounts for nearly all of its forecasting gain, but its registered placebo clause fails, so the mediation is not established. A zero-fee natural experiment leaves reversion unchanged and lowers impact. On October 10, 2025, reported descriptively, the summed response was elevated all day but did not warn of anything volatility missed in the final hours. We therefore read the evidence as showing that thin books are risky because their prices revert slowly relative to impact, which is what the statistic measures, not as a test of a deeper mechanism.

3.9.6 Every registered test

Of twenty-five registered primaries, fifteen pass, nine fail and one was not tested because the test it depended on failed (Table 3.4).

How to read the passes. The program was sequential: each registration protected one test from specification search on its own test data, but which test to run next was chosen after the earlier results were known. The passing Binance tests share training days and are not independent replications; they vary one thing at a time (symbols, days, baseline), and the US test shares none of their data. Every feature uses only minutes before the forecast hour, and a scrambling audit leaves every feature unchanged when later data are permuted. The gains are modest on top of a strong baseline, a 15% increase in the explained variance of the next hour's excursion at most; they are not a new predictor that replaces volatility.

Table 3.4: All pre-registered tests

Test (specification hash)	Sample	Primary statistic [CI]	Result	Opened
BTCUSDT futures, MAE | $\hat{C}_3$ (`d9521218`)	2025–26	ΔR^2 +0.018 [−0.100, +0.132]	fail	1
BTCUSDT spot, MAE | amplification (`b1fd7f8e`)	2024	ΔR^2 −0.023 [−0.056, −0.002]	fail	3
Thin books, 10 symbols (`5910b6b4`)	2025–26	ΔR^2 +1.216 [+0.273, +7.987]	pass	2
New thin symbols, 10 (`6aa8e2c0`)	2025–26	ΔR^2 +0.845 [+0.303, +6.371]	pass	1
Fresh days, 20 thin, impact controls (`65acda36`)	2025–26	ΔR^2 +1.583 [+0.588, +3.237]	pass	1
Position sizing, same specification	2025–26	$\Delta \log$ TLR −0.017 [−0.045, +0.011]	fail	1
Bybit, 14 coins, depth-free	2025–26	ΔR^2 −0.000 [−0.001, +0.001]	fail	1
Beyond volatility forecasting, 20 thin (`71983f1a`)	2025–26	ΔR^2 +2.787 [+1.425, +4.944]	pass	1
Certified guard, position sizing (`b9a56af4`)	2025–26	$\Delta \log$ TLR −0.024 [−0.051, −0.0004]	pass	1
Bybit with order-book depth, 6 coins (`ec2145fa`)	2025–26	ΔR^2 +0.033 [+0.019, +0.053]	pass	1
Thin US stocks, 23, depth-free (`f5800a69`)	2025–26	ΔR^2 +0.499 [+0.272, +0.779]	pass	1
Untouched days, frozen model (`69849c96`)	2025–26	ΔR^2 +4.196 [+1.551, +8.233]	pass	1
Untouched days, guard vs static (same)	2025–26	$\Delta \log$ TLR −0.084 [−0.173, −0.011]	pass	1
Untouched symbols, 5 thin (`0523e476`)	2025–26	ΔR^2 +0.727 [+0.377, +1.428]	pass	1
Untouched symbols, beyond VPIN (same)	2025–26	ΔR^2 +0.562 [+0.274, +1.159]	pass	1
Untouched symbols, guard vs static (same)	2025–26	$\Delta \log$ TLR −0.256 [−0.497, −0.021]	pass	1
New venue, Coinbase, 6 thin (`3233c52b`)	2025–26	ΔR^2 −0.092 [−0.407, +0.158]	fail	1
Coinbase, real minus placebo (same)	2025–26	+0.013 [−0.331, +0.298]	not tested	1
OI liquidation forecast, 19 thin (`2b3794e3`)	2025–26	ΔR^2_{LIQ} −0.795 [−1.529, −0.179]	fail	1
OI liquidation mediation, 19 thin (same)	2025–26	Share −6.8% [−10.2%, −3.1%]	fail	1
Non-resetting 18h guard capacity (`d4169377`)	2025–26	$\Delta\overline{N}$ (k$) +332.8 [−20.4, +1180.8]	fail	1
Capped 18h throttle liq. rate (`698f9119`)	2025–26	ΔLiq −7.41 [−11.45, −4.15] pp	pass	1
Capped 18h throttle ES_{95} (same)	2025–26	ΔES_{95} (k$) −26.4 [−53.1, −6.3]	pass	1
Persistence forecast, 19 thin (`8ebd67f3`)	2025–26	ΔR^2_{pers} +3.713 [+1.884, +6.035]	pass	1
Persistence mediation + placebo (same)	2025–26	Share 97.6% [96.6%, 98.3%]	fail	1

ΔR^2 in percentage points. "Opened" counts executions of the test cell; the repeated openings used unchanged code and specifications (Sections 3.9.2 and 3.9). Intervals are 95%, except 97.5% for the two fresh-day primaries and the two Part 15 primaries.

Table 3.5: Simulation tests of the information-safety results

Test	Question	Result
T01	Can markets agree on early responses yet differ in risk?	Example 3.5.1 in rational arithmetic: lags 0–2 equal, summed responses 1.92 and 5.59
T02	Does Proposition 3.5.2 bound the capacity a safe rule gives up?	Attained exactly with identical data (0.324 at $\beta = 0.05$); respected in 9 of 9 noisy designs by the certificate rule, whose unsafe-admission rate is at most 3.2%
T03	Is certifying risk cheaper than identifying cohorts?	Table 3.1: slope -6.06 in $\Delta\eta$ for identification, $+0.23$ for certification
T04	Does the certificate cover at its nominal level?	18 designs: coverage at or above nominal in all (minimum 93.6% at nominal 90%); median excess 3–15% with memory bound 0.6–0.9, about 230% at 0.97
T05	Is the corrected gate exact?	2,000 cases, horizons 1–10, every sign path enumerated: per-period worst case equals $V \sum \lvert g \rvert$ and total-budget worst case equals $V \max \lvert g \rvert$ in all; the peak gain understates the per-period worst case in 1,800
T06	Is the closed-form guard globally optimal?	500 economies: equal to brute force over 1,001 levels from the state matrices in all; safe sets always intervals; 28 infeasible states flagged
T07	What does disclosure add?	Table 3.2
T08	Does the guard depend on the channel?	Momentum learners, fixed-rule traders, depth and slow dealer reversion calibrated to $S = 5$: identical guard capacity (0.20); one-period margins differ (-0.34 to -0.20)
T09	What breaks the certificate?	Coverage at nominal 95%: as assumed 100%; delayed feedback 100%; contrarian cohorts 100%; depth channel 92.8%; memory longer than declared 86.3%; feedback 20% stronger after certification 34.7%
T10	Are the window endpoints proved?	Certified by verified polynomial sign conditions for every window; outside the 60-digit endpoints by at most 1.5×10^{-8} relative
T11	What if the tap noise is correlated?	(O1) certificate covers 74–76% (exact value 75.1%); t-certificate 94–95%; Hotelling 100%; dependent experiments 82–85%
T12	Should rebates use certified bounds?	No: a 10% overstatement of S costs 2.4–2.8% of first-best welfare, a 50% overstatement 70–80%
T13	What if admission changes the dynamics?	Safe sets $[0, 0.035] \cup [0.386, 1]$ (closed form) and $[0, 0.053] \cup [0.178, 1]$ (book, certified); no certified level unsafe in 433 sampled economies
T14	Does sequential probing keep its error level?	No wrong decision in 7,000 runs; 3.7 to 107 probes against 130 for a fixed design

All tests are deterministic given their seeds and run without data (`research_extension/information_safety/run_all.py` → `results/T*.json`, about six minutes). Further checks: the joint per-period and total-budget worst case equals its linear-programming value in 300 random cases (T05b), and the balance-sheet step keeps the reserve in all 17,551 sampled states that satisfy the guard inequality (T12b). Kind: recomputed.

3.10 Tests of the information-safety results

The theorems of Sections 3.5–3.7 concern declared contracts, which market data cannot deliver. They are therefore tested in simulations with a known truth (Table 3.5).

The simulations confirm what the theorems claim and locate what they do not (Table 3.5). The certificate is valid where its contract holds, including with delayed feedback and contrarian

cohorts, and conservative when the declared memory is long. It fails in two ways that matter for practice: when memory is longer than declared, which is the case Theorem 3.5.3(iii) warns about, and when feedback strengthens after certification, in which case a stale certificate covers the new risk only a third of the time. A guard must therefore be re-certified as conditions change; it cannot be set once. The remaining tests check the extensions: the window endpoints are proved rather than computed (T10), certificates that do not assume independent taps keep their level when the taps are correlated while the (O1) certificate does not (T11), non-monotone admission is certified by branch and bound (T13), and sequential probing keeps its error level (T14).

3.11 Policy implications

Rule 15c3-5 requires pre-trade controls that prevent orders exceeding credit or capital thresholds (SEC, 2010); static caps and daily value-at-risk limits do not see transient dynamics. Lemma 3.6.1 names the quantity a solvency-motivated control should bound, inventory times the shock envelope times the summed absolute impulse response, and peak- or eigenvalue-based limits are unsafe or blind exactly where they matter. The evidence says where such a control matters (thin books, under about \$0.4M of depth within 1% of the mid) and what it can do with public data: correct a volatility limit modestly, not replace it. Because the summed response is certified without identifying the traders, a supervisor or venue can set limits, or price a reversion rebate in proportion to $S^2/\tilde{\kappa}$, from an observable. Guards should use certified upper bounds; rebates should use point estimates.

3.12 Conclusion

The theory answers four questions. A book amplifies when depletion outweighs impact-weighted reversion; cumulative transient risk is long-run impact over reversion net of momentum; the guard on the summed response is the largest safe admission rule and needs no identification of the traders; and the information that certifies a safe decision is cheap in exactly the directions in which identification is expensive. The evidence is mixed in an informative way: the deepest book does not amplify and the statistic is useless there, while in thin books it forecasts the size of adverse excursions beyond every baseline we tried, at a size that depends on the venue, and as a practical control it adds to volatility only modestly. Message-level data from a regulated market, data on liquidity-provider identities, and a market observed near the stability boundary would settle what remains open.

Data, code and AI use. All data are public and every number is produced by archived code from them; specifications, hashes and unlock counters are in the replication package. The authors used Claude (Anthropic) to write and debug analysis code, check derivations and draft

and edit text; every number comes from code the authors executed, and the authors take full responsibility for the content.

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3.13 Appendix: Proofs

Proposition 3.2.1. With one cohort, $J = \left(\begin{smallmatrix} a & \kappa q \\ \eta a & 1-\eta+\eta\kappa q \end{smallmatrix}\right)$, so $\mathrm{tr}J = a + 1 - \eta + \eta\kappa q$ and $\det J = a(1-\eta)$. The Jury conditions for a real 2×2 matrix are $|\det J| < 1$ and $1 \pm \mathrm{tr}J + \det J > 0$. Substituting $a = 1 - \kappa\varphi_f$: $1 - \mathrm{tr}J + \det J = \eta(1 - a - \kappa q) = \eta\kappa(\varphi_f - q)$, positive iff $q < \varphi_f$; and $1 + \mathrm{tr}J + \det J = (4 - 2\eta) - \kappa[(2-\eta)\varphi_f - \eta q]$, positive iff the stated bound on κ holds (automatically when $(2-\eta)\varphi_f \le \eta q$).

Proposition 3.2.2. Equal merged measures give equal G, hence equal H, and the impulse coefficients are the Laurent coefficients of H (Chapter 2, Theorem on merged equivalence). Under $z \mapsto Tz$, $c^\top J^k b$ is unchanged because $CT^{-1}(TJT^{-1})^k Tb = CJ^k b$, while the row norm $\|CT^{-1}(TJT^{-1})^k\|$ depends on T. The displayed price path is the convolution of inputs with $\{g_k\}$, and its worst case over the ℓ_∞ ball is $\bar{u}\sum|g|$ by Hölder.

Proposition 3.2.3. (i) Taking z-transforms of $y_{i,t} = (1-\eta_i)y_{i,t-1} + \eta_i x_t$, $I_{i,t} = (1-\delta_{\mathrm{inv}})I_{i,t-1} + u_{i,t}$ and $u_{i,t} = q_i y_{i,t} - \psi_i I_{i,t-1}$ gives $Y_i(z) = \frac{\eta_i z}{z-1+\eta_i}X(z)$ and $(z - 1 + \delta_{\mathrm{inv}} + \psi_i)I_i(z) = q_i z Y_i(z)$, whence $U_i(z) = \frac{z-1+\delta_{\mathrm{inv}}}{z-1+\delta_{\mathrm{inv}}+\psi_i} q_i Y_i(z)$ and $G_{\mathrm{inv}}(z) = \sum_i U_i(z)/X(z)$ has the stated form. Evaluating at $z = 1$ when the augmented operator is Schur stable gives $G_{\mathrm{inv}}(1) = \sum_i \frac{\delta_{\mathrm{inv}}}{\delta_{\mathrm{inv}}+\psi_i} q_i = \Phi_{\mathrm{eff}}(\psi)$

and $S(\psi) = H_{\text{inv}}(1) = \kappa/(1 - a - \kappa\Phi_{\text{eff}}(\psi))$. (ii) Write $r_i = 1 - \eta_i$ and $\rho_i = 1 - \delta_{\text{inv}} - \psi_i$. When $0 \le \psi_i \le 1 - \delta_{\text{inv}}$ and $\delta_{\text{inv}} \ge \eta_i$, we have $0 \le \rho_i \le 1 - \delta_{\text{inv}} \le r_i < 1$. Expanding $U_i(z)/X(z) = q_i\eta_i \frac{z}{z-r_i}\big(1 - \frac{\psi_i}{z-\rho_i}\big)$ in powers of z^{-1}, the m-th open-loop feedback tap of cohort i is $k_{i,0} = q_i\eta_i \ge 0$ and, for $m \ge 1$, $k_{i,m} = q_i\eta_i\big(r_i^m - \psi_i \sum_{j=0}^{m-1} r_i^{m-1-j}\rho_i^j\big) \ge q_i\eta_i r_i^{m-1}\big(r_i - \frac{\psi_i}{1-\rho_i/r_i}\big) \ge 0$ because $\frac{\psi_i}{r_i-\rho_i} = \frac{\psi_i}{\delta_{\text{inv}}-\eta_i+\psi_i} \le 1 \le r_i$ whenever $\psi_i \le (1 - \delta_{\text{inv}})$ and $\delta_{\text{inv}} \ge \eta_i$ with $r_i^m \ge \psi_i \frac{r_i^m - \rho_i^m}{r_i - \rho_i}$ (since $\frac{\psi_i}{r_i-\rho_i} \le 1$). Nonnegative open-loop taps $k_m(\psi) \ge 0$ with $a + \kappa \sum_m k_m(\psi) = a + \kappa\Phi_{\text{eff}}(\psi) < 1$ generate a nonnegative closed-loop renewal sequence $h_t(\psi) \ge 0$, so $\|h(\psi)\|_1 = \sum_{t\ge 0} h_t(\psi) = S(\psi)$.

Theorem 3.3.1. (i) $\sup_t \|CJ^t z_0\| \le \sup_t \|CJ^t\| \|z_0\|$, with equality approached by aligning z_0 with the top right singular vector of CJ^{t^*} at the maximizing t^*. (ii) The Kreiss matrix theorem with Spijker's sharp constant gives $\mathcal{K}(J) \le \sup_t \|J^t\| \le en\mathcal{K}(J)$; substitute in (i). (iii) Take $J_\theta = \left(\begin{smallmatrix} \rho & \theta \\ 0 & \rho/2 \end{smallmatrix}\right)$; then $\rho(J_\theta) = \rho$ and $(1, 0)J_\theta e_2 = \theta$, so $M_C \ge |\theta|$ and $r^*_{\text{exact}} \le B/|\theta| \to 0$.

Proposition 3.6.5. (i) In the positive cohort economy $J(\alpha) \ge 0$ entrywise, so for any $z_0 = (x_0, y_0^\top)^\top$, $|e_x^\top J(\alpha)^t z_0| \le e_x^\top J(\alpha)^t (|x_0|, |y_0|^\top)^\top \le e_x^\top J(\alpha)^t (|x_0|, \bar{y}_0 \mathbf{1}^\top)^\top$, with equality when $y_0 = \text{sign}(x_0)\bar{y}_0\mathbf{1}$. Under the price-first unforced recursion $x_t = ax_{t-1} + \alpha\kappa \sum_i q_i y_{i,t-1}$ $(t \ge 1)$ and $y_{i,t} = (1-\eta_i)y_{i,t-1} + \eta_i x_t$ $(t \ge 1)$ with initial state $(|x_0|, \bar{y}_0\mathbf{1}^\top)$ in the unit-sum memory coordinate (where the initial cohort state $y_{i,0}$ enters with unit cumulative weight $\sum_{m\ge 0}(1-\eta_i)^m\eta_i = 1$, i.e. $\tilde{y}_{i,0} = \eta_i y_{i,0}$ bounded by $\eta_i\bar{y}_0$), summing over $t \ge 1$ gives $\sum_{t\ge 0} q_i y_{i,t} = q_i\bar{y}_0 + q_i \sum_{t\ge 1} x_t$, and hence $(1 - a - \alpha\kappa\Phi) \sum_{t\ge 1} x_t = a|x_0| + \alpha\kappa\Phi\bar{y}_0$, proving (3.11). (ii) Adding the forced worst-case shock loss $|I|\frac{\kappa\bar{u}}{1-a-\alpha\kappa\Phi}$ to $|I|$ times (3.11) and requiring the total loss to be at most $\mathcal{B}$ at $\Phi = \Phi^+$ rearranges to $\alpha\kappa\Phi^+(\mathcal{B} + |I|\bar{y}_0) \le (1 - a)\mathcal{B} - |I|(a|x_0| + \kappa\bar{u})$, which gives (3.12). (iii) Subtracting $\alpha^*_{\text{drift}} = c_I/[\kappa\Phi^+(1 + |I|\bar{y}_0/\mathcal{B})]$ from $\alpha^* = c_I/(\kappa\Phi^+)$ at $x_0 = 0$ gives (3.13).

Theorem 3.5.3. The recursion $h_{t+1} = ah_t + \kappa \sum_{m\le t} k_m h_{t-m}$ with $h_0 = \kappa \ne 0$ gives k_t from $h_0, \dots, h_{t+1}$, so the window determines $k_0, \dots, k_{n-1}$ and nothing more about M: any nonnegative measure on $[0, \bar{r}]$ with these moments, written as atoms (r_i, w_i) or as a limit of such, is the merged measure of a positive economy with $\eta_i = 1 - r_i$ and $q_i = w_i/\eta_i$ that produces the same window. (i) The compatible measures form a convex, weak-* compact set, and $\Phi = \int f\, dM$ is linear and continuous on it (f is continuous on $[0, \bar{r}]$), so its range is a closed interval $[\Phi_-, \Phi_+]$. Since $f^{(n)}(r) = n!/(1 - r)^{n+1} > 0$ on $[0, \bar{r}]$, $(1, r, \dots, r^{n-1}, f)$ is an extended complete Chebyshev system, and by the Markov–Krein theorem (Krein and Nudel'man, 1977, Theorems IV.1.1 and IV.6.1) the maximum is attained at the upper and the minimum at the lower principal representation, which have at most $\lfloor n/2 \rfloor + 1$ atoms. The theorem applies to moment vectors in the interior of the moment space, where both principal representations exist and are unique; a vector on the boundary has exactly one representing measure, so the interval collapses to a point. $S = \kappa/(1 - a - \kappa\Phi)$ is increasing in Φ on $\Phi < (1 - a)/\kappa$, which gives the interval of summed responses, unbounded when $\Phi_+ \ge (1 - a)/\kappa$ because by convexity every $\Phi \in [\Phi_-, \Phi_+]$ is attained. (ii) If μ, ν share moments up to order $n - 1$, then $\int f\, d(\mu - \nu) = \int (f - p)\, d(\mu - \nu)$

for every polynomial p of degree $n-1$, so $|\int f\,d(\mu-\nu)| \le 2k_0 \inf_p \|f-p\|_\infty = 2k_0 E_{n-1}$. With $r=\bar r(1+x)/2$, $f=(2/\bar r)/(x_0-x)$ on $[-1,1]$ with $x_0=2/\bar r-1>1$, and Chebyshev's formula gives $E_m(1/(x_0-x))=\varrho^m/(x_0^2-1)$ (Achieser, 1956); we checked it against a linear-programming minimax fit for $m\le 8$. Expanding $x_0=1+2(1-\bar r)/\bar r$ gives the asymptotics. (iii) Let the moment vector c be interior to the moment cone of $[0,1]$. Interior points have an open neighbourhood in the cone, and $m(r_0)=(1,r_0,\dots,r_0^{n-1})$ has norm at most $\sqrt{n}$, so for some $\epsilon>0$, $c-\epsilon m(r_0)$ is interior for every $r_0\in[0,1]$ and is represented by its lower principal representation, which has no atom at 1. Adding $\epsilon\delta_{r_0}$ gives a measure on $[0,1)$ with moments c and $\Phi\ge\epsilon/(1-r_0)\to\infty$ as $r_0\to 1$; by convexity every larger Φ below $(1-a)/\kappa$ is attained by a stable economy, so S is unbounded.

Theorem 3.5.4. (i) The flow impulse enters the stacked state as $b=\kappa(1,\eta_1,\dots,\eta_K)^\top$, and $h_t=e_x^\top J^t b\le\|J^t b\|_\infty\le\|J\|_\infty^t\|b\|_\infty$ because $J\ge 0$. The price row of J sums to $a+\kappa\Phi\le\bar s$ and cohort row i to $1-\eta_i(1-a-\kappa\Phi)\le 1-(1-\bar r)(1-\bar s)=\bar\lambda$, so $h_t\le\kappa\bar\lambda^t$ and $\sum_{t\ge T}h_t\le\kappa\bar\lambda^T/(1-\bar\lambda)$. $\sum_{t<T}\bar h_t$ is Gaussian with mean $\sum_{t<T}h_t$ and variance $T\sigma^2/N$, so $\sum_{t<T}\bar h_t+z_{1-\gamma}\sigma\sqrt{T/N}\ge\sum_{t<T}h_t$ with probability $1-\gamma$, and adding the tail bound gives validity. On the two-sided event of probability $1-2\gamma$, $\hat S^+-S\le 2z_{1-\gamma}\sigma\sqrt{T/N}+\kappa\bar\lambda^T/(1-\bar\lambda)$; T_ϵ makes the second term at most $\epsilon/2$ and the stated N makes the first at most $\epsilon/2$. (ii) Let θ_1 raise the cohort weight of θ_0 by Δ chosen so that $S_1-S_0=2\epsilon$; then $\|h_1-h_0\|_2=2C_0\epsilon(1+o(1))$. Validity at θ_1 gives $P_1(\hat S\ge S_1)\ge 1-\gamma$, hence $P_0(\hat S-S_0\ge 2\epsilon)\ge 1-\gamma-\mathrm{TV}(P_0,P_1)$. For N independent Gaussian responses, $\mathrm{KL}=N\|h_1-h_0\|_2^2/(2\sigma^2)$, and Pinsker's inequality gives $\mathrm{TV}\le\sqrt{N}\,C_0\epsilon(1+o(1))/\sigma$. An overshoot probability below $1/2$ therefore needs $\mathrm{TV}>1/2-\gamma$, which is the stated bound. (iv) Write $M=\sum_i q_i(1-r_i)\delta_{r_i}$ for the merged measure, $k_m=\int r^m dM$ for the open-loop kernel and $\mathcal{M}_0$ for the measures on $[0,\bar r]$ of mass at most $B_0=(1-\bar r)(\bar s-a)/\kappa$; economies with merged measure in $\mathcal{M}_0$ lie in the class, and finite-atom measures are weak-$*$ dense in $\mathcal{M}_0$. *Step 1 (closed loop to open loop).* The closed-loop transforms satisfy $H=\kappa/(1-az-\kappa zK)$, so $H_1-H_0=\kappa^{-1}zH_1H_0(K_1-K_0)$ and, by Young's inequality, $\|h_1-h_0\|_2\le\kappa^{-1}S_1S_0\|k_1-k_0\|_2\le C_1\|k_1-k_0\|_2$ on the class, for any number of recorded responses. *Step 2 (two points).* If two economies have $S_1>S_0+\epsilon$, validity at the first and the overshoot bound at the second make $\{\hat S^+\le S_0+\epsilon\}$ have probability at least $1-\gamma$ under P_0 and at most γ under P_1, so the total variation is at least $1-2\gamma$; for Gaussian responses this needs $\|h_1-h_0\|_2\ge 2z_{1-\gamma}\sigma/\sqrt{N}$. Hence $\omega_S(2z_{1-\gamma}\sigma/(C_1\sqrt{N}))\le\epsilon$, where $\omega_S(\delta)$ is the largest difference in S between two economies with $\|k_1-k_0\|_2\le\delta$. Since $\partial S/\partial\Phi=\kappa^2/(1-a-\kappa\Phi)^2\ge\kappa^2$, $\omega_S\ge\kappa^2\omega_\Phi$. *Step 3 (duality).* For a depth L, $\omega_\Phi(\delta)$ over $\mathcal{M}_0$ with the first L kernel values equals $\min_{c\in\mathbb{R}^L}\{\delta\|c\|_2+B_0\psi(g-p_c)\}$, with $g(r)=1/(1-r)$, $p_c(r)=\sum_{t<L}c_t r^t$ and $\psi(h)=(\sup h)_+ + (-\inf h)_+$ on $[0,\bar r]$: the inequality $\le$ is direct, and equality follows from Sion's minimax theorem because $\mathcal{M}_0^2$ is weak-$*$ compact (Donoho, 1994); the modulus for all kernel values is the limit as $L\to\infty$, by compactness. *Step 4 (a pole forces large coefficients).* If $\|c\|_2\le R$ and $\eta=\sup_{[0,\bar r]}|g-p_c|$, then $\log(1/\eta)\le K_0\max\{R^2,(1-\bar r)^{-1}\}$ with $K_0=8\log 6\sqrt{1-\bar r}\log\rho_*$, $\rho_*=(1+\sqrt{1-\bar r})^2/\bar r$. Indeed $e=g-p_c$ is

analytic on the disc of radius $\rho = 1 - s/2$, $s = \min\{1/(4R^2), (1-\bar{r})/4\}$, with $|e(1-s)| \geq 1/(2s)$ and $|e| \leq 3/s$ on $|z| = \rho$ because $|p_c(z)| \leq R(1-|z|)^{-1/2}$; the two-constants theorem on the disc slit along $[0, \bar{r}]$, with the harmonic measure of the slit bounded below by an annulus comparison through the Joukowski map, gives the claim. *Step 5.* For every c, either $\eta \geq \delta\sqrt{\log(1/\delta)}$, or $\log(1/\eta) > \frac{1}{2}\log(1/\delta)$ and Step 4 forces $\|c\|_2 \geq \sqrt{\log(1/\delta)/(2K_0)}$ once $\log(1/\delta) \geq 2K_0/(1-\bar{r})$; so $\omega_\Phi(\delta) \geq \min\{B_0, (2K_0)^{-1/2}\}\delta\sqrt{\log(1/\delta)}$ for small δ, and Step 2 gives $N \geq c\,\sigma^2\epsilon^{-2}\log(1/\epsilon)$. (v) The identification rates are Chapter 2's Theorems on the K-cohort lower bound, the matching upper bound and general memory.

Lemma 3.6.3. On the event that $\mathcal{C}(D)$ contains the true economy, the supremum over $\mathcal{C}(D)$ dominates the true worst-case loss for every action, including one selected from D; the event has probability at least $1 - \gamma$.

Theorem 3.6.4. (i) For $c_I \geq 0$, $\alpha \mapsto S(\alpha) = \kappa/(1-a-\kappa\alpha\Phi)$ is increasing, and so is the worst-case loss $|I|\bar{u}S(\alpha)$, which is also increasing in Φ; the robust constraint is therefore the constraint at Φ^+, and $|I|\bar{u}\kappa/(1-a-\kappa\alpha\Phi^+) \leq \mathcal{B}$ is equivalent to $\alpha \leq c_I/(\kappa\Phi^+)$. (ii) is Lemma 3.6.3. (iii) At $\Phi = \Phi^+$ the constraint binds at α^* when $\alpha^* < 1$. (iv) At $\alpha = 0$, $S = \kappa/(1-a)$. (v) Upper: α^* is Lipschitz in Φ^+ with constant $c_I/(\kappa\Phi^2_{\min})$ and Φ^+ is Lipschitz in $\hat{S}^+$ with constant $1/S^2_{\min}$, so Theorem 3.5.4(i) with $T \asymp \log N$ gives the rate. Lower: take two single-cohort economies in the class whose weights differ by $\Delta = c\,\sigma/\sqrt{N}$; their oracle capacities differ by a constant times Δ, and TV $\leq \sqrt{N}\|\partial h/\partial q\|_2\Delta/(2\sigma)$, which is below $(1-\beta)/2$ for small c; Proposition 3.5.2 gives an expected loss of at least $(1-\beta)/2$ times the capacity difference.

Theorem 3.7.1. (i) If O' reveals everything O reveals, every economy compatible with O' data is compatible with O data, so suprema over $\Theta_{O'}$ are at most those over Θ_O, and Theorem 3.6.4(i) is monotone in Φ^+. (ii) α^* is strictly decreasing in Φ^+ when $c_I > 0$ and $\alpha^* < 1$, and $\Phi^+_{O'} < \Phi^+_O$ exactly when no economy with $\Phi = \Phi^+_O$ remains compatible (the supremum is attained, by Theorem 3.5.3(i)). (iii) follows from Theorem 3.5.3(iii). (iv) If O identifies Φ, $\Phi^+_O = \Phi$ and no contract can raise α^*. (v) The delta method applied to $(w, r) \mapsto w/(1-r)$, whose gradient is $(1-r)^{-1}(1, \Phi_i)$, gives the standard error. If each certificate is valid at level $1 - \gamma/2$, their minimum is valid at level $1-\gamma$ by the union bound and is no larger than either, so the admitted level is at least the larger of the two admitted levels.

Chapter 4

Certify, don't identify: a theory of safety without identification

This chapter is the third paper of the thesis and the candidate's own work. It places the results of Chapters 2 and 3 in one framework, proves when a question about a market's unknown agents is cheap or impossible to certify, and extends the safety results to agents that are nonlinear and keep learning. Results proved here are marked PROVED IN THIS THESIS; results proved in Chapters 2–3 are marked accordingly; the conjecture is marked as such.

Summary. Regulators ask to know which algorithms trade in a market; we ask which questions about those algorithms can be answered from market data at all, and at what cost. Every question an outside observer can ask is a functional of the agents' merged measure of memory rates and weights, observed through noisy moments. For linear questions we determine the number of experiments needed to certify the answer, up to a constant factor in accuracy, by a modulus of continuity whose dual is a problem of polynomial approximation with penalized coefficients. A question is certifiable at the parametric rate exactly when its kernel has square-summable power-series coefficients; questions singular at infinite memory cannot be certified without a memory bound. Safety depends on the agents only through their total feedback weight, which sits on the boundary: it costs exactly $\sigma^2\epsilon^{-2}\log(1/\epsilon)$ experiments, and the logarithm cannot be removed, whereas identifying the agents costs polynomially or logarithmically more. The summed-gain guard remains safe for agents whose rules are nonlinear and whose memories and weights change over time, given only a bound on their total sensitivity, and when agents keep learning a certificate lasts for its safety margin divided by their learning speed. The safety bound extends to several assets with positive cross-impact, sharply. A discontinuous guard can fail to admit a strategic equilibrium.

4.1 Markets with adaptive agents

Definition 4.1.1 (Cohort economy)**.** A state $x_t \in \mathbb{R}$, the displacement of the price from its fundamental value, evolves as

$$x_{t+1} = a\,x_t + \kappa\,(u_t + w_{t+1}), \qquad u_t = \sum_{i=1}^{K} q_i\,\varphi_i(y_{i,t}), \qquad y_{i,t} = r_i\,y_{i,t-1} + (1-r_i)\,x_t, \tag{4.1}$$

with persistence $a \in [0,1)$, impact $\kappa > 0$, exogenous shocks w_t, and K cohorts of agents with weights $q_i \geq 0$, memory rates $r_i \in [0,1)$ and response rules φ_i. The economy is *linear* when $\varphi_i(y) = y$ for all i.

The exponential state $y_{i,t}$ is the reduced form of steady-state Kalman filtering and of regularized online gradient descent (Lemma 2.2.1); Section 4.12 relaxes linearity and stationarity. In the linear economy the aggregate response is a linear filter of past states,

$$u_t = \sum_{m\geq 0} k_m\,x_{t-m}, \qquad k_m = \int r^m\,dM(r), \qquad M = \sum_{i=1}^{K} q_i(1-r_i)\,\delta_{r_i},$$

so the agents enter the observable dynamics only through the *merged measure* M, a positive measure on $[0,1)$, and through its moments k_m. Cohorts that share a memory rate are indistinguishable, whatever their identities (Theorem 2.3.1). We write $\mathcal{M}(\bar r, B)$ for the positive measures on $[0,\bar r]$ of total mass at most B; $\bar r < 1$ is a *memory bound*.

4.2 Questions are functionals of the merged measure

Every question an outside observer can answer from the observable dynamics is a functional F of M (and of the known plant parameters a, κ). Three families matter.

Identification. $F(M) = M$ itself, in a metric such as the Wasserstein distance W_1, or the atoms (r_i, q_i) when M has K atoms. Chapter 2 studies these.

Safety. For bounded shocks $|w_t| \leq \bar w$, the worst-case transient loss of a position I is $|I|\,\bar w\,\|h\|_1$, where h is the closed-loop impulse response, and a pre-trade guard on $\|h\|_1$ admits every safe action and no unsafe one (Lemma 3.6.1). In the linear economy with $a \geq 0$ all h_t are nonnegative, so $\|h\|_1$ equals the summed response

$$S(M) = \frac{\kappa}{1 - a - \kappa\,\Phi(M)}, \qquad \Phi(M) = \sum_{m\geq 0} k_m = \int \frac{dM(r)}{1-r} = \sum_i q_i, \tag{4.2}$$

whenever $a + \kappa\Phi < 1$ (Theorem 3.4.3). Safety therefore depends on the agents only through one linear functional of M, the total feedback weight Φ.

Other operational questions. The response at a given lag, $k_m = \int r^m dM$; the discounted feedback $\int (1-\beta r)^{-1} dM$; the peak of the impulse response. The first two are linear in M; the peak is not, and a guard built on it admits unsafe actions (Chapter 3).

4.3 Observation contracts and the cost of a question

Assumption 4.3.1 (Reset contract)**.** The observer can run N independent reset experiments. Each returns the first L aggregate response taps with independent Gaussian noise, $\hat{k}_m = k_m + \sigma\xi_m$, $\xi_m \sim N(0,1)$, $m = 0, \dots, L-1$.

This is the controlled-probe contract with observed aggregate flow (contract I_1 of Chapter 2). Chapter 3 works with the closed-loop taps h_t instead; for the summed response the two contracts give the same rate (Theorem 3.5.4). Correlated tap noise, passive data and tagged disclosure are other contracts, treated in Chapters 2 and 3.

Definition 4.3.2 (Cost of certification)**.** For a functional F, a class $\mathcal{M}$, accuracy $\epsilon > 0$ and level $1-\alpha$, $N_{\mathrm{cert}}(\epsilon; F, \mathcal{M})$ is the smallest N for which some interval built from the data has half-width at most ϵ and covers $F(M)$ with probability at least $1-\alpha$ for every $M \in \mathcal{M}$.

4.4 The certification–identification dichotomy

Consider a linear question $F_g(M) = \int g\, dM$, where g is continuous on $[0, \bar{r}]$; in the examples $g(r) = \sum_{t\geq 0} a_t r^t$ converges absolutely there. Write $k_L(M) = (k_0, \dots, k_{L-1})$ for the first L taps and $p_c(r) = \sum_{t<L} c_t r^t$ for the polynomial with coefficient vector $c \in \mathbb{R}^L$. Truncating the Taylor series gives a first measure of how hard the question is. Its *tail at depth* T and its *coefficient energy* are

$$\tau_g(T) = B \sup_{0\leq r\leq \bar{r}} \Big| \sum_{t\geq T} a_t r^t \Big|, \qquad A_g(T) = \sum_{t<T} a_t^2,$$

and its *truncation depth* is $T_g(\epsilon) = \min\{T \geq 1 : \tau_g(T) \leq \epsilon/2\}$; $E_T(g) = \min_{\deg p < T} \max_{[0,\bar{r}]} |g-p|$ is the error of the best polynomial approximation of degree below T. The quantity that decides the cost exactly is the *modulus of continuity* of the question with respect to the observed taps,

$$\omega_L(\delta) = \sup\big\{ F_g(M_1) - F_g(M_0) :\ M_0, M_1 \in \mathcal{M}(\bar{r}, B),\ \|k_L(M_1) - k_L(M_0)\|_2 \leq \delta \big\},$$

the largest difference in the answer between two economies whose observable taps differ by at most δ. For a function h on $[0, \bar{r}]$ put $\psi(h) = (\sup h)_+ + (-\inf h)_+$; $\psi(h) \geq \|h\|_\infty$, with equality when h changes sign.

Theorem 4.4.1 (Certification of linear questions)**.** *Under Assumption 4.3.1, with z_β the standard normal β-quantile and $\alpha \in (0, 1/2)$:*

(i) Upper bound. *If* $L \geq T_g(\epsilon)$, *then* $N_{\mathrm{cert}}(\epsilon; F_g, \mathcal{M}(\bar{r}, B)) \leq \lceil 4z^2_{1-\alpha/2}\sigma^2 A_g(T_g(\epsilon))\,\epsilon^{-2}\rceil$. PROVED IN THIS THESIS

(ii) Truncation is necessary. *For every* T *there are* $M, M' \in \mathcal{M}(\bar{r}, B)$ *whose first* T *taps coincide and with* $|F_g(M) - F_g(M')| \geq 2B\,E_T(g)$. *Hence any certificate that uses only the first* T *taps has worst-case half-width at least* $B\,E_T(g)$. PROVED IN THIS THESIS

(iii) Long memory makes certification impossible. *Without a memory bound, that is on the positive measures on* $[0,1)$ *of mass at most* B, *if* $\limsup_{r\uparrow 1} |g(r)|\sqrt{1-r} = \infty$ *then for every level* $1-\alpha$ *with* $\alpha < 1/4$ *no finite* N *certifies* F_g *to any finite accuracy.* PROVED IN THIS THESIS

(iv) Exact cost. *For every depth* L, *the modulus has the dual form*

$$\omega_L(\delta) = \min_{c\in\mathbb{R}^L} \{\delta\|c\|_2 + B\,\psi(g - p_c)\} \qquad (\delta > 0),$$

and $\omega_L(0) = 2B\,E_L(g)$. *If* N *experiments certify* F_g *to half-width* ϵ *at level* $1-\alpha$, *then* $\omega_L(2z_{1-\alpha}\sigma/\sqrt{N}) \leq 2\epsilon$. *Conversely, if* $\omega_L(2z_{1-\alpha/2}\sigma/\sqrt{N}) \leq 2\epsilon$, *an affine interval* $\sum_t c_t\bar{k}_t + b \pm \epsilon$ *certifies* F_g *with* N *experiments. Writing* $N_\omega(\epsilon) = \min\{N : \omega_L(2z_{1-\alpha}\sigma/\sqrt{N}) \leq 2\epsilon\}$ *and* $\lambda_\alpha = z_{1-\alpha/2}/z_{1-\alpha}$,

$$N_\omega(\epsilon) \;\leq\; N_{\mathrm{cert}}(\epsilon; F_g, \mathcal{M}(\bar{r}, B)) \;\leq\; N_\omega(\epsilon/\lambda_\alpha).$$

PROVED IN THIS THESIS

Part (iv) replaces the matching lower bound in terms of coefficient energy that earlier drafts stated as a target. That statement is false (Remark 4.4.6); the modulus is the right quantity, and it determines the cost of every linear question up to the factor λ_α in accuracy ($\lambda_{0.05} \approx 1.19$). The structure of part (iv), a modulus that governs both a lower bound and an affine interval, is the classical theory of estimating a linear functional over a convex class (Ibragimov and Has'minskii, 1984; Donoho, 1994). What is specific to markets is the dual form: the cost of a question about the agents is the value of a problem of polynomial approximation of g in which the polynomial's coefficients are penalized, because each coefficient multiplies a noisy tap.

Proof of (i)–(iii). (i) Let $\bar{k}_t$ be the average of the N replicates of tap t and $T = T_g(\epsilon)$. The estimator $\hat{F} = \sum_{t<T} a_t\bar{k}_t$ has mean $\sum_{t<T} a_t k_t$ and variance $\sigma^2 A_g(T)/N$. Its bias is $|\sum_{t\geq T} a_t k_t| = |\int \sum_{t\geq T} a_t r^t\, dM(r)| \leq \tau_g(T) \leq \epsilon/2$ for every $M \in \mathcal{M}(\bar{r}, B)$. The interval $\hat{F} \pm (\epsilon/2 + z_{1-\alpha/2}\sigma\sqrt{A_g(T)/N})$ therefore covers $F_g(M)$ with probability at least $1-\alpha$, and its half-width is at most ϵ once $z_{1-\alpha/2}\sigma\sqrt{A_g(T)/N} \leq \epsilon/2$.

(ii) By duality for best uniform approximation, $E_T(g) = \max\{\int g\, d\nu\}$ over signed measures ν on $[0, \bar{r}]$ with total variation at most one that annihilate every polynomial of degree below T. Take a maximizer and write $\nu = \nu^+ - \nu^-$. Orthogonality to constants gives $\nu^+([0,\bar{r}]) =$

$\nu^-([0,\bar r]) =: c \le 1/2$. Set $M = (B/c)\nu^+$ and $M' = (B/c)\nu^-$: both lie in $\mathcal{M}(\bar r, B)$, their first T moments coincide, and $F_g(M) - F_g(M') = (B/c)E_T(g) \ge 2B\,E_T(g)$. Two measures that produce identical data cannot both be covered by an interval of half-width below half their distance.

(iii) Compare $M_0 = 0$ with $M_1 = c\,\delta_r$. Under M_1 the taps are $k_t = c\,r^t$, so the Kullback–Leibler divergence between the two data distributions over any number of taps is at most $Nc^2/(2\sigma^2(1-r^2))$. Choose $c = \sigma\sqrt{(1-r^2)/N}$, which is at most B for r near one; the divergence is then at most $1/2$, so by Pinsker's inequality the total variation distance between the two data distributions is at most $1/2$. The functional differs by $c\,|g(r)| = \sigma\sqrt{(1-r^2)/N}\,|g(r)| \ge \sigma\sqrt{(1-r)/N}\,|g(r)|$, which is unbounded along a sequence $r \uparrow 1$ by assumption. Suppose an interval of half-width ϵ covered F_g with probability at least $1-\alpha$ under both economies. Once the two values differ by more than 2ϵ, the event that the interval contains $F_g(M_0)$ has probability at least $1-\alpha$ under M_0 and at most α under M_1, so the total variation distance is at least $1-2\alpha > 1/2$, a contradiction. □

Proof of (iv). Duality. For any $c \in \mathbb{R}^L$ and any admissible pair, with $h = g - p_c$,

$$F_g(M_1) - F_g(M_0) = \int h\,dM_1 - \int h\,dM_0 + \langle c, k_L(M_1) - k_L(M_0)\rangle \le B\psi(h) + \delta\|c\|_2,$$

because $\int h\,dM_1 \le B(\sup h)_+$ and $-\int h\,dM_0 \le B(-\inf h)_+$ for measures of mass at most B. So $\omega_L(\delta)$ is at most the minimum. For the reverse inequality write $\mu = (M_0, M_1)$ and

$$\Lambda(\mu, c) = \int h\,d(M_1 - M_0) + \delta\|c\|_2, \qquad h = g - p_c.$$

Since $\inf_c\{\delta\|c\|_2 - \langle c, v\rangle\}$ equals 0 if $\|v\|_2 \le \delta$ and $-\infty$ otherwise, $\omega_L(\delta) = \sup_\mu \inf_c \Lambda(\mu, c)$. The set $\mathcal{M}(\bar r, B)^2$ is convex and compact in the weak-$*$ topology, Λ is affine and weak-$*$ continuous in μ (because h is continuous) and convex and continuous in c, so Sion's minimax theorem (Sion, 1958) gives $\omega_L(\delta) = \inf_c \sup_\mu \Lambda(\mu, c)$. The inner supremum is $B\psi(g - p_c) + \delta\|c\|_2$, attained by putting mass B at a maximizer of h (if $\sup h > 0$) and at a minimizer (if $\inf h < 0$). For $\delta > 0$ the objective is convex, continuous and coercive in c, so the infimum is a minimum. At $\delta = 0$ the same argument gives $\omega_L(0) = \inf_c B\psi(g - p_c)$; since adding a constant to p_c can make $g - p_c$ change sign and then $\psi = \sup - \inf$, the infimum is B times the smallest oscillation of $g - p$, that is $2B\,E_L(g)$, which is part (ii) with equality.

Lower bound. The averages $\bar k = (\bar k_0, \dots, \bar k_{L-1})$ are sufficient and distributed as $N(k_L(M), (\sigma^2/N)I_L)$, so the total variation distance between the data distributions of two economies is $2\Phi_{\mathcal{N}}(\sqrt{N}\|\Delta k\|_2/(2\sigma)) - 1$, with $\Phi_{\mathcal{N}}$ the normal distribution function. Take an admissible pair with $\|k_L(M_1) - k_L(M_0)\|_2 < 2z_{1-\alpha}\sigma/\sqrt{N}$; the distance is then below $1 - 2\alpha$. If the two answers differed by more than 2ϵ, an interval of half-width ϵ covering each with probability $1-\alpha$ would separate the two data distributions with total variation at least $1 - 2\alpha$, as in (iii). Hence $\omega_L(t) \le 2\epsilon$

for every $t < 2z_{1-\alpha}\sigma/\sqrt{N}$. The modulus is concave (the admissible pairs form a convex set and the constraint is a norm), finite and nondecreasing, hence continuous on $(0,\infty)$, so $\omega_L(2z_{1-\alpha}\sigma/\sqrt{N}) \le 2\epsilon$.

Upper bound. Put $\delta = 2z_{1-\alpha/2}\sigma/\sqrt{N}$, let c attain the minimum in the dual form and $h = g - p_c$. The interval $\sum_t c_t \bar{k}_t + b \pm \epsilon$ with $b = B[(\sup h)_+ - (-\inf h)_+]/2$ has bias at most $B\psi(h)/2$ for every economy, by the bounds in the duality step, and its noise is normal with standard deviation $\sigma\|c\|_2/\sqrt{N}$. It covers F_g with probability at least $1-\alpha$ whenever $\epsilon \ge B\psi(h)/2 + z_{1-\alpha/2}\sigma\|c\|_2/\sqrt{N} = \omega_L(\delta)/2$.

Sandwich. The left inequality restates the lower bound. Concavity and $\omega_L(0) \ge 0$ give $\omega_L(\lambda t) \le \lambda\,\omega_L(t)$ for $\lambda \ge 1$. If $\omega_L(2z_{1-\alpha}\sigma/\sqrt{N}) \le 2\epsilon/\lambda_\alpha$, then $\omega_L(2z_{1-\alpha/2}\sigma/\sqrt{N}) \le 2\epsilon$, and the upper bound applies. □

What the theorem says. The cost of answering a linear question is the cost of approximating g, on the memory range $[0,\bar{r}]$, by a polynomial whose coefficients are cheap to estimate. Truncating the Taylor series is one such polynomial and gives part (i); it is often not the best one. Questions singular at infinite memory, faster than $(1-r)^{-1/2}$, cannot be certified without a memory bound (part (iii)). The number of agents and the spacing of their memories do not enter. Two consequences make the picture precise.

Proposition 4.4.2 (Which questions are parametric)**.** *Let $\omega_\infty(\delta) = \inf_L \omega_L(\delta)$, the modulus when enough taps are observed. Then $\sup_{0<\delta\le 1}\omega_\infty(\delta)/\delta < \infty$, so that $N_{\mathrm{cert}} = O(\sigma^2\epsilon^{-2})$ for observation depths L large enough, if and only if g agrees on $[0,\bar{r}]$ with a power series $\sum_t a_t r^t$ whose coefficients are square summable. When g is given by a power series on $[0,\bar{r}]$ with $\bar{r} > 0$, this is the condition $\sum_t a_t^2 < \infty$ on its Taylor coefficients.* PROVED IN THIS THESIS

Proof. If $a \in \ell^2$, take $c = (a_0,\dots,a_{L-1})$: by Cauchy–Schwarz $\psi(g-p_c) \le 2\sup_{r\le\bar{r}}|\sum_{t\ge L} a_t r^t| \le 2\|a\|_2\,\bar{r}^L/\sqrt{1-\bar{r}^2}$, so $\omega_L(\delta) \le \delta\|a\|_2 + 2B\|a\|_2\bar{r}^L/\sqrt{1-\bar{r}^2}$ and $\omega_\infty(\delta) \le \delta\|a\|_2$. Conversely, suppose $\omega_\infty(\delta) \le C\delta$ for $\delta \in (0,1]$. For $\delta_n = 1/n$ choose L_n and $c^{(n)} \in \mathbb{R}^{L_n}$ with $\delta_n\|c^{(n)}\|_2 + B\psi(g-p_{c^{(n)}}) \le 2C\delta_n$. Then $\|c^{(n)}\|_2 \le 2C$ and $\|g-p_{c^{(n)}}\|_\infty \le 2C\delta_n/B \to 0$. Padded with zeros, the $c^{(n)}$ are bounded in ℓ^2, so a subsequence converges weakly to some $a \in \ell^2$. For each $r \in [0,\bar{r}]$ the sequence $(r^t)_t$ lies in ℓ^2, hence $p_{c^{(n)}}(r) = \langle c^{(n)}, (r^t)\rangle \to \sum_t a_t r^t$ along the subsequence, while $p_{c^{(n)}}(r) \to g(r)$. Uniqueness of power series that agree on an interval gives the last sentence. □

The square-summable power series are the restrictions to $[0,\bar{r}]$ of the Hardy space H^2 of the unit disc, whose functions grow at most like $(1-r)^{-1/2}$ as $r \uparrow 1$. That is exactly the growth beyond which part (iii) rules out certification without a memory bound: below the threshold a memory bound makes a question parametric or nearly so, above it nothing helps.

The single tap $g(r) = r^m$ and the discounted feedback $g(r) = (1-\beta r)^{-1}$ with $\beta < 1$ have square-summable coefficients and are parametric. The total feedback Φ, the quantity that safety depends on, has $a_t \equiv 1$: it sits exactly on the boundary, and it pays exactly one logarithm.

Theorem 4.4.3 (The safety question costs exactly one logarithm)**.** *Let $g(r) = 1/(1-r)$, so that $F_g = \Phi$, and put*

$$\rho_* = \frac{(1+\sqrt{1-\bar r})^2}{\bar r}, \qquad K = 8\log 6\,\sqrt{1-\bar r}\,\log\rho_*, \qquad c = \min\{B, (2K)^{-1/2}\}.$$

For every $\delta \le \delta_0 = e^{-2K/(1-\bar r)}$ and every depth L, $\omega_L(\delta) \ge c\,\delta\sqrt{\log(1/\delta)}$. For every $\delta \le e^{-1}$ and $L \ge \log(1/\delta)/\log(1/\bar r)$,

$$\omega_L(\delta) \le \delta\sqrt{1 + \log(1/\delta)/\log(1/\bar r)} + B\delta/(1-\bar r).$$

Hence $c \le \liminf_{\delta\to 0}\omega_L(\delta)/(\delta\sqrt{\log(1/\delta)}) \le \limsup_{\delta\to 0}\omega_L(\delta)/(\delta\sqrt{\log(1/\delta)}) \le (\log(1/\bar r))^{-1/2}$ for deep enough observation, and $N_{\mathrm{cert}}(\epsilon;\Phi,\mathcal{M}(\bar r,B)) \asymp \sigma^2\epsilon^{-2}\log(1/\epsilon)$ as $\epsilon\to 0$. The same holds for the summed response $S = \kappa/(1-a-\kappa\Phi)$ whenever $a+\kappa B/(1-\bar r) < 1$. The logarithmic factor in the certification cost of Chapter 3 cannot be removed. PROVED IN THIS THESIS

The lower bound rests on a lemma about polynomials: a polynomial that approximates $1/(1-r)$ closely on the memory range must have large coefficients, because $1/(1-z)$ has a pole at $z=1$ while a polynomial with coefficient norm R grows at most like $R(1-|z|)^{-1/2}$ inside the disc. The constant $\log\rho_*$ is the Green's function of the complement of $[0,\bar r]$ with pole at infinity, evaluated at the singularity $z=1$: the further the memory range stays from one, the larger it is.

Lemma 4.4.4. *With ρ_* and K as in Theorem 4.4.3, every polynomial $p(r) = \sum_{t<L} c_t r^t$ with $\|c\|_2 \le R$ satisfies*

$$\log\frac{1}{\eta} \le K\max\{R^2, (1-\bar r)^{-1}\}, \qquad \eta = \sup_{0\le r\le\bar r}\Big|\frac{1}{1-r} - p(r)\Big|.$$

Proof. Put $e(z) = 1/(1-z) - p(z)$ and $E = [0,\bar r]$. Let $s = \min\{1/(4R^2), (1-\bar r)/4\}$, $x = 1-s$ and $\rho = 1-s/2$, so that $\bar r < x < \rho < 1$. By Cauchy–Schwarz $|p(z)| \le R(1-|z|^2)^{-1/2} \le R(1-|z|)^{-1/2}$ in the disc. At $z=x$ this is at most $R/\sqrt{s} \le 1/(2s)$, so $|e(x)| \ge 1/s - 1/(2s) = 1/(2s)$. On the circle $|z|=\rho$, $|e(z)| \le 2/s + R\sqrt{2/s} \le 3/s$. The function $\log|e|$ is subharmonic on the slit disc $D_\rho \setminus E$, at most $\log\eta$ on E and at most $\log(3/s)$ on $|z|=\rho$, so by the two-constants theorem (Ransford, 1995, Theorem 4.3.7)

$$\log\frac{1}{2s} \le \log|e(x)| \le \varpi\log\eta + (1-\varpi)\log\frac{3}{s},$$

where ϖ is the harmonic measure of E in $D_\rho\setminus E$ seen from x. Rearranging, $\varpi\log(1/\eta) \le \log 6$.

It remains to bound ϖ from below explicitly. The affine map $z \mapsto (2z-\bar r)/\bar r$ sends E to $[-1,1]$, x to $u = 2x/\bar r - 1$ and D_ρ to the disc of radius $2\rho/\bar r$ centred at -1, which contains the disc D_{R_*} centred at 0 with $R_* = 2\rho/\bar r - 1 \in (u, u_1]$, $u_1 = 2/\bar r - 1$. Harmonic measure of $[-1,1]$

only decreases when the outer domain shrinks, first to D_{R_*} and then to the interior of the ellipse with foci ± 1 and major semi-axis R_*, which lies in D_{R_*}. The Joukowski map $\zeta \mapsto (\zeta + \zeta^{-1})/2$ carries the annulus $1 < |\zeta| < R_* + \sqrt{R_*^2 - 1}$ onto that ellipse minus $[-1, 1]$, and in the annulus the harmonic measure of the inner circle is explicit. Hence

$$\varpi \geq \frac{\operatorname{arccosh} R_* - \operatorname{arccosh} u}{\operatorname{arccosh} R_*} \geq \frac{(R_* - u)/\sqrt{R_*^2 - 1}}{\operatorname{arccosh} u_1} = \frac{s/\bar{r}}{(2\sqrt{1-\bar{r}}/\bar{r}) \log \rho_*} = \frac{s}{2\sqrt{1-\bar{r}} \log \rho_*},$$

using $\operatorname{arccosh} R_* - \operatorname{arccosh} u = \int_u^{R_*} (v^2 - 1)^{-1/2} dv$, $R_* - u = 2(\rho - x)/\bar{r} = s/\bar{r}$, $\sqrt{u_1^2 - 1} = 2\sqrt{1-\bar{r}}/\bar{r}$ and $\operatorname{arccosh} u_1 = \log \rho_*$. So $\log(1/\eta) \leq 2 \log 6 \sqrt{1-\bar{r}} \log \rho_* / s$, and $1/s = \max\{4R^2, 4/(1-\bar{r})\}$. □

Proof of Theorem 4.4.3. Upper bound. Take $c = (1, \dots, 1)$ of length $T = \lceil \log(1/\delta)/\log(1/\bar{r}) \rceil \leq L$. Then $g - p_c = r^T/(1-r) \geq 0$, so $\psi(g - p_c) = \bar{r}^T/(1-\bar{r}) \leq \delta/(1-\bar{r})$, $\|c\|_2 = \sqrt{T} \leq \sqrt{1 + \log(1/\delta)/\log(1/\bar{r})}$, and the bound follows from the dual form.

Lower bound. By part (iv) it suffices to bound $\delta\|c\|_2 + B\psi(g - p_c)$ below for every c. Let $R = \|c\|_2$ and $\eta = \|g - p_c\|_\infty \leq \psi(g - p_c)$. If $\eta \geq \delta\sqrt{\log(1/\delta)}$, the objective is at least $B\delta\sqrt{\log(1/\delta)}$. Otherwise $\eta < \sqrt{\delta}$, so $\log(1/\eta) > \frac{1}{2}\log(1/\delta)$, and Lemma 4.4.4 gives $K \max\{R^2, (1-\bar{r})^{-1}\} > \frac{1}{2}\log(1/\delta) \geq K/(1-\bar{r})$ for $\delta \leq \delta_0$. The maximum is therefore R^2, so $R \geq \sqrt{\log(1/\delta)/(2K)}$ and the objective is at least $\delta\sqrt{\log(1/\delta)/(2K)}$.

Cost. By part (iv), N certifies Φ to half-width ϵ only if $\omega_L(2z_{1-\alpha}\sigma/\sqrt{N}) \leq 2\epsilon$, which with the lower bound forces $N \geq c'\sigma^2\epsilon^{-2}\log(1/\epsilon)$ for small ϵ; the upper bound and part (iv) give a matching $N \leq C'\sigma^2\epsilon^{-2}\log(1/\epsilon)$. On $\mathcal{M}(\bar{r}, B)$, $\Phi \leq B/(1-\bar{r})$, and when $a + \kappa B/(1-\bar{r}) < 1$ the map $\Phi \mapsto S$ is increasing with derivative $\kappa^2/(1 - a - \kappa\Phi)^2$ bounded above and below, so intervals for Φ and for S convert into each other with constant changes in accuracy. □

The constants are explicit but not sharp. For $\bar{r} = 0.5$ the bracket for the limiting ratio $\omega_L(\delta)/(\delta\sqrt{\log(1/\delta)})$ is $[0.17, 1.20]$ (with $B \geq 0.17$), and for $\bar{r} = 0.9$ it is $[0.41, 3.08]$; the computed ratios at $\delta = 10^{-5}$, 1.15 and 2.99, lie near the top of each bracket. The upper constant is the Taylor rate $(\log(1/\bar{r}))^{-1/2}$, and the lower constant loses mainly in the comparison of harmonic measures. Closing the gap is open (Chapter 6).

Corollary 4.4.5 (The rates of this thesis as cases of one picture)**.** *(a)* Safety with a memory bound. $N_{\mathrm{cert}}(\epsilon; \Phi) \asymp \sigma^2\epsilon^{-2}\log(1/\epsilon)$, *and the same for S (Theorem 4.4.3). Part (i) alone gives the upper bound $O(\sigma^2\epsilon^{-2}\log(1/\epsilon)/\log(1/\bar{r}))$, the rate of Theorem 3.5.4; Theorem 4.4.3 shows that its logarithm is necessary.* PROVED IN THIS THESIS

(b) Safety without a memory bound. $g(r) = (1-r)^{-1}$ *satisfies the condition of (iii): no finite number of experiments certifies the risk, as Theorem 3.5.3(iii) shows for finite windows.* PROVED IN CH. 3

(c) A single response tap, and discounted feedback. *For $g(r) = r^m$, $\omega_L(\delta) \le \delta$; for $g(r) = (1-\beta r)^{-1}$ with $\beta < 1$, $\omega_\infty(\delta) \le \delta/\sqrt{1-\beta^2}$. Both are parametric (Proposition 4.4.2).* PROVED IN THIS THESIS

(d) Identifying K cohorts. *Recovering the atoms is not a linear question. It requires moments up to order $2K-1$ to be resolved against moment-matched alternatives whose higher moments differ by $O((\Delta\eta)^{2K-1})$, which costs $(\Delta\eta)^{-(4K-2)}$ experiments (Theorems 2.6.4 and 2.6.5).* PROVED IN CH. 2

(e) Identifying a general memory measure. *Recovering M in W_1 means certifying F_g uniformly over all 1-Lipschitz g. Their best approximations by degree-T polynomials improve only as $1/T$, and the coefficients of those polynomials grow geometrically in T, so accuracy improves only logarithmically in N: the rate $1/\log N$ of Theorem 2.6.8.* PROVED IN CH. 2

The link in (e) between coefficient growth and the logarithmic rate is a heuristic reading of Chapter 2's proof through the dual form of part (iv), not a separate result here. The ladder is now exact at its first rungs: a single tap and any discounted feedback are parametric, the undiscounted total feedback that safety depends on costs exactly one extra logarithm, questions singular faster than $(1-r)^{-1/2}$ cannot be certified without a memory bound, and identification costs polynomially or logarithmically more.

Remark 4.4.6 (Why coefficient energy is the wrong lower-bound quantity). Earlier drafts stated, as a target, a lower bound $N_{\mathrm{cert}} \ge c\sigma^2 A_g(c' T_g(\epsilon))\epsilon^{-2}$ with constants uniform over a regular coefficient class. It is false. Fix $\epsilon_0 > 0$, $\beta \in (0,1)$, $\eta > 0$ with $\eta/(1-\beta\bar{r}) < 1/2$, and set $g_m(r) = (\epsilon_0/B)[(1-r/\bar{r})^m + \eta/(1-\beta r)]$. Its coefficients are eventually positive with polynomial growth; on $[0,\bar{r}]$ it takes values in an interval of length below $3\epsilon_0/(2B)$, so a constant polynomial gives $\psi(g_m - p_c) < 3\epsilon_0/(2B)$ with $\|c\|_2$ bounded and the question is certified to accuracy ϵ_0 with no data. Yet $T_{g_m}(\epsilon_0) = m+1$ and $A_{g_m}(c' T_{g_m}(\epsilon_0))\epsilon_0^{-2} \ge B^{-2}(m/\bar{r} - \eta\beta)^2 \to \infty$, because the alternating binomial tail $\sum_{t=T}^{m}(-1)^t\binom{m}{t} = (-1)^T\binom{m-1}{T-1}$ keeps $\tau_{g_m}(T) > \epsilon_0/2$ for $T \le m$. Taylor coefficients can be large while a different polynomial approximates the question cheaply; part (iv) measures the cheapest one.

Numerical certificates. The script `checks/check_modulus.py` solves the primal (pairs of measures on a grid) and the dual (coefficient vectors) of part (iv) for the total feedback, for discounted feedback with $\beta = 0.8$ and for the tap r^3, at $\bar{r} \in \{0.5, 0.9\}$ and δ from 10^{-1} to 10^{-5}, and stores both. Without any solver, the check rescales the stored measures to exact feasibility (a lower bound on ω_L) and evaluates the stored coefficients over the whole interval with a curvature allowance between grid points (an upper bound, by weak duality). The brackets agree to a relative 4×10^{-4} or better. Over four decades of δ, $\omega_L(\delta)/\delta$ settles at the values $\|a\|_2$ of Proposition 4.4.2 for the two parametric questions ($1/\sqrt{1-\beta^2} = 5/3$ for the discounted question and 1 for the tap) and keeps growing for Φ (from 6.6 to 10.1 at $\bar{r} = 0.9$), while

$\omega_L(\delta)/(\delta\sqrt{\log(1/\delta)})$ stays between 1.1 and 4.4. The check also verifies $\omega_L(0) = 2B\,E_L(g)$ at small depths. These computations illustrate the theorems; they are not part of their proofs.

4.5 Questions that are not linear

The peak of the impulse response is the question an operator might ask first, and Chapter 3 shows that a guard built on it admits unsafe actions. The framework explains both facts. Write $h_1, h_2, \dots$ for the response of the price displacement to a unit shock in the linear economy, $h_1 = \kappa$ and $h_{t+1} = a h_t + \kappa \sum_{m=0}^{t-1} k_m h_{t-m}$, so that $\sum_t h_t = S$.

Proposition 4.5.1 (Finitely dependent questions are parametric)**.** *Let F be any functional on a class $\mathcal{M}$ of economies. (a) The lower bound of Theorem 4.4.1(iv) holds for F with ω_L replaced by its own modulus $\omega_L^F(\delta) = \sup\{F(M_1) - F(M_0) : \|k_L(M_1) - k_L(M_0)\|_2 \le \delta\}$; linearity is not used. (b) If $F(M) = \Psi(k_0(M), \dots, k_{T-1}(M))$ for a function Ψ that is Λ-Lipschitz in the sup norm on the box $\mathcal{B} = \prod_{m<T}[\underline{k}_m, \bar{k}_m]$ containing the taps of every economy in $\mathcal{M}$, and $L \ge T$, then*

$$N_{\mathrm{cert}}(\epsilon; F, \mathcal{M}) \le \lceil \Lambda^2 z^2_{1-\alpha/(2T)} \sigma^2 \epsilon^{-2} \rceil .$$

PROVED IN THIS THESIS

Proof. (a) The two-point argument in the proof of part (iv) compares the data distributions of two economies and the values of F at them; it never uses linearity. (b) Clip each averaged tap to $[\underline{k}_m, \bar{k}_m]$ and evaluate Ψ there. Clipping to an interval that contains the true tap does not increase the error, so $|\Psi(\text{clipped } \bar{k}) - F(M)| \le \Lambda \max_{m<T} |\bar{k}_m - k_m|$. By the union bound, $\max_m |\bar{k}_m - k_m| \le z_{1-\alpha/(2T)}\sigma/\sqrt{N}$ with probability at least $1-\alpha$, so the interval with that half-width times Λ covers F. □

Corollary 4.5.2 (The peak is cheap and uninformative)**.** *On the stable class, $a \ge 0$ and $a + \kappa B/(1-\bar{r}) < 1$, every economy in $\mathcal{M}(\bar{r}, B)$ has $0 \le h_t \le h_1 = \kappa$ for all t: the peak response equals κ whatever the agents, and it is certified with no data. The summed response S, which is what a position's worst-case loss depends on (Theorem 4.12.2), varies over the class from $\kappa/(1-a)$ to $\kappa/(1-a-\kappa B/(1-\bar{r}))$ and costs $\sigma^2\epsilon^{-2}\log(1/\epsilon)$ experiments (Theorem 4.4.3). More generally, under any observation contract in which the closed-loop taps h_t are observed and decay geometrically, $h_t \le C\lambda^{-t}$ with $\lambda > 1$, uniformly over the class, and the peak is at least $h_1 \ge c_0 > 0$, the peak is attained before $T_* = 1 + \log(C/c_0)/\log\lambda$ and is a maximum of T_* observed taps, hence parametric by Proposition 4.5.1(b) with $\Lambda = 1$.* PROVED IN THIS THESIS

Proof. On the stable class $\sum_m k_m = \Phi \le B/(1-\bar{r})$, so $a + \kappa\sum_{m<t} k_m < 1$ for every t. If $0 \le h_s \le \kappa$ for $s \le t$, then $0 \le h_{t+1} \le (a + \kappa\sum_{m<t} k_m)\kappa \le \kappa$, and induction from $h_1 = \kappa$ gives the first claim. The range of S follows from $0 \le \Phi \le B/(1-\bar{r})$ with both ends attained in the class (mass B at $\bar{r}$, and the zero measure). For the general claim, $h_t < c_0 \le \max_s h_s$ once $C\lambda^{-t} < c_0$. □

The ladder therefore has a rung below the parametric one. The peak in the cohort economy is known before any data arrive, yet a guard on it controls nothing; the summed response costs one logarithm more than a parametric question and is the quantity that loss depends on. The cheapest question is the wrong one, and the right one is only slightly more expensive. The locality and boundedness of the impulse-response recursion are checked in Lean (`CurrentMM.thesis_peak_le_first`, `CurrentMM.thesis_peak_locality`, `CurrentMM.thesis_peak_monotone`).

4.6 Horizon-limited amplification and natural experiments

The summed response S is expensive because of its tail: it needs a memory bound and costs a logarithm (Theorem 4.4.3). A supervisor who cares about losses over a fixed horizon need not pay for the tail. And a market supplies its own reset experiments whenever an outside event moves fundamental value faster than the price can follow. Both observations were suggested by Mansi Bhavsar in comments on an earlier draft; this section makes them precise.

Write $h_1, h_2, \dots$ for the impulse response of the linear economy, as in Section 4.5, and

$$S_H = \sum_{s=1}^{H} h_s, \qquad \|h\|_{1,H} = \sum_{s=1}^{H} |h_s|.$$

Proposition 4.6.1 (Horizon-limited amplification)**.** *Let the linear economy start at rest, $x_0 = 0$ and $y_{i,0} = 0$.*

(i) For shocks with $|w_t| \le \bar{w}$, $\max_{1\le t\le H} |x_t| \le \bar{w}\,\|h\|_{1,H}$, and some admissible shock sequence attains it. If $a \ge 0$ and every $k_m \ge 0$, then $\|h\|_{1,H} = S_H$.

(ii) Suppose each of N independent experiments returns $h_1, \dots, h_T$, $T \ge H$, with independent $N(0, \sigma^2)$ errors per tap (contract (O1) of Chapter 3). Then $\hat{S}_H \pm z_{1-\alpha/2}\,\sigma\sqrt{H/N}$, with $\hat{S}_H$ the average over experiments of the summed first H taps, covers S_H with probability $1-\alpha$ for every economy, with no memory bound; half-width ϵ needs $N = \lceil z^2_{1-\alpha/2}\sigma^2 H\epsilon^{-2}\rceil$ experiments.

(iii) On the stable class of Corollary 4.5.2, $S_H \uparrow S$ as $H \to \infty$ uniformly over $\mathcal{M}(\bar{r}, B)$. Without a memory bound, $\sup(S - S_H) = \infty$ over the stable economies for every H.

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Proof. (i) A shock w_τ moves $x_{\tau+s-1}$ by $h_s w_\tau$, so $x_t = \sum_{\tau=1}^{t} h_{t-\tau+1} w_\tau$ and $|x_t| \le \bar{w}\sum_{s\le t}|h_s| \le \bar{w}\|h\|_{1,H}$ for $t \le H$. With $w_{H-s+1} = \bar{w}\,\mathrm{sign}(h_s)$ for $s \le H$, $x_H = \bar{w}\|h\|_{1,H}$. If $a \ge 0$ and $k_m \ge 0$, the recursion $h_{t+1} = a h_t + \kappa \sum_{m<t} k_m h_{t-m}$ with $h_1 = \kappa > 0$ keeps every $h_t \ge 0$. (ii) The summed first H taps of one experiment are Gaussian with mean S_H and variance $H\sigma^2$, independently across experiments, so their average has standard deviation $\sigma\sqrt{H/N}$. (iii) Each h_t is a polynomial in $k_0, \dots, k_{t-2}$, and each $k_m = \int r^m\, dM$ and $\Phi = \int (1-r)^{-1} dM$ is weak-* continuous

on $\mathcal{M}(\bar{r}, B)$, because r^m and $(1-r)^{-1}$ are continuous on $[0, \bar{r}]$; so S_H and $S = \kappa/(1-a-\kappa\Phi)$ are continuous on the stable class, which is weak-* compact. There $h_t \geq 0$ (Corollary 4.5.2), so S_H increases to S pointwise, and Dini's theorem makes the convergence uniform. Without a memory bound, S_H depends on M only through $k_0, \ldots, k_{H-2}$, and Theorem 3.5.3(iii) gives stable economies with these moments and arbitrarily large S. □

Part (ii) is the parametric rate with no logarithm and no memory bound, because the horizon-limited question depends on finitely many observed taps (compare Proposition 4.5.1(b)). What a memory bound buys is part (iii), the passage from a horizon to forever. A supervisor whose capital is at risk over a known horizon, a margin period or a trading day, should certify S_H for that horizon and leave S to the theory of Section 4.4.

Proposition 4.6.2 (Natural experiments)**.** *In the linear economy with $a + \kappa\Phi < 1$, suppose an outside event at time 0 moves the fundamental value so that the price is displaced by $x_0 = d \neq 0$, and adds no order flow, $w_t = 0$ for $t \geq 1$. Then:*

(i) $x_t = d\, h_{t+1}/\kappa$ for all $t \geq 0$: the displacement follows the economy's own impulse response, started one step early;

(ii) $\sum_{t\geq 0} x_t/d = S/\kappa = 1/M$, where $M = 1-a-\kappa\Phi$ is the stability margin, and $\sum_{t<H} x_t/d = S_H/\kappa$;

(iii) if the fundamental value moves by βe and the price by a share $p_0 \in [0,1)$ of that move within period 0, so that $d = -(1-p_0)\beta e$, then the share of the long-run move still missing after t periods, $G_t = 1 - b_t/\beta$ with b_t the cumulative price move per unit of e, satisfies $\sum_{t\geq 0} G_t = (1-p_0)/M$;

(iv) for every horizon $H \geq 1$, with $R_n = \sum_{m\geq n} k_m$ the tail of the response kernel,

$$x_0 - x_H = M \sum_{t<H} x_t + \kappa \sum_{t<H} x_t\, R_{H-t},$$

so the finite-horizon margin $M_H = (x_0 - x_H)/\sum_{t<H} x_t$ exceeds M by a memory term; in the positive class with memory bound $\bar{r}$ and $d > 0$, $0 \leq M_H - M \leq \kappa\Phi \sum_{t<H} x_t \bar{r}^{\,H-t} / \sum_{t<H} x_t$, which tends to zero as $H \to \infty$.

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Proof. (i) With $u_t = \sum_{m\geq 0} k_m x_{t-m}$ and $x_t = 0$ for $t < 0$, $x_{t+1} = a x_t + \kappa \sum_{m=0}^{t} k_m x_{t-m}$. Set $g_t = x_t/d$, so $g_0 = 1$ and $g_{t+1} = a g_t + \kappa \sum_{m=0}^{t} k_m g_{t-m}$. The sequence $\hat{g}_t = h_{t+1}/\kappa$ has $\hat{g}_0 = 1$ and, from $h_{t+2} = a h_{t+1} + \kappa \sum_{m=0}^{t} k_m h_{t+1-m}$, the same recursion; the two sequences coincide. (ii) $\sum_t h_t = S = \kappa/M$ by equation (4.2). (iii) $x_t = (b_t - \beta)e$ for $t \geq 0$, so $G_t = (\beta - b_t)/\beta = (1-p_0)\, x_t/x_0$, and part (ii) gives the sum. (iv) Summing $x_{t+1} - x_t = (a-1)x_t + \kappa \sum_{m=0}^{t} k_m x_{t-m}$

over $t < H$ and exchanging the double sum, $\sum_{t<H}\sum_{m\le t} k_m x_{t-m} = \sum_{s<H} x_s \sum_{m<H-s} k_m = \sum_{s<H} x_s(\Phi - R_{H-s})$, which gives the identity. In the positive class $x_t \ge 0$ and $R_n = \int r^n(1-r)^{-1}dM \le \bar{r}^{\,n}\Phi$, which gives the bound; it tends to zero because $\sum_t x_t < \infty$ and $\bar{r}^{\,H-t} \to 0$ for each fixed t (dominated convergence). □

The proposition turns an outside shock into a reset experiment. A small market whose fundamental value moves with a larger one (an altcoin with bitcoin, a stock with its index future) is displaced every time the larger market jumps; its catch-up path is h, its catch-up area over a horizon is S_H/κ, and the area over all horizons is the reciprocal of the stability margin. Over a finite horizon only part (iv) is available, and it overstates the margin by a memory term: certifying the margin itself again needs a memory bound, while the horizon-limited area does not. None of these requires separating reversion from feedback or one cohort from another: they are functionals of the response, which is what the theory says is cheap. Splitting the catch-up path into exponential modes is identification, which Chapter 2 says is not. Two things can break the correspondence, and both can be checked in data: an event that also moves the small market's order flow directly ($w_t \neq 0$), and a small market that leads the large one. A catch-up path that changes sign, an overshoot, is impossible when all agents follow the price and points to contrarian agents, which the cohort model of Definition 4.1.1 excludes; Proposition 4.6.3 makes this precise. Section 6.1.6 reports how far public data are from this contract. Part (i) and the finite-horizon identity of part (iv) of Proposition 4.6.2, and part (i) of Proposition 4.6.1, are checked in Lean (`CurrentMM.thesis_natural_displacement`, `CurrentMM.thesis_natural_finite_margin`, `CurrentMM.thesis_horizon_worst_case`).

Proposition 4.6.3 (An overshoot certifies contrarian feedback)**.** *Allow signed weights $q_i \in \mathbb{R}$ in Definition 4.1.1 (a cohort with $q_i < 0$ is contrarian), so that $k_m = \sum_i q_i(1-r_i)r_i^m$ may be negative, and keep $a \in [0,1)$ and $\kappa > 0$. After an outside event as in Proposition 4.6.2, with $x_0 = d \neq 0$ and $w_t = 0$ for $t \ge 1$:*

(i) if $k_m \ge 0$ for every m, in particular if every $q_i \ge 0$, then $d\,x_t \ge 0$ for every t: the displacement never changes sign. The missing share G_t of Proposition 4.6.2(iii) then satisfies $G_t G_0 \ge 0$ for every t, and every horizon-limited area $A_H = \sum_{t<H} G_t$ has the sign of G_0;

(ii) consequently, if $G_s G_0 < 0$ for some s (an overshoot), then $k_m < 0$ for some m: some cohort is contrarian. The conclusion uses only the sign of the observed path and identifies no cohort, memory rate or weight;

(iii) conversely, for every $a \in [0,1)$ and $\kappa > 0$, one contrarian cohort with memory rate $r = 0$ and weight $q \in \big(-(1+a)/\kappa,\, -a/\kappa\big)$ gives a stable economy whose path $x_t = d\,(a+\kappa q)^t$ alternates in sign, with margin $M = 1 - a - \kappa q > 1 - a$;

(iv) the identities of Proposition 4.6.2(i) and (iv) hold for signed kernels. If moreover $\sum_m |k_m| < \infty$ *and the economy is stable, in the sense that* $f(z) = 1 - az - \kappa z \sum_m k_m z^m$ *has no zero in the closed unit disc, then* $M = f(1) > 0$ *and* $\sum_{t\geq 0} x_t/d = 1/M > 0$*: the total catch-up area has the sign of* d*, so an overshoot must be paid back later.*

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Proof. (i) $d\,x_0 = d^2 \geq 0$, and if $d\,x_s \geq 0$ for all $s \leq t$, then $d\,x_{t+1} = a\,(d\,x_t) + \kappa \sum_{m=0}^{t} k_m\,(d\,x_{t-m}) \geq 0$. In the proof of Proposition 4.6.2(iii), $x_t = (b_t - \beta)e$, so $G_t = -x_t/(\beta e)$ and $G_t G_0 = d\,x_t/(\beta e)^2 \geq 0$. (ii) is the contrapositive of (i). (iii) With $r = 0$ the kernel is $k_0 = q$ and $k_m = 0$ for $m \geq 1$, so $x_{t+1} = (a + \kappa q)x_t$ with $a + \kappa q \in (-1, 0)$; $f(z) = 1 - (a + \kappa q)z$ vanishes only at $z = 1/(a + \kappa q)$, outside the unit disc; and $M = 1 - a - \kappa q > 1 - a$ because $q < 0$. (iv) The proofs of Proposition 4.6.2(i) and (iv) use no sign condition. Multiplying the recursion by z^{t+1} and summing, $X(z) = \sum_t x_t z^t$ satisfies $f(z)X(z) = d$. The power series f converges absolutely on the closed disc and has no zero there, so by Wiener's lemma (absolutely convergent power series without zeros on the closed disc are invertible among such series) $1/f$ converges absolutely on the closed disc; hence $\sum_t |x_t| < \infty$ and $\sum_t x_t = X(1) = d/f(1)$. On $[0, 1]$ the function f is real and continuous, $f(0) = 1$, and it has no zero, so $f(1) > 0$; and $f(1) = 1 - a - \kappa\Phi = M$. □

The proposition gives a property of the agents that public data can certify without identifying anyone: whether some of them trade against the price. A pooled overshoot after outside shocks certifies contrarian feedback; it says nothing about which agents, how many, or how strong. Part (iv) adds a check on the model itself. In a stable linear economy an overshoot can make the catch-up area of the wrong sign over a short horizon, but not over all horizons; if the area stays of the wrong sign up to the end of an observation window, either the level the price settles at within the window is not its long-run level, so that adjustment continues beyond the window, or the linear model fails. Test O1 (Section 6.1) asks whether the pooled overshoot exists after scheduled US releases; it was registered before its data were downloaded, and this proposition was added before its result was known. O1 found no overshoot (Section 6.1.4): on the 2022–2024 releases the catch-up was gradual and of one sign, as in the positive class, so these data give no evidence of contrarian feedback. Parts (i)–(iii) are checked in Lean (`CurrentMM.thesis_overshoot_positive_class`, `CurrentMM.thesis_overshoot_certifies_contrarian`, `CurrentMM.thesis_overshoot_realizable`, `CurrentMM.thesis_overshoot_alternates`).

4.7 Passive data: certifying safety costs the inverse margin

The theorems above use reset experiments. Public data are passive: the observer sees the market run, with no experiment and no reset. Chapter 2 shows that passive data identify the agents

at the same polynomial cost as active probing (Theorem 2.6.10). The next theorem gives the cost of the safety question under the same contract. Certifying safety from passive data is the classical problem of rejecting a unit root, and its cost is set by the margin, not by the agents.

Theorem 4.7.1 (Passive certification of safety)**.** *Observe the displacement passively, $x_{t+1} = a\,x_t + \kappa \sum_{m\geq 0} k_m x_{t-m} + \kappa w_{t+1}$ with independent $N(0, s^2)$ shocks, for $t = 0, \dots, T-1$ from the common starting value $x_0 = 0$. Let $\mu \in (0, 1]$.*

(i) Lower bound. *Take two economies with one cohort of memory rate zero, the same $a \in [0, 1-\mu]$ and $\kappa > 0$, and weights $q, q' \geq 0$ with $a + \kappa q = 1 - \mu$ (margin $M = \mu$, safe) and $a + \kappa q' = 1 + \mu$ (margin $-\mu$, unsafe). The Kullback–Leibler divergence between their distributions of $(x_1, \dots, x_T)$ is at most $2T\mu$. Any procedure that decides the sign of the margin with error at most $\alpha < 1/2$ under both economies therefore needs $T \geq (1-2\alpha)^2/\mu$.*

(ii) Upper bound. *In the economies with one cohort of memory rate zero, $x_{t+1} = \rho x_t + \kappa w_{t+1}$ with $\rho = 1 - M$, and the least-squares estimate $\hat\rho$ gives $\hat M = 1 - \hat\rho$. For fixed $\rho \in (-1, 1)$, $\sqrt{T}(\hat M - M) \to N(0, M(2-M))$; along sequences $M_T \to 0$ with $TM_T \to \infty$, $(T/2M_T)^{1/2}(\hat M - M_T) \to N(0, 1)$. The interval $\hat M \pm z_{1-\alpha/2}(2\hat M/T)^{1/2}$ therefore decides the sign with asymptotic error at most α once $TM > 2z^2_{1-\alpha/2}$: the cost is of order $1/M$.*

(iii) The passive dichotomy. *Under the same passive contract, separating K cohorts whose memory rates are $\Delta\eta$ apart needs $T = \Omega((\Delta\eta)^{-(4K-2)})$ whatever the margin (Theorem 2.6.10), while deciding the safety of a market at margin μ needs T of order μ^{-1} in the economies of parts (i)–(ii), and no fewer in any class that contains them. For a market at a fixed margin, the ratio of the two costs grows without bound as the cohorts become similar.*

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Proof. (i) Both economies are $x_{t+1} = \rho_j x_t + e_{t+1}$ with $e_{t+1} = \kappa w_{t+1} \sim N(0, \kappa^2 s^2)$, $\rho_0 = 1 - \mu$ and $\rho_1 = 1 + \mu$. The Gaussian likelihood factorizes over t, so the divergence is $\sum_{t<T} \mathbb{E}_0[(\rho_0 - \rho_1)^2 x_t^2]/(2\kappa^2 s^2) = \frac{1}{2}(2\mu)^2 \sum_{t<T} v_t$, where $v_t = \mathbb{E}_0 x_t^2/(\kappa^2 s^2)$ satisfies $v_0 = 0$ and $v_{t+1} = \rho_0^2 v_t + 1$. Then $v_t \leq 1/(1-\rho_0^2) \leq 1/\mu$, because $1 - (1-\mu)^2 = \mu(2-\mu) \geq \mu$, so the divergence is at most $2T\mu$. By Pinsker's inequality the total variation distance between the two distributions is at most $(T\mu)^{1/2}$. A procedure whose error is at most α under both economies separates them with total variation at least $1 - 2\alpha$, so $(T\mu)^{1/2} \geq 1 - 2\alpha$. (ii) For fixed $|\rho| < 1$ the least-squares estimate satisfies $\sqrt{T}(\hat\rho - \rho) \to N(0, 1-\rho^2)$ (Mann and Wald, 1943), and $1 - \rho^2 = M(2-M)$. Along $\rho_T = 1 - M_T$ with $M_T \to 0$ and $TM_T \to \infty$, the moderate-deviation limit theory of Phillips and Magdalinos (2007) gives $(T/2M_T)^{1/2}(\hat\rho - \rho_T) \to N(0, 1)$; since $\hat M/M_T \to 1$ in probability, the studentized interval has asymptotic coverage $1 - \alpha$, and it excludes zero once $M > z_{1-\alpha/2}(2M/T)^{1/2}$. (iii) combines (i), (ii) and Theorem 2.6.10. □

Part (i) is the reason unit-root tests have power only when $T(1-\rho)$ is of order one or larger (Elliott, Rothenberg and Stock, 1996): a market close to the boundary cannot be certified safe from a short passive record, whoever its agents are. Part (iii) is the passive form of the certification–identification dichotomy. The cost of the safety question depends on how close the market is to the boundary; the cost of identification depends on how similar the agents are; and the two are unrelated. It also gives a prediction for data. For a market at margin M the standardized margin grows like $(TM/2)^{1/2}$, so the share of markets whose safety is decided should rise with the length of the record, while the share whose memories are identified should barely move. In public data the second half of this prediction holds and the first holds across markets but not market by market, because the margin itself drifts over months (Section 6.1.6); the theorem's stationarity is the assumption that data strain. The variance recursion and the divergence bound of part (i) are checked in Lean (`CurrentMM.thesis_passive_variance_le_inv`, `CurrentMM.thesis_passive_kl_sum`, `CurrentMM.thesis_passive_kl_margin`).

Part (ii) concerns economies whose closed loop has a single root. The next theorem extends both bounds to economies with any number of cohorts, provided their memory is finite: the kernel vanishes beyond L lags, as it does in the estimators of Tests C2, C3 and F1, and its signs are unrestricted, so contrarian cohorts (Section 4.9) are included.

Theorem 4.7.2 (Passive certification with several cohorts)**.** *Let $k_m = 0$ for $m \geq L$, with $L \geq 2$, and write the passive economy of Theorem 4.7.1 as $x_{t+1} = \sum_{i=0}^{L-1} \phi_i x_{t-i} + e_{t+1}$, where $\phi_0 = a + \kappa k_0$, $\phi_i = \kappa k_i$ for $i \geq 1$, the $e_t = \kappa w_t$ are independent $N(0, \sigma^2)$ with $\sigma = \kappa s$, and $x_t = 0$ for $t \leq 0$. The closed-loop polynomial is $f(z) = 1 - \sum_{i<L} \phi_i z^{i+1}$, and the margin is $M = 1 - a - \kappa\Phi = f(1)$. Suppose $f(z) = (1-\rho z)g(z)$, where g has real coefficients, $g(0) = 1$, degree $p = L-1$, and no zeros in $|z| \leq 1+\gamma$ for some $\gamma > 0$; then $g(1) > 0$ and $M = (1-\rho)g(1)$.*

(i) Lower bound. *For $0 < \mu \leq g(1)$, the economies $f_\pm(z) = (1 - \rho_\pm z)g(z)$ with $\rho_\pm = 1 \mp \mu/g(1)$ have margins $\pm\mu$, and the Kullback–Leibler divergence between their distributions of $(x_1, \ldots, x_T)$ is at most $2T\mu/g(1)$. Any procedure that decides the sign of the margin with error at most $\alpha < 1/2$ under both needs $T \geq (1-2\alpha)^2 g(1)/\mu$.*

(ii) Upper bound. *Let $\rho \in [0, 1)$, so that the economy is stationary and safe. In the exact augmented regression*

$$x_{t+1} - x_t = -M\, x_t + \textstyle\sum_{i=1}^{p} c_i (x_{t+1-i} - x_{t-i}) + e_{t+1},$$

the least-squares estimate $\hat{M}$ (minus the coefficient on x_t) satisfies $\sqrt{T}(\hat{M} - M) \to N(0, V)$ with

$$V \leq C_{p,\gamma}\, M, \qquad C_{p,\gamma} = 4p^2(p+1)(1+\pi^2)\, 4^p \Big(\frac{1+\gamma}{\gamma}\Big)^p.$$

The studentized interval $\hat{M} \pm z_{1-\alpha/2}(\hat{V}/T)^{1/2}$ therefore has asymptotic coverage $1-\alpha$ and half-width of order $(C_{p,\gamma} M/T)^{1/2}$, which is below M once $T > z_{1-\alpha/2}^2 C_{p,\gamma}/M$: the cost is

again of order $1/M$, with a constant set by the memory length and the root gap and not by how similar the cohorts are.

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Proof. (i) g is real on $[0,1]$, equals 1 at 0 and has no zero there, so $g(1) > 0$, and $f_\pm(1) = (1-\rho_\pm)g(1) = \pm\mu$. Both economies are finite-memory economies: with $\kappa > 0$ fixed, any real ϕ is obtained from $k_i = \phi_i/\kappa$ for $i \geq 1$ and $a = \phi_0 - \kappa k_0$. The Gaussian likelihood factorizes, and the divergence is $\sum_{t<T} \mathbb{E}_+[((f_+ - f_-)(\mathcal{L})x_{t+1})^2]/(2\sigma^2)$, where $\mathcal{L}$ is the lag operator. Since $(f_+ - f_-)(z) = (\rho_- - \rho_+)z\,g(z)$, this is $(2\mu/g(1))^2 \sum_{t<T} \mathbb{E}_+ y_t^2/(2\sigma^2)$ with $y_t = g(\mathcal{L})x_t$. Under f_+, $(1-\rho_+\mathcal{L})y_{t+1} = e_{t+1}$ with $y_0 = 0$, so $\mathbb{E}_+ y_t^2 = \sigma^2 v_t$ with $v_0 = 0$, $v_{t+1} = \rho_+^2 v_t + 1$, and $v_t \leq 1/(1-\rho_+^2)$. With $1-\rho_+^2 \geq 1-\rho_+ = \mu/g(1)$, the divergence is at most $(2\mu/g(1))^2 T/(2\mu/g(1)) = 2T\mu/g(1)$. Pinsker's inequality finishes as in Theorem 4.7.1(i).

(ii) The regression is an invertible linear reparametrization of the autoregression of x_{t+1} on $x_t, \ldots, x_{t-p}$, whose roots, $1/\rho$ and those of g, lie outside the unit circle. By Mann and Wald (1943), $\sqrt{T}$ times the error of the least-squares coefficients is asymptotically normal with covariance $\sigma^2\Gamma^{-1}$, where Γ is the covariance matrix of the regressors $(x_t, \Delta x_t, \ldots, \Delta x_{t-p+1})$, and the entry of Γ^{-1} for x_t is $1/\tau^2$, with τ^2 the residual variance of x_t after projection on the lagged differences. Any such residual is $h(\mathcal{L})x_t$ with $h(z) = 1 - \sum_{i=1}^{p} c_i(z^{i-1} - z^i)$, a polynomial of degree at most p with $h(1) = 1$. With $\delta = 1-\rho \in (0,1]$ and the spectral density $s_x(\lambda) = \sigma^2/(2\pi|f(e^{-i\lambda})|^2)$,

$$\tau^2 = \min_h \int_{-\pi}^{\pi} |h(e^{-i\lambda})|^2 s_x(\lambda)\,d\lambda \;\geq\; \frac{\sigma^2}{2\pi G^2} \min_h \int_{-\pi}^{\pi} \frac{|h(e^{-i\lambda})|^2}{\delta^2+\lambda^2}\,d\lambda,$$

where $G = \max_{|z|=1}|g(z)|$, because $|1-\rho e^{-i\lambda}|^2 = \delta^2 + 2\rho(1-\cos\lambda) \leq \delta^2+\lambda^2$. Write $H(\lambda) = h(e^{-i\lambda})$ and $I = \int |H|^2/(\delta^2+\lambda^2)$. Since $\delta^2+\lambda^2 \leq 1+\pi^2$, $\int|H|^2 \leq (1+\pi^2)I$; a polynomial of degree p satisfies $\max|H|^2 \leq (p+1)\int|H|^2/(2\pi)$ by Cauchy–Schwarz, and $\max|H'| \leq p\max|H|$ by Bernstein's inequality, so $\max|H'| \leq pB\sqrt{I}$ with $B^2 = (p+1)(1+\pi^2)/(2\pi)$. Let $r = \min(\pi, 1/(2pB\sqrt{I}))$. On $|\lambda| \leq r$, $|H(\lambda)| \geq 1 - |\lambda|pB\sqrt{I} \geq 1/2$, so $I \geq \frac{1}{4}\int_{-r}^{r} d\lambda/(\delta^2+\lambda^2) = \arctan(r/\delta)/(2\delta)$. If $r \geq \delta$ this is at least $\pi/(8\delta)$; if $r < \delta$, then $r = 1/(2pB\sqrt{I})$ because $\delta \leq 1 < \pi$, and $I > 1/(4p^2B^2\delta^2)$. In both cases $I \geq c/\delta$ with $c = \min(\pi/8, 1/(4p^2B^2)) = 1/(4p^2B^2)$, using $\delta \leq 1$. Hence $\tau^2 \geq \sigma^2 c/(2\pi G^2\delta)$, and with $\delta = M/g(1)$,

$$V = \sigma^2/\tau^2 \leq 2\pi G^2 M/(c\,g(1)) = 4p^2(p+1)(1+\pi^2)\,G^2M/g(1).$$

Finally, $g(z) = \prod_{j\leq p}(1 - z/r_j)$ over its roots, $|r_j| \geq 1+\gamma$, so $G \leq (1+1/(1+\gamma))^p \leq 2^p$ and $g(1) = |g(1)| \geq \prod_j(1-1/|r_j|) \geq (\gamma/(1+\gamma))^p$. The studentized interval is consistent because $\hat{V} \to V$ in probability. □

Remark 4.7.3 (What remains open). Theorem 4.7.2 covers finite memory, which is the model the estimators of the tests fit, and its upper bound is a pointwise limit for each fixed safe economy.

Section 4.8 takes up the two steps beyond it that concern the theory: geometric memory, where a memory bound turns out to be necessary (Theorem 4.8.1), and bounds uniform as $M \to 0$ (Theorem 4.8.2). The third step, the estimator of Tests C2, C3, F1 and S1, which pools a price equation and a flow equation rather than regressing the displacement alone, is covered by Theorem 4.8.3 for fixed economies. The constant $C_{p,\gamma}$ is crude. For two economies with $p = 2$ (roots of g at 2 and -3, and at $1.5 \pm i$) and $\rho \in \{0.9, 0.97, 0.99\}$, the exact V computed from the spectral density, and the value estimated by simulation with $T = 2 \times 10^5$, satisfy $V/M \approx 2g(1)$ within 15%, against $C_{p,\gamma}$ in the thousands. With $V \approx 2g(1)M$ the upper bound needs T of order $2z^2_{1-\alpha/2}g(1)/M$, and the lower bound $(1-2\alpha)^2 g(1)/M$: the two match in their dependence on $g(1)$ and M. The divergence bound, the factor bounds and the case analysis of part (ii) are checked in Lean (`CurrentMM.thesis_pm_kl`, `CurrentMM.thesis_pm_prod_lower`, `CurrentMM.thesis_pm_christoffel_cases`, `CurrentMM.thesis_pm_variance`).

Public data strain the theorem's stationarity: estimated margins drift from quarter to quarter while their sign stays put (Section 6.1.6). The next proposition replaces stationarity by a bound on how fast the margin can move, in the reduced form in which each day contributes an unbiased, noisy reading of that day's margin.

Proposition 4.7.4 (Certifying a drifting margin)**.** *On day t the observer has $m_t = M_t + e_t$, with e_t independent $N(0, s^2)$, and the margin moves by at most $d > 0$ per day, $|M_{t+1} - M_t| \le d$. Let $z = z_{1-\alpha/2}$.*

(i) Upper bound. *The average $\bar m_W$ of the last W readings before day T has bias at most $d(W+1)/2$ for M_T and standard deviation $s/\sqrt{W}$, so $\bar m_W \pm (d(W+1)/2 + zs/\sqrt{W})$ covers M_T with probability at least $1-\alpha$, and excludes zero whenever $|M_T| > d(W+1) + 2zs/\sqrt{W}$. Since $dW + 2zs/\sqrt{W} \ge 3(zs)^{2/3}d^{1/3}$, with equality at $W^* = (zs/d)^{2/3}$, the window W^* certifies the sign of every margin with $|M_T| > 3(zs)^{2/3}d^{1/3} + d$.*

(ii) Lower bound. *Let $\mu \ge d$. The constant path $M_t \equiv \mu$ and the path that equals μ until day $T-k$ and then falls linearly to $-\mu$ at day T, with $k = \lceil 2\mu/d \rceil$, both move by at most d per day; the Kullback–Leibler divergence between their readings is at most $6\mu^3/(ds^2)$. Hence if $\mu < (ds^2/12)^{1/3}$, no procedure, using any number of past readings, decides the sign of M_T with error at most $\alpha < 1/4$ under both paths.*

(iii) The drift floor. *Together, the smallest margin whose sign passive data can certify is of order $(s^2 d)^{1/3}$, between $(1/12)^{1/3}$ and $3z^{2/3}$ times $(s^2 d)^{1/3}$ when $d \le (s^2 d)^{1/3}$; the best record length is of order $(s/d)^{2/3}$ days. As $d \to 0$ the floor vanishes and the record length grows without bound, which is the stationary regime of Theorem 4.7.1.*

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Proof. (i) $|M_{T-1-j} - M_T| \le d(j+1)$, so the bias of the average over $j < W$ is at most $d\sum_{j<W}(j+1)/W = d(W+1)/2$; the noise is $N(0, s^2/W)$. The interval excludes zero whenever

$|M_T|$ exceeds twice its half-width. With $u = \sqrt{W}$, $a = du^2$ and $b = c = zs/u$, the inequality $27abc \le (a+b+c)^3$ for non-negative numbers gives $(dW + 2zs/\sqrt{W})^3 \ge 27d(zs)^2$, with equality when $a = b = c$, that is $W = (zs/d)^{2/3}$. (ii) The two paths agree before day $T-k$ and differ by $2\mu j/k$ on day $T-k+j$, $j = 1, \dots, k$; their slope $2\mu/k \le d$. The divergence is $\sum_j (2\mu j/k)^2/(2s^2) \le (2\mu/k)^2 k^3/(2s^2) = 2\mu^2 k/s^2$, and $k \le 2\mu/d + 1 \le 3\mu/d$, so it is at most $6\mu^3/(ds^2)$. By Pinsker's inequality the total variation distance is at most $(3\mu^3/(ds^2))^{1/2}$, below $1/2$ when $\mu^3 < ds^2/12$, while a procedure with error at most $\alpha < 1/4$ under both paths separates them with total variation at least $1 - 2\alpha > 1/2$. (iii) combines (i) and (ii). □

The proposition turns the drift found in data from a defect into a result: a slowly drifting margin can still be certified, from a record of the right length, unless it is closer to the boundary than a floor set by the noise and the drift; no amount of history removes that floor. Its constants come from data: with the per-day noise and the quarter-to-quarter drift of the sixth pilot, the best record length is about one quarter (Section 6.1.6). The reduced form treats each day's reading as an independent unbiased estimate; the estimator of Tests C2, C3 and F1 pools the days' sufficient statistics instead, which has the same first-order behaviour. The window bound, the bias sum and the divergence bound are checked in Lean (`CurrentMM.thesis_drift_bias`, `CurrentMM.thesis_drift_window`, `CurrentMM.thesis_drift_squares`).

Corollary 4.7.5 (What the closed loop adds)**.** *In the positive class, let an interval certify the feedback weight Φ to half-width ϵ_Φ, and let $M = 1 - a - \kappa\Phi > 0$. For $\epsilon \in (0, 1/2]$: if $\epsilon_\Phi \le \epsilon M/(2\kappa)$, the image of the interval under $\Phi \mapsto S = \kappa/(1 - a - \kappa\Phi)$ certifies S to relative accuracy ϵ; conversely, any interval that certifies S to relative accuracy ϵ yields one that certifies Φ to half-width $2\epsilon M/\kappa$. Certifying the amplification S to relative accuracy ϵ therefore costs as much as certifying Φ to accuracy of order $\epsilon M/\kappa$, that is $\sigma^2\kappa^2 M^{-2}\epsilon^{-2}\log(\kappa/(\epsilon M))$ experiments up to constants (Theorem 4.4.3), while the cost of identifying the agents does not depend on M at all.* PROVED IN THIS THESIS

Proof. S is increasing in Φ on $\Phi < (1-a)/\kappa$, so intervals map to intervals. For Φ' with $|\Phi' - \Phi| \le \epsilon M/(2\kappa)$, $M' = 1 - a - \kappa\Phi' \ge M(1 - \epsilon/2)$ and $(S' - S)/S = \kappa(\Phi' - \Phi)/M'$, so $|S' - S|/S \le \kappa\,\epsilon M/(2\kappa)\,/\,(M(1-\epsilon/2)) \le \epsilon$. Conversely, $|S' - S| \le \epsilon S$ gives $\kappa|\Phi' - \Phi| \le \epsilon M' \le \epsilon(M + \kappa|\Phi' - \Phi|)$, hence $\kappa|\Phi' - \Phi| \le \epsilon M/(1-\epsilon) \le 2\epsilon M$. The cost statement substitutes accuracy $\epsilon M/\kappa$ into the rate of Theorem 4.4.3; the identification bounds of Chapter 2 depend on the spacing of memory rates and not on a, κ or Φ beyond constants. □

The corollary is the answer to the objection that the closed loop is a reparametrization of a known moment problem. For identification it is: the map from the merged measure to the observed response is smooth with constants that do not depend on the spacing of the agents, and the barrier is that of the moment problem. For safety it is not: the loop turns the linear question about Φ into a question about S whose cost grows as the market nears the boundary, like M^{-2} for relative accuracy from reset experiments and like M^{-1} for the sign of the margin

from passive data (Theorem 4.7.1). The economics of the problem, how close the market is to instability, enters the cost of certification and only of certification.

4.8 Uniform certification and the price of memory

This section completes the passive theory in three directions. It shows that without a bound on the agents' memory no procedure can certify safety from passive data, however safe the market; that with finite memory the cost $g(1)/M$ holds uniformly as the margin vanishes; and that the quantity which measures worst-case loss when agents are contrarian, the horizon-limited absolute response, is certified from passive data at the parametric rate.

Theorem 4.8.1 (Hidden slow feedback)**.** *Observe the passive economy of Theorem 4.7.1 for $t < T$ from $x_0 = 0$. Let $\mathcal{E}_0$ have one cohort of memory rate zero with $a + \kappa q_0 = 1 - c$, $c \in (0, 1]$, so that its margin is c; and for $r \in [0, 1)$ and $\mu > 0$ let $\mathcal{E}_r$ add to it a momentum cohort with memory rate r and weight $q_1 = (c + \mu)/\kappa$, so that its margin is $-\mu$.*

(i) The Kullback–Leibler divergence between the distributions of $(x_1, \dots, x_T)$ under $\mathcal{E}_0$ and $\mathcal{E}_r$ is at most $(c + \mu)^2(1 - r)^2 T^3/(2c)$.

(ii) No certificate without a memory bound. *For every T and every procedure, $\sup_{r<1} \Pr_{\mathcal{E}_r}(\text{certify safe}) \geq \Pr_{\mathcal{E}_0}(\text{certify safe})$. A procedure that certifies an unsafe market with probability at most α certifies the safe market $\mathcal{E}_0$, whose margin may be as large as one, with probability at most α: on the positive economies without a memory bound, passive certification has power at most its size.*

(iii) The price of a memory bound. *On economies with memory bound $\bar{r}$, deciding the sign of the margin with error at most $\alpha < 1/2$ under $\mathcal{E}_0$ and $\mathcal{E}_{\bar{r}}$ needs*

$$T \;\geq\; \Big(\frac{2(1 - 2\alpha)\sqrt{c}}{(c + \mu)(1 - \bar{r})}\Big)^{2/3}.$$

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Proof. (i) Under $\mathcal{E}_r$ the kernel is $k_m = q_0 \mathbf{1}\{m = 0\} + q_1(1 - r)r^m$, so the conditional means of x_{t+1} under the two economies differ by $D_t = \kappa q_1(1 - r) \sum_{m \leq t} r^m x_{t-m}$, and the divergence is $\sum_{t<T} \mathbb{E}_0 D_t^2/(2\kappa^2 s^2)$. By Cauchy–Schwarz with $r^{2m} \leq 1$, $(\sum_{m \leq t} r^m x_{t-m})^2 \leq (t+1) \sum_{m \leq t} x_{t-m}^2$. Under $\mathcal{E}_0$, $x_{t+1} = (1 - c)x_t + \kappa w_{t+1}$, so $\mathbb{E}_0 x_s^2 \leq \kappa^2 s^2/(1 - (1 - c)^2) \leq \kappa^2 s^2/c$. Hence $\mathbb{E}_0 D_t^2 \leq (c+\mu)^2(1-r)^2(t+1)^2\kappa^2 s^2/c$, and the divergence is at most $(c+\mu)^2(1-r)^2 \sum_{t<T}(t+1)^2/(2c) \leq (c + \mu)^2(1 - r)^2 T^3/(2c)$. (ii) For any event A, $\Pr_{\mathcal{E}_r}(A) \geq \Pr_{\mathcal{E}_0}(A) - \mathrm{TV}$, and by Pinsker's inequality $\mathrm{TV} \leq (\mathrm{KL}/2)^{1/2} \to 0$ as $r \uparrow 1$ for fixed T. (iii) A procedure with error at most α under both separates them with total variation at least $1 - 2\alpha$, so $2(1 - 2\alpha)^2 \leq \mathrm{KL} \leq (c + \mu)^2(1 - \bar{r})^2 T^3/(2c)$. □

The theorem is the market form of a classical result: unit-root tests have power no greater than size against all stationary processes (Blough, 1992; Faust, 1996; Pötscher, 2002). Its economic content is specific. What hides instability from a passive record is a slow momentum cohort: agents who trade on long averages of the price contribute little to each period's response and a large amount to the total feedback, so a market can be unstable without leaving any trace in a finite record. Simulation makes the point concrete: with $c = \mu = 0.2$ and $T = 1000$, an augmented Dickey–Fuller certificate at the 5% level certified the safe economy in 100% of 400 replications and the unsafe economy with memory rate 0.999 in 100%. Certification from passive data therefore needs a bound on memory, and the bound has a price of order $(1-\bar{r})^{-2/3}$ in record length. This is what disclosure must supply for safety: not the identity of the agents, but a bound on how slowly the largest of them can trade.

With a memory bound, and in particular with finite memory, certification is uniform as the margin vanishes. Write $\mathcal{C}(p, \gamma, B)$ for the finite-memory economies of Theorem 4.7.2 with $f(z) = (1-\rho z)g(z)$, $\rho \in [0,1]$, and g of degree p, without zeros in $|z| \le 1+\gamma$, with coefficients bounded by B. The certificate declares the market safe when the augmented Dickey–Fuller t-statistic for the coefficient on x_t in the regression of Theorem 4.7.2(ii) lies below the Dickey–Fuller critical value c_α (Dickey and Fuller, 1979).

Theorem 4.8.2 (Uniform certification with finite memory). *(i)* Size, uniformly. $\lim_{T\to\infty} \sup \Pr(\textit{certify safe})$ α, *the supremum over the economies of* $\mathcal{C}(p, \gamma, B)$ *on the boundary* $M = 0$ *(*$\rho = 1$*); for every fixed explosive economy (*$M < 0$*) the probability tends to zero.*

(ii) Power, uniformly. *For every* $\epsilon > 0$ *there is* $C_\epsilon < \infty$*, depending only on* α *and* ϵ*, such that* $\liminf_{T\to\infty} \inf \Pr(\textit{certify safe}) \ge 1-\epsilon$*, the infimum over the economies of* $\mathcal{C}(p, \gamma, B)$ *with* $TM/g(1) \ge C_\epsilon$*.*

(iii) Sharpness. *With Theorem 4.7.2(i), the record length needed to certify safety at margin* M *is of order* $g(1)/M$ *uniformly over* $\mathcal{C}(p, \gamma, B)$*: no procedure does better than* $(1-2\alpha)^2 g(1)/M$*, and the certificate succeeds once* $T \ge C_\epsilon g(1)/M$*.*

PROVED IN THIS THESIS *(given the uniform local-to-unity theory cited in the proof)*

Proof. Write $\rho = 1-\theta/T$, so that $\theta = T(1-\rho) = TM/g(1) \ge 0$. In the exact augmented regression the lagged differences are a stationary process whose spectral density is bounded above and below on $\mathcal{C}(p, \gamma, B)$, and the t-statistic converges in distribution to $\tau_\theta = \big(\int J_\theta \, dW\big) \big/ \big(\int J_\theta^2\big)^{1/2} - \theta\big(\int J_\theta^2\big)^{1/2}$, where J_θ is the Ornstein–Uhlenbeck process $dJ_\theta = -\theta J_\theta \, dt + dW$ on $[0,1]$ (Phillips, 1987; Chan and Wei, 1987). The convergence is uniform over $\theta \in [0, \infty)$ and over the stationary nuisance in a compact set (Mikusheva, 2007, 2012), and $\tau_\theta \to -\infty$ in probability as $\theta \to \infty$. (i) At $\theta = 0$, τ_0 is the Dickey–Fuller distribution, so the probability that the statistic falls below c_α tends to α uniformly. For fixed $\rho > 1$ the coefficient on x_t is positive and its t-statistic diverges to $+\infty$. (ii) Choose C_ϵ with $\Pr(\tau_\theta < c_\alpha) \ge 1-\epsilon/2$ for all $\theta \ge C_\epsilon$, which the divergence of

τ_θ allows; uniform convergence gives the bound for T large. (iii) combines (ii) with the lower bound of Theorem 4.7.2(i). □

In simulation the certification rate depends on the margin only through $\theta = TM/g(1)$, as the theorem says: for two economies with different g and for $T = 500$ and 2000, the 5% certificate succeeded in 4% to 7% of replications at $\theta = 0$, 68% to 78% at $\theta = 10$, and in at least 100% at $\theta = 40$. For cohorts with geometric memory and a memory bound $\bar{r} < 1$, the closed loop is an autoregressive moving average whose moving-average roots $1/r_i$ lie outside the unit circle, and the augmented regression with a number of lags that grows with T (more slowly than $T^{1/3}$) keeps the Dickey–Fuller limit for each fixed economy (Said and Dickey, 1984); Theorem 4.8.1 shows that the memory bound cannot be dropped, and a version uniform over the memory structure is open.

The theorems so far use the displacement alone. The estimator of Tests C2, C3, F1 and S1 also uses the order flow: it regresses the change in the displacement on the contemporaneous flow and the lagged displacement, which gives the impact κ and the reversion, and the flow on lags of the displacement, which gives the kernel. The next theorem shows that this pooled estimator certifies the margin at the same cost.

Theorem 4.8.3 (The pooled price-and-flow estimator)**.** *Observe the displacement x_t and the order flow f_t of a finite-memory economy with $L = p + 1$ lags,*

$$f_{t+1} = \sum_{m<L} k_m x_{t-m} + w_{t+1}, \qquad x_{t+1} = a\, x_t + \kappa f_{t+1} + \varepsilon_{t+1},$$

with $w_t \sim N(0, s_w^2)$ and $\varepsilon_t \sim N(0, s_\varepsilon^2)$ independent of each other and over time, $\kappa > 0$, and $x_t = 0$ for $t \le 0$: the price noise ε_{t+1} is independent of the period's flow. Then x is the finite-memory economy of Theorem 4.7.2 with innovation variance $\sigma^2 = \kappa^2 s_w^2 + s_\varepsilon^2$ and margin $M = 1 - a - \kappa\Phi = f(1)$; suppose f satisfies the conditions there. The pooled estimator is $\hat{M} = -\hat{\beta} - \hat{\kappa}\sum_m \hat{k}_m$, with $(\hat{\kappa}, \hat{\beta})$ from the least-squares regression of $x_{t+1} - x_t$ on f_{t+1} and x_t, and $\hat{k}$ from that of f_{t+1} on $x_t, \dots, x_{t-p}$.

(i) Lower bound with the flow observed. *For $\mu \in (0, 1]$, two economies with one cohort of memory rate zero, the same a and κ, and margins $\pm\mu$ have divergence at most $2T\mu(1+R)$ over T observations of (x_t, f_t), with $R = s_\varepsilon^2/(\kappa^2 s_w^2)$; deciding the sign with error at most $\alpha < 1/2$ under both needs $T \ge (1-2\alpha)^2/(\mu(1+R))$.*

(ii) Upper bound. *For $\rho \in [0, 1)$, $\sqrt{T}(\hat{M} - M) \to N(0, V_{\text{pool}})$ with $V_{\text{pool}} \le 4C_{p,\gamma}M$, where $C_{p,\gamma}$ is the constant of Theorem 4.7.2(ii).*

Observing the flow therefore changes the constant, not the order: the record length needed to decide the sign of the margin is of order $1/M$ for the estimator the tests use. PROVED IN THIS THESIS

Proof. (i) With one cohort of memory rate zero, $f_{t+1} = qx_t + w_{t+1}$ and $x_{t+1} = ax_t + \kappa f_{t+1} + \varepsilon_{t+1}$. The joint density of (f_{t+1}, x_{t+1}) given the past factorizes into the flow equation and the price equation given the flow; the second is the same under both economies, so the divergence is $\sum_{t<T} \mathbb{E}_+(q_+ - q_-)^2 x_t^2/(2s_w^2)$ with $q_+ - q_- = -2\mu/\kappa$. Under the safe economy $x_{t+1} = (1-\mu)x_t + \kappa w_{t+1} + \varepsilon_{t+1}$, so $\mathbb{E}_+ x_t^2 \le \sigma^2/\mu$, and the divergence is at most $(2\mu/\kappa)^2 T\sigma^2/(2s_w^2\mu) = 2T\mu(1+R)$. Pinsker's inequality finishes as before.

(ii) Write $\hat{M} - M = -(\hat{\beta} + \hat{\kappa}\Phi + M) - \hat{\kappa}(\hat{\Phi} - \Phi)$ with $\hat{\Phi} = \sum_m \hat{k}_m$. Both regressions are equations of a stationary Gaussian vector autoregression in (x_t, f_t), and their regressors are uncorrelated with their errors (w_{t+1} is independent of the past; ε_{t+1} is independent of f_{t+1} and the past), so the estimators are jointly asymptotically normal (Mann and Wald, 1943) and $\hat{\kappa} \to \kappa$.

The flow equation. Reparametrize its regressors as x_t and the lagged differences $\Delta x_t, \dots, \Delta x_{t-p+1}$; the coefficient on x_t is Φ, so the asymptotic variance of $\hat{\Phi}$ is s_w^2/τ^2, with τ^2 the residual variance of x_t after projection on the lagged differences. The proof of Theorem 4.7.2(ii) gives $\tau^2 \ge \sigma^2 c/(2\pi G^2 \delta)$ with $\delta = 1 - \rho$, so $\kappa^2 s_w^2/\tau^2 \le 2\pi G^2 \delta/c$, using $\kappa^2 s_w^2 \le \sigma^2$.

The price equation. With $\tilde{w}_{t+1} = f_{t+1} - \Phi x_t$, the price equation reads $x_{t+1} - x_t = \kappa \tilde{w}_{t+1} - Mx_t + \varepsilon_{t+1}$, a reparametrization of the regression on f_{t+1} and x_t in which the coefficient on x_t is $\beta + \kappa\Phi = -M$. So $\hat{\beta} + \hat{\kappa}\Phi + M$ has asymptotic variance $s_\varepsilon^2/\tau_\varepsilon^2$, with τ_ε^2 the residual variance of x_t after projection on $\tilde{w}_{t+1}$. Since $\tilde{w}_{t+1} = w_{t+1} - \sum_m k_m(x_t - x_{t-m})$ lies in the span of w_{t+1} and the lagged differences, and w_{t+1} is independent of x_t and of them, $\tau_\varepsilon^2 \ge \tau^2$, and $s_\varepsilon^2/\tau_\varepsilon^2 \le 2\pi G^2\delta/c$, using $s_\varepsilon^2 \le \sigma^2$.

Together. By $(u+v)^2 \le 2u^2 + 2v^2$, $V_{\text{pool}} \le 2(2\pi G^2\delta/c + 2\pi G^2\delta/c) = 8\pi G^2\delta/c$, and with $\delta = M/g(1)$ and the bounds on G and $g(1)$ of Theorem 4.7.2, $V_{\text{pool}} \le 4C_{p,\gamma} M$. □

Two assumptions carry the theorem to the data, and both are stated as limits. The price noise must be independent of the period's flow, the usual identifying assumption of impact regressions; in five-minute bars flow and price noise can be simultaneous. And the tests measure the displacement from an exponential moving average of the price, a proxy for the distance from the fundamental value that the theorem uses. In simulation of an economy with $p = 2$, $\kappa = 0.5$, $s_w = 1$ and $s_\varepsilon = 0.5$, $T\,\mathrm{Var}(\hat{M})/M$ was 1.36 to 1.46 for ρ from 0.9 to 0.99: the pooled variance is proportional to the margin, as part (ii) says, with a constant near $2g(1)$ rather than the crude bound. The variance combination, the divergence identity and the record-length bound are checked in Lean (`CurrentMM.thesis_pooled_sq`, `CurrentMM.thesis_pooled_var`, `CurrentMM.thesis_pooled_kl`, `CurrentMM.thesis_pooled_record`).

The margin decides whether a market is stable. When agents can be contrarian it no longer measures worst-case loss (Proposition 4.9.1); the horizon-limited absolute response does. For a finite-memory economy write $\psi_j(\phi)$ for the coefficients of $1/f$, so that the response of the displacement to a unit innovation j periods earlier is ψ_j, and let $\Psi_H(\phi) = \sum_{j<H} |\psi_j(\phi)|$. A

shock sequence bounded by k innovation standard deviations moves the displacement by at most $k\sigma\Psi_H$ within H periods, and some sequence attains it (Proposition 4.6.1(i)).

Theorem 4.8.4 (Passive certification of worst-case loss)**.** *Let $\mathcal{K}$ be a compact set of finite-memory economies with L lags, signed kernels allowed, whose closed-loop roots lie outside $|z| \le 1+\gamma$, and let $\hat{\phi}$ be the least-squares estimate from T passive observations, with $\hat{\Sigma} = \hat{\sigma}^2\hat{\Gamma}^{-1}$ its estimated asymptotic covariance.*

(i) Upper bound. *Let G_H be a Lipschitz constant of Ψ_H on a neighbourhood of $\mathcal{K}$, for instance $\sup \sum_{j<H} \|\nabla\psi_j\|_2$. The interval $\Psi_H(\hat{\phi}) \pm G_H\big(\chi^2_{L,1-\alpha}\lambda_{\max}(\hat{\Sigma})/T\big)^{1/2}$ covers $\Psi_H(\phi)$ with probability at least $1-\alpha+o(1)$, uniformly over $\mathcal{K}$; its half-width is of order $T^{-1/2}$. No sign condition and no identification of the cohorts is needed.*

(ii) Lower bound. *For $H \ge 2$ and the memory-zero economies $x_{t+1} = \rho x_t + e_{t+1}$, $\Psi_H = \sum_{j<H}\rho^j$ is strictly increasing in $\rho \in [0,1)$, and any interval with coverage at least $1-\alpha$ at both ρ and $\rho+\delta T^{-1/2}$ has expected half-width at least of order $T^{-1/2}$: the rate is sharp.*

(iii) What the summed response misses. *$|\sum_j \psi_j| = 1/|f(1)| = 1/M$ can be far below Ψ_∞; for one contrarian cohort with memory rate zero, $\Psi_\infty/(1/M) = (1+q_0)/(1-q_0)$ is unbounded (Proposition 4.9.1(ii)).*

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Proof. (i) $\sqrt{T}(\hat{\phi}-\phi) \to N(0, \sigma^2\Gamma^{-1})$ uniformly over compact sets of stationary autoregressions (Mann and Wald, 1943; Mikusheva, 2007), and $\hat{\Sigma}$ is uniformly consistent, so $T(\hat{\phi}-\phi)^\top\hat{\Sigma}^{-1}(\hat{\phi}-\phi) \le \chi^2_{L,1-\alpha}$ with probability at least $1-\alpha+o(1)$; on that event $\|\hat{\phi}-\phi\|_2 \le (\chi^2_{L,1-\alpha}\lambda_{\max}(\hat{\Sigma})/T)^{1/2}$, which is eventually inside the neighbourhood, and $|\Psi_H(\hat{\phi}) - \Psi_H(\phi)| \le \sum_{j<H}|\psi_j(\hat{\phi}) - \psi_j(\phi)| \le G_H\|\hat{\phi}-\phi\|_2$, because each ψ_j is a polynomial in ϕ and $|\,|u|-|v|\,| \le |u-v|$. (ii) Monotonicity: each ρ^j is non-decreasing and ρ^1 strictly increasing. The two autoregressions with roots ρ and $\rho' = \rho+\delta T^{-1/2}$, $\rho' < 1$, have divergence at most $\delta^2/(2(1-\rho'^2))$ over T observations by the computation of Theorem 4.7.1(i), so their total variation distance is bounded away from one, while $\Psi_H(\rho') - \Psi_H(\rho) \ge \delta T^{-1/2}$. An interval that covers both values with probability $1-\alpha$ must, on an event of probability bounded away from zero under one of them, contain both, and so has half-width at least $\delta T^{-1/2}/2$ there. (iii) $\sum_j \psi_j z^j = 1/f(z)$ at $z=1$; the example is that of Proposition 4.9.1(ii). □

In a simulated contrarian economy ($L=3$, $\phi = (0.5, 0.3, -0.4)$, $H=12$, $T=2000$) the interval of part (i) covered $\Psi_H = 2.68$ in 100% of replications, with median half-width 1.00; the conservative Lipschitz constant makes it wide. The summed response of that economy, $1/M = 1.67$, understates the absolute response by 38%. Test R1 (Section 6.1) asked whether this quantity, estimated from public data, forecasts losses where the margin did not. Losses scaled with it within markets, at about half the predicted rate, but beyond volatility it forecast

smaller losses, not larger, and at a one-hour horizon the estimated responses rarely changed sign, so the data do not yet distinguish Ψ_H from the summed response. The Cauchy–Schwarz step and the divergence and record-length bounds of Theorem 4.8.1, and the monotonicity of Theorem 4.8.4(ii), are checked in Lean (`CurrentMM.thesis_hidden_cs`, `CurrentMM.thesis_hidden_kl`, `CurrentMM.thesis_hidden_record`, `CurrentMM.thesis_absresp_mono`).

4.9 Contrarian agents

The cohort model of Definition 4.1.1 has non-negative weights: every agent follows the price. The data say otherwise. In the exploration and in Tests C2 and C3 the estimated feedback was contrarian in most symbols (Section 6.1), and Proposition 4.6.3 shows how contrarian agents can be certified from an overshoot. The next proposition says what changes for certification when weights may be negative: linear questions cost at most a constant factor more, but the safety question changes, because the summed response no longer measures the worst-case loss.

Proposition 4.9.1 (Contrarian agents)**.** *Let the merged measure M be a signed measure on $[0,\bar r]$ with total variation $|M|\le B$, so that $k_m=\int r^m\,dM$ may be negative, and write $\omega_L^{\pm}$ for the modulus of Theorem 4.4.1 taken over these measures.*

(i) Linear questions. *$\omega_L^{\pm}(\delta)=\min_{c\in\mathbb{R}^L}\{\delta\|c\|_2+2B\|g-p_c\|_\infty\}$, and $\omega_L(\delta)\le\omega_L^{\pm}(\delta)\le 2\,\omega_L(\delta)$. Theorem 4.4.1(iv) holds with $\omega_L^{\pm}$ in place of ω_L; in particular, accuracy ϵ over signed measures costs no more experiments than accuracy $\epsilon/2$ over positive ones, and Proposition 4.4.2 and the one-logarithm cost of the feedback weight Φ hold unchanged up to constants.*

(ii) The summed response no longer measures risk. *With negative weights the impulse response can change sign, the worst-case loss is $|I|\,\bar w\,\|h\|_1\ge|I|\,\bar w\,|S|$, and the ratio $\|h\|_1/|S|$ is unbounded on stable economies with memory bound $\bar r$: one contrarian cohort with memory rate zero and weight $q=-(a+q_0)/\kappa$ (total variation $(a+q_0)/\kappa$), so that $a+\kappa q=-q_0$ with $q_0\in[0,1)$, has $h_t=\kappa(-q_0)^{t-1}$, partial sums of h in $[0,2\kappa]$, and $\|h\|_{1,H}=\kappa(1-q_0^H)/(1-q_0)$.*

(iii) What can be certified. *The horizon-limited worst case $\bar w\|h\|_{1,H}$ of Proposition 4.6.1(i) is certified without a memory bound and without a sign condition: under contract (O1), with $\bar h$ the average of N experiments, the interval $\|\bar h\|_{1,H}\pm\sigma\sqrt{H/N}\,(\sqrt{H}+\sqrt{2\log(1/\alpha)})$ covers $\|h\|_{1,H}$ with probability at least $1-\alpha$, so half-width ϵ needs N of order $\sigma^2H^2\epsilon^{-2}$, against $\sigma^2H\epsilon^{-2}$ in the positive class.*

(iv) Forever needs more than a memory bound. *If $B\ge(1+a)/\kappa$, then on the stable signed economies with memory bound $\bar r$, total variation at most B and $|S|\le\kappa$, $\sup\|h\|_1=\infty$; certifying $\|h\|_1$ itself requires a bound on the closed loop's distance from instability, not only on memory.*

PROVED IN THIS THESIS

Proof. (i) For a continuous h on $[0,\bar r]$, $\sup\{\int h\,dM : |M| \le B\} = B\|h\|_\infty$, attained by a point mass of sign sign $h(r^*)$ at a maximizer r^* of $|h|$. The duality step in the proof of Theorem 4.4.1(iv) then gives the upper bound $\delta\|c\|_2 + 2B\|g - p_c\|_\infty$ for every c; the signed measures of total variation at most B on $[0,\bar r]$ form a convex weak-* compact set (Banach–Alaoglu), Λ is affine and weak-* continuous in the pair and convex in c, so Sion's theorem gives equality. The lower bound, the affine interval and the sandwich in the proof of part (iv) use only the Gaussian data, concavity of the modulus and the duality, and carry over verbatim. Since $\|h\|_\infty \le \psi(h) \le 2\|h\|_\infty$, for every c the signed objective lies between the positive one and twice it, which gives $\omega_L \le \omega_L^{\pm} \le 2\omega_L$; and if $\omega_L(t) \le \epsilon$ then $\omega_L^{\pm}(t) \le 2\epsilon$. (ii) With $r = 0$ the kernel is $k_0 = q$, so $h_1 = \kappa$ and $h_{t+1} = (a + \kappa q)h_t$. The partial sums $\kappa\sum_{t<H}(-q_0)^t = \kappa(1 - (-q_0)^H)/(1 + q_0)$ lie in $[0, 2\kappa]$, while $\sum_{t<H} q_0^t = (1 - q_0^H)/(1 - q_0)$. (iii) $\big|\|\bar h\|_{1,H} - \|h\|_{1,H}\big| \le \|\bar h - h\|_{1,H} \le \sqrt{H}\,\|\bar h - h\|_{2,H}$, and $\sqrt{N}\|\bar h - h\|_{2,H}/\sigma$ is the norm of a standard Gaussian vector in $\mathbb{R}^H$, which exceeds $\sqrt{H} + \sqrt{2\log(1/\alpha)}$ with probability at most α by Gaussian concentration of Lipschitz functions. (iv) In the example of (ii), $|S| = \kappa/(1 + q_0) \le \kappa$ and $\|h\|_1 = \kappa/(1 - q_0) \to \infty$ as $q_0 \uparrow 1$; the economy is stable for every $q_0 < 1$ and has memory rate $0 \le \bar r$. □

For the guard of Chapter 3 this means a change of statistic, not of principle. With contrarian agents, the quantity to certify is the horizon-limited absolute response $\|h\|_{1,H}$, which is still a functional of the observed response and still needs no identification of the agents; it costs a factor of H more than in the positive class. The registered tests decided the stability margin, which is the right question for whether the market is stable; it is not the worst-case loss when contrarian feedback makes the response oscillate. Parts (i) and (ii) are checked in Lean in their finite-dimensional form (`CurrentMM.thesis_signed_psi_bounds`, `CurrentMM.thesis_signed_modulus_sandwich`, `CurrentMM.thesis_signed_alternating_sum`, `CurrentMM.thesis_signed_abs_sum`).

4.10 Worked examples: how many experiments

Table 4.1 puts numbers on the theory. Each entry is the number of reset experiments, with tap noise $\sigma = 1$, mass bound $B = 1$ and level 95%, below which no interval can certify the question (Theorem 4.4.1(iv), lower bound), at which the affine interval of part (iv) certifies it, and at which the Taylor interval of part (i) does. The first two come from the modulus and are certified to within 1% by stored primal and dual solutions; the third is the closed form of part (i).

Three things stand out. The affine interval is never more than about 1.4 times the lower bound, as part (iv) guarantees, so for these questions the cost is known almost exactly. The Taylor interval of part (i) is three to fifteen times more expensive than necessary: it reserves half of the accuracy for a worst-case truncation bias and weights every retained coefficient equally, while the optimal interval trades bias against noise and shrinks the coefficients that multiply

Table 4.1: Reset experiments needed to certify three questions

Question	$\bar{r}$	ϵ	No interval below	Affine interval	Taylor interval
Total feedback Φ	0.5	0.1	1,750	2,520	9,220
Total feedback Φ	0.5	0.01	240,000	345,000	1,390,000
Discounted, $\beta = 0.8$	0.5	0.1	711	1,020	3,560
Discounted, $\beta = 0.8$	0.5	0.01	73,800	106,000	409,000
Single tap k_3	0.5	0.1	0	0	1,540
Single tap k_3	0.5	0.01	10,700	15,500	154,000
Total feedback Φ	0.9	0.1	14,100	20,300	78,400
Total feedback Φ	0.9	0.01	1,830,000	2,630,000	11,300,000
Discounted, $\beta = 0.8$	0.9	0.1	748	1,080	4,260
Discounted, $\beta = 0.8$	0.9	0.01	74,800	108,000	427,000
Single tap k_3	0.9	0.1	70	101	1,540
Single tap k_3	0.9	0.01	11,700	16,900	154,000

$\sigma = 1$, $B = 1$, $\alpha = 0.05$, observation depth L with $\bar{r}^L \le 10^{-12}$. "No interval below" is the lower bound of Theorem 4.4.1(iv), rounded down; the other two columns are achievable numbers, rounded up. The gap between the first two columns is the factor $\lambda^2_{0.05} \approx 1.42$ of part (iv), up to the outward rounding. A zero means that the question's range over the whole class is narrower than 2ϵ, so the midpoint of that range certifies it with no data.

low-information taps. And the total feedback costs far more than a single tap or a discounted sum, more so when memory is long ($\bar{r} = 0.9$): this is the logarithm of Theorem 4.4.3 at work, multiplied by the size of Φ itself.

4.11 Disclosure changes the contract, not the question

Disclosure regimes are observation contracts. Class-tagged flow reveals each class's own moments, which turns identification into an ordinary estimation problem and is worth a factor $(\Delta\eta)^{-(4K-4)}$ in samples (Theorem 2.8.2). For safety the value of disclosure is different: it raises safe capacity exactly when it rules out the extremal compatible economy, and it is worth most when memory is long, where Theorem 4.4.1(iii) says public data certify nothing (Theorem 3.7.1). Supervision that wants safety should therefore ask for the disclosures that bound memory or reveal Φ, not for identities.

4.12 A robust safety bound: nonlinear and changing agents

Papers 1 and 2 work with agents whose rules are linear filters with fixed memory and weight. Real agents cap positions, trade only beyond thresholds, adapt their gains, and keep learning while the operator watches. This section and the next show that the safety side survives all of this under one condition: a bound on the agents' total sensitivity.

Return to the economy of Definition 4.1.1 and let everything about the agents vary over time.

Assumption 4.12.1 (Bounded total sensitivity)**.** For each cohort i and period t, the memory rate $r_i(t) \in [0, 1)$ is arbitrary, the weight satisfies $|q_i(t)| \le \bar{q}_i$, and the response rule satisfies the sector condition $|\varphi_{i,t}(y)| \le s_i|y|$ for all y. Write $\bar{\Phi} = \sum_i \bar{q}_i s_i$ for the total sensitivity.

The sector condition covers saturating momentum ($s_i = 1$ for tanh), position limits, threshold and dead-zone rules ($y\mathbf{1}\{|y| > \tau\}$), and any rule whose response never exceeds a multiple of its signal. It does not require the rule to be smooth, monotone, stationary or known.

Theorem 4.12.2 (Robust safety bound)**.** *Under Assumption 4.12.1, if* $|a| + \kappa\bar{\Phi} < 1$ *and the economy starts at rest, then for every shock sequence with* $|w_t| \le \bar{w}$*,*

$$\sup_{t\ge 0} |x_t| \;\le\; \bar{w}\,\frac{\kappa}{1 - |a| - \kappa\bar{\Phi}}.$$

When $a \ge 0$*, this is the worst-case loss bound of the linear stationary economy with weights* $\bar{q}_i s_i$*, so the summed-gain guard of Chapter 3 computed at total sensitivity* $\bar{\Phi}$ *is safe for every economy satisfying the assumption.* PROVED IN THIS THESIS

Proof. Let $Z_t = \max_{s\le t} |x_s|$. Each memory state is a convex combination of its previous value and the current state, so $|y_{i,t}| \le r_i(t)|y_{i,t-1}| + (1 - r_i(t))|x_t| \le \max(|y_{i,t-1}|, |x_t|)$, and by induction from $y_{i,0} = 0$, $|y_{i,t}| \le Z_t$. Then

$$|x_{t+1}| \le |a|\,|x_t| + \kappa\Big(\sum_i |q_i(t)|\,|\varphi_{i,t}(y_{i,t})| + |w_{t+1}|\Big) \le (|a| + \kappa\bar{\Phi})\,Z_t + \kappa\bar{w}.$$

Let $Z^* = \kappa\bar{w}/(1 - |a| - \kappa\bar{\Phi})$. If $Z_t \le Z^*$ then $|x_{t+1}| \le (|a| + \kappa\bar{\Phi})Z^* + \kappa\bar{w} = Z^*$, so $Z_{t+1} \le Z^*$; and $Z_0 = 0$. For the second claim, with $a \ge 0$ the linear stationary economy with weights $\bar{q}_i s_i$ is a positive system, its impulse response is non-negative, and its worst-case peak response to shocks bounded by $\bar{w}$ is $\bar{w}\sum_t h_t = \bar{w}\,\kappa/(1 - a - \kappa\bar{\Phi})$ (equation (4.2)). The bound of the theorem therefore coincides with the quantity the guard controls (Lemma 3.6.1). □

What the theorem says. An operator who wants a safe admission level does not need to know how many agents there are, what their memories are, whether their rules are linear, or whether they change over time. It needs one number, an upper bound on the agents' total sensitivity $\bar{\Phi}$. The bound is tight: when the agents are linear, stationary, positive and at their maximal weights, it is attained. Nonlinearity, memory changes and negative feedback can only make the market safer than the bound says.

Remark 4.12.3 (What the theorem does not cover). Rules that amplify their signal, $|\varphi(y)| > s|y|$ for the declared s, break the bound; so do changes in the market's own persistence a or impact κ, which the operator must certify separately (Chapter 3, Section 3.5). The bound controls the peak of the price path, which is what a fixed position's loss depends on; dynamic positions are handled by the balance-sheet guard of Chapter 3.

Proposition 4.12.4 (Several assets)**.** *Let $x_t \in \mathbb{R}^d$ evolve as $x_{t+1} = Ax_t + \mathcal{K}\big(\sum_i Q_i(t)\varphi_{i,t}(y_{i,t}) + w_{t+1}\big)$, with memory states $y_{i,t} = R_i(t)y_{i,t-1} + (I - R_i(t))x_t$, $R_i(t)$ diagonal with entries in $[0,1)$, sector conditions $|\varphi_{i,t}(y)| \le s_i|y|$ componentwise, and $\|Q_i(t)\|_\infty \le \bar{q}_i$. If $\|A\|_\infty + \|\mathcal{K}\|_\infty \sum_i \bar{q}_i s_i < 1$, then $\sup_t \|x_t\|_\infty \le \bar{w}\,\|\mathcal{K}\|_\infty/(1 - \|A\|_\infty - \|\mathcal{K}\|_\infty \sum_i \bar{q}_i s_i)$ for $\|w_t\|_\infty \le \bar{w}$.* PROVED IN THIS THESIS

Proof. The proof of Theorem 4.12.2 goes through with absolute values replaced by the sup norm and scalars by induced sup norms: each component of $y_{i,t}$ is a convex combination of its past value and the corresponding component of x_t. □

The induced-norm bound discards the pattern of cross-impact. For positive cross-impact, the following comparison gives the sharp replacement.

Proposition 4.12.5 (Positive cross-impact comparison)**.** *Assume the economy starts at rest, $A, \mathcal{K}, \overline{Q}_i$ are entrywise non-negative, $|Q_i(t)| \le \overline{Q}_i$ entrywise, and $|\varphi_{i,t}(y)| \le S_i|y|$ for a non-negative diagonal matrix S_i. Memory matrices $R_i(t)$ are diagonal with entries in $[0,1)$. Put $G = A + \mathcal{K}\sum_i \overline{Q}_i S_i$. If $\rho(G) < 1$, then for every $\|w_t\|_\infty \le \bar{w}$,*

$$\sup_t \|x_t\|_\infty \le \bar{w}\,\big\|(I - G)^{-1}\mathcal{K}\big\|_\infty.$$

This bound is sharp over the stated class. PROVED IN THIS THESIS

Proof. The Neumann series $\sum_{j\ge 0} G^j = (I - G)^{-1}$ has non-negative entries. Write $\mathbf{1}$ for the vector of ones and set $z = \bar{w}(I - G)^{-1}\mathcal{K}\mathbf{1} \ge 0$. Then $Gz + \bar{w}\mathcal{K}\mathbf{1} = z$. Suppose $|x_t| \le z$ and $|y_{i,t}| \le z$ componentwise. The plant and the sector bounds give

$$|x_{t+1}| \le Az + \mathcal{K}\Big(\sum_i \overline{Q}_i S_i z + \bar{w}\mathbf{1}\Big) = z.$$

Each next memory coordinate is a convex combination of its previous value and the corresponding next state coordinate, so its absolute value is also bounded by the corresponding coordinate of z. Induction from rest proves the componentwise bound for all times. Because the resolvent times $\mathcal{K}$ is non-negative, its induced sup norm is its maximum row sum, and $\|z\|_\infty$ equals the displayed bound.

For sharpness, choose $R_i(t) = 0$, $Q_i(t) = \overline{Q}_i$, $\varphi_{i,t}(y) = S_i y$, and $w_t = \bar{w}\mathbf{1}$. The state then satisfies $x_{t+1} = Gx_t + \bar{w}\mathcal{K}\mathbf{1}$, so from rest $x_t = \bar{w}\sum_{j<t} G^j\mathcal{K}\mathbf{1}$ increases componentwise to z. Its supremum norm therefore approaches the bound. Sharpness means a supremum over the admitted model class, not attainment at a finite time in every economy. □

Registered numerical validation. The specification and its hash were pushed before the seeded instances were generated, and the unlock counter was advanced before the single opening. The joint primary required both componentwise enclosure for nonlinear, time-varying cohorts and convergence to the bound in the maximal positive linear instance. It passed in 128 of 128

registered random instances. The generated matrices, tolerances and diagnostics are recorded in `checks/R_multiasset.json`. This is a numerical check of the proved statement under the registered construction, not evidence for it.

4.13 Certificates expire: learning agents

Theorem 4.12.2 allows weights to change, provided they stay below their bounds. When agents learn, the bounds themselves are the issue: a certificate issued today says nothing about weights next month. The theorem turns this into a computable lifetime.

Corollary 4.13.1 (Certificate lifetime)**.** *Suppose a certificate at time* 0 *establishes* $|q_i(0)| \le q_i^+$ *for each cohort, and the guard's loss budget is met for every total sensitivity up to* $\Phi_{\max}$*, that is* $\kappa\bar{w}/(1-a-\kappa\Phi_{\max})$ *times the position equals the budget. If each weight changes by at most* γ_i *per period,* $|q_i(t+1)-q_i(t)| \le \gamma_i$*, and* $\sum_i q_i^+ s_i \le \Phi_{\max}$ *with* $\sum_i \gamma_i s_i > 0$*, then the guard remains safe for at least*

$$T_{\text{valid}} = \left\lfloor \frac{\Phi_{\max} - \sum_i q_i^+ s_i}{\sum_i \gamma_i s_i} \right\rfloor$$

periods under the sector bounds. PROVED IN THIS THESIS

Proof. On $[0, T_{\text{valid}}]$, $|q_i(t)| \le q_i^+ + \gamma_i T_{\text{valid}} =: \bar{q}_i$ and $\sum_i \bar{q}_i s_i \le \Phi_{\max}$. Theorem 4.12.2 applies on that horizon, since its induction uses only periods up to the current one. □

The lifetime is the certificate's safety margin divided by the agents' speed of learning. This is a guaranteed lower bound, not a claim that loss first crosses the budget at this time. Realized safety can last much longer if learning slows, changes sign, or stops below the sensitivity budget. When $\sum_i \gamma_i s_i = 0$, the sensitivity envelope does not grow and the same certificate remains valid for every horizon under the stated sector and plant assumptions; the displayed quotient is not used. If the initial sensitivity bound already exceeds $\Phi_{\max}$, this argument supplies no valid initial certificate. Recertification frequency can therefore be based on the certified sensitivity margin and the learning-speed bound, while additional information about the realized learning path may justify a longer horizon.

Formal scope. The finite-cohort, finite-horizon safety induction is checked by `CurrentMM.thesis_sector_safety`. The drift, floor and sensitivity calculations are checked by `CurrentMM.thesis_lifetime_envelope` and its helper lemmas. The deterministic certification arithmetic is checked by `CurrentMM.thesis_variance_radius` and `CurrentMM.thesis_interval_half_width`; the new bias lemma covers finite positive atomic measures. For several assets, `CurrentMM.thesis_multiasset_comparison` checks the comparison step of Proposition 4.12.5 for any number of assets and cohorts, nonlinear time-varying sector-bounded responses, entry-wise weight bounds and all times: every non-negative supersolution bounds every state and

memory coordinate; `CurrentMM.thesis_multiasset_linear_monotone` checks that the maximal positive linear economy rises from rest and stays below every supersolution. The Neumann series that produces the resolvent supersolution and the limit in the sharpness argument are written proofs. For finite atomic measures, `CurrentMM.thesis_modulus_weak_duality` and `CurrentMM.thesis_modulus_affine_bias` check the deterministic half of Theorem 4.4.1(iv): the weak-duality bound on the modulus and the bias of the affine interval. These declarations do not formalize Gaussian coverage, a general-measure integration theorem, the minimax step and the lower bound of part (iv), or Theorem 4.4.3, which rest on Sion's minimax theorem and on potential theory. The coverage register records these boundaries separately.

Identification under learning, by contrast. Chapter 2, Section 2.7, shows that trained online learners obey the same δ^{-4} law for separating memory rates when the competing populations' weights match, and that populations whose learned weights differ are told apart by those weights instead. Learning therefore leaves identification as hard as before while making certification only perishable, not harder per renewal.

4.14 Markets of trading algorithms that learn

Markets in which learning algorithms, including reinforcement-learning and language-model agents, trade with each other are where the results of this chapter matter most. Two of them apply directly. Certification of safety needs only a bound on total sensitivity (Theorem 4.12.2); and the certificate lasts for its safety margin divided by the agents' learning speed (Corollary 4.13.1). The second result gives a supervisory rule with no counterpart in identity-based regimes: recertify at a frequency set by how fast the agents update.

A registered simulation with learning agents. Corollary 4.13.1 is a guarantee, so a simulation cannot confirm it; what a simulation can show is whether it is informative when agents learn from their own trading. We registered one (specification hash `58c32f9b...`, pushed with its runner in a commit of its own before the run). The economy is that of Definition 4.1.1 with $a = 0.3$, $\kappa = 0.4$ and four cohorts with memory rates $0.2, 0.5, 0.7, 0.85$, each following the saturating momentum rule $\varphi(y) = \tanh y$ (sector bound 1). Each cohort learns its weight online by a gradient step on its own trading profit, clipped to the declared learning rate γ, so weights rise or fall but never move by more than γ per period. Weights start at 0.05 (total sensitivity 0.2), the budget is $\Phi_{\max} = 1.4$ (so the loss budget is $\kappa\bar{w}/(1 - a - \kappa\Phi_{\max}) = 2.86$ for $\bar{w} = 1$), and $T_{\text{valid}} = \lfloor 1.2/(4\gamma) \rfloor$ is 300, 100, 30 and 10 periods for $\gamma = 0.001, 0.003, 0.01, 0.03$. Shocks are bounded by $\bar{w}$: in the adversarial regime they push in the direction of the price displacement nine times in ten; in the random regime they are uniform. There are 25 runs for each regime and learning rate, each lasting five lifetimes.

The registered check (P1) held in 200 of 200 runs: up to T_{valid} every price path stayed within the loss budget and total sensitivity within $\Phi_{\max}$, as the corollary guarantees. The informative primary (P2) asked whether learning actually exhausts the sensitivity budget within three lifetimes in at least half of the adversarial runs. It passed: it did in 98 of 100 runs. Table 4.2 shows that under adversarial shocks the budget is exhausted after 1.3 to 1.8 lifetimes, and the realized loss first exceeds its budget after four to five. So the lifetime bound is close to tight when trend-following pays, and a certificate that is not renewed does eventually admit losses beyond its budget. Under random shocks trend-following loses money on average, weights decay towards zero and no run comes near the budget: there the bound is conservative, because it charges every period for the fastest possible learning.

Table 4.2: Registered simulation with learning agents: certificate lifetime and realized loss

Shocks	γ	T_{valid}	Budget exhausted	Median time	Loss breached	Median time	Median margin
Adversarial	0.001	300	25/25	1.27	25/25	3.97	2.7
Adversarial	0.003	100	25/25	1.30	25/25	3.99	2.7
Adversarial	0.010	30	25/25	1.33	25/25	4.03	2.9
Adversarial	0.030	10	25/25	1.80	17/25	4.70	3.7
Random	0.001	300	0/25	—	0/25	—	5.4
Random	0.003	100	0/25	—	0/25	—	5.8
Random	0.010	30	0/25	—	0/25	—	6.2
Random	0.030	10	0/25	—	0/25	—	7.1

25 runs per row, horizon $5T_{\text{valid}}$. "Budget exhausted": runs in which total sensitivity exceeds $\Phi_{\max}$; "Loss breached": runs in which $|x_t|$ exceeds the loss budget. Median times are in units of T_{valid}, over the runs in which the event occurs. "Median margin": loss budget divided by the largest $|x_t|$ up to T_{valid}. Source: `checks/R_multiagent_v2.json`. An earlier simulation, whose weight updates could only increase and so could not test anything, is withdrawn (Chapter 6).

4.15 Agents who respond to the guard

A guard changes the environment it certifies. If agents learn that the operator throttles admission when measured sensitivity is high, they may change how they trade.

Conjecture 4.15.1 (Guard-consistent equilibrium)**.** *Suppose each cohort chooses its weight $q_i \in [0, \bar{q}_i]$ to maximize expected trading profit net of a convex cost, given an admission rule that throttles the dealer's position when certified total sensitivity exceeds $\Phi_{\max}$, and given the equilibrium reversion supplied by dealers (Proposition 3.8.1). Then (i) an equilibrium exists; (ii) in every equilibrium the guard is safe; and (iii) equilibrium total sensitivity is weakly lower than without the guard.* CONJECTURE

Remark 4.15.2 (Counterexample to existence under a discontinuous guard). Part (i) of the conjecture is false under the stated assumptions. Consider one cohort with $q \in [0, 1]$, a perfect certificate $\Phi = q$, and admitted dealer position $I(q) = 1$ for $q \leq 1/2$ and $I(q) = 0$ for $q > 1/2$. This is an admission rule that throttles above the certified threshold. Let expected trading

profit, as a function of weight and admitted position, be

$$\pi(q, I) = (1 - I)(1 - q), \qquad c(q) = 0.$$

The profit primitive is continuous in (q, I) and the zero cost is convex. After composition with the guard, utility is zero at $q \leq 1/2$ and equals $1 - q$ at $q > 1/2$. If $q \leq 1/2$, the admissible deviation $q' = 3/4$ strictly increases utility. If $q > 1/2$, the admissible deviation $q' = (q + 1/2)/2$ lies strictly between $1/2$ and q and also strictly increases utility. Thus no weight maximizes utility: the one-player game has no pure equilibrium despite its compact strategy set, convex cost, and exact sensitivity certificate.

Randomization does not restore existence. Utility is strictly below $1/2$ everywhere, so its expectation under any probability distribution is strictly below $1/2$; a pure weight sufficiently close to the threshold from above earns more. The universal profitable-deviation statement and nonexistence of a pure maximizer are formalized as `CurrentMM.thesis_strategic_profitable_deviation` and `CurrentMM.thesis_strategic_no_equilibrium`. The probability-integral step for arbitrary mixed strategies is a written argument, not part of those Lean declarations. Exact rational diagnostics are recorded separately in `checks/R_strategic.json`.

The failure concerns equilibrium existence, not the safety invariant. For example take $a = 0$, $\kappa = 1/4$, and $\bar{w} = 1$. On admitted profiles $q \leq 1/2$, unit position loss is bounded by $2/7$; on throttled profiles the position is zero. Every admitted decision therefore respects this budget, although the agents' game has no equilibrium. A revised strategic theorem needs assumptions that rule out the best-response discontinuity. Convex cost alone also supplies no ordering of equilibrium sensitivity with and without a guard; that comparative-static claim remains open under suitable additional payoff restrictions.

4.16 Relation to the statistics of functionals

The certification problem of this chapter is an instance of estimating a functional of an unknown measure from indirect, noisy observations, and three literatures meet in it.

Linear functionals over convex classes. For a linear functional of an unknown element of a convex set observed with Gaussian noise, the minimax risk of affine estimators is within a small constant of the minimax risk over all estimators, and both are governed by a modulus of continuity (Ibragimov and Has'minskii, 1984; Donoho, 1994); fixed-width and adaptive confidence intervals follow the same geometry (Cai and Low, 2004), and the optimal affine rule can be computed by convex programming (Juditsky and Nemirovski, 2009). Theorem 4.4.1(iv) applies this principle to the moment observations of a closed-loop market. What is specific here is the dual form: the modulus is the value of a polynomial approximation problem with an ℓ^2 penalty on coefficients, a K-functional in the sense of approximation theory (DeVore and Lorentz, 1993,

Chapter 6), which turns the question "how many experiments?" into "how well can g be approximated by a power series that is cheap to estimate?". Proposition 4.4.2 identifies the questions with parametric cost as those whose kernels lie in the Hardy space H^2 of the disc (Duren, 1970), and Theorem 4.4.3 computes the cost at the boundary of that space using potential theory.

Moments, mixtures and super-resolution. Recovering a measure from its first moments is the problem behind finite mixture estimation, whose optimal rate for k atoms is $n^{-1/(4k-2)}$ (Heinrich and Kahn, 2018; Wu and Yang, 2020), and behind super-resolution, where the conditioning of Vandermonde systems sets the cost of resolving nearby spikes (Candès and Fernandez-Granda, 2014; Moitra, 2015). Chapter 2 shows that identifying a market's agents is a problem of this kind. The contrast drawn here, that a smooth functional of the same measure can be far cheaper than the measure itself, parallels the estimation of smooth versus non-smooth functionals of a density or a regression function (Lepski et al., 1999; Cai and Low, 2011). The difference is the observation scheme: here the data are noisy moments of a measure on a memory interval, so smoothness is measured by analytic continuation towards $r = 1$ rather than by derivatives.

Control and identification. Positive-system comparison arguments (Farina and Rinaldi, 2000; Rantzer, 2015) and integral quadratic constraints (Megretski and Rantzer, 1997) give the robust safety bounds of Sections 4.12 and 4.13; system identification (Ljung, 1999) and partial identification (Manski, 2003) supply the stance that a decision can be determined even when the structure is not.

4.17 What the framework does not cover

The framework is linear in the observable layer and assumes the reset contract for its sharpest statements. Sections 4.12 and 4.13 extend the safety results to sector-bounded nonlinear and to learning agents; strategic equilibrium existence fails without additional assumptions (Remark 4.15.2); existence for suitably regularized guards remains open. Passive observational data are slower but obey the same polynomial order for identification (Theorem 2.6.10); a finite-sample theory of certification from observational order-book data, with dependence and non-stationarity, is not developed here, and the empirical certificates of Chapter 3 are bootstrap diagnostics rather than guarantees.

Chapter 5

AI agents in the loop: supervising the market, not the models

Supervisors are now asking how markets and banks should be governed when trading and other decisions are delegated to AI agents. In October 2026 the Financial Stability Board issued sound practices for AI adoption that centre on governance inside each firm, risk-based triggers for human approval before an agent acts, and concentration risk from shared vendors (FSB, 2026); US banking agencies asked banks how they monitor and audit their agents and how they would shut them down (Reuters, 2026); and the Bank for International Settlements warned that institutions using similar models, data or decision rules may act at the same time and amplify price moves, and that supervisors should look beyond individual institutions to common exposures (Zhang, 2026). The instruments on the table act on one firm or one agent at a time. The risk that the last warning names is a property of the loop that many agents form with prices. This chapter applies the thesis's central result to that gap: the stability of the loop can be certified from public data without inspecting any agent, and the one thing the certificate needs from the agents' owners is a bound on their memory.

5.1 What the theory already says about AI agents

Four results of the earlier chapters apply to AI agents as they stand.

Their internals are not needed, and cannot be learned from the market. Identifying the agents from market data costs a number of observations that grows polynomially in the inverse of their similarity (Chapter 2), and passive observation does not help (Theorem 2.6.10). An AI agent is harder to identify than the exponential-memory cohorts of the theorems, not easier: its rule is high-dimensional, it changes when the model is updated, and, in the Financial Stability Board's words, it does not reason like a human.

Nonlinear and changing agents are covered. The robust bound of Section 4.12 holds for agents whose rules are nonlinear and whose memories and weights change over time, given a bound on their total sensitivity, and when agents keep learning a certificate lasts for its margin divided by their learning speed (Corollary 4.13.1). These are the properties that distinguish an AI agent from a fixed algorithm.

The system-level quantity is certifiable from public data. The margin is decided from prices and anonymous flow with a record of order $1/M$, for any number of cohorts with finite memory, uniformly as the margin vanishes, and for the pooled estimator used in practice (Theorems 4.7.2, 4.8.2 and 4.8.3).

The owners must disclose a memory bound. Without one, no passive certificate exists: a slow momentum cohort can make a market unstable without leaving a trace in a finite record (Theorem 4.8.1).

5.2 Monoculture

Agents built on one model and fed the same data respond alike. In the cohort model they share a memory rate, so they merge into a single cohort whose weight is the sum of theirs (Theorem 2.3.1). The next proposition makes the consequence for stability explicit.

Proposition 5.2.1 (Monoculture)**.** *Let N agents trade on a market with persistence a and impact κ. A share $s \in [0,1]$ of them run a common model with response q_D; the others have mean response $\bar{q}$.*

(i) The margin is linear in the adoption share: $M(s) = M(0) - s\,\kappa N(q_D - \bar{q})$, with $M(0) = 1 - a - \kappa N \bar{q}$.

(ii) If $q_D > \bar{q}$ and $M(0) > 0$, the market is stable for $s < s^*$ and unstable for $s > s^*$, where $s^* = M(0)/(\kappa N(q_D - \bar{q}))$.

(iii) The adopters form one cohort of the merged measure: adoption lowers the number of distinct cohorts, which makes identification easier, while it lowers the margin, which makes certification urgent.

(iv) If each adopter's decisions carry noise of variance σ^2 and adopters of one model share it, the adopters' noise contributes variance $(sN)^2\sigma^2$ rather than $sN\sigma^2$: correlated decisions enlarge the shocks that the loop amplifies by a factor equal to the number of adopters.

PROVED IN THIS THESIS

Proof. (i) The total feedback is $\Phi = N((1-s)\bar{q} + sq_D)$ and $M = 1 - a - \kappa\Phi$. (ii) $M(s)$ is affine and decreasing in s with zero at s^*. (iii) Agents with a common model have a common memory rate, and cohorts with a common rate merge (Theorem 2.3.1). (iv) The sum of n copies of one noise has variance $n^2\sigma^2$; the sum of n independent ones has variance $n\sigma^2$. □

Part (ii) is static: it freezes the other agents. When they learn, they respond to the price path that the adopters create, and the threshold can move. Whether it moves up, because the others learn to trade against the adopters, or down, because they learn to follow them, is a question about learning dynamics that the proposition does not answer; Test B1 answers it in simulation. The linear threshold, the sign changes and the noise identity are checked in Lean (`CurrentMM.thesis_mono_margin`, `CurrentMM.thesis_mono_threshold`, `CurrentMM.thesis_mono_below`, `CurrentMM.thesis_mono_above`, `CurrentMM.thesis_mono_common_noise`).

5.3 A two-number disclosure and a loop-level trigger

The results suggest a supervisory instrument with two parts.

What deployers disclose. The owner of an AI trading agent discloses two numbers: a bound s_i on the agent's sensitivity to the price displacement (the sector constant of Assumption 4.12.1) and a bound $\bar{r}_i$ on its memory, the longest horizon over which its response depends on past prices. Neither reveals the model, its data or its strategy. The sum of the sensitivity bounds certifies safety by itself through the robust bound of Section 4.12, conservatively; the memory bound is what makes the public-data certificate possible (Theorem 4.8.1) and what the disclosed estimator of Test B1 uses.

What the supervisor computes. From public prices and anonymous flow the supervisor estimates the margin daily on a rolling window, with the pooled estimator augmented by the disclosed memory bound, and acts on two signals: the loss of the certificate that the margin exceeds a buffer, and a projected time to the boundary, the current margin divided by its recent rate of decline. A pause or a review is triggered by the loop, not by any one firm. Firm-level kill switches stop one agent; a trigger on the margin addresses the risk that many agents create together.

How it maps onto current guidance. The Financial Stability Board's practice on human oversight asks for risk-based triggers before an agent acts; the trigger here is a market-level input to those triggers. Its practice on third-party dependencies asks firms to oversee concentration on vendors; Proposition 5.2.1 says what that concentration does to stability and at what adoption share it becomes dangerous. The US agencies' question of how a bank would shut its agents down has a market-level counterpart: when should all of them slow down?

5.4 Test B1: learning agents and a vendor model

Design. The market is the closed loop of the thesis on five-minute bars, with 50 agents. Independent learners each have their own memory and learn their response online; with reverting prices they learn to trade against moves. A vendor model is a pre-trained momentum product with one memory rate; every agent that adopts it responds identically. Adoption ramps from none to 80% of the agents between days 24 and 96 of a 120-day run, in two versions: a fast vendor model (memory rate 0.9 per bar) and a slow one (0.995, a half-life of about eleven hours); in the control nobody adopts. The observer sees only the price displacement and the anonymous flow and runs the pooled estimator of Theorem 4.8.3 daily on a ten-day window, alone (standard) or with the flow equation augmented by the signal at a disclosed memory bound of 0.995 (disclosed). A crisis is the first day on which the displacement exceeds ten times its standard deviation before adoption. The warning rules were fixed in an exploratory pilot on other seeds: the certificate of a positive margin is lost; the certificate of a margin above a buffer of 0.1 is lost; or the projected time to the boundary, the estimate divided by its rate of decline over five days, falls below ten days. Test B1 was registered, with its code and four primaries, before its seeds (1001–1100, three scenarios, 300 runs) were run, and it ran once.

Result. All four primaries passed. In the 100 control runs the standard estimator raised no false alarm under any rule (0 for the time-to-boundary rule). Every one of the 200 runs with adoption ended in a crisis, and the time-to-boundary rule warned before it in 193 (96%), with a median lead of 13.0 days (11.5 for the fast vendor model, 17.5 for the slow one). The level of the margin alone gave no warning: the standard certificate was lost on the day of the crisis (median lead 0 days). With the slow vendor model, the disclosed estimator's buffer warning came earlier than the standard one in 100 of 100 runs and later in 0, a median gain of 17.0 days (median leads 19 and 1 days). And in all 200 runs the crisis came at an adoption share below the static threshold of Proposition 5.2.1, computed with the learners frozen when adoption began: a median share of 0.46 against a median threshold of 0.82.

What the secondaries show. Disclosure has a price in false alarms. The disclosed estimator is noisier: in the controls its buffer rule fired in 17 of 100 runs, its lost-certificate rule in 7, and its time-to-boundary rule in 99, so only the standard estimator's trend and the disclosed estimator's buffer are usable together, and the gain of Primary 3 is earned partly by a rule that would false-alarm about one time in six. The standard estimator's daily estimate differed from the true margin by a median of 0.20, mostly as a lag of a few days behind a margin that falls quickly.

Reading. Three findings matter for supervision. The certificate's level is not an early warning, but its trend is: a margin falling toward zero is visible in public data about two weeks before the crisis, with no false alarms in markets where nothing changes. A disclosed memory bound

buys more warning when the shared model is slow, which is the case Theorem 4.8.1 says prices alone cannot see, at a cost in false alarms that a supervisor would have to weigh. And learning makes monoculture dangerous earlier than static analysis predicts: as the adopters push prices, the independent learners stop trading against the move and start following it, so the market crosses the boundary at about half the adoption share at which a vendor-by-vendor assessment, freezing everyone else, would put it. That last finding is the strongest argument for supervising the loop rather than the models: the danger is created by the interaction, and an assessment of any one model, however careful, misses it.

5.5 What this chapter does not show

The agents of Test B1 are simple learning rules and a fixed momentum product, not language-model agents or the systems banks deploy; adoption is imposed rather than chosen; the observer sees the true displacement rather than a moving-average proxy; and the warning rules and the crisis definition were chosen in an exploratory pilot before registration. The empirical chapters show that the certificate behaves as the theory says in real crypto markets, but those markets were far from the boundary and nothing in their data says which traders are AI agents. The chapter therefore offers a framework and an instrument with simulated evidence, not evidence about AI agents in real markets; Test H1, with participant-level data, is the first step toward the latter.

Chapter 6

Conclusion and open problems

This thesis asked whether safety in a market with adaptive agents requires knowing who the agents are. For the markets studied here, it does not. Prices reveal the agents only through a merged measure of their memories and weights, and recovering the agents from that measure costs trading capital of order δ^{-4}, experiments of order $(\Delta\eta)^{-(4K-2)}$, and with general memory an accuracy that improves only logarithmically (Chapter 2). Safety depends on the agents only through their total feedback weight, which is certified at the parametric rate under a memory bound and cannot be certified without one; a guard built on that certificate is the largest safe admission rule (Chapter 3). The theory of Chapter 4 places these results in one picture, in which the cost of a question is set by its behaviour at long memory, and shows that the safety bound needs only the agents' total sensitivity, even when their rules are nonlinear and their memories and weights change, and that learning makes certificates perishable at a computable rate. In data, disclosed trader classes reveal what prices do not, and registered tests replicate this on new days. The safety statistic forecasts adverse excursions in thin Binance books but not on spot venues, and the registered tests find that, with enough days, public data decide the safety question in new markets whose agents they do not identify, but that this decision rests mainly on price reversion, that the summed response itself is no more precise than the separation of memories, and that a guard estimated from public data does not reduce realized tail losses.

The policy message is direct. Supervisors who want to control transient risk should ask for the information that certifies safety, a bound on memory and on total sensitivity, renewed as often as the agents learn, rather than for the identities of the agents. Identity-based regimes may serve other purposes, such as enforcement and attribution, but in the model of this thesis they are neither necessary nor efficient for safety. The evidence adds a practical point: passive observation of prices delivers neither kind of knowledge reliably, and disclosure is what makes either attainable.

6.1 The registered tests

The empirical work of Chapters 2 and 3 tests each paper's own predictions. Twelve further tests were registered to test the thesis as a whole: each specification was pushed to the public repository in a commit of its own before any of its data were downloaded. Ten were run once each on 24 and 25 October 2026, the last two, S1 and R1, registered on 25 October after the first eight results were known; and two are prospective: H1, on participant-level data that did not exist at registration, opens from 10 December 2026, and E2 cannot run before 2 February 2027. Of the twenty-three primary hypotheses tested, eleven passed and twelve failed, and one was not tested because the hypothesis it depended on failed (Table 6.1). The results are reported as they came out. Explanations found afterwards are labelled post hoc, as Rule 3 of Section 6.1.8 requires.

6.1.1 What the earlier evidence establishes

Prices do not reveal the agents. In 30 Binance perpetuals, trader-class positions restrict a persistence that prices and flow leave essentially unrestricted (with a one-hour price-only kernel the intervals are 28 to 124 times wider; at a matched horizon the gap is smaller but present in every symbol, on the original and on 30 new symbols, Section 6.1), and in a pre-registered test the CFTC's trader-class reports do the same in all eight regulated futures markets examined (Sections 2.8.3 and 2.8.4). This is the identification half of the thesis in data: the merged measure leaves class memories essentially unrestricted.

The safety statistic carries information. In thin Binance books, six-hour estimates of the amplification margin forecast the size of the next hour's adverse excursion out of sample in every pre-registered forecasting test, beyond a full volatility-forecasting baseline and VPIN; the effect is about a fifth as large in thin US stocks, negligible on Bybit and absent on Coinbase (Section 3.9.2 and Table 3.4).

What it does not establish. The forecasting result is reduced form. Tests of a leverage or liquidation mechanism fail (Section 3.9.5), and the follow-up that points to slow reversion is close to a check of how the statistic is built (Section 3.9.5). As a risk control, the certified guard improves on a static limit but not, at matched exposure, on a volatility cap (Section 3.9.4). And from passive data the certificate itself is not always precise: in Paper 1's registered test the summed response was significantly different from zero in only 5 of 30 thin markets (Section 2.8.3). Test E1 checks the thesis's central claim directly.

6.1.2 Test E1: certify versus identify on fresh data

Design. The summed response is a linear functional of the merged measure and the theory says it is cheap to certify, while the agents' memories are expensive to identify. In passive data both come from one regression of five-minute order flow on twelve lagged returns: the sum of its coefficients is $\hat{\Phi}$, and two exponentials fitted to the coefficients give the price-only memory rates. Published trader-class positions give class-tagged rates. Thirty Binance USD-M perpetuals on 105 fresh dates (the 4th, 5th, 18th, 19th and 23rd of each month, January 2025 to September 2026); each symbol gives I_{cert}, the width of the day-bootstrap interval for $\hat{\Phi}$ relative to $|\hat{\Phi}|$, and I_{id}, the same for the separation of the price-only rates. Primary 1: $I_{\text{id}} \geq 10\, I_{\text{cert}}$; Primary 2: price-only rate intervals at least ten times wider than class-tagged ones; each in at least 75% of symbols with a Holm-corrected binomial test. The specification (hash `0b6c31c0...`) called Primary 1 a risky prediction.

Result. Primary 2 passed: price-only intervals were at least ten times wider than class-tagged ones in all 30 symbols, with a median ratio of 74.9, a replication of Paper 1 on 105 new days. Primary 1 failed: it held in 10 of 30 symbols (Holm-adjusted $p = 0.98$). The median I_{cert} was 2.67 and the median I_{id} 2.38: relative to its size, the certificate was no more precise than the separation of memories, and the summed response was significantly different from zero in only 9 symbols.

Post hoc reading. There are two reasons, one of design and one of substance. The price-only rates were fitted on $[0, 0.99]$, so the interval for their separation cannot be wider than 0.99. In 28 of the 30 symbols at least one rate interval reaches a bound of the fit, which means the data do not restrict that rate, yet I_{id} is then capped near one: 16 symbols have $I_{\text{id}} < 2.5$. The relative-width measure therefore cannot register the failure of identification that Primary 2 shows directly. And $\hat{\Phi}$ is small relative to its sampling error in passive five-minute data, as Paper 1's earlier test had warned. Neither reason contradicts the theorems, which concern controlled experiments; both limit what passive data can show. The test failed as registered.

6.1.3 Test E2: a prospective test of the mechanism

Idea. Every pre-registered test so far used archived data. A prospective test removes any doubt about the order of registration and data: the data do not exist when the specification is written.

Design. The original specification (pushed on 23 October 2026) asks, for 20 thin perpetuals over at least three months after registration, whether six-hour $\hat{C}_3$ forecasts next-hour liquidation volume beyond volatility, impact and lagged liquidations (Primary 1), and whether adding realized next-hour liquidations to the excursion model removes at least half of $\hat{C}_3$'s forecasting

gain (Primary 2). Binance's forced-liquidation stream, which it named as the source, cannot be recorded from the United States, and the public archive contains no liquidation files. An addendum, pushed on its own before any post-registration file was downloaded, keeps the hypotheses and replaces liquidation volume by forced deleveraging measured from open interest: the open interest destroyed during five-minute moves larger than twice their recent standard deviation. It also fixes the window (24 October 2026 to 31 January 2027), the split into 50 training and 50 test dates, and the estimators, which are those of the thin-book tests of Section 3.9.2. The price of the substitution is stated in advance: E2 can no longer test the liquidation channel with direct liquidation data, which was its advantage over the open-interest proxy of Section 3.9.5.

Status. The analysis code and its notebook are committed with the addendum and refuse to run before 2 February 2027.

6.1.4 Tests D1, G1, V1, C2, O1, C3, F1, S1 and R1

D1: does tagging restore the parametric rate as data grow? Theorem 2.8.2 predicts that class-tagged data make memory an ordinary estimation problem, while prices and flow alone leave it far harder. On 58 untouched dates for the same 30 perpetuals, with nested subsamples of 14, 28 and all days, Primary 1 asked that the class-tagged interval width fall with a slope between -0.75 and -0.25 in $\log n$, and Primary 2 that the advantage of tagging grow with slope at least 0.2, each in at least 75% of symbols. *Result.* Primary 2 passed: the advantage grew in 22 of 29 symbols, exactly the number required (median slope 0.34, Holm-adjusted $p = 0.008$). Primary 1 failed: 20 of 29 symbols were in range, two short of the 22 required, although its binomial p-value was 0.031; class-tagged intervals shrank more slowly than the parametric rate, with a median slope of -0.32 against $-1/2$. *Post hoc reading.* The 58 dates span 21 months. If class memory drifts, days from new periods do not shrink an interval at the parametric rate, and the specification named this as the informative failure. The exploration of Section 6.1.6 finds such drift.

G1: does the guard beat a volatility limit in stress hours? Chapter 3 finds no advantage over a volatility limit at matched exposure over all hours. The theory concerns worst-case loss, so G1 asked the question in stress hours, those whose six-hour volatility is in the top decile of the symbol's training distribution, for 20 thin perpetuals. *Result.* Both primaries failed. The training data chose $\theta = 0$: the training tail-loss ratio rose steadily with the weight on amplification, from 383 at $\theta = 0$ to 461 at $\theta = 1$, so the amplification-adjusted limit coincided with the volatility limit and the difference was exactly zero. In 2557 stress hours on 46 test dates the guard's tail-loss ratio was +4.5% higher than the volatility limit's (log difference +0.044, 95% interval $[-0.14, 0.23]$, $p = 0.71$). Over all 16,560 test hours, a secondary comparison registered as a replication of Chapter 3, it was 1.52 times higher (log difference +0.42, interval [0.35,0.48]).

Reading. The guard's theorems bound worst-case loss under a certified operator; G1 shows that, estimated from six hours of public data and used to size positions, it does not reduce realized tail losses, in stress hours or otherwise. Post hoc: sizing divides by the product of two noisy six-hour estimates, which enlarges positions whenever the estimate is spuriously low.

V1: the thin-book effect on a new venue. The thin-book forecasting result is large on Binance, small in US stocks and absent on Coinbase. A thin-book explanation predicts an effect in the thinnest pairs of any venue; an explanation specific to leveraged perpetuals predicts none on spot. V1 tested the depth-free statistic on Kraken spot, a venue never used in this project. *Result.* Primary 1 failed. Of the 20 thin pairs fixed by the universe rule, only 3 met the inclusion rule of 12 valid training and 20 valid test days; on their 2,319 test hours the statistic added nothing out of sample ($\Delta R^2 = -0.05$ points, 95% interval [−0.29,0.19], Clark–West $p = 0.36$), and Primary 2 was not tested. With three pairs the test had little power. Its registered reading holds: the result is consistent with the Coinbase result and with an explanation specific to perpetual futures, and it identifies no mechanism.

C2: certify the margin, identify the memories, on new symbols. Designed after the exploration of Section 6.1.6, C2 drew 30 Binance perpetuals that appear nowhere else in this project (seeded random order, written down before any data were downloaded) and used 105 dates each. The margin $M = 1-a-\kappa\Phi$ was estimated with the displacement kernel; it counted as decided if its day-bootstrap interval excluded zero, and the memories counted as identified under the corrected criterion. A third primary repeated the horizon-matched comparison of Pilot 3: the width of the interval of log half-life for class-tagged persistence against the price-only slow memory from a 96-lag (eight-hour) kernel. *Result.* Primary 1 failed by one symbol: the margin was decided in 22 of 30, all on the safe side, against 23 required (Holm-adjusted $p = 0.008$). Primary 2 passed: 22 symbols had a decided margin and unidentified memories, 1 the reverse ($p = 6 \times 10^{-6}$), and the memories were identified in 1 symbol. Primary 3 passed: the tagged interval was narrower in 29 of 29 evaluable symbols ($p = 6 \times 10^{-9}$), by a median factor of 7.3 and at least 2.6, and the price-only interval reached a bound of the fit in 26; the pilot's 7.5 was replicated. *What the secondaries show.* As in the pilot, the decision rests mainly on reversion: without the feedback term it was the same in 26 of 30, and the feedback was contrarian in 14. The registered jackknife correction changes the reading of Primary 3. The uncorrected tagged persistence is 0.985 per five minutes (half-life 3.8 hours); corrected for the within-day bias, the median is 1.0008, and the corrected interval lies below one in only 7 of 87 class fits. Within a day, class positioning ratios are close to a random walk, and the hours-long half-lives of the uncorrected estimator are largely produced by removing each day's mean. Primary 3 stands as registered, but what it shows is that positioning data restrict a persistence parameter that prices cannot bound, not that they measure a learning memory: slow turnover of positions produces the same persistence.

O1: do small perpetuals overshoot after scheduled releases? Pilot 2b found that, after the 2025–2026 releases, the point estimates of the 120-second area were negative, an overshoot, in most symbols, and Proposition 4.6.3, added before O1's result was known, says that a genuine overshoot certifies contrarian feedback. O1 asked the question once, pooled across the pilot's 30 perpetuals with the releases resampled jointly, on the 96 CPI, Employment Situation and FOMC releases of 2022–2024, which the pilot had not used. *Validity.* Both registered conditions held: the pooled pre-release drift was -0.056 of the long-run response (interval $[-0.17,0.03]$), and bitcoin's moves at pseudo-events twenty minutes earlier were 6% as large as at the releases. *Result.* The primary failed: the pooled area was $\bar{A}_{120} = 27.4$ seconds (interval [11,39], one-sided $p = 0.995$), positive, not negative. Only 2 of 30 symbols had a negative point estimate, none significantly. *What the secondaries show.* In 2022–2024 the response had the shape the natural-experiment design assumes: a median 37% of the long-run response (an altcoin move of 1.32 times bitcoin's) was in place within ten seconds, against 94% in the pilot's 2025–2026 releases, and the rest followed gradually. The two-sided horizon-limited bound $|A_{120}| < 60$ seconds held in 28 of 30 symbols, and the memories were identified in 2. *Reading.* The overshoot of the pilot is not a stable feature of these markets: it does not appear in the earlier releases, so it is either specific to 2025–2026 or noise, and these data give no evidence of contrarian feedback. Post hoc, and not tested here: the share absorbed within ten seconds rose from about a third to almost all between the two periods, consistent with faster algorithmic arbitrage between bitcoin and small perpetuals.

C3: is the margin decided once the sample is long enough? C2's first primary failed by one symbol with 105 days per symbol, while the theory says the certificate's precision improves at the parametric rate. C3 asked the same question once, on 30 further symbols never used in this project and 430 days each, a sample chosen in advance by a power calculation from C2's own per-symbol results. The estimators were C2's; one rule was added after C2 drew a symbol whose price did not move: a day is valid only if at least half of its five-minute returns are non-zero, and a symbol is evaluable with at least 200 valid days. *Result.* Both primaries passed. 5 of the 30 symbols had fewer than 200 valid days and were excluded under that rule. In the remaining 25, with a median of 430 valid days, the margin was decided in 23, all on the safe side, against 19 required (Holm-adjusted $p = 1 \times 10^{-5}$); 22 symbols had a decided margin and unidentified memories and none the reverse ($p = 5 \times 10^{-7}$), and the memories were identified in 1. Counting the excluded symbols as undecided, 23 of 30 would still meet the 75% threshold. *What the secondaries show.* On each symbol's first 105 valid days the margin was decided in 20 of 25, so the gain came with the sample, as the theory predicts. The decision still rests mainly on reversion: without the feedback term it was the same in 22 of 25, and the feedback was contrarian in 16. C3 shows that the safety question is decidable from public data in these markets while the agents are not identified; it does not show that the feedback term is measured precisely.

F1: certification and risk on a third set of new symbols. F1 was designed after C3 and after the exploratory Pilots 4 and 4b on C3's spent symbols (Section 6.1.6), and asked three questions once on 30 further perpetuals never used in this project, with C3's 430 dates, valid-day rule and estimators. Primaries 1 and 2 repeat C3's. Primary 3 asks whether the certified quantity matters for risk: whether the margin estimated on the previous 20 valid days forecasts the next valid day's largest adverse move out of sample, beyond the previous day's and the trailing 20 days' realized variance, with symbol fixed effects fitted on the first 60% of each symbol's observations and the Clark–West loss differentials averaged within dates. *Execution.* The first run stopped while drawing the universe, before any symbol was chosen: the archive lists a perpetual whose name is written in Chinese characters, which the universe filter accepted and for which the download code cannot form a web address. An addendum, pushed on its own before any data were downloaded, requires names to be ASCII and changes nothing else; the run then continued from the same registration, with the unlock counter still at one. *Result.* Primaries 1 and 2 passed; Primary 3 failed. 4 of the 30 symbols had fewer than 200 valid days and were excluded. In the remaining 26, with a median of 430 valid days, the margin was decided in 24, all on the safe side, against 20 required (Holm-adjusted $p = 1 \times 10^{-5}$), and 23 symbols had a decided margin and unidentified memories against 0 the reverse ($p = 4 \times 10^{-7}$); the memories were identified in 1. The risk forecast had the predicted sign, a lower margin forecasting a larger adverse move (pooled coefficient -1.13), and a gain of 0.42 points of R^2 on 286 test dates, but the date-averaged Clark–West statistic was $t = 1.51$ ($p = 0.065$), short of the registered threshold. *What the secondaries show.* For next-day realized variance the same forecast gave $t = 2.92$ ($p = 0.002$) with the winsorized margin and $t = 2.89$ with its within-symbol rank; for the adverse move with the rank, $t = 1.38$. These are secondaries and were not the registered target. On the first 60, 120, 240 and all valid days the margin was decided in 20, 23, 22 and 22 symbols and the memories identified in 1, 1, 2 and 1; on the first 105 days it was decided in 23. The standardized margin grew with the sample in only 11 of 26 symbols, as in Pilot 4, which is why that prediction was not registered. As in C2 and C3 the decision rests mainly on reversion: without the feedback term it was the same in 22 of 26, and the feedback was contrarian in 14. *Reading.* The certification result is now replicated on a third set of new symbols. The link from the certificate to realized risk is not established by a registered test: the adverse-move forecast that Pilot 4b supported without date clustering does not survive the registered date-averaged test, and the variance forecast that does is a secondary result, to be registered before it is claimed.

S1: does the margin measure fragility? C2, C3 and F1 certify markets that are far from the boundary, which shows stability but not that the margin detects danger, and F1's forecast of adverse moves failed while its variance forecast, a secondary, did not. S1 was registered on 25 October 2026, after F1's results and before Pilot 7's were seen, on 40 further perpetuals never used in this project and every day from January 2025 to September 2026. Its four pri-

maries, Holm-corrected, asked whether on the 20 worst market-wide days the symbols with a lower prior margin had larger adverse moves, controlling for volatility; whether a lower margin forecasts next-day realized variance out of sample; whether it still does when the branching ratio of a Hawkes model of trade arrivals, the standard measure of self-excitation in markets (Filimonov and Sornette, 2012; Hardiman, Bercot and Bouchaud, 2013), estimated by the INAR approximation of Kirchner (2016), is added to both models; and whether it forecasts next-day extreme moves. *Result.* Primaries 2 and 3 passed; Primaries 1 and 4 failed. 6 of the 40 symbols had fewer than 200 valid days and were excluded. In the remaining 34, on 444 test dates, a lower margin forecast higher next-day variance (coefficient -1.57, Clark–West $t = 4.82$, Holm-adjusted $p = 3 \times 10^{-6}$, gain 0.64 points of R^2), and the forecast was unchanged with the branching ratio in both models ($t = 4.84$, $p = 3 \times 10^{-6}$). On the stress days the mean partial correlation between the prior margin and the day's adverse move was 0.000, negative on only 8 of 20 days, and the forecast of extreme moves fell short ($t = 1.54$, raw $p = 0.062$, Holm $p = 0.12$). *What the secondaries show.* The branching ratio alone did not forecast variance ($t = -0.54$; median $\hat{n} = 0.80$). With its within-symbol rank the margin forecast variance with $t = 4.58$ and extreme moves with $t = 1.94$; the forecast of next-day adverse moves that failed as F1's primary gave $t = 3.62$ here, a secondary. Before the stress days the cross-sectional median margin sat at the 45th percentile of all dates, and on 10 October 2025 the partial correlation was -0.11. The branching ratio did rank stress-day damage, but with the sign opposite to the criticality reading: symbols with a higher prior $\hat{n}$ had smaller adverse moves (mean partial correlation -0.11, $t = -2.86$, negative on 13 of 20 days). *Reading.* The margin carries information about the next day's variance that the standard measure of self-excitation does not, on new symbols and as registered; it does not tell which markets a market-wide shock will hit hardest, and it does not forecast extreme moves at the registered level. The certificate measures how turbulent the coming day will be, not where a crash will land.

R1: does the absolute response measure risk? F1 and Pilot 7 found that the margin, though decided, does not forecast losses, and Theorem 4.8.4 says that with contrarian agents the quantity that bounds worst-case loss is the horizon-limited absolute response $\Psi_H = \sum_{n \le H} |h_n|$, not $S = \kappa / M$. R1 was registered on 25 October 2026 on 40 further perpetuals, drawn only after S1's universe was committed (the two share no symbol). Ψ_H, with H one hour, was computed from the response implied by C3's estimates on the previous 20 valid days. *Result.* Primary 1 passed; Primaries 2 and 3 failed. 4 symbols had fewer than 200 valid days. In the remaining 36, within symbols, the next day's log adverse move rose with log Ψ_H, instrumented by its value from the 20 days before and controlling for the shock size: elasticity 0.59 (standard error 0.09, $t = 6.9$; first-stage R^2 0.67; ordinary least squares 0.22), positive as predicted though below the theory's value of one. Out of sample, with the volatility terms in the model, adding Ψ_H improved the forecast ($t = 4.13$; 0.77 points of R^2), but its coefficient was negative (-0.037), and so it was with the margin in both models (-0.040, $t = 4.05$): the registered prediction required

a positive coefficient, and both primaries fail. *What the secondaries show.* The elasticity for realized variance was 1.07 (0.11), against two in theory. It was the same in the 17 symbols with contrarian estimated feedback (0.59) and the 19 with momentum feedback (0.60), the same for horizons of half an hour and two hours (0.61, 0.62), and the same for the signed sum $|S_H|$ (0.59): the median ratio $\Psi_H/|S_H|$ was 1.00, so at a one-hour horizon the estimated responses rarely change sign. The amplification $S = \kappa/M$, whose elasticity was negative in Pilot 7, had elasticity 0.46 (0.12) here. On S1's stress-day rule Ψ_H did not rank damage (mean partial correlation -0.03). *Reading.* The size of the estimated closed-loop response does scale with the next day's losses within markets, as the linear theory predicts in sign, at about half the predicted rate, on new symbols and as registered. It does not separate the signed theory from the positive one: estimated responses are of one sign at the horizon used, and S scales too once the shock control excludes impact (Pilot 7's control multiplied the flow shock by the estimated impact, which may explain its negative sign; that explanation is post hoc). And beyond recent volatility a larger response forecasts a smaller adverse move, not a larger one, so it does not add the risk information the theory claims. *Exploratory robustness, after the result.* On R1's committed rows the elasticity was 0.97 in 2025 and 0.08 in 2026, fell to 0.29 (s.e. 0.15) with the volatility terms in the regression, and the estimated impact $\hat{\kappa}$ alone had elasticity 0.70: the registered result mostly says that days of higher estimated impact are followed by larger moves. With Pilot 7's control the elasticity of S on these rows was -0.08, with the shock alone 0.46, which supports the post hoc reading of Pilot 7's sign. S1's variance forecast, by contrast, kept a Clark–West t between 2.32 and 5.52 across clustering units, training shares, winsorization, calendar years, the exclusion of October 2025 and volatility halves, and between 4.40 and 5.37 when any one market was removed.

H1: real agents (prospective, opens 10 December 2026) Every test above uses anonymous prices and flow or class-level ratios. Hyperliquid, a perpetual-futures exchange whose order book runs on a public blockchain, publishes every trade with the buyer's and the seller's wallet address. H1 was registered on 25 October 2026 before any of its data existed: a collector records every trade in 30 markets (ranked 11 to 40 by volume when collection starts), and the analysis opens once at least 20 markets have 30 days with near-complete recording. For each of the ten largest accounts in a market it estimates the account's memory, the persistence of its position, and its response to the price displacement. The primaries ask whether in at least 75% of markets two of these accounts have distinguishable memories, and whether prices and anonymous flow fail to separate memories where the accounts' own records succeed. It is the first test of the identification theory against real agents rather than proxies; its limits, stated at registration, are that addresses are accounts, not firms or algorithms, and that positions are reconstructed only from the start of collection.

6.1.5 What the tests change

The tests confirm the identification half of the thesis on new data: prices and order flow leave memories unrestricted, also at a matched horizon and on symbols never used before, while class positioning data restrict a persistence parameter that prices cannot bound, with an advantage that grows with the sample. Whether that persistence is a learning memory is not established: corrected for its within-day bias it is close to a random walk (C2). On three sets of new symbols the decision a guard needs was reached where the memories were not identified (22 symbols against 1 in C2, 22 against 0 in C3, 23 against 0 in F1), and with 430 days the margin was decided in 23 of 25 symbols (C3) and 24 of 26 (F1), after falling one symbol short with 105 days (C2). This is the empirical form of the safety half, within limits: the decision rests mainly on reversion, so the data certify that these markets are far from the boundary without measuring the feedback precisely; the summed response itself was no more precise than the separation of memories (E1); a guard estimated from public data did not reduce realized tail losses (G1); the 20-day margin did not pass the registered tests of forecasting the next day's largest adverse move (F1) or extreme moves (S1), and it did not rank damage on market-wide stress days (S1). It did forecast next-day realized variance on new symbols, beyond the Hawkes branching ratio, as registered (S1), and the size of the estimated closed-loop response scaled with next-day losses within markets, at about half the predicted rate, though beyond volatility it forecast smaller losses, not larger (R1). The theorems of Chapters 3 and 4 are untouched, because they concern declared contracts (reset experiments, exact windows, audited disclosure) and passive records are none of these. What changes is the claim the thesis makes about public data. Identification from prices fails, as predicted. Certification from prices works, with enough days, where the market is far from the boundary (C2, C3, F1); near the boundary the feedback term would have to be measured, which passive data have not done. The information that makes either attainable is disclosure, or an outside shock that plays the role of an experiment (Proposition 4.6.2).

6.1.6 Exploration after the tests

After the four tests, their downloaded days, which are spent for confirmation, were explored to find a design that public data can decide, so that the next confirmatory test can be fixed before its data are seen. Nothing in this subsection is confirmatory evidence; it is reported because it shapes what comes next.

The decision a guard needs is decidable; the separation of memories is not. The theory's safety question is the sign of the margin $M = 1 - a - \kappa\Phi$ (equation (4.2)). Estimated with the feedback kernel acting on the price displacement, as in Definition 4.1.1, rather than on returns as in E1, the day-bootstrap interval for the margin excluded zero in 29 of 30 symbols, all on the safe side; with E1's return kernel it did in 11. With the identification criterion corrected, so that a separation counts only if its interval excludes zero and neither rate interval touches a

bound of the fit, memories were identified in 0 of 30. On random subsamples the margin was decided in 16, 22 and 28 symbols with 10, 20 and 40 days, and the memories in no more than 2.

The decision rests mostly on reversion. The feedback term $\kappa\Phi$ was typically about a quarter of the size of the reversion term (median ratio 0.24) and negative, that is contrarian, in 22 of 30 symbols; with it removed the decision was the same in 28 of 30. These markets are certifiably far from the boundary, and the data do not measure the feedback itself precisely. The sign matters: the non-negativity of Definition 4.1.1 excludes contrarian agents, a limitation Mansi Bhavsar pointed out in comments on an earlier draft, and here contrarian feedback is the common case.

Class memory drifts. The top-position memory rate differed across calendar quarters (heterogeneity test at 5%) in 13 of 30 symbols, against 1.5 expected by chance. Within quarters, 24 of the tagged-width slopes fall in D1's range, against 18 across the whole period, although the median slope within quarters (-0.34) is still flatter than $-1/2$.

Natural experiments: bitcoin moves are not outside shocks. Proposition 4.6.2 turns a displacement that adds no order flow into a reset experiment. A first pilot used every one-minute bitcoin move as such a displacement of 30 small perpetuals. The altcoins absorbed a median 91% of a move within the minute, but both placebos failed: the pre-move drift was consistent with zero in only 10 of 30 symbols and the reverse regression in 9. The altcoins slightly lead bitcoin, so an ordinary bitcoin move is not an outside shock to them, and the design was dropped.

Scheduled releases are outside shocks, and the response overshoots. A second pilot used the 56 US releases whose timing is fixed in advance and whose content is unknown until the release second: every CPI and Employment Situation release (8:30 ET) and every FOMC statement (2:00 ET) from January 2025 to October 2026, at one-second resolution. Here the design holds. Before the release the altcoins did not drift towards the coming news (the interval covered zero in 30 of 30 symbols, median size 3.6% of the full response), and at pseudo-events twenty minutes earlier bitcoin's moves were 9% as large as at the release. The response was not the gradual catch-up the margin estimator assumes. A median 94% of the long-run response (an altcoin move of 1.45 times bitcoin's) was in place within ten seconds, and the point estimates then overshoot the long-run level and revert over minutes: the 120-second area A_{120} of Proposition 4.6.1 is negative in 26 of 30 symbols, though its interval excludes zero in only 2. Three consequences follow. The registered-style reading "upper bound of A_{120} below 60 seconds" holds in all 30 symbols, but only because the area is negative; the two-sided bound $|A_{120}| < 60$ seconds holds in 9. The finite-horizon margin of Proposition 4.6.2(iv) needs a positive area and can be computed in only 4 symbols. And the memories are identified in 1 of 30, as the theory predicts. Interval widths shrink with the number of releases at slope -0.45 in $\log n$, close to

the parametric $-1/2$. By Proposition 4.6.3, a genuine overshoot requires contrarian feedback, and the area up to the plateau should then turn positive; it is negative in 28 of 30 symbols (interval below zero in 3), which, if confirmed, would mean that adjustment continues beyond the fifteen-minute window. Because all symbols share the same releases, the overshoot in 26 of 30 symbols is not 30 independent findings. The registered test of it, O1, on the 2022–2024 releases, found no overshoot (Section 6.1.4).

The disclosure result at a matched horizon. Paper 1's price-only route regresses five-minute flow on twelve lagged returns, a one-hour window, while class-tagged half-lives are several hours. A third pilot refits the price-only kernel with 12, 48 and 96 lags (one, four and eight hours) on the E1v2 data and compares both routes in one unit, the half-life. With twelve lags the price-only slow half-life is a median 11 minutes and below half an hour in 22 of 30 symbols: the one-hour kernel cannot see memories of hours. With 96 lags it is a median 5.6 hours, against a tagged median of 4.2 hours, and within a factor of two of the tagged value in 12 symbols. The tagged intervals stay narrower in all 30 symbols at every horizon (smallest ratio 1.4 at eight hours), but the median ratio of interval widths in log half-life falls from 11.5 to 9.0 to 7.5 as the kernel lengthens, and the price-only interval still reaches a bound of the fit in 25 of 30 symbols at eight hours. The robust statement is therefore not a precision ratio but a failure of identification: at a matched horizon prices do not bound the slow memory in most symbols. Within-day demeaning biases the tagged rates down; the registered correction in C2 shows how much, and kernels longer than a day do not fit this design.

The learning curve and the value of the certificate. A fourth pilot used C3's spent data to design a flagship test. Theorem 4.7.1 predicts that, with a longer record, more markets have a decided margin while their memories stay unidentified. On the first 30, 60, 120, 240 and all valid days of the 25 evaluable symbols, the margin was decided in 17, 19, 20, 20 and 23 (one symbol has fewer than 240 days), and the memories were identified in 2, 1, 0, 2 and 1. Across symbols this is the predicted pattern. Symbol by symbol it is not: the standardized margin, which the theorem says should grow like the square root of the sample, grew in only 11 of 25 symbols (median slope -0.02 in $\log n$), because the estimated margin moves between 2025 and 2026; the stationarity behind the theorem fails at this horizon, as D1's drift already suggested. The pilot also asked whether the certificate carries information about risk. In rolling 20-day windows, a lower estimated margin went with a higher next-day realized variance (panel t =-2.9; t =-2.0 with 60-day windows) and, more weakly, a larger adverse move (t =-1.7), the effect coming from the reversion part of the margin rather than from feedback; but with a separate forecasting model per symbol, adding the margin lowered out-of-sample accuracy in every specification (change in R^2 from -2.9 to -1.0 points). With one pooled coefficient across symbols, fitted on the first 60% of each symbol's days, the forecast improved out of sample in all four pre-declared variants (Clark–West t from 3.6 to 4.9, not clustered by date; gains in R^2

of 0.17 to 0.37 points; negative coefficient in 4 of 4), and this design became the third primary of Test F1. A fifth pilot found the same direction an order of magnitude faster: the margin estimated on each six-hour block forecast the next hour's largest adverse move out of sample in 29 symbols and 31,782 blocks (Clark–West $t = 2.36$ with the differentials averaged within dates; $t = 2.26$ with the margin's rank), though not realized variance ($t = 0.74$), and the gain was tiny (0.03 points of R^2). A sixth pilot certified the margin quarter by quarter: in 169 symbol-quarters of 27 symbols, about sixty days each, the margin was decided in 135, never on the unsafe side (0), and in every quarter for 7 symbols; its level differed across quarters (heterogeneity at 5%) in 12 symbols, with non-overlapping decided intervals in 5. The sign of the margin is stable on the time scale on which the theorem's stationarity is reasonable; its level is not. Proposition 4.7.4 turns this into numbers: with the median per-day noise $s = 0.093$ and drift $d = 0.00047$ per day estimated from these quarters, the best record length is about 53 days, close to one quarter, and the floor below which no record certifies the sign lies between 0.007 and 0.075, against a median margin of about 0.04: most of these markets sit above the lower end of the floor and below the upper one, which is why roughly four quarters in five are decided.

The size of the effect, and one crash. A seventh pilot, also on C3's spent symbols, asked two questions that the theory answers with numbers. In the positive class the worst-case transient loss is proportional to $S = \kappa/M$ at a given shock, so within a symbol the next day's log adverse move should rise one-for-one with $\log S$, and log realized variance two-for-one. In 7639 symbol-days, with the 20-day estimate of $\log S$ instrumented by the estimate from the 20 days before (first-stage R^2 0.14) and the day's shock size controlled, the elasticities were -0.13 (standard error 0.03) for the adverse move and -0.21 (0.04) for variance; ordinary least squares gave -0.08 and -0.13. They are small, and negative, not 1 and 2. On 10 October 2025, the day with the largest median adverse move in the sample, the Spearman correlation across 25 symbols between the margin of the previous 20 days and that day's adverse move was +0.48: the symbols with the lower margin moved less, not more, and the cross-sectional median margin before the crash was at the 34th percentile of all dates, not unusually low. *Post hoc reading.* The elasticity prediction belongs to the positive class, in which S is the norm of the response; with contrarian feedback, the common case in these data, S no longer measures risk (Proposition 4.9.1), so the failure of the scaling is what the signed theory allows; it is not evidence for it. Test R1 later found a positive elasticity for S on new symbols with a shock control that excludes the estimated impact, so Pilot 7's sign may come from its control (Section 6.1.4). A single crash day is one cross-section. Test S1 was registered before these results were seen, and its first primary asks the stress-day question on 20 days and new symbols; this pilot makes a pass unlikely.

Next. Tests C2, O1 and C3 were registered and run; their results are reported with the registered tests above. What remains open is the feedback term itself: every decision so far rests mainly on reversion, and a market near the boundary would need the feedback measured, from

disclosure or from outside shocks.

6.1.7 Withdrawn results

Two results reported in an earlier draft of this thesis are withdrawn. The first was presented as a run of Test E1 in 38 markets. It did not estimate anything from fresh data: it read results archived from earlier tests and used hard-coded and invented intervals, and its design compared quantities in different units. The specification it ran under is superseded by the one above, and none of its numbers appears in this thesis. The second was a simulation of learning agents whose weights, by construction, could only increase; its "realized lifetimes" measured that construction and could not have fallen below the bound they were said to test. Section 4.14 replaces it with a registered simulation in which weights follow the agents' trading profits. The code and output of both are kept in the repository, marked as withdrawn, so that the record is complete.

6.1.8 Rules for every new test

1. The specification, its hash and an unlock counter are pushed to the public repository in a commit of their own, so the platform records when they became public, before any test data are downloaded or recorded.

2. Each test has at most two primaries with a pre-stated pass rule and a pre-stated correction across them.

3. A failed primary is reported as failed. An explanation found afterwards is labelled post hoc, and any test that follows from it is labelled as registered after the failure.

4. Every number in the thesis is produced by code that writes a result file, and an automated check compares the printed numbers with those files.

5. A result found to be wrong after it was reported is withdrawn in the text, with the reason, and its files are kept and marked rather than deleted.

6.2 Open problems

1. *Nonlinear questions.* Theorem 4.4.1(iv) gives the exact cost of every linear question, and Proposition 4.5.1 settles questions that depend on finitely many taps, such as the peak. Extend the exact theory to questions that depend nonlinearly on infinitely many taps, and make the link to the $1/\log N$ rate of identification (Corollary 4.4.5(e)) a theorem rather than a reading of Chapter 2's proof.

2. *Constants.* Theorem 4.4.3 gives explicit constants, but its lower constant is far below its upper one (0.41 against 3.08 at $\bar{r} = 0.9$). Find the limit of $\omega_L(\delta)/(\delta\sqrt{\log(1/\delta)})$; the computed brackets sit near the upper constant.

3. *Minimal disclosure for safety.* Characterize the least informative disclosure under which a given safety question becomes certifiable from finite data. Theorem 4.4.1(iii) says a memory bound is necessary for the summed response; is it sufficient in a sense that can be made operational?

4. *Strategic agents.* Remark 4.15.2 refutes existence under a discontinuous admission rule. Determine sufficient assumptions for existence under a regularized guard and for the claimed comparison of equilibrium sensitivity.

5. *Observational certification.* Develop finite-sample certificates from observational order-book data with dependence and non-stationarity, replacing the bootstrap diagnostics of Chapter 3. Test E1 found passive five-minute data far from the reset contract of Theorem 4.4.1; natural experiments (Proposition 4.6.2) are the nearest substitute, and their sampling theory under dependent shocks is open. Theorems 4.7.1, 4.7.2 and 4.8.2 settle the cost of passive certification for one closed-loop root and, uniformly as the margin vanishes, for any number of cohorts with finite memory; Theorem 4.8.1 shows that without a memory bound no passive certificate exists. The pooled price-and-flow estimator that the tests use attains the same order (Theorem 4.8.3). What remains is a version uniform over geometric memory structures and over the pooled estimator's nuisance parameters, and the gap between the moving-average displacement the tests use and the distance from the fundamental value the theory uses.

6. *The evidence.* Run Test E2 after 2 February 2027; measure the feedback term itself, not only the margin, from disclosure or outside shocks; and extend the cohort model to contrarian agents, which the exploration finds to be common.

Table 6.1: The registered tests and their results

Test	Primary hypothesis	Result
E1	1. Certifying the summed response is at least 10 times more precise than separating memories	failed: 10 of 30 symbols (23 needed)
	2. Price-only memory intervals are at least 10 times wider than class-tagged ones	passed: 30 of 30
D1	1. Class-tagged interval width falls with slope in $[-0.75, -0.25]$	failed: 20 of 29 (22 needed)
	2. The advantage of tagging grows with the sample	passed: 22 of 29
G1	1. Amplification-adjusted volatility limit beats a volatility limit in stress hours	failed: identical rules ($\theta = 0$)
	2. The summed-response guard beats a volatility limit in stress hours	failed: tail-loss ratio +4.5% higher
V1	1. The depth-free statistic forecasts excursions in thin Kraken pairs	failed: $\Delta R^2 = -0.05$ points
	2. It beats an order-splitting placebo	not tested
C2	1. The safety margin is decided in at least 75% of 30 new symbols	failed: 22 of 30 (23 needed)
	2. Decided but not identified outnumbers identified but not decided	passed: 22 against 1
	3. At a matched eight-hour horizon, class-tagged persistence is tighter than price-only memory	passed: 29 of 29 (22 needed)
C3	1. With 430 days, the safety margin is decided in at least 75% of new symbols	passed: 23 of 25 (19 needed)
	2. Decided but not identified outnumbers identified but not decided	passed: 22 against 0
F1	1. The safety margin is decided in at least 75% of new symbols	passed: 24 of 26 (20 needed)
	2. Decided but not identified outnumbers identified but not decided	passed: 23 against 0
	3. The 20-day margin forecasts the next day's largest adverse move out of sample	failed: Clark–West $t = 1.51$, $p = 0.065$
S1	1. On market stress days, symbols with a lower prior margin are hit harder	failed: mean partial correlation 0.000
	2. A lower margin forecasts next-day realized variance out of sample	passed: Clark–West $t = 4.82$
	3. It does so beyond the Hawkes branching ratio	passed: $t = 4.84$
	4. A lower margin forecasts next-day extreme moves	failed: $t = 1.54$, Holm $p = 0.12$
R1	1. Next-day adverse moves rise with the estimated absolute response Ψ_H	passed: elasticity 0.59 (s.e. 0.09)
	2. Ψ_H forecasts next-day adverse moves out of sample, with a positive coefficient	failed: coefficient -0.037 ($t = 4.13$)
	3. It does so beyond the margin	failed: coefficient -0.040 ($t = 4.05$)
H1	1. Real accounts have distinct memories that their own records reveal	prospective; opens 10 December 2026
	2. Prices and anonymous flow do not reveal them	prospective; opens 10 December 2026
E2	Prospective test of the forced-deleveraging channel	runs after 2 February 2027
O1	After scheduled US macro releases (2022–2024), small perpetuals overshoot (pooled area $\bar{A}_{120} < 0$)	failed: $\bar{A}_{120} = 27.4$ s, interval [11,39]

Each primary needed its pre-stated share of symbols (E1, D1) or interval and p-value (G1, V1), with Holm's correction across the primaries of a test, or a fixed sequence for V1. Sources: the committed result files of each test, checked against their registered specification hashes.

Appendix A

Status of results and notation across chapters

A.1 Status register

Result	Location	Status
Identification (Chapter 2)		
Merged rate–weight equivalence	Thm 2.3.1	proved
Executed quantity $\Theta(\delta^{-2})$	Thm 2.4.1	proved
Capital $\Theta(\delta^{-4})$	Thm 2.4.3	proved
Spectral round-trip loss bound; adaptive probing; finite-horizon dual	Thms 2.5.2, 2.5.3, 2.5.4	proved
K cohorts: $(\Delta\eta)^{-(4K-2)}$, lower and upper	Thms 2.6.4, 2.6.5	proved
Merged memory distribution at rate $N^{-1/(4K-2)}$ without separation	Thm 2.6.6	proved
General memory: $\Theta(1/\log N)$	Thm 2.6.8	proved
Passive data: same polynomial order	Thm 2.6.10	proved
Tagged flow removes the barrier; value of disclosure	Prop. 2.8.1, Thm 2.8.2	proved
Spectral blind spot	Thm 2.9.1	proved
Trained learners: δ^{-4} when weights match	Sec. 2.7	simulation
Class disclosure restricts persistence (Binance, CFTC)	Secs. 2.8.3, 2.8.4	data, pre-registered
Safety (Chapter 3)		
Closed form of S	Thm 3.4.3	proved
Exact overshoot condition	Thm 3.4.2	proved

Result	Location	Status
Spectral stability is not solvency	Thm 3.3.1	proved
Summed-gain guard is maximal	Lemma 3.6.1	proved
Invariance to cohort splits	Prop. 3.2.2	proved
Finite-window certification and its limit	Thm 3.5.3	proved
Certification cost $\Theta(\sigma^2\epsilon^{-2}\log(1/\epsilon))$, logarithm necessary	Thm 3.5.4	proved
Optimal uncertainty-aware guard	Thm 3.6.4	proved
Value of disclosure for safe capacity	Thm 3.7.1	proved
Equilibrium under-supply of reversion	Prop. 3.8.1	proved
Carried inventory; hidden drift	Props. 3.2.3, 3.6.5	proved
Thin-book forecasting; guard backtests; mechanism tests	Secs. 3.9.2, 3.10	data, pre-registered
Paper 3 (Chapter 4)		
Upper bound on certification cost for linear questions	Thm 4.4.1(i)	proved here
Truncation necessity; impossibility at infinite memory	Thm 4.4.1(ii)–(iii)	proved here
Exact cost of linear questions (modulus, duality, sandwich)	Thm 4.4.1(iv)	proved here
Parametric iff square-summable kernel coefficients	Prop. 4.4.2	proved here
Safety question costs exactly $\sigma^2\epsilon^{-2}\log(1/\epsilon)$, explicit constants	Thm 4.4.3, Lemma 4.4.4	proved here
Finitely dependent nonlinear questions are parametric; the peak is uninformative	Prop. 4.5.1, Cor. 4.5.2	proved here; recursion in Lean
Worked examples: experiments needed	Sec. 4.10	computed, certified
Horizon-limited amplification	Prop. 4.6.1	proved
Natural experiments and the stability margin	Prop. 4.6.2	proved
An overshoot certifies contrarian feedback	Prop. 4.6.3	proved; (i)–(iii) in Lean
Passive certification costs the inverse margin	Thm. 4.7.1	proved (one root); bound in Lean
Passive certification, several cohorts, finite memory	Thm. 4.7.2	proved; Lean: arithmetic steps
Hidden slow feedback: a memory bound is necessary	Thm. 4.8.1	proved; Lean: arithmetic steps

Result	Location	Status
Uniform certification with finite memory	Thm. 4.8.2	proved given cited uniform limit theory
Passive certificate of the horizon-limited absolute response	Thm. 4.8.4	proved; Lean: monotonicity step
Pooled price-and-flow estimator of the margin	Thm. 4.8.3	proved for fixed economies; Lean: arithmetic steps
Monoculture threshold	Prop. 5.2.1	proved; Lean: arithmetic steps
Learning-agent market simulation (B1)	Sec. 5.4	registered; run once; P1–P4 passed (disclosed rule false-alarms in 17% of controls)
Passive certification uniform over geometric memory and the pooled estimator's nuisance	Remark 4.7.3, Sec. 4.8	open problem
Contrarian agents	Prop. 4.9.1	proved; finite parts in Lean
What the closed loop adds	Cor. 4.7.5	proved
Certifying a drifting margin	Prop. 4.7.4	proved; bounds in Lean
Intensity tied to information quality	Prop. 2.4.7	proved
Uniform coefficient-energy lower bound (earlier target)	Remark 4.4.6	false; replaced by (iv)
Robust safety bound (nonlinear, time-varying agents)	Thm 4.12.2	proved; finite-horizon Lean
Several assets (induced-norm bound)	Prop. 4.12.4	proved here
Positive cross-impact sharp bound	Prop. 4.12.5	proved; comparison step in Lean
Certificate lifetime	Cor. 4.13.1	proved; drift/floor Lean
Guard-consistent equilibrium	Remark 4.15.2	existence refuted; pure case Lean
Learning-agent simulation (B3v2)	Sec. 4.14	simulation, registered

Result	Location	Status
Certify versus identify (E1)	Sec. 6.1.2	run: Primary 1 failed, Primary 2 passed
Fragility test (S1)	Sec. 6.1.4	run: primaries 2 and 3 passed (variance, beyond the branching ratio), 1 and 4 failed
Absolute-response test (R1)	Sec. 6.1.4	run: primary 1 passed (elasticity 0.59), 2 and 3 failed (negative coefficient)
Participant-level test on Hyperliquid (H1)	Sec. 6.1.4	prospective; opens 10 Dec 2026
Prospective mechanism test (E2)	Sec. 6.1.3	registered with addendum; opens 2 Feb 2027
Tagging scaling (D1)	Sec. 6.1.4	run: Primary 1 failed, Primary 2 passed
Stress-hour guard (G1), new venue (V1)	Sec. 6.1.4	run: failed
Margin versus memories on new symbols (C2)	Sec. 6.1.4	run: Primary 1 failed, Primaries 2 and 3 passed
Release overshoot (O1)	Sec. 6.1.4	run: failed (no overshoot)
Margin decision with more days (C3)	Sec. 6.1.4	run: both primaries passed
Certification and risk forecast on new symbols (F1)	Sec. 6.1.4	run: primaries 1 and 2 passed, 3 failed
Exploration of the spent days	Sec. 6.1.6	exploratory
Earlier E1 run and earlier simulation	Sec. 6.1.7	withdrawn

A.2 Notation across chapters

Chapter 4	Chapter 2	Chapter 3
memory rate r_i	$1 - \eta_i$	$1 - \eta_i$
weight q_i	q_i	q_i
persistence a	a	$a = 1 - m$ (reversion m)
impact κ	κ	κ, $\tilde{\kappa}$ with depth
merged measure $M = \sum_i q_i(1 - r_i)\delta_{r_i}$	$M = \sum_i q_i \eta_i \delta_{1-\eta_i}$	taps k_m of M
total weight $\Phi = \int (1 - r)^{-1} dM$	$\sum_i q_i$	q (aggregate)
summed response $S = \kappa/(1 - a - \kappa\Phi)$	—	$S = \tilde{\kappa}/(m - q\tilde{\kappa})$

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